Inverse-bremsstrahlung heating of KrF laser gases

Afterglow reheating as it stands, and the six notes behind it

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This compendium collects six notes on the heating of the free electrons of KrF laser gases by inverse bremsstrahlung, and asks of them one question: can an electromagnetic wave that reheats the electrons of the afterglow, after the electron beam has stopped, make the laser more efficient? Part I answers it as the notes stand. The heating half of the proposal is verified in silico. At the afterglow's 10^{20} m^{-3} a wave of 0.64 to 3.8 THz is absorbed through 10 cm of gas, where microwaves heat a layer millimetres deep, the absorption coefficient is known to 1.6 % in its kernel and to about 20 % in krypton's cross section at the afterglow's 0.63 eV, and 1.4×10^3 to 1.5×10^4 $\mathrm{W\,cm^{-2}}$ holds the electrons, far from the nonlinear regimes of Langdon and Silin. The efficiency half is open. A THz wave would form 1.4 to 1.8 times the electron beam's KrF^* per absorbed joule, but that count leaves out the electron collisions of Kr^* and KrF^* on which the sign of the gain rests, and no note follows the KrF^* to laser energy or to the wall plug. The nonlinearity of inverse bremsstrahlung and the laser's own field are invalidated as levers. Part I ends with the measurements and the calculation that would settle the rest, and Part II reproduces the six notes in full.

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Part I

Afterglow reheating: where the case stands

Chapter 1

The question and the verdict

1.1 The proposal

(1) An electron beam pumps a KrF laser gas by making ion pairs and excited atoms, and KrF^* forms from them, much of it through the harpoon reaction of excited krypton with F_2. When the beam stops, the electrons it leaves behind cool and are lost to attachment and recombination. This is the afterglow. Part II describes it in the neon-buffered fill of an electron-beam-pumped laser, $Ne/Kr/F_2 = 0.948/0.05/0.002$ at 2 atm, by electrons at 0.63 eV and 10^{20} m^{-3}, against 1.5 eV and 2.72×10^{21} m^{-3} with the beam on (Sec. 6.5). The afterglow holds 27 times fewer electrons, and colder ones.
(2) Reheating sends an electromagnetic wave into the afterglow. The wave heats the free electrons by inverse bremsstrahlung (IB), the hotter electrons excite krypton again, and the excited krypton forms KrF^* while F_2 lasts. The question of this compendium is whether that can make the laser more efficient, that is, raise the laser energy per unit of electrical energy.
(3) It can only if five links hold (Fig. 1.1):

- L1 *Coupling.* The wave enters the gas and is absorbed through the depth of the gain medium, rather than reflected or absorbed in a thin skin.
- L2 *Absorption.* The absorption coefficient that decides L1 is known.
- L3 *Heating.* An available intensity raises the electron temperature, in a regime that the kinetic equation of the electrons describes.
- L4 *Partition.* The energy the electrons take from the wave forms KrF^* at a useful efficiency.
- L5 *Laser and wall plug.* Laser photons extract the KrF^*, and the source of the wave is efficient enough for the gain to pay.

(4) A link is called *verified* here when the calculations of Part II establish it, with their inputs tested against measurements where the notes cite such tests. Nothing in Part II measures reheating itself. A link or a lever is *invalidated* when those calculations rule it out, and *open* when they do not settle it, in which case Chapter 3 names what would.

1.2 The verdict

(1) The heating half of the proposal stands, and the efficiency half is open. Coupling, absorption and heating are verified in silico, with one constraint that shapes the proposal: only the THz

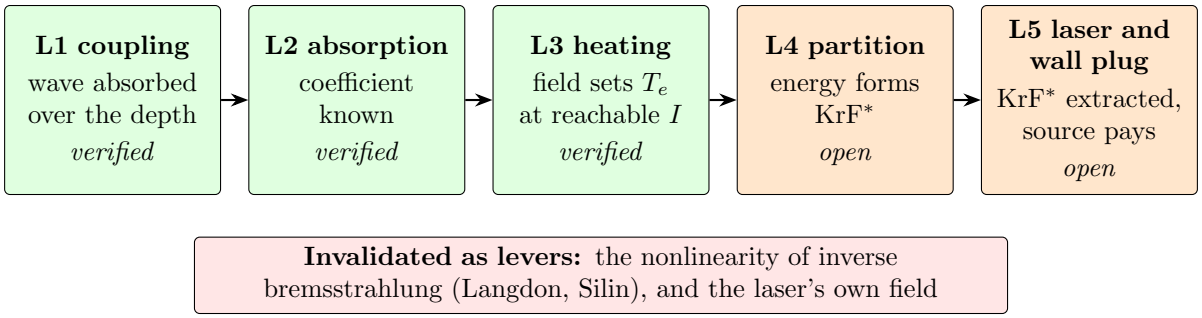


Figure 1.1: The chain from the wave to the laser energy, with the standing of each link in Part II. The THz band carries L1. The open links hinge on the electron collisions of Kr* and KrF*, the extraction of the KrF* and the efficiency of the source.

band heats the afterglow through centimetres, while microwaves heat a layer millimetres deep. The partition favours the field. At the afterglow's density a wave far above the collision frequency forms KrF* at 1.4 to 1.8 times the electron beam's rate per absorbed joule, but that count leaves out the electron collisions with Kr* and KrF* on which the sign of the gain rests, and no note carries the KrF* to laser energy or to the wall plug. Two levers are invalidated: the nonlinearity of IB, and the laser's own field. Table 1.1 collects the verdict, and Chapter 2 gives the evidence link by link.

1.3 Reading Part II

(1) Part II is six notes, one chapter each, in their reading order. Chapter 4 derives the best frequency for krypton gases and settles THz against GHz (L1, L2). Chapter 5 follows the excited atoms inside the electrons' balances and marches a pulse (L1, L4). Chapter 6 locates the lasers and the sources against the nonlinearities of IB and counts what field pumping buys (L3, L4, L5). Chapter 7 tests the absorption model against measured absorption in laser plasmas (L2). Chapter 8 derives the quantum theory of IB on ions and Chapter 9 that of absorption on atoms at THz frequencies (L2, L3).
(2) Each chapter keeps its note's text, notation and references, with its own numbering inside the chapter. Chapters 4, 7, 8 and 9 use Gaussian units and Chapters 5 and 6 SI units. Electron densities are in cm^{-3} in Chapters 4, 5 and 7 and in m^{-3} in Chapters 6 and 9, and 10^{14} cm^{-3} is 10^{20} m^{-3}. "These notes" in a chapter means that chapter.

Table 1.1: Where afterglow reheating stands, link by link. The numbers are for the afterglow of Chapter 6, 10^{20} m^{-3}, unless stated otherwise.

link	standing	what Part II establishes	where
L1 coupling	verified	10 cm is absorbed at 0.64 to 3.8 THz, 2.6 cm at 0.67 THz, and a microwave within millimetres. Reflection does not bind.	Secs. 4.5, 6.5, Tables 4.2, 5.3
L2 absorption	verified	The classical kernel is within 1.6 % of the quantum one at 0.3 to 4 THz. Krypton's cross section is tested to a few per cent above 0.95 eV and to about 20 % at 0.63 eV. Ions matter only in a cold afterglow.	Secs. 9.9, 4.8.2, 9.10
L3 heating	verified	1.4×10^3 to 1.5×10^4 W cm^{-2} hold the electrons at 0.63 eV, 5.8 kW cm^{-3}, far below Langdon's and Silin's thresholds.	Sec. 6.5, Table 6.2
nonlinearity, laser's field	invalidated	The lasing field is 7.1×10^3 times below the heating threshold, and Langdon's regime needs 33 to 80 times it.	Secs. 6.1, 6.5
L4 partition	open	0.274 to 0.352 KrF* per absorbed joule at high frequency against the beam's 0.198, without the electron collisions of Kr* and KrF* or the 5s to 5p cycle.	Tables 6.3, 6.4, Secs. 6.8, 5.3
L5 laser, wall plug	open	No note follows the photons. The source must reach 0.56 to 0.72 of the beam's efficiency times the ratio of the two extractions.	Secs. 5.9, 6.8, Eq. (2.1)

Chapter 2

The evidence, link by link

2.1 L1: coupling

(1) A wave heats a depth L of gas best at the frequency whose absorption length is L, provided the gas reflects no more than 5 % of it, Eq. (4.1). Far above the collision and plasma frequencies the absorption coefficient falls as f^{-2}, Eq. (4.6), and the frequency that matches the depth grows as the square root of the electron density, the collision frequency and the depth, Eq. (4.23). The afterglow's fewer and colder electrons therefore need a lower frequency than the gas with the beam on.
(2) At the afterglow's density, 10^{14} cm^{-3}, a depth of 10 cm calls for 1.1, 2.2 and 3.8 THz in pure krypton at 0.1, 0.5 and 2 atm (Table 4.2), and for 0.64 to 2.4 THz in the Ne fill and 1 to 3.5 THz in the Ar fill from 0.1 to 2 atm (Table 5.3), with the electrons heated by 300 kW cm^{-3} in both tables. In the afterglow itself, at 0.63 eV in the Ne fill, a 0.67 THz wave is absorbed in 2.6 cm (Sec. 6.5). There the electrons' collision frequency is 0.12 of the wave's angular frequency (Table 6.2) and the cutoff lies at 0.090 THz (Sec. 4.4.1), so the absorption coefficient falls as f^{-2}, and 10 cm takes $(10/2.6)^{1/2} = 2.0$ times that frequency, 1.3 THz.
(3) Microwaves do not reach that depth. In the afterglow a 2.45 GHz wave lies far below the collision frequency, $\langle\nu_m\rangle/\omega = 32$ (Table 6.2), and a 170 GHz wave heats a layer 2.5 mm deep (Sec. 6.5). In krypton at 10^{14} cm^{-3} and 0.5 atm, where the collisions outrun the cutoff, a GHz wave does enter the gas, 43 % of it reflected at 2.45 GHz and 1 % at 95 GHz, but it is absorbed in 1.8 to 4.2 mm, and over the pressures and both gases in 0.6 to 8.3 mm, so that the front millimetres take all of it (Sec. 4.5).
(4) The cutoff does not bind in the afterglow. The plasma frequency is 0.090 THz at 10^{14} cm^{-3} (Sec. 4.4.1), where with the beam on it is 0.47 THz (Sec. 6.6).

Standing: verified, with a band. A wave of 0.64 to 3.8 THz, 1.3 THz in the Ne fill's afterglow, deposits its power through 10 cm of gas at the afterglow's density. A 0.67 THz wave deposits it through 2.6 cm, and a microwave in a layer millimetres deep. Volumetric reheating is a THz proposal.

2.2 L2: absorption

(1) The absorption coefficient is the Drude form with the collision frequency $\nu_{\text{eff}} = \nu_{ea} + \nu_T\Lambda$, Eq. (4.17), the atoms entering through the energy-weighted average of their momentum-transfer collision frequency, Eq. (4.12), and the ions through the Spitzer frequency and the Gaunt factor.

Three things decide whether it is right: the kernel of the atoms, the cross sections that the kernel takes, and the term of the ions.
(2) *The kernel.* The classical kernel treats the electron as a particle that absorbs continuously. Against the quantum rate between exact scattering states it is within 1.6 % in either fill from 0.3 to 4 THz at every electron temperature from 300 K to 1.08 eV (Sec. 9.9), so the absorption lengths of L1 stand. It fails only far above that band: at 28.3 THz and 300 K the quantum absorption is 10 % below it in the neon fill and 28 % below it in the argon fill.
(3) *The cross sections.* Krypton's momentum-transfer cross section is tested by the drift velocities it predicts. Between 0.95 and 2.6 eV the computed drift velocity is 0.96 to 1.02 times Nakamura's measurements [1] and 0.84 to 0.97 times those of Pack, Voshall and Phelps [2] (Sec. 4.8.2). Below 0.95 eV only the latter exist, and across the Ramsauer minimum the computed drift velocity is 0.80 to 1.20 times them. Since the drift velocity weights the inverse of the cross section over the same electrons, the cross section is tested there to about 20 %, and the afterglow's 0.63 eV electrons sit in that minimum, which lies at 0.5 eV (Sec. 4.2.2).
(4) *The ions.* The ions matter only in a cold afterglow. While the electrons stay at 0.63 eV the ions catch up with the atoms only above 10^{22} m^{-3}. At 300 K they take over at 1 THz above 7.0×10^{19} m^{-3} in the neon fill and 1.5×10^{20} m^{-3} in the argon fill with the classical Gaunt factor, and at 10^{20} m^{-3} they absorb 1.4 times as much as the atoms in the neon fill (Sec. 9.10). There the collisions are classical, and the ions' term rests on an estimate, the larger of Scheuer's classical Gaunt factor and one. The theory of that term is tested elsewhere. The quantum theory gives the Coulomb logarithm with no cutoff put in by hand (Sec. 8.13, Eq. (4.16)), and with no fitted parameter the ladder of Chapter 7 gives 0.96 ± 0.04 of the measured absorption of ten nitrogen–hydrogen laser plasmas of Turnbull et al. [3, 4] and 1.19 ± 0.06 of four hydrogen ones (Eq. (7.30)). Those plasmas, at temperatures such as the 0.36 keV of the hydrogen point of Sec. 7.1, are far from the afterglow.
(5) *The solvers.* Two independently written solvers of the two-term equation agree within 0.2 % in the electrons' mean energy and in the absorbed power (Sec. 5.8), and the kinetic solver of Chapter 6 reproduces Langdon's absorption fit within 1.8 % (Sec. 6.4).

Standing: verified where the atoms absorb. The kernel holds to 1.6 % in the THz band, and krypton's cross section is tested to a few per cent for electrons above 0.95 eV and to about 20 % at the afterglow's 0.63 eV. In a thermalized afterglow above 10^{20} m^{-3} the ions take over, and their term rests there on a classical estimate.

2.3 L3: heating

(1) In a laser gas the field takes over the electron temperature once its heating exceeds the electrons' collisional losses, at the intensity I_{heat} of Eq. (6.3), below which a wave crosses a weakly ionized plasma without changing it appreciably [5]. In the Ne fill's afterglow the electrons lose 3.6×10^8 eV s^{-1} each at 0.63 eV, and I_{heat} is 2.8×10^2 W cm^{-2} at 2.45 GHz, 1.4×10^3 W cm^{-2} at 170 GHz and 1.5×10^4 W cm^{-2} at 0.67 THz (Table 6.2). Gyrotrons deliver 1 MW at 170 GHz [6] and 0.21 MW at 0.67 THz [7] (Fig. 6.2), and the second holds 1.5×10^4 W cm^{-2} over 14 cm^2.
(2) The power per volume is small against the beam's. To hold the afterglow's electrons at 0.63 eV the field must give each 3.6×10^8 eV s^{-1}, 5.8 kW cm^{-3} at 10^{20} m^{-3}, and to hold them at the beam-on 1.5 eV 1.1×10^9 eV s^{-1} (Table 6.2), 18 kW cm^{-3}. These are 1.0 and 2.9 % of the 0.6 MW cm^{-3} of beam that makes the beam-on density in a full kinetic model of the laser (Sec. 6.6).
(3) The heating stays in the regime the kinetic equation describes. Langdon's flattening of the distribution needs 33 to 51 times I_{heat} in the afterglow, and Silin's regime a further 11.5 times

(Sec. 6.5). At every point of the scans the quiver velocity is at most 0.13 of the thermal one (Sec. 6.3). For electrons whose energy far exceeds the photon's, the exchange of several photons changes how the energy is gained but not how fast, so the classical kernel holds at any intensity, and with elastic losses alone a 0.5 eV rise of the electron temperature in pure krypton needs 53 $\mathrm{W\,cm^{-2}}$ at 0.3 THz and 585 $\mathrm{W\,cm^{-2}}$ at 1 THz (Sec. 9.7).
(4) The density answers to the heating. Heated hard, the afterglow ionizes itself. At 10^{14} $\mathrm{cm^{-3}}$ and 300 $\mathrm{kW\,cm^{-3}}$ the density e-folds in 2.4 to 5.2 ns in pure krypton and in krypton with F_2 at 0.1 atm (Sec. 4.6.2), and the Ne fill multiplies its electrons more than e-fold in 1 μs at every pressure at that density (Table 5.3). Above the ionization balance the reheated afterglow becomes a discharge sustained by the field. A self-sustained discharge forms 0.337 to 0.354 $\mathrm{KrF^*}$ per absorbed joule (Sec. 6.8), and the notes do not address its stability. Below that balance the F_2 removes the electrons. In krypton with 2 % F_2 attachment takes them in 0.1 to 17 ns (Sec. 4.6.4), so reheating acts on electrons that turn over in nanoseconds.

Standing: verified. 1.4×10^3 to 1.5×10^4 $\mathrm{W\,cm^{-2}}$ hold the afterglow's electrons at their temperature, and a few per cent of the beam's power per volume holds them at the beam-on temperature, in a regime that the two-term kinetic equation and the classical kernel describe.

2.4 What is invalidated

(1) *The laser's own field.* At 10 $\mathrm{MW\,cm^{-2}}$, a few saturation intensities, the KrF field is 7.1×10^3 times below I_{heat} in the Ne fill and 1.5×10^4 times in the Ar fill (Sec. 6.5). The laser's light does not reheat its own electrons.
(2) *The nonlinearity of IB.* An auxiliary field reaches only the nonlinearity of a weakly ionized gas, in which the field sets the electron temperature, long before Langdon's flattening [8] or Silin's saturation, which need 33 to 80 and 380 to 920 times more intensity (Sec. 6.1). The nonlinearity of IB is not by itself a lever on the efficiency of a gas laser (Sec. 6.10).
(3) *The frequency as a lever, with the beam on.* A field far above the collision frequency heats each electron in proportion to its collision frequency and gives a distribution whose tail reaches krypton's 9.9 eV levels, where a DC field heats the slow electrons most (Sec. 6.6). At the beam-on density the electron–electron collisions supply most of that tail anyway, and what is left of the frequency's advantage is 2 to 7 % where the field's power exceeds the beam's (Sec. 6.10). In the afterglow, with 27 times fewer electrons, the lever survives, which L4 takes up.

2.5 L4: partition

(1) The formation efficiency η_{form} of Eq. (6.16) counts the $\mathrm{KrF^*}$ that a partition of the absorbed power forms per absorbed energy, at the photon energy of 4.997 eV, with the branchings of a KrF kinetic network: 0.87 for $\mathrm{Kr^*}$ through the harpoon in the Ne fill, 0.81 in the Ar fill, and 0.68 through the ions. The electron beam forms 0.198 in the Ne fill by the same count once its sub-excitation electrons are credited, 0.175 without them, and a full kinetic model of the laser gives 0.194 (Sec. 6.8).
(2) At the afterglow's density the frequency still matters. At 10^{20} $\mathrm{m^{-3}}$ a field far above the collision frequency, 28.3 THz in the scans, gives krypton 66.4 % of the absorbed power in the Ne fill at the lowest power, against 42.7 % from a DC field of the same absorbed power, a ratio of 1.556, and the ratio is 1.596 in the Ar fill (Table 6.3). With electrons sustained by the beam it is 1.709 and 1.899 (Table 6.4). In the afterglow a 2.45 GHz wave is a DC field for this purpose, $\langle\nu_m\rangle/\omega = 32$, while a THz wave lies above the collision frequency, $\langle\nu_m\rangle/\omega = 0.12$ already at 0.67 THz (Table 6.2).

(3) The high frequency forms 0.274 to 0.352 KrF^* per absorbed joule in the Ne fill at 10^{20} m^{-3}: 0.274 and 0.351 at the powers of 2 and 5 Td with the growth model (Table 6.3), and 0.298 and 0.352 for the field's marginal power with beam-sustained electrons (Table 6.4). That is 1.4 to 1.8 times the beam's 0.198. A DC field forms 0.175 to 0.326 there. In the Ar fill the high frequency forms 0.342 to 0.375 against 0.101 to 0.208.
(4) The count leaves out what a reheated afterglow does most.

- *The electron collisions of Kr^* and KrF^*.* At the beam-on density the stepwise ionization $e + Kr^* \rightarrow Kr^+ + 2e$ runs at 0.39 to 1.63 of the harpoon rate in the Ne fill, at the rate of Kannari et al. [9], and lowers the KrF^* yield of a Kr^* by 4.8 to 11.8 %, which lowers field pumping, mostly through Kr^*, more than beam pumping, mostly through ions. In a full kinetic model of microwave co-pumping the sign of the laser energy that the field gains rests on that one rate, and the electron quench of KrF^* is a candidate second (Sec. 6.8). No measurement of the stepwise ionization was found (Sec. 5.3). In the afterglow these collisions weigh more, since Kr^* accumulates once its harpoon partner is gone, 10 and 8 times in the pulses of Sec. 5.7.

- *The 5s to 5p cycle.* The electrons lift Kr^* to 5p with the measured cross sections of Jung et al. [10], and in the krypton fill at 1 atm and 10^{15} cm^{-3}, at 10 THz, the 5p manifold returns 84 % by collisions with the gas, 15 % superelastically and 1.2 % by radiation (Sec. 5.3). Net of what they give back, the excited atoms take 0.007 to 2.9 times the absorbed power from the electrons in the Ne fill and up to 23.5 times in the krypton fill (Sec. 5.5). A faster collisional return ends more of the cycle's energy in the gas (Sec. 5.6), and one measured total loss rate of a 5p level bounds it [11].

- *The F_2.* Kr^* forms KrF^* only while F_2 lasts, and the electrons burn it by attachment, 3×10^8 to 6×10^8 molecules per electron per second in the pulses of Sec. 5.7, where half of it is gone at 134 ns at 10^{15} cm^{-3} and at 11 ns at 10^{16} cm^{-3}. The afterglow inherits what the pump leaves.

Standing: open. Per absorbed joule a THz wave in the afterglow would form 1.4 to 1.8 times the KrF^* of the beam, and a microwave less. The count leaves out the stepwise ionization, the electron quench of KrF^* and the 5s to 5p cycle, which grow with the electron temperature and with Kr^*, and whether Kr^* becomes KrF^* at all depends on the F_2 the pump leaves.

2.6 L5: the laser and the wall plug

(1) KrF^* formed in the afterglow is laser energy only if photons extract it before it is lost. No note follows the photons. The excimers' own kinetics, the photons, the gas temperature, the atomic fluorine and the walls are left to a time-domain model of the laser (Sec. 5.9).
(2) Field pumping beats beam pumping at the wall plug when $\eta_{\text{src}}\eta_{\text{form},f} > \eta_{\text{beam}}\eta_{\text{form},b}$, that is when the source's efficiency exceeds 0.57 to 0.65 of the efficiency with which the pulsed power puts energy into the gas through the foil, for the field's power at the beam-on density (Sec. 6.8). In the afterglow the KrF^* must also be extracted. With η_x the fraction of the KrF^* that leaves as laser light, the condition is

$$\eta_{\text{src}}\,\eta_{\text{form},f}\,\eta_{x,f} > \eta_{\text{beam}}\,\eta_{\text{form},b}\,\eta_{x,b}, \tag{2.1}$$

with f for the field and b for the beam. With the afterglow's $\eta_{\text{form},f} = 0.274$ to 0.352 against 0.198, the source must reach 0.56 to 0.72 of η_{beam} times $\eta_{x,b}/\eta_{x,f}$. No note computes either η_x.

(3) The sources that the notes cite are the 1 MW gyrotron at 170 GHz and the 0.21 MW gyrotron at 0.67 THz. In the afterglow the first heats 2.5 mm and the second 2.6 cm (L1). The notes cite no source of comparable power at the 1.3 THz that heats 10 cm of the Ne fill's afterglow.

Standing: open. Whether reheating raises the laser's efficiency depends on the extraction of the afterglow's KrF^* and on the efficiency of a THz source, neither of which Part II computes.

Chapter 3

What would settle it

(1) The open links hinge on two rates, one return rate and one calculation. In the order of their cost:

1. *The stepwise ionization* $e + \mathrm{Kr}^* \to \mathrm{Kr}^+ + 2e$ *and the electron quench of KrF**. They decide the sign of the field's gain (L4). A THz pulse sets the electron temperature on demand and its transmission reads the density and the temperature back (Secs. 6.1 and 6.10). With elastic losses alone a 0.5 eV rise in krypton needs 53 $\mathrm{W\,cm^{-2}}$ at 0.3 THz (Sec. 9.7).

2. *The collisional return of 5p to 5s in krypton, level by level* [12]. It is the largest uncertainty of the best frequency at the dense krypton points, 13 % in it and 19 % in the electron temperature (Sec. 5.6), and it decides how much of the 5s to 5p cycle's energy ends in the gas (L4).

3. *Drift velocities in krypton below 0.95 eV*, across the Ramsauer minimum where the afterglow's electrons sit and which one experiment covers (Sec. 4.8.2). They would test the absorption at 0.63 eV to the few per cent of the warmer electrons (L2).

4. *A time-dependent kinetic model of the laser with the THz absorption computed from its running state*, the electron collisions of the excited atoms of Table 5.2, the burn of the $\mathrm{F_2}$ and the extraction, run in pairs with and without the field and with the uncertain rates bracketed (Secs. 5.9 and 6.10). It gives η_x and the laser energy per absorbed joule, the two numbers that L5 lacks.

5. *The wall plug*, Eq. (2.1), with those numbers and the efficiency of a source near 1.3 THz.

(2) The proposal is invalidated if the run of item 4 finds the laser energy that the field gains negative over the measured range of the two rates of item 1, if the pump leaves too little $\mathrm{F_2}$ for the reheated Kr^* to form KrF^*, or if no source meets Eq. (2.1) with the extraction that the run finds.

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Part II

The notes

Chapter 4

Detailed derivation of the best frequency for inverse-bremsstrahlung heating of krypton laser gases, with a worked example

The note of October 12, 2026.

These notes derive, with every intermediate result written out, the frequency at which an electromagnetic wave heats the free electrons of a krypton laser gas best by inverse bremsstrahlung, for two gases, pure krypton and krypton with 2 % fluorine, at electron densities of 10^{14} to 10^{16} cm^{-3}, pressures of 0.1 to 2 atm, and 100 to 500 kW/cm^3 deposited for a microsecond. The absorption coefficient is the Drude form with a collision frequency built from the momentum-transfer cross sections of the atoms through the two-term kinetic equation, and from the exact free–free Gaunt factor for the ions. The electron temperature follows from the power balance, the best frequency from the depth of the gas, the choice between the GHz and the THz ranges from the absorption across the spectrum, and the uncertainty of the result from the envelope against a kinetic solution, from the electron–ion term against the measurements of Turnbull et al., and from the cross sections. A worked example at one operating point carries the whole chain to a recommended frequency. Every boxed result and every printed number is checked numerically by a script, which Appendix 4.A describes.

For notation, Gaussian units are used in the derivations, as in [1], and practical units in the numbers. The electron has charge $-e$ and mass m. The atoms have density N and fractions x_g, the electrons density n_e and temperature T_e, the ions density $n_i = n_e$ and charge e. The wave has angular frequency $\omega = 2\pi f$ and the field $E_0 \cos \omega t$, linearly polarized. The thermal velocity is $v_t = (k_B T_e/m)^{1/2}$ and the plasma frequency $\omega_p = (4\pi n_e e^2/m)^{1/2}$. An electron energy is $\varepsilon = mv^2/2$. A boxed equation is a result that is used later. An equation of another document is cited by its own number followed by the reference, as in Eq. (11.9) [1], and an equation number without a reference is one of these notes.

4.1 The question and what best means

(1) A wave is to deposit a power P of 100 to 500 kW/cm^3 for a microsecond into the free electrons of a laser gas, of density $n_e = 10^{14}$ to 10^{16} cm^{-3}, at $p = 0.1$ to 2 atm and 300 K, where the gas

density is $N = 2.45 \times 10^{19}\, p$ atoms per cm^3 per atmosphere. The electrons absorb the wave by inverse bremsstrahlung, while they collide with the atoms and, far less often, with the ions. Which frequency does this best?

(2) A wave below the plasma frequency is reflected. A wave just above it is absorbed in a thin sheet at the front of the gas. A wave far above it crosses the gas and leaves. The best frequency is therefore the one whose absorption length matches the depth L of the gas, provided that the gas reflects little of it:

$$\boxed{f_{\text{best}}(L) = \max\big(f_A, f_R\big), \qquad \alpha(f_A)\, L = 1, \qquad R(f_R) = 5\,\%,} \tag{4.1}$$

with α the absorption coefficient of the intensity and R the reflection of the front surface at normal incidence, f_R the lowest frequency that reflects no more than 5 %. At f_A one pass absorbs $1 - e^{-1} = 63\,\%$ of what enters and the deposition at the back is e^{-1} of that at the front. Asking for $\alpha L = q$ instead moves f_A by $q^{-1/2}$ where $\alpha \propto f^{-2}$. The reference depth is $L = 10$ cm, and Sec. 4.4 shows $f_A \propto L^{1/2}$.

(3) Two gases are compared: pure Kr, and Kr with 2 % F_2. The momentum-transfer and inelastic cross sections are the Biagi set for krypton [2] and Morgan's set for F_2 [3].

(4) The plan. Section 4.2 derives the absorption coefficient and the collision frequency of the atoms and of the ions. Section 4.3 derives the electron temperature from the power balance, since what the atoms scatter depends on it. Section 4.4 derives the best frequency in closed form. Section 4.5 follows the absorption across the spectrum and settles the GHz range against the THz range. Section 4.6 gives the results for the two gases with the particle balance. Section 4.7 checks the Maxwellian envelope against a two-term kinetic solution, Sec. 4.8 derives the uncertainty of every entry, and Sec. 4.9 works one operating point through from the gas density to the recommended frequency.

4.2 The absorption coefficient

4.2.1 The Drude model

This subsection follows [1, Sec. 11.1.1], in whose convention a perturbation varies as $e^{-i\omega t}$ and the physical quantity is the real part.

(1) The electron velocity obeys Newton's equation with a friction force that stands for the collisions,

$$-i\omega m\mathbf{v} = -e\mathbf{E} - \nu m\mathbf{v} \quad \longrightarrow \quad \mathbf{v} = -\frac{e\mathbf{E}}{m(\nu - i\omega)} = -\frac{ie\mathbf{E}}{m(\omega + i\nu)}, \tag{4.2}$$

which is Eq. (11.7) [1], with $\nu - i\omega = -i(\omega + i\nu)$ in the last step.

(2) The current density and the conductivity are

$$\mathbf{j} = -en_e\mathbf{v} = \frac{ie^2 n_e}{m(\omega + i\nu)}\,\mathbf{E}, \qquad \sigma = \frac{ie^2 n_e}{m(\omega + i\nu)} = \frac{e^2 n_e}{m(\nu - i\omega)}, \tag{4.3}$$

and with the dielectric function $\varepsilon = 1 + (4\pi i/\omega)\sigma$ of Eq. (11.4) [1],

$$\boxed{\varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega(\omega + i\nu)} = 1 - \frac{\omega_p^2}{\omega^2 + \nu^2} + i\,\frac{\omega_p^2\,\nu}{\omega(\omega^2 + \nu^2)}.} \tag{4.4}$$

(3) A plane wave $e^{i(kz-\omega t)}$ has $k = (\omega/c)\sqrt{\varepsilon}$. With $\sqrt{\varepsilon} = n_r + i\kappa$ the intensity falls as $e^{-2\,\mathrm{Im}\,k\,z}$, and the reflection of a sharp boundary at normal incidence is that of the Fresnel formula,

$$\boxed{\alpha = \frac{2\omega}{c}\,\kappa, \qquad R = \left|\frac{1-\sqrt{\varepsilon}}{1+\sqrt{\varepsilon}}\right|^2.} \tag{4.5}$$

Where the absorption is weak, $\mathrm{Im}\,\varepsilon \ll \mathrm{Re}\,\varepsilon$, one has $2\kappa \simeq \mathrm{Im}\,\varepsilon/n_r$ with $n_r \simeq (\mathrm{Re}\,\varepsilon)^{1/2}$, so that

$$\boxed{\alpha = \frac{\omega}{c}\,\frac{\mathrm{Im}\,\varepsilon}{(\mathrm{Re}\,\varepsilon)^{1/2}} = \frac{\nu}{c}\,\frac{\omega_p^2}{\omega^2}\Big(1 - \frac{\omega_p^2}{\omega^2}\Big)^{-1/2}\Big[1 + O\Big(\frac{\nu^2}{\omega^2}\Big)\Big],} \tag{4.6}$$

which is Eq. (11.12) [1]. Equation (4.5) is used in the numbers and Eq. (4.6) in the closed forms.

(4) The power absorbed per unit volume is the cycle average of $\mathbf{j}\cdot\mathbf{E}$,

$$\boxed{P = \tfrac{1}{2}\,\mathrm{Re}\,\sigma\,E_0^2 = \frac{n_e e^2}{2m}\,\frac{\nu}{\omega^2+\nu^2}\,E_0^2.} \tag{4.7}$$

Everything that distinguishes one model of inverse bremsstrahlung from another sits in ν, which the next two subsections build from the collisions with the atoms and with the ions.

4.2.2 The collision frequency of the atoms

(1) In a weakly ionized gas the electrons collide with atoms. The momentum-transfer collision frequency of an electron of speed v is $\nu_m(v) = N\sigma_m(v)v$, with σ_m the momentum-transfer cross section of the mixture, $\sigma_m = \sum_g x_g\sigma_{m,g}$. It depends on v, strongly so near a Ramsauer minimum, and the single ν of Eq. (4.2) has to be an average of it with the right weight.

(2) The weight follows from the two-term kinetic equation [4]. Write the velocity distribution as $f_0(v) + f_1(v)\cos\theta$, with θ the angle to the field and $\int 4\pi v^2 f_0\,\mathrm{d}v = n_e$. The anisotropic part obeys, for the field $\mathrm{Re}[E_0\hat{\mathbf{e}}e^{-i\omega t}]$,

$$-i\omega f_1 + \nu_m(v)\,f_1 = \frac{eE_0}{m}\,\frac{\partial f_0}{\partial v} \quad\longrightarrow\quad f_1 = \frac{eE_0}{m}\,\frac{\partial_v f_0}{\nu_m(v) - i\omega}, \tag{4.8}$$

where the collisions relax f_1 at the rate ν_m and leave f_0 alone.

(3) The current is the first moment of f_1,

$$j = -e\int v\cos\theta\,f_1\cos\theta\,\mathrm{d}^3v = -\frac{4\pi e}{3}\int_0^\infty v^3 f_1\,\mathrm{d}v = -\frac{4\pi e^2 E_0}{3m}\int_0^\infty \frac{v^3\,\partial_v f_0}{\nu_m(v) - i\omega}\,\mathrm{d}v, \tag{4.9}$$

and its real part, which carries the absorption, is

$$\mathrm{Re}\,\sigma = -\frac{4\pi e^2}{3m}\int_0^\infty \frac{v^3\nu_m(v)}{\nu_m^2(v) + \omega^2}\,\partial_v f_0\,\mathrm{d}v. \tag{4.10}$$

(4) Compare with the Drude form, $\mathrm{Re}\,\sigma = n_e e^2\nu/[m(\nu^2+\omega^2)]$ from Eq. (4.3). For $\omega \gg \nu_m$ both denominators are ω^2 and the comparison defines the effective collision frequency of the atoms,

$$\nu_{ea} = -\frac{4\pi}{3n_e}\int_0^\infty v^3 \nu_m(v)\,\partial_v f_0\,\mathrm{d}v. \tag{4.11}$$

For a Maxwellian, $f_0 \propto e^{-v^2/2v_t^2}$ and $\partial_v f_0 = -(v/v_t^2)f_0$, so that

$$\boxed{\nu_{ea} = \frac{\langle \nu_m v^2\rangle}{3v_t^2} = N\Big\langle \sigma_m v\,\frac{2\varepsilon}{3k_BT_e}\Big\rangle,} \tag{4.12}$$

the average over the Maxwellian with the energy as a weight: the field heats the fast electrons most. For a constant σ_m this is 4/3 of the Drude mean $\langle\nu_m\rangle$, and for the two gases of these notes it is 1.72 to 2.23 times $\langle\nu_m\rangle$ at the cells of Table 4.2 where $T_e \geq 0.5$ eV, since σ_m of krypton climbs steeply above its Ramsauer minimum at 0.5 eV. By the low-energy theorem [5, 6] this kernel is exact for photons of energy small against the electron's, and the atom enters only through σ_m.
(5) The same collisions take energy from the electrons. An electron of energy ε that scatters elastically through the angle χ on an atom of mass $M \gg m$ at rest loses $2(m/M)\varepsilon(1-\cos\chi)$, and the average of $1-\cos\chi$ over the angular distribution defines σ_m, so that the loss rate is $(2m/M)\nu_m(v)\varepsilon$ per electron. Over the Maxwellian, with the gas at T_g giving back k_BT_g for k_BT_e,

$$\boxed{P_{\mathrm{el}} = \sum_g x_g\,\frac{2m}{M_g}\,N\langle\sigma_{m,g} v\,\varepsilon\rangle\Big(1-\frac{T_g}{T_e}\Big).} \tag{4.13}$$

4.2.3 The collision frequency of the ions

(1) For electrons on ions of charge e the Drude collision frequency is $\nu_{ei} = \nu_T\Lambda$ with the Spitzer frequency

$$\boxed{\nu_T = \frac{4\sqrt{2\pi}}{3}\,\frac{n_i e^4}{m^{1/2}(k_BT_e)^{3/2}} = 2.91\times10^{-12}\,\frac{n_i\,[\mathrm{m}^{-3}]}{(k_BT_e\,[\mathrm{eV}])^{3/2}}\ \mathrm{s}^{-1},} \tag{4.14}$$

which is Eq. (10.133) [1] without its logarithm, and Λ holds the physics of the Coulomb collision.

(2) The quantum theory of inverse bremsstrahlung gives Λ as a Gaunt factor. The absorption, net of stimulated emission, of a Maxwellian is

$$\boxed{\Lambda = \frac{\pi}{\sqrt{3}}\,\langle g\rangle(\gamma^2,u)\,\frac{1-e^{-u}}{u}, \qquad u = \frac{\hbar\omega}{k_BT_e}, \quad \gamma^2 = \frac{\mathrm{Ry}}{k_BT_e},} \tag{4.15}$$

with $\langle g\rangle$ the thermally averaged free–free Gaunt factor of the exact Coulomb wave functions [7], tabulated by van Hoof et al. [8] against γ^2 and u. Its Born limit, $\gamma^2 \to 0$, is

$$\Lambda_1(u) = \frac{1-e^{-u}}{u}\,e^{u/2}K_0\Big(\frac{u}{2}\Big) \simeq \ln\frac{4k_BT_e}{e^\gamma\hbar\omega} = \ln\frac{b_{\max}}{b_{\min}}, \qquad b_{\max} = \frac{v_t}{\omega}, \quad b_{\min} = \frac{e^\gamma}{4}\,\frac{\hbar}{mv_t}, \tag{4.16}$$

for $u \ll 1$, with γ Euler's constant in the exponent: the upper cutoff is the distance an electron travels in a radian of the period, the dynamic cutoff of Dawson and Oberman [9], which Turnbull

et al. measured to be the right one [10], and not the Debye length.

(3) The Born limit needs $\eta_t = e^2/\hbar v_t \ll 1$. Here $k_B T_e$ is about an electron volt and $\eta_t = 3.2$ to 5.3 at the warm cells: the collisions are classical, and in the soft-photon limit the exact logarithm of one electron is the Born one plus Bloch's term $-\gamma - \mathrm{Re}\,\psi(1+i\eta)$, which for $\eta \gg 1$ is $-\gamma - \ln\eta$ and turns the lower cutoff into the classical $e^\gamma e^2/2mv^2$. The tabulated $\langle g\rangle$ contains this: at the warm cells $\Lambda = 2.5$ to 5.6, where the Born logarithm Λ_1 would give 3.0 to 7.1.

(4) The two kinds of scatterer add, since electron–ion and electron–atom collisions are separate events and electron–electron collisions radiate no dipole:

$$\boxed{\nu_{\mathrm{eff}} = \nu_{ea} + \nu_T \Lambda.} \tag{4.17}$$

At the cells of Table 4.2 with $T_e \geq 0.5$ eV, $\nu_{\mathrm{eff}} = 2.7\times10^{11}$ to 5.7×10^{12} s^{-1}, and the ions' share of it is below 2 % at 10^{15} cm^{-3} and 3 to 26 % at 10^{16} cm^{-3}, the most at 0.1 atm.

4.2.4 What is left out, and its size

(1) Screening by the electrons and the correlations of the ions change Λ by a few per cent at most at these densities. (2) The quantum reduction of the atoms' term at a finite photon energy is below 1 % for $u \leq 0.12$, which holds at every warm cell. (3) Langdon's flattening of the distribution by the ion heating has the parameter $\alpha_L = v_{\mathrm{os}}^2/v_t^2$ with the quiver velocity $v_{\mathrm{os}} = eE_0/m\omega$; from Eq. (4.7) at $\omega \gg \nu$,

$$\boxed{\alpha_L = \frac{v_{\mathrm{os}}^2}{v_t^2} = \frac{e^2E_0^2}{m^2\omega^2 v_t^2} = \frac{2\,P/n_e}{\nu_{\mathrm{eff}}\,k_B T_e},} \tag{4.18}$$

where the factor $1 + \nu^2/\omega^2$ left out is below 1.06 at the warm cells. It is at most 3×10^{-3} in Table 4.2 wherever the ions carry more than 2 % of the absorption, and Langdon's factor [12] then changes the absorption by less than 1 % at the warm cells. (4) The reflection is that of a sharp boundary, an upper bound for a graded one.

In Summary: The absorption coefficient is the Drude form (4.5) with the collision frequency (4.17): the atoms through the energy-weighted average (4.12) of their momentum-transfer collision frequency, exact for soft photons, and the ions through the Spitzer frequency (4.14) and the exact Gaunt factor (4.15), whose cutoffs are the dynamic one and the classical one. Both depend on T_e, which the next section fixes.

4.3 The electron temperature

(1) The field delivers P/n_e to each electron, and in a steady state the electrons lose it by elastic recoil, Eq. (4.13), and by the inelastic collisions with threshold energies ε_j:

$$\boxed{\frac{P}{n_e} = P_{\mathrm{el}}(T_e) + N\sum_j x_j k_j(T_e)\,\varepsilon_j,} \tag{4.19}$$

which fixes T_e for a Maxwellian. The electrons that attachment removes leave with their own energy and are replaced by the source that holds the density, so attachment is not a loss in Eq. (4.19). Per electron, P/n_e runs from 6.2×10^7 eV/s at 10^{16} cm^{-3} and 100 kW/cm^3 to 3.1×10^{10} eV/s at 10^{14} cm^{-3} and 500 kW/cm^3.

(2) The rate coefficient of a process of cross section $\sigma_j(\varepsilon)$ is $k_j = \langle \sigma_j v \rangle$. With the Maxwellian written in the energy, $F(\varepsilon) = \frac{2}{\sqrt{\pi}}(k_B T_e)^{-3/2} e^{-\varepsilon/k_B T_e}$ normalized by $\int_0^\infty \varepsilon^{1/2} F \, \mathrm{d}\varepsilon = 1$, and $v = \gamma_e \varepsilon^{1/2}$ with $\gamma_e = (2/m)^{1/2}$,

$$\boxed{k_j = \int_0^\infty \sigma_j(\varepsilon)\, v\, \varepsilon^{1/2} F(\varepsilon)\, \mathrm{d}\varepsilon = \gamma_e \int_0^\infty \varepsilon\, \sigma_j(\varepsilon)\, F(\varepsilon)\, \mathrm{d}\varepsilon,} \tag{4.20}$$

and the same weight εF gives Eq. (4.12) as $\nu_{ea} = N\gamma_e \int \varepsilon \sigma_m (2\varepsilon/3k_B T_e) F \, \mathrm{d}\varepsilon$ and Eq. (4.13) with $\varepsilon^2 \sigma_m F$.

(3) In pure Kr the recoil takes 0.1 to 24 % of the power and the 5s excitation of krypton at 9.9 eV the rest, so T_e sits where the Maxwellian tail reaches 10 eV: 0.95 to 2.6 eV at 300 kW/cm^3. With 2 % F_2 the vibrational channels (0.1 to 0.4 eV) and the dissociative ones (3.2 and 4.3 eV) take 1 to 13 % of the losses at 10^{14} cm^{-3}, 8 to 74 % at 10^{15} and 45 to 100 % at 10^{16}, where the power per electron is least, and lower T_e to 1.05 eV at 10^{15} cm^{-3} and 2 atm and, at 10^{16} cm^{-3}, to 1.23 eV at 0.1 atm and to 0.19 and 0.088 eV at 0.5 and 2 atm.

(4) At the two cells where F_2 holds T_e at 0.19 and 0.088 eV, in and below the Ramsauer minimum of krypton, the ions carry 79 to 92 % of ν_{eff}, u reaches 0.30 and 1.0, outside the soft-photon range of the atoms' term, and no source could hold the electrons (Sec. 4.6.2). Their entries in the tables are formal.

(5) The field itself follows from Eq. (4.7) once $\alpha L = 1$ ties the frequency to the collision frequency. With Eq. (4.6), $\nu/(\omega^2 + \nu^2) = c\, n_r/(\omega_p^2 L)$ at f_A, and Eq. (4.7) becomes

$$\boxed{E_0 = \left(\frac{8\pi P L}{c\, n_r}\right)^{1/2} = 47.5 \text{ kV/cm} \left(\frac{P}{300 \text{ kW/cm}^3} \frac{L}{10 \text{ cm}}\right)^{1/2} n_r^{-1/2},} \tag{4.21}$$

the same at every cell but for n_r, since the absorbed power per volume is the intensity over the absorption length: the intensity is PL in that sense, 3 MW/cm^2 at 300 kW/cm^3 and 10 cm.

In Summary: The power balance (4.19) with the Maxwellian rates (4.20) sets T_e at 0.95 to 2.6 eV at the warm cells of the two gases. The 5s level of krypton takes the power in pure Kr, and F_2's low channels take over where the power per electron is least and cool the electrons into the Ramsauer minimum.

4.4 The best frequency

4.4.1 The reflection limit

(1) For $\nu \ll \omega$ the dielectric function (4.4) is real, $\varepsilon = 1 - \omega_p^2/\omega^2$, and above the cutoff $\sqrt{\varepsilon} = n_r$ is real too, so that Eq. (4.5) gives $R = [(1 - n_r)/(1 + n_r)]^2$. Solving for $R = 5\,\%$,

$$n_r = \frac{1 - \sqrt{R}}{1 + \sqrt{R}} = 0.6345, \qquad \boxed{\frac{f_R}{f_{pe}} = \left(1 - n_r^2\right)^{-1/2} = 1.294 \quad (\nu \ll \omega_p),} \tag{4.22}$$

with the cutoff $f_{pe} = \omega_p/2\pi = 8.98 \text{ kHz}\,(n_e/\text{cm}^{-3})^{1/2}$, which is 0.090, 0.28 and 0.90 THz at the three densities. So $f_R \leq 1.16$ THz, at 10^{16} cm^{-3}. Where ν exceeds ω_p, at 10^{14} cm^{-3} from 0.5 atm up, the collisions damp the reflection and f_R falls to 0.03 THz and below, far under f_A.

4.4.2 The absorption length

(1) Well above the cutoff and the collision frequency Eq. (4.6) is $\alpha = \omega_p^2 \nu_{\rm eff}/(c\omega^2)$, and $\alpha L = 1$ gives

$$\boxed{f_A = \frac{1}{2\pi}\Big(\frac{\omega_p^2\,\nu_{\rm eff}\,L}{c}\Big)^{1/2} = 5.2\ \text{THz}\Big(\frac{n_e}{10^{15}\ \text{cm}^{-3}}\,\frac{\nu_{\rm eff}}{10^{12}\ \text{s}^{-1}}\,\frac{L}{10\ \text{cm}}\Big)^{1/2},} \tag{4.23}$$

within 3.1 % of the exact root of $\alpha L = 1$ with Eq. (4.5) at every one of the 54 points computed below. At every one of them $f_A > f_R$: the absorption length, not the cutoff, sets the frequency.
(2) Since $\nu_{\rm eff} \simeq \nu_{ea} \propto N\sigma_m(T_e)$, the best frequency grows as the square root of the electron density, of the gas density, and of the depth. The exponents come out below 1/2 in the fits of Sec. 4.6.3, because the hotter, thinner cells scatter more per atom: T_e falls with $n_e p$ through Eq. (4.19), and σ_m with it.

In Summary: The best frequency is Eq. (4.23), the frequency whose absorption length is the depth of the gas, bounded below by the reflection limit (4.22), which never binds here.

4.5 The spectrum: GHz or THz?

(1) Before the best frequency is read off, the whole spectrum is worth a look, since the microwave sources sit in the GHz range. With each cell's $\nu_{\rm eff}$ and ω_p, Eqs. (4.4) and (4.5) give $\alpha(f)$ and $R(f)$ from 1 GHz to 100 THz, the ions' logarithm taken at each frequency. Three regimes appear. Below the collision frequency the wave enters as a collisional skin: with $\omega \ll \nu$ and $\omega_p^2 \gg \omega\nu$, Eq. (4.4) is dominated by its imaginary part, $\varepsilon \simeq i\omega_p^2/\omega\nu$, so that $\sqrt{\varepsilon} \simeq (1+i)(\omega_p^2/2\omega\nu)^{1/2}$ and

$$\boxed{\alpha \simeq \frac{\omega_p}{c}\Big(\frac{2\omega}{\nu_{\rm eff}}\Big)^{1/2} \qquad (\omega \ll \nu_{\rm eff}),} \tag{4.24}$$

rising as $f^{1/2}$: at 2.45 GHz this law reproduces the exact α within a factor 0.98 to 1.14 at the 18 cells. Near the larger of the collision frequency and the cutoff, α peaks, at 0.05 to 0.50 THz over the 18 cells, and the absorption length is shortest there, 0.03 to 1.7 mm for pure Kr at 0.5 atm. Above both, the gas is transparent, α falls as f^{-2} by Eq. (4.6), and $R \to 0$. Here $\nu_{\rm eff}/2\pi = 0.009$ to 0.91 THz and $f_{pe} = 0.090$ to 0.90 THz, so the boundary between the two bands is where the physics turns: the GHz band lies on the skin side and the THz band on the transparent side.

(2) Table 4.1 and Fig. 4.1 give pure Kr at 0.5 atm: the absorption length $1/\alpha$, the reflection R, and the fraction $A = (1-R)(1-e^{-\alpha L})$ of the incident power deposited in the 10 cm.
(3) *GHz.* At 10^{15} and 10^{16} cm^{-3} the gas is a mirror: it reflects 31 to 94 % between 2.45 and 95 GHz at 0.5 atm, 8 to 96 % over all pressures, and absorbs the remainder within 0.04 to 1.1 mm of the front. At 10^{14} cm^{-3} the collisions outrun the cutoff and the wave gets in (43 % reflected at 2.45 GHz and 1 % at 95 GHz), but it is absorbed in 1.8 to 4.2 mm (0.6 to 8.3 mm over the pressures and both gases): the front millimetres take all of it and the rest of the gas nothing, since the deposition falls by $e^{\alpha L}$, at least 10^5, from front to back.

THz. The gas reflects at most 15 % (1 THz at 10^{16} cm^{-3}) and nothing above 3 THz, and the absorption length runs from 0.2 mm to 2 m, through the depth of the gas at 2.2, 5.7 and 15.4 THz for the three densities at 0.5 atm, and at 1.1 to 26.5 THz over the 18 cells. At 3 THz the 10 cm deposits 41, 97 and 100 % of the incident power. Per electron the heating at a given intensity goes as $\nu_{\rm eff}/(\omega^2 + \nu_{\rm eff}^2)$ by Eq. (4.7), largest at GHz, where every electron heats as in a

Table 4.1: Pure Kr at 0.5 atm, from 2.45 GHz to 30 THz: the absorption length, the reflection in per cent, and the fraction of the incident power deposited in $L = 10$ cm, in per cent, at the three densities.

	10^{14} cm^{-3}			10^{15} cm^{-3}			10^{16} cm^{-3}		
f	$1/\alpha$	R	A	$1/\alpha$	R	A	$1/\alpha$	R	A
2.45 GHz	4.2 mm	43	57	1.1 mm	80	20	0.31 mm	94	6
10 GHz	2.4 mm	17	83	0.52 mm	65	35	0.15 mm	88	12
30 GHz	1.8 mm	4	96	0.30 mm	49	51	0.08 mm	82	18
95 GHz	1.8 mm	1	99	0.17 mm	31	69	0.04 mm	75	25
300 GHz	3.5 mm	0	100	0.26 mm	7	93	0.03 mm	70	30
1 THz	2.2 cm	0	99	3.0 mm	0	100	0.19 mm	13	87
3 THz	18.7 cm	0	41	2.7 cm	0	97	3.5 mm	0	100
10 THz	2 m	0	5	30.5 cm	0	28	4.2 cm	0	91
30 THz	19 m	0	1	3 m	0	4	38.1 cm	0	23

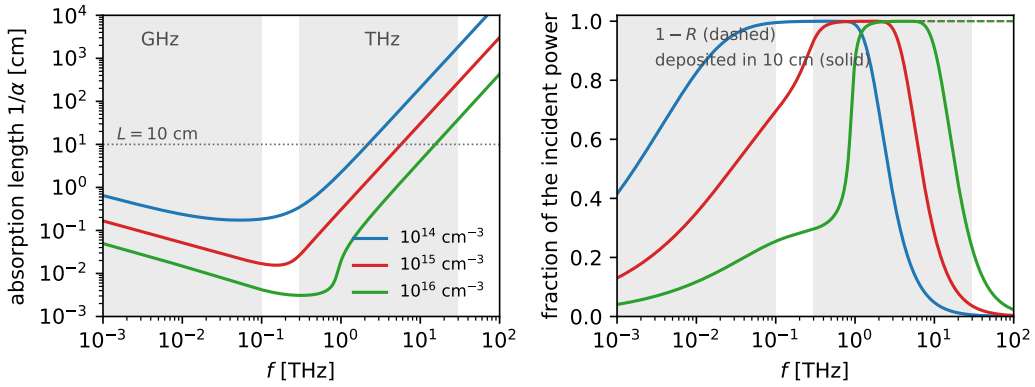


Figure 4.1: Pure Kr at 0.5 atm. Left: the absorption length against frequency for the three densities, with the depth $L = 10$ cm dotted; the shaded bands are the GHz range (1 to 100 GHz) and the THz range (0.3 to 30 THz). Right: the fraction of the incident power that enters the gas, $1 - R$ (dashed), and the fraction deposited in the 10 cm (solid).

static field, and that is exactly why the skin is thin: the first electrons take the whole wave. The GHz range would be the right one only for a gas a millimetre thick at 10^{14} cm^{-3} and 0.5 atm or more, where 95 GHz enters with 1 % reflection and is absorbed in 1.8 to 5.4 mm, or at densities below 10^{12} cm^{-3}, where the cutoff falls to 9 GHz. Kr/F_2 changes the absorption lengths by at most a factor 1.3 at 95 GHz and 3 THz and the reflections by at most 9 points.

In Summary: For a gas deeper than a few millimetres the THz range is better: it enters the gas and deposits over centimetres, where the GHz range reflects or skins. The best frequency of Eq. (4.23) lies on the transparent side of the peak, in the THz range at every cell.

4.6 The two gases

4.6.1 The best frequencies

Table 4.2 gives f_{best} at $L = 10$ cm and 300 kW/cm^3 for the two gases, with T_e from Eq. (4.19). The flags come from the particle balance of Sec. 4.6.2.

Table 4.2: The best frequency in THz at $L = 10$ cm and 300 kW/cm^3, with T_e in eV in parentheses, for the two gases. ^{a}: the field's own ionization outruns the losses and n_e grows during the pulse. ^{s}: the losses outrun it and a source must hold n_e.

n_e [cm^{-3}]	0.1 atm	0.5 atm	2 atm
pure Kr			
10^{14}	1.1^a (2.6)	2.2^a (2.0)	3.8^a (1.6)
10^{15}	3.0^s (1.8)	5.7^s (1.4)	10.1^s (1.2)
10^{16}	8.7^s (1.3)	15.4^s (1.1)	26.5^s (1.0)
Kr + 2 % F_2			
10^{14}	1.1^a (2.6)	2.2^s (1.9)	3.8^s (1.6)
10^{15}	3.0^s (1.7)	5.6^s (1.4)	9.2^s (1.1)
10^{16}	8.5^s (1.2)	13.8^s (0.2)	21.8^s (0.1)

4.6.2 The particle balance

(1) The density is not the field's to choose. Direct ionization by the electrons, at the rate $\nu_{\text{ion}} = N\sum_j x_j k_{\text{ion},j}$ from the thresholds at 14.0 eV (Kr) and 15.7 eV (F_2), makes electrons. Attachment to F_2 at $\nu_{\text{att}} = N x_{F_2} k_{\text{att}}$ removes them, and so does recombination. For the recombination every cation is taken as Kr_2^+ with the coefficient of Shiu and Biondi [16],

$$\alpha_{\text{rec}} = 1.6\times10^{-6}\Big(\frac{300\text{ K}}{T_e}\Big)^{0.55}\ \text{cm}^3/\text{s}, \qquad \nu_{\text{rec}} = \alpha_{\text{rec}}\, n_e, \tag{4.25}$$

measured from 300 to 8400 K, 0.72 eV, and extrapolated here. The net rate is $\nu_{\text{net}} = \nu_{\text{ion}} - \nu_{\text{att}} - \nu_{\text{rec}}$.

(2) Where $\nu_{\text{net}} > 0$ the density e-folds in $1/\nu_{\text{net}}$, which is 2.4 to 5.2 ns at the four cells marked ^{a} (pure Kr at 10^{14} cm^{-3}, and Kr/F_2 at 10^{14} cm^{-3} and 0.1 atm). Where $\nu_{\text{net}} < 0$ a source must supply $-\nu_{\text{net}} n_e$ electrons per second, at the least at the ionization energy of krypton,

$$\boxed{P_{\text{src}} \geq n_e\,(\nu_{\text{att}} + \nu_{\text{rec}} - \nu_{\text{ion}}) \times 14.0\text{ eV}.} \tag{4.26}$$

In pure Kr, where recombination is the loss, that is 300 to 410 kW/cm^3 at 10^{15} and 41 to 49 MW/cm^3 at 10^{16} cm^{-3}. With 2 % F_2 the loss is attachment at 5.8×10^7 to 1.3×10^{10} s^{-1}, and the source needs 32 and 430 kW/cm^3 at 10^{14} cm^{-3} and 0.5 and 2 atm, 0.5 to 7.8 MW/cm^3 at 10^{15} and 46 to 470 MW/cm^3 at 10^{16} cm^{-3}. Neither gas holds 10^{16} cm^{-3} at these powers, and Kr/F_2 does not hold 10^{15} either.

(3) Pure Kr left to itself settles where its ionization balances recombination, $\nu_{\text{ion}}(T_e(P/n_e)) = \alpha_{\text{rec}}(T_e)\, n_e$, in $1/\nu_{\text{ion}} = 15$ to 19 ns: at $n_e = 4.5$, 3.6 and 2.9×10^{14} cm^{-3} for 0.1, 0.5 and 2 atm ($T_e = 2.0$, 1.6 and 1.4 eV), where $f_{\text{best}} = 2.1$, 3.7 and 6.1 THz. Over 100 to 500 kW/cm^3 that is 1.6 to 6×10^{14} cm^{-3} and 1.5 to 7.2 THz.

4.6.3 The power laws

Fitted to the nine cells of each gas at 300 kW/cm^3,

$$\boxed{f_{\text{best}} \approx C \left(\frac{n_e}{10^{15}\ \text{cm}^{-3}}\right)^b \left(\frac{p}{1\ \text{atm}}\right)^c \left(\frac{L}{10\ \text{cm}}\right)^{1/2},} \tag{4.27}$$

with $C = 7.6$ THz, $b = 0.43$, $c = 0.40$ for pure Kr within 5.1 %, and $C = 7.1$ THz, $b = 0.41$, $c = 0.37$ for Kr/F_2 within 8.6 %. From 100 to 500 kW/cm^3 the best frequency rises by at most a factor 1.16 in pure Kr, through T_e: at 10^{15} cm^{-3} and 0.5 atm it is 4.4, 7.1 and 8.7 THz for $T_e = 1$, 2 and 4 eV. In Kr/F_2 the power moves it by up to a factor 3.4, at 10^{15} cm^{-3} and 2 atm, since at 100 kW/cm^3 the F_2 losses already hold the electrons at 0.16 eV there.

4.6.4 Fluorine against pure krypton

(1) At a given T_e the 2 % F_2 adds at most 4.6 % to the kernel (4.12), so the frequency is the same wherever the krypton excitation takes the power: Kr/F_2 is within 2 % of pure Kr at six cells ($T_e \geq 1.2$ eV) and 9 to 18 % below it at 10^{15} cm^{-3} and 2 atm and at 10^{16} cm^{-3} and 0.5 and 2 atm, where the F_2 losses hold the electrons in the Ramsauer minimum.

(2) What F_2 changes is the closure. Attachment removes the electrons in 0.1 to 17 ns, so a source must hold the density nearly everywhere, and from 10^{15} cm^{-3} up the source's power exceeds the field's and its electrons set T_e (Secs. 4.7 and 4.8).

In Summary: Pure Kr wants 1.1 to 3.8 THz at 10^{14} cm^{-3}, 3.0 to 10.1 at 10^{15} and 8.7 to 26.5 THz at 10^{16}, and 2 % F_2 changes the frequency only where its losses cool the electrons. The density is the source's choice wherever F_2 is present, and neither gas holds 10^{16} cm^{-3} at these powers.

4.7 The kinetic check

(1) The envelope takes the electrons as Maxwellian. A two-term kinetic solution [4] of every cell, with the same cross sections, the elastic collisions with the ions, electron–electron collisions in Landau's form, and the temporal-growth model in which the electrons that attachment removes or ionization adds are replaced or removed with the shape of the distribution, gives T_e as two thirds of the mean energy and the absorption from the complex mobility at the envelope's f_{best}, from which the kinetic f_A follows with $\alpha \propto f^{-2}$.

(2) In pure Kr the kinetic T_e is 0.94 to 1.35 times the envelope's and the kinetic f_{best} 0.89 to 1.23 times it: the inelastic collisions deplete the tail that a Maxwellian keeps, so the bulk must be hotter to lose the same power, most at 10^{15} cm^{-3}. With 2 % F_2 the same holds at the four cells where the field outweighs attachment, with f_{best} at 0.96 to 1.12 times the envelope. At the other five, 10^{15} cm^{-3} from 0.5 atm up and all of 10^{16}, attachment takes 0.6 to 25 times the field's power from the electrons, so the source that replaces them sets T_e, and the growth model is no longer a field-heated plasma.

In Summary: The Maxwellian envelope is good to −11 to +23 % in frequency where the field sets the temperature. Where the source does, the temperature, and with it the frequency, is the source's.

4.8 The uncertainty of the result

4.8.1 Propagation

(1) From Eq. (4.23), $f_{\text{best}} \propto \nu_{\text{eff}}^{1/2}$, so that a fractional change of the collision frequency moves the frequency by half of it, and a change of the ions' term alone by half of its share $s_i = \nu_{ei}/\nu_{\text{eff}}$:

$$\boxed{\frac{\delta f_{\text{best}}}{f_{\text{best}}} = \frac{1}{2}\,\frac{\delta\nu_{\text{eff}}}{\nu_{\text{eff}}}, \qquad \frac{\delta f_{\text{best}}}{f_{\text{best}}} = \frac{s_i}{2}\,\frac{\delta\nu_{ei}}{\nu_{ei}}.} \tag{4.28}$$

A change of T_e acts through $\sigma_m(T_e)$: between 1 and 2 eV in pure Kr, $f_{\text{best}} \propto T_e^{0.70}$, and between 2 and 4 eV, $T_e^{0.31}$.

(2) Each cell gets a central value and one uncertainty, $f_{\text{best}} \pm \delta$. With the bands $[m_-, m_+]$, $[i_-, i_+]$, $[e_-, e_+]$ and $[x_-, x_+]$ of the four sources below as factors on the envelope's value f_{env},

$$\boxed{\begin{aligned} f_c &= f_{\text{env}}\,\frac{m_- + m_+}{2}\,\frac{i_- + i_+}{2}, \\ \frac{\delta}{f_c} &= \left[\left(\frac{m_+ - m_-}{2}\right)^2 + \left(\frac{i_+ - i_-}{2}\right)^2 + \left(\frac{e_+ - e_-}{2}\right)^2 + \left(\frac{x_+ - x_-}{2}\right)^2\right]^{1/2}, \end{aligned}} \tag{4.29}$$

the half-widths added in quadrature as independent errors.

4.8.2 The sources

(M) *The envelope against the kinetic solution.* The two models of Sec. 4.7 bracket the answer: $m_\mp$ are the smaller and the larger of 1 and $f_{\text{kin}}/f_{\text{env}}$. The half-width is 0.2 to 12 % in pure Kr. At the five source-set cells of Kr/F_2 the band is instead f_{best} over $T_e = 1$ to 4 eV, 0.79 to 2.5 times the envelope, and its half-width, 27 to 62 %, stands in for (M).

(I) *The electron–ion term against the measurements.* Turnbull et al. measured absorption fractions of 1 to 4 % in hydrogen and nitrogen–hydrogen plasmas with an absolute uncertainty of (−0.12 %, +0.07 %), found their best shots predicted to better than 10 %, and their models 16 % above the data on average without the fitted screening factor [10, 11]. The electron–ion absorption of Eqs. (4.14) and (4.15), with the dynamic cutoff and the exact Gaunt factor and no fitted factor, is 1.19 to 1.20 times their four hydrogen measurements. Taking the measurements as right within 10 %, the ion term is 0.75 to 0.92 of the computed one, and Eq. (4.28) with the midpoint 0.84 gives $i_\mp$: the half-width is 0.0 to 1.0 % in pure Kr, the most at 10^{16} cm^{-3} and 0.1 atm, and up to 4.3 % at the cold cells, where the ions carry most of the absorption.

(E) *The momentum-transfer cross section against swarm experiments.* The atoms carry the absorption, so the cross section σ_m of krypton is the input that matters, and the electron drift velocity W in a DC field is its experimental test: for $\omega \gg \nu$ the absorption weights σ_m as Eq. (4.12) does, and the drift velocity weights $1/\sigma_m$ over the same electron energies, so that $\delta W/W \simeq -\delta\sigma_m/\sigma_m$. The two-term equation [4] with the same Biagi set gives W at each field strength of two measurements in pure krypton, Pack, Voshall and Phelps [13] at 300 K, as compiled by Dutton [14], and Nakamura [15], read from the figure of the published abstract to about 3 %. Each measured point is placed at the electron temperature $T_e = \frac{2}{3}\langle\varepsilon\rangle$ of the computed swarm. Between 0.95 and 2.6 eV, the range of the warm cells, the computed W is 0.96 to 1.02

times Nakamura's (rms 1.8 %, 13 points) and 0.84 to 0.97 times Pack's (rms 8.4 %, 7 points), and the two experiments differ from each other by as much. Below that range only Pack's data exist, and across the Ramsauer minimum the computed W is 0.80 to 1.20 times them. For each cell the points within ±25 % of its T_e give the rms deviation of each experiment, and their quadratic mean d weights the two experiments equally. Since only the atoms' share $1 - s_i$ of ν_{eff} depends on σ_m, and $f_{\text{best}} \propto \nu_{\text{eff}}^{1/2}$, the half-width is

$$\boxed{\frac{e_+ - e_-}{2} = (1 - s_i)\,\frac{d}{2},} \tag{4.30}$$

1.4 to 4.0 % in pure Kr. The same deviation, taken as an uncertainty of σ_m, replaces an assumed one: had σ_m been taken as good to 10 %, Eq. (4.28) and the cells recomputed with $\sigma_m \times 0.9$ and $\times 1.1$ give 3.7 to 5.0 % in pure Kr. Adding both would count the cross section twice, so Eq. (4.29) carries (E), and the 10 % figure stands as a bound that the swarm data do not reach.
(X) *The inelastic cross sections of krypton*, taken as good to 20 %, move T_e through Eq. (4.19) and with it f_{best} by 1.0 to 1.7 % in pure Kr.
(D) *The recombination coefficient*, ±10 % as stated [16], moves the self-sustained density of pure Kr by −4.0 to +4.6 % and its frequency by ∓1.9 %. Its extrapolation beyond 0.72 eV is not bounded.

4.8.3 The result

Table 4.3 gives $f_{\text{best}} \pm \delta$ at the 18 cells with the four contributions. In pure Kr δ/f_{best} is 2 to 12 %, and (M) and (E) together carry most of δ^2 at eight of the nine cells. In Kr/F_2 it is 3 to 7 % at the four field-set cells and 27 to 62 % at the five source-set ones, where the temperature the source chooses is the uncertainty.

Table 4.3: The best frequency with its uncertainty at every cell of Table 4.2, and the four contributions to δ in per cent. ^{t}: the source sets T_e, and the half-width of f_{best} over $T_e = 1$ to 4 eV stands in for (M).

gas	n_e [cm^{-3}]	p [atm]	$f_{\text{best}} \pm \delta$ [THz]	δ [%]	(M)	(I)	(E)	(X)
Kr	10^{14}	0.1	1.1 ± 0.1	5	3	0.0	4.0	1.4
Kr	10^{14}	0.5	2.1 ± 0.1	5	4	0.0	2.6	1.6
Kr	10^{14}	2	3.6 ± 0.2	6	5	0.0	2.6	1.6
Kr	10^{15}	0.1	3.1 ± 0.2	6	5	0.1	2.9	1.7
Kr	10^{15}	0.5	6.2 ± 0.6	9	9	0.0	1.7	1.7
Kr	10^{15}	2	11.3 ± 1.3	12	12	0.0	1.9	1.7
Kr	10^{16}	0.1	8.5 ± 0.2	2	0	1.0	1.4	1.0
Kr	10^{16}	0.5	15.8 ± 0.6	4	3	0.4	1.6	1.4
Kr	10^{16}	2	27.9 ± 1.7	6	5	0.1	2.0	1.4
Kr/F_2	10^{14}	0.1	1.1 ± 0.1	5	2	0.0	4.0	1.4
Kr/F_2	10^{14}	0.5	2.2 ± 0.1	3	1	0.0	2.6	1.6
Kr/F_2	10^{14}	2	4.0 ± 0.2	5	5	0.0	1.9	1.5
Kr/F_2	10^{15}	0.1	3.1 ± 0.2	7	6	0.1	2.9	1.6
Kr/F_2	10^{15}	0.5	6.6 ± 2.5	38	38^t	0.0	1.7	1.5
Kr/F_2	10^{15}	2	13.1 ± 6.1	47	47^t	0.0	1.8	1.1
Kr/F_2	10^{16}	0.1	10.0 ± 2.7	27	27^t	1.1	1.4	0.7
Kr/F_2	10^{16}	0.5	19.5 ± 9.2	47	47^t	4.3	0.4	0.0
Kr/F_2	10^{16}	2	38.9 ± 24.1	62	62^t	3.7	2.1	0.0

In Summary: The uncertainty of the best frequency is Eq. (4.29) from the Maxwellian envelope against the kinetic solution, the electron–ion term against Turnbull's data, the momentum-transfer cross section against the swarm measurements, and the inelastic cross sections. It is 2 to 12 % in pure Kr, since the atoms carry the absorption, their cross section reproduces the measured drift velocities to a few per cent, and the ion term that Turnbull's measurements test contributes at most 1 %, and it is the source's temperature wherever F_2 makes a source necessary.

4.9 A worked example

Pure Kr at $n_e = 10^{15}$ cm^{-3}, $p = 0.5$ atm, $P = 300$ kW/cm^3 and $L = 10$ cm, the central cell of Table 4.2, carried through from the gas density to the recommended frequency.

(1) *The gas and the power per electron.* $N = 2.45 \times 10^{19} \times 0.5 = 1.22 \times 10^{19}$ atoms per cm^3, and $P/n_e = 3 \times 10^5$ W cm$^{-3}/10^{15}$ cm$^{-3} = 1.87 \times 10^9$ eV/s per electron.

(2) *The temperature.* Equation (4.19) balances at $k_B T_e = 1.42$ eV: the recoil of Eq. (4.13) takes 3.3×10^7 eV/s, 1.8 %, and krypton's levels take 1.84×10^9 eV/s, the rest.

(3) *The collision frequency.* The kernel (4.12) gives $\nu_{ea} = 1.22 \times 10^{12}$ s^{-1}, 2.10 times the Drude mean 5.8×10^{11} s^{-1}. For the ions, Eq. (4.14) gives $\nu_T = 1.71 \times 10^9$ s^{-1}, $\eta_t = 4.4$ puts the collisions in the classical regime, $\Lambda = 3.65$ where the Born logarithm would give 4.90, and $\nu_{ei} = 6.3 \times 10^9$ s^{-1}, 0.5 % of $\nu_{\text{eff}} = 1.22 \times 10^{12}$ s^{-1}. So $\nu_{\text{eff}}/2\pi = 0.19$ THz.

(4) *The cutoff and the reflection limit.* $f_{pe} = 8.98$ kHz $\times (10^{15})^{1/2} = 0.28$ THz, and $f_R = 1.294 f_{pe} = 0.37$ THz without the collisions, 0.32 THz with them in Eq. (4.5).

(5) *The absorption length.* Equation (4.23) with $n_e/10^{15} = 1$, $\nu_{\text{eff}}/10^{12} = 1.22$ and $L/10$ cm $= 1$ gives $f_A = 5.2 \times (1.22)^{1/2} = 5.73$ THz, and the exact root of $\alpha L = 1$ is 5.73 THz. So $f_{\text{best}} = 5.7$ THz, far above f_R. There $u = 0.017$ and $\alpha_L = 2.2 \times 10^{-3}$, so the soft-photon kernel and the absence of Langdon's flattening both hold.

(6) *The field.* Equation (4.21) gives $E_0 = 47.6$ kV/cm, an intensity of 3 MW/cm^2.

(7) *GHz instead?* At 95 GHz, $\nu_{\text{eff}}/\omega = 2.1$: the wave is in the collisional skin, with an absorption length of 0.17 mm and 31 % reflected. At 2.45 GHz, 1.1 mm and 80 % reflected. At 1 THz the gas is already transparent, $R = 0$, but the absorption length is 3.0 mm and the front centimetre takes everything. At 3 THz it is 2.7 cm. The depth of the gas is matched on the transparent side, at 5.7 THz.

(8) *The density.* Direct ionization runs at $\nu_{\text{ion}} = 1.6 \times 10^7$ s^{-1} against recombination at 1.8×10^8 s^{-1} (5.7 ns per electron), so this density decays unless a source supplies at least 360 kW/cm^3 by Eq. (4.26), more than the field. Left to itself at this pressure and power, pure Kr settles at 3.6×10^{14} cm^{-3} and $k_B T_e = 1.62$ eV in 17 ns, where $f_{\text{best}} = 3.7$ THz.

(9) *The uncertainty.* The kinetic solution of Sec. 4.7 gives $k_B T_e = 1.80$ eV and $f_{\text{best}} = 6.7$ THz, 1.17 times the envelope, so (M) contributes a half-width of 8.7 %; (I) 0.02 %, since the ions carry 0.5 %; (E) 1.7 %, from the 9 swarm points within ±25 % of 1.42 eV, where the computed drift

velocity deviates from Nakamura's by 1.2 % rms and from Pack's by 4.6 %, so that $d = 3.3\,\%$; (X) 1.7 %. Equation (4.29) gives $\delta/f_{\text{best}} = 9.0\,\%$ about the midpoint:

$$\boxed{f_{\text{best}} = 6.2 \pm 0.6 \text{ THz.}} \tag{4.31}$$

(10) *With 2 % F_2 at the same point.* $k_B T_e = 1.37$ eV, F_2's channels taking 26 % of the losses, and $f_{\text{best}} = 5.6$ THz, 2 % below pure krypton's. But attachment runs at 6.1×10^8 s^{-1}, 1.6 ns per electron, the source must supply at least 1.7 MW/cm^3, six times the field's power, and in the kinetic solution it takes 0.60 of the field's power from the electrons, so it sets T_e and the frequency is 6.6 ± 2.5 THz, 38 %.

In Summary: At 10^{15} cm^{-3}, 0.5 atm and 300 kW/cm^3 in a 10 cm gas, the best frequency is 6.2 ± 0.6 THz, Eq. (4.31). The GHz range is out: 31 to 80 % reflected and the rest absorbed in 0.2 to 1 mm. The density is not free: pure Kr at this density needs a source of 360 kW/cm^3 or settles at 3.6×10^{14} cm^{-3}, where it wants 3.7 THz, and with F_2 the source needs 1.7 MW/cm^3 and sets the temperature.

4.10 Summary

In Summary: The best frequency for inverse-bremsstrahlung heating of a krypton laser gas is the one whose absorption length matches the depth of the gas, Eq. (4.1), and it is Eq. (4.23): the square root of the electron density, of the collision frequency and of the depth. The collision frequency is that of the atoms, Eq. (4.12), which the ions supplement by a few per cent at 10^{15} cm^{-3} and by up to a quarter at 10^{16}, and it depends on the electron temperature that the power balance (4.19) sets, 0.95 to 2.6 eV where the field sets it. Pure Kr wants 1.1 ± 0.1 THz at 10^{14} cm^{-3} and 0.1 atm, rising as $(n_e p)^{0.4}$ and $L^{1/2}$ to 27.9 ± 1.7 THz at 10^{16} cm^{-3} and 2 atm, with 2 to 12 % uncertainty, and left to itself it runs at 1.6 to 6×10^{14} cm^{-3} and wants 1.5 to 7 THz. The THz range is the right one: across the spectrum the absorption peaks at 0.05 to 0.50 THz, between the bands, and the GHz range either reflects, 31 to 94 % from 10^{15} cm^{-3} up, or is absorbed in a millimetre skin, while the THz range enters the gas and deposits over centimetres. 2 % F_2 leaves the frequency alone wherever the krypton excitation takes the power. But F_2 attaches the electrons in 0.1 to 17 ns, so a source sets the density and, from 10^{15} cm^{-3} up, the temperature, and the frequency is then known only to the source's temperature, 27 to 62 %. Neither gas holds 10^{16} cm^{-3} at 100 to 500 kW/cm^3. At the worked example, pure Kr at 10^{15} cm^{-3} and 0.5 atm, the answer is 6.2 ± 0.6 THz.

4.A The numerical checks

A check script verifies every boxed result of these notes, in their order, by an independent numerical route: the Drude optics against the exact complex refractive index, the energy-weighted kernel (4.12) against the velocity-space integral (4.11) by quadrature and against the 4/3 of a constant cross section, the rate (4.20) and the recoil loss (4.13) against quadrature and against the closed form of a step cross section, the Spitzer prefactor, the Born and classical limits of the logarithm, the reflection limit (4.22), the closed form (4.23) against the exact root, the field (4.21) and Langdon's parameter (4.18) against the solved cells, the particle balance, the fits (4.27), the collisional-skin law (4.24), the swarm comparison (4.30) with the drift velocities recomputed and the measured values read from their data files, the combination (4.29), and every step of the worked example of Sec. 4.9.

It reproduces every number printed in these notes to its last printed digit from the stored results of the calculation of the cells and of the swarm comparison, checks the quoted figures of Turnbull et al. against the text of their papers, and checks the logical statements. It runs with `numpy` and `scipy` under Python 3 in a few seconds from those stored results, which the scripts behind them take about four minutes to produce. Table 4.4 is written by the script itself. A quantitative check passes when its deviation is below the tolerance stated in the script, a printed number when the script reproduces it to its last printed digit, and a logical condition when it holds.

Table 4.4: The numerical checks of these notes, by the equation, table or section they verify. The second column counts all checks of the row. The third gives the largest deviation among its quantitative checks, relative, or absolute for quantities near zero, and is empty when the row has none. The fourth counts the numbers printed in these notes that the script reproduces to their last printed digit, and the fifth the logical conditions.

Equation	Checks	Largest deviation	Printed	Logical
Sec. 4.1	2		1	1
Eq. (4.1)	5		2	3
Eq. (4.4)	2	0		1
Eq. (4.5)	10	1.3×10^{-9}	6	2
Eq. (4.6)	2	1.0×10^{-4}		
Eq. (4.7)	1	0		
Eq. (4.3)	1	0		
Eq. (4.12)	8	1.7×10^{-7}	5	
Sec. 4.2.2	1		1	
Eq. (4.13)	4	1.7×10^{-7}	2	1
Eq. (4.14)	5	1.3×10^{-3}	2	
Eq. (4.16)	3	1.1×10^{-9}	1	
Eq. (4.15)	4	4.2×10^{-3}	1	1
Sec. 4.2.3	7	2.7×10^{-6}	6	
Eq. (4.17)	12		9	3
Sec. 4.2.4	2		1	1
Eq. (4.18)	4	5.7×10^{-2}	2	1
Eq. (4.19)	6	1.3×10^{-13}	3	1
Eq. (4.20)	3	8.3×10^{-5}		
Sec. 4.3	21		19	2
Eq. (4.21)	6	2.7×10^{-10}	4	1
Eq. (4.22)	10	1.0×10^{-13}	7	1
Sec. 4.4.1	2		1	1
Eq. (4.23)	8	2.2×10^{-3}	5	2
Sec. 4.5	42		36	6
Eq. (4.24)	5	0	3	1
Table 4.1	10			10
Table 4.2	19			19
Eq. (4.25)	6		3	3
Sec. 4.6.2	36		33	3
Eq. (4.26)	4		1	3
Eq. (4.27)	9	0		3

Equation	Checks	Largest deviation	Printed	Logical
Sec. 4.6.3	7		6	1
Sec. 4.6.4	7		5	2
Sec. 4.10	15		14	1
Sec. 4.6	1			1
Sec. 4.7	16		13	3
Eq. (4.28)	3	3.9×10^{-3}	2	
Eq. (4.29)	11		8	3
Sec. 4.8.2	29		21	8
Eq. (4.30)	23	0	11	8
Sec. 4.8.3	7		6	1
Table 4.3	19			19
Sec. 4.9	22	0	16	4
Eq. (4.31)	5	0	2	1
Appendix 4.A	4			4
Total	429		258	126

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Chapter 5

The best frequency for inverse bremsstrahlung in excimer laser gases: a krypton fill, the measured 5s to 5p rate, and the pulse

The note of October 12, 2026.

5.1 The question and the answer

A wave of frequency f heats the electrons of an excimer laser gas by inverse bremsstrahlung. To deposit a power density P evenly through a layer of depth L, it has to be absorbed over that depth and not reflected at the surface. The best frequency f_{best} is the frequency that does both, defined in Section 5.2. It depends on the electrons through their temperature and density and on the atoms and ions they collide with, and the field that heats the electrons also excites the gas. The excited atoms are ionized by a second electron at under a third of the energy the ground state needs, they ionize each other and the krypton, and they trade energy with the electrons both ways. This note follows them inside both balances of the electrons, the balance of their number and the balance of their energy, at every point of a table of three fills, two buffered and one of krypton with 0.5 % F_2, five pressures and five electron densities, 75 points in all, and it does four things the earlier form of the calculation could not.

First, the excitation of krypton's 5s levels to 5p, the rate that carries most of the cooling of the electrons, is taken from the measured cross sections of Jung et al. [11, 12] rather than from a discharge model's fit, and the 5p manifold radiates to 5s at the measured lifetimes [29]. Second, the krypton fill is added, with no buffer gas. Its best frequency lies 1.04 to 1.20 times the argon fill's at the same pressure and density, 1.05 to 8.8 THz at 0.1 atm and 3.8 to 26.9 THz at 2 atm, with its electrons at 0.94 to 2.44 eV, and the beam holds the density at 11 of its 25 points. With the measured cross section the neon fill lies at 0.933 to 0.997 and the argon fill at 0.646 to 0.995 of the calculation without excited atoms. The fit in place of the measurement raises the best frequency by up to 5 %, the 5p rung's electron collisions left out raise it by up to 1.46 times, and the return of 5p to 5s by collisions, raised from the estimate used here to the one measured total loss rate of a 5p level [28], lowers it by up to 13 % at the dense krypton points and the electron temperature there by 19 %: that return rate is now the largest uncertainty of the answer. Third, the calculation

is marched in time through a pulse of 1 μs at three points, with the F_2 burning and the density following its own balance. The steady table is the state of the first 100 ns at 10^{15} cm^{-3} and of the first 10 ns at 10^{16} cm^{-3}: the attachment that holds the density also burns the F_2, which is gone in 0.36 to 0.38 μs at 10^{15} cm^{-3} and in 35 ns at 10^{16} cm^{-3}, after which the density runs up 3.7 to 6.8 times to where the beam's pairs balance recombination, the electrons cool to 0.53 to 0.54 eV and the best frequency moves, by -18 and -6 % at 10^{15} cm^{-3} and by 2.8 times at 10^{16} cm^{-3}. Fourth, the kinetic solver is checked against a second solver of the same equation, written independently for a time-domain model of the KrF laser: at seven states from 10^{12} to 10^{16} cm^{-3} the two agree within 0.2 % in the electrons' mean energy and in the wave's absorbed power, and the two full models, started from the same cold seed under the same wave, decay the plasma at rates 1 % apart with electrons 0.3 % apart in temperature.

5.2 The gas, the electrons and the best frequency

The table. The three fills are Ne/Kr/F_2 in the mole fractions 0.948/0.05/0.002, Ar/Kr/F_2 in 0.597/0.4/0.003 and Kr/F_2 in 0.995/0.005, at 300 K and pressures of 0.1, 0.2, 0.5, 1 and 2 atm. The electron density n_e is 10^{14}, 3×10^{14}, 10^{15}, 3×10^{15} or 10^{16} cm^{-3}, and the field deposits $P = 300$ kW/cm^3 in a layer of depth $L = 10$ cm. At each of these 75 points the frequency runs over 13 values, five per decade from 0.1 to 25.1 THz.

The electrons. The electrons' energy distribution $F(\epsilon)$ solves the two-term Boltzmann equation in the oscillating field [1], with the cross sections of Biagi's database for neon, argon, krypton and F_2 [2], electron–electron collisions, and collisions with the cations through the Coulomb cross section $\pi\,(e^2/4\pi\epsilon_0\epsilon)^2\Lambda(\epsilon,\omega)$. The logarithm Λ is the quantum one of Turnbull et al. for the photon energy $\hbar\omega$ [3, 4], cut off at the ion-sphere radius of the cations. The field's amplitude is set so that the electrons absorb P. The electrons' complex mobility $\mu(\omega)$ gives the conductivity $\sigma = n_e e\mu$ and the dielectric function $\varepsilon = 1 + i\sigma/\epsilon_0\omega$, and from them the absorption coefficient and the reflection at normal incidence,

$$\alpha = \frac{2\omega}{c}\,\mathrm{Im}\sqrt{\varepsilon}, \qquad R = \left|\frac{1-\sqrt{\varepsilon}}{1+\sqrt{\varepsilon}}\right|^2 . \tag{5.1}$$

α falls with f once ω exceeds the collision and plasma frequencies. f_A is the frequency at which $\alpha L = 1$, so that the wave is absorbed over the layer, and f_R the lowest at which $R \le 0.05$. The best frequency is

$$f_{\mathrm{best}} = \max\left(f_A,\, f_R\right), \tag{5.2}$$

both found by interpolating $\ln\alpha$ and $\ln R$ in $\ln f$ between the frequencies of the scan.

The ions and how the density is held. Electrons are lost to attachment on F_2 and to dissociative recombination on molecular cations. The two buffered fills have five cations each, the atomic and the molecular ion of the buffer gas, the heteronuclear ion of the buffer with krypton, Kr^+ and Kr_2^+, and the krypton fill has Kr^+ and Kr_2^+. They are made by the field's ionization of each gas and by an electron beam, they convert into one another, and they recombine with the electrons and with F^-, through the reactions of Table 5.10 in Appendix 5.B. The recombination with F^- turns over with pressure, as the three-body capture calculations that Kannari et al. review do [7], and the other charges screen each pair, which multiplies it here by $S_c = 1/(1 + R_c/\lambda_D)$ with $R_c = 2e^2/(3\cdot 4\pi\epsilon_0 kT_g)$ and λ_D the Debye length of all the charges at the gas temperature T_g. Where the ionization falls short of the losses, the beam makes up the difference and holds n_e.

Table 5.1: The levels whose energy reaches the electrons, as formation energies above the ground-state atoms, in eV.

species	energy	origin
Kr^*	9.915	lowest 5s level, 9.9152, [6]
Kr^{**}	11.300	lowest 5p level, 11.3035, [6]
Kr^+	13.996	ionization limit, 13.9996, [6]
Ar^*	11.548	lowest 4s level, 11.5484, [6]
Ar^+	15.760	ionization limit, 15.7596, [6]
Ne^*	16.619	lowest 3s level, 16.6191, [6]
Ne^+	21.560	ionization limit, 21.5645, [6]
Kr_2^+	12.846	Kr^+ less the binding 1.15 measured, [19]
Ar_2^+	14.530	Ar^+ less the binding 1.23 measured, [19]
Ar_2^*	10.868	taken
Kr_2^*	9.200	taken
Ne_2^*	16.100	taken
Ne_2^+	20.200	taken
$ArKr^*$	9.500	taken
$ArKr^+$	13.500	taken
$NeKr^+$	13.700	taken

Its electrons enter with the energy spectrum of the slow secondaries of a fast beam, whose mean is 4.506 eV, and they are marked b where they bring more than a third of the absorbed power. Where the ionization exceeds the losses, the field multiplies the electrons at the net rate g, and the point is marked a where g exceeds one e-fold in the pulse of 1 μs. A point marked a is not steady over the pulse, and its best frequency is that of the density it passes through.

5.3 The excited atoms

The levels. The field excites krypton into its four 5s levels and into 5p and the levels above. They are followed as two species, Kr^* for the 5s levels and Kr^{**} for 5p and above, each standing at the energy of its lowest level, 9.915 and 11.30 eV, so that no excitation booked into a species creates energy. The buffer gas is followed the same way, Ar^* at argon's lowest 4s level, 11.548 eV, and Ne^* at neon's lowest 3s level, 16.619 eV, and four excited dimers form from these atoms, mostly in three-body collisions, Kr_2^*, Ar_2^*, $ArKr^*$ and Ne_2^*. The krypton fill carries Kr^*, Kr^{**} and Kr_2^*. Every energy is a formation energy above the ground-state atoms, Table 5.1. Those of the atoms lie within 0.005 eV of the levels and ionization limits of the NIST database [6], those of Kr_2^+ and Ar_2^+ follow from the measured binding of the two ions [19], and the others are taken as the table gives them.

The field makes Kr^* and Kr^{**} through the excitation cross sections of the Boltzmann equation, the 5s group into Kr^* and every higher group into Kr^{**}, and the buffer's excitations into its own species. The beam shares its energy between the gases by their stopping powers for fast electrons, from Bethe's formula at 600 keV with mean excitation energies of 137, 188, 352 and 115 eV for Ne, Ar, Kr and F_2, and makes one ion pair per 36.6 eV in neon [23], 26.2 eV in argon and 32 eV in F_2 [7], and 24.2 eV in krypton, where Kannari et al. print 24.3. With each pair it makes 0.305 excited atoms in neon and 0.284 in argon, the energies per pair of Lee et al. over their energies per excited atom, 120 and 93 eV [23], and 0.28 in krypton [7]. The recombination of the molecular cations leaves an excited atom behind. The excited atoms are lost to F_2, which turns Kr^* into KrF^* at 8.47×10^{-10} cm^3 s^{-1}, the mean of three measurements, with a yield of 0.90 [13, 14], to the

three-body formation of dimers, to radiation, to Penning ionization, and to the electrons. Tables 5.9 and 5.10 of Appendix 5.B list every reaction between heavy particles with its rate and its source, and Table 5.2 the electron collisions. Each species stands at the steady state of its balance, solved for all of them at once at the electrons' distribution. Where the field multiplies the electrons at the rate g, every density grows with them, the ions as the excited atoms, and each species is diluted at gn on top of its losses.

Table 5.2: The electron collisions of the excited atoms as the Boltzmann equation carries them. The rate is the source's in $\mathrm{cm^3\,s^{-1}}$ with T_e in eV, E_0 the onset of the cross section of Eq. (5.3) in eV, and Δ the energy the electron loses, negative where it gains. The standing says whether the rate is measured, printed by its source from elsewhere, calculated, the value of a related process, estimated, or detailed balance.

reaction	rate	E_0	Δ	standing	source
$e + \mathrm{Kr^*} \to \mathrm{Kr^+} + 2\,e$	$7.00 \times 10^{-8}\,T_e^{0.68} e^{-4.04/T_e}$	4.081	+4.081	calculated	the first fits the calculation of the second, [7, 9]
$e + \mathrm{Kr^*} \to e + \mathrm{Kr^{**}}$	measured cross section, Sec. 3	1.385	+1.385	measured	the two metastable 5s levels into the ten 5p levels, lumped over the 5s manifold 5:3:1:3 (Sec. 3), [11, 12]
$e + \mathrm{Kr^{**}} \to e + \mathrm{Kr^*}$	Eq. (5.4), $g_l/g_u = 1/3$	0.000	-1.385	detailed balance	of the measured excitation, [5]
$e + \mathrm{Kr^{**}} \to \mathrm{Kr^+} + 2\,e$	$2.00 \times 10^{-7}\,T_e^{0.62} e^{-2.53/T_e}$	2.696	+2.696	calculated	[7, 9]
$e + \mathrm{Kr^*} \to e + \mathrm{Kr}$	Eq. (5.4), $g_l/g_u = 1/12$	0.000	-9.915	detailed balance	of the 5s excitation, [5, 2]
$e + \mathrm{Ar^*} \to \mathrm{Ar^+} + 2\,e$	$1.56{\times}10^{-7}\,T_e^{0.71} e^{-4.212/T_e}$	4.212	+4.212	analog	the law R9 of the source for its higher excited argon, at the 4s binding energy, [25]
$e + \mathrm{Ar^*} \to e + \mathrm{Ar}$	Eq. (5.4), $g_l/g_u = 1/12$	0.000	-11.548	detailed balance	[5, 2]
$e + \mathrm{Ar^*} \to e + \mathrm{Ar}$	Eq. (5.4), $g_l/g_u = 1/12$	0.000	-11.623	detailed balance	[5, 2]
$e + \mathrm{Ar^*} \to e + \mathrm{Ar}$	Eq. (5.4), $g_l/g_u = 1/12$	0.000	-11.723	detailed balance	[5, 2]
$e + \mathrm{Ar^*} \to e + \mathrm{Ar}$	Eq. (5.4), $g_l/g_u = 1/12$	0.000	-11.828	detailed balance	[5, 2]
$\mathrm{ArKr^*} + e \to \mathrm{ArKr^+} + 2\,e$	$9.00 \times 10^{-8}\,T_e^{0.7} e^{-3.6/T_e}$	4.000	+4.000	printed	[7]
$\mathrm{ArKr^*} + e \to \mathrm{Ar} + \mathrm{Kr} + e$	1.0×10^{-7}	0	-9.500	estimated	estimated there, [7]
$e + \mathrm{Ne^*} \to \mathrm{Ne^+} + 2\,e$	$3.04 \times 10^{-8}\,T_e^{0.54} e^{-7.5/T_e}$	7.500	+4.941	printed	[23]
$e + \mathrm{Ne_2^*} \to \mathrm{Ne_2^+} + 2\,e$	$9.75 \times 10^{-9}\,T_e^{0.71} e^{-3.4/T_e}$	4.100	+4.100	printed	[25]

The 5s to 5p excitation is measured. Jung et al. measured the cross sections for electron-impact excitation out of krypton's two metastable 5s levels, $1s_5$ and $1s_3$ in Paschen's notation, into each of the ten 5p levels, from 2 to 300 eV, with uncertainties of 35 % for $1s_5$ and 40 % for $1s_3$ [11, 12]. The cross section used here for $\mathrm{Kr^*} \to \mathrm{Kr^{**}}$ is their sum over the ten 5p levels, lumped over the 5s manifold by statistical weight, $1s_5 : 1s_4 : 1s_3 : 1s_2 = 5 : 3 : 1 : 3$, with each unmeasured resonance level taken as its core partner, $1s_4$ as $1s_5$ and $1s_2$ as $1s_3$, since the core-preserving cross sections

scale with the optical oscillator strengths. Each level's cross section is zero below its threshold, rises linearly to the first tabulated point, follows the table log-log between the points, and falls as $1/\epsilon$ past the last. Its onset on the ladder is the gap between the lumped levels, 1.385 eV. On Maxwellians this cross section gives 1.36 times the rate of the discharge model's fit that Kannari et al. print [7, 10] at the coldest electrons of the table, 0.87 eV, and 0.92 times at the hottest, 4.69 eV, differences at or inside the measurement's own uncertainty. The fit is kept as the upper end of a band, Section 5.6.

The 5p manifold returns to 5s three ways. Kr^{**} gives its energy back to the electrons in superelastic collisions, the detailed balance of the excitation, and it returns to 5s by radiation and in collisions with the gas. The ten 5p levels radiate to the 5s levels in the near infrared, on lines that end on the 5s levels and are not trapped, with the lifetimes that Dzierzega et al. derive from the beam-gas-laser measurements of six of the levels, 23.1 to 38.1 ns [29]. The lumped manifold decays at the $2J+1$-weighted mean of their rates, one lifetime of 29.0 ns, and the 1.385 eV leaves as light. The return by collisions is the estimate of 1.0×10^{-10} $\mathrm{cm^3\,s^{-1}}$ that Kannari et al. mark as such for Kr^{**} + Ar [7], used for every partner. One measurement in krypton exists: Gazeli et al. measured the total loss rate of the $5p'[3/2]_2$ level in krypton at 300 K, $(1.7 \pm 0.2) \times 10^{-10}$ $\mathrm{cm^3\,s^{-1}}$, without resolving its product [28]. It bounds the return of the lumped manifold from above and is run as the upper end of a band. The level-resolved rates of Whitehead et al. [30] would settle it and are not used here. At the krypton fill at 1 atm and 10^{15} $\mathrm{cm^{-3}}$, at 10 THz, the manifold returns 84 % by collisions, 15 % superelastically and 1.2 % by radiation.

Stepwise ionization has no measurement. No measurement of the ionization of krypton's 5s levels by electrons was found, as a cross section or as a rate. Kannari et al. print $7.00 \times 10^{-8}\, T_e^{0.68} e^{-4.04/T_e}$ $\mathrm{cm^3\,s^{-1}}$ for it from Vriens's binary-encounter calculation [7, 9], and Johnson and Hunter print 4.8×10^{-8} from the same calculation [8]. This note takes Kannari's fit. The stepwise ionization of Ar^* has only a transplanted law, that of Chen et al. for the higher of their two excited argon species [25] at argon's 4s binding energy. In the calculation with Kannari's 5s to 5p fit, a stepwise rate 2.1 to 2.3 times Kannari's, or none at all, moved the best frequency by less than 1.5 %, so the answer does not rest on this rate and no band is run on it here.

5.4 The excited atoms inside the electrons' balances

Every electron collision of an excited atom is a process of the Boltzmann equation. Each excited species enters the Boltzmann equation as a gas of its own, at the mole fraction its balance gives, with every collision of an electron with it that Table 5.2 lists. A rate fitted as $k = A\,T^B e^{-E/T}$ becomes the cross section

$$\sigma(\epsilon) = C\,\frac{(\epsilon - E_0)^{B+1/2}}{\epsilon} \quad (\epsilon > E_0), \qquad C = \frac{A\sqrt{\pi}}{2\gamma\,\Gamma(B+3/2)}, \tag{5.3}$$

with $\gamma = (2e/m_e)^{1/2}$, whose rate over a Maxwellian of temperature T is exactly $A\,T^B e^{-E_0/T}$. The electron loses in it the energy Δ of the reaction on the ladder of formation energies of Table 5.1. An electron below Δ cannot pay it, so the onset E_0 is the larger of the fit's E and Δ. Where the fit's E lies above Δ the printed rate is kept and the electron loses Δ. Where it lies below, the onset moves up to Δ, which lowers the rate on a Maxwellian by $e^{-(\Delta - E)/T}$. A constant rate k is the cross section $k/\gamma\sqrt{\epsilon}$, and a measured cross section enters as it is, above the ladder's gap.

De-excitation by detailed balance. In a superelastic collision the electron gains the energy Δ of the level it empties. Its cross section follows from that of the excitation it reverses by detailed balance [5],

$$\sigma_{\downarrow}(\epsilon) = \frac{g_l}{g_u}\,\frac{\epsilon+\Delta}{\epsilon}\,\sigma_{\uparrow}(\epsilon+\Delta), \tag{5.4}$$

with g_l and g_u the statistical weights of the lower and the upper level. Over a Maxwellian its rate is $(g_l/g_u)\,e^{\Delta/T}$ times that of the excitation, and when the electrons are in equilibrium with the excited atoms the two flows cancel. The Boltzmann operator places the electron Δ above the energy it had, so that the equation is no longer solved from the top of the energy grid down and is solved as one linear system.

The electrons that the excited atoms set free. Penning ionization, between two excited atoms or dimers or between one of them and ground-state krypton, releases an electron with the energy that the reaction leaves on the ladder, from 2.1 to 12.0 eV. These electrons enter the Boltzmann equation as a source at their release energy, each in a band 0.2 eV wide.

The two closures. The Boltzmann equation is solved in steady state with a source of electrons. Where the beam holds the density, the source is the beam's slow secondaries and the Penning electrons, mixed in proportion to their rates, and the rate of injection per electron is what the losses leave over,

$$s = N\,(k_{\mathrm{att}} + k_{\mathrm{rec}} - k_{\mathrm{ion}}) = \frac{S_{\mathrm{beam}} + S_{\mathrm{pen}}}{n_e}, \tag{5.5}$$

with N the gas density and k_{ion} the ionization of every gas of the equation, the excited ones among them. Where the field multiplies the electrons at the rate g, the Penning electrons are the source and g enters as a loss from every energy in proportion to the electrons there, the dilution that growth brings, set so that

$$N\,(k_{\mathrm{att}} + k_{\mathrm{rec}} - k_{\mathrm{ion}}) + g = \frac{S_{\mathrm{pen}}}{n_e}. \tag{5.6}$$

Without a source the second closure is the growth model of the two-term equation [1]. It leaves out the share of the growth term in the anisotropic part of the distribution, which is of the order of g/ν_m, the growth rate over the momentum-transfer frequency, below 3×10^{-3} at every growing point. The cations follow the same particle balance, with the excited atoms' stepwise electrons added to the ionization of their ground gas and the dimers' and the Penning electrons to their own cations.

The loop. Each round solves the Boltzmann equation with the excited atoms at their current mole fractions, then their balance with the rates over the new distribution, then the ions, and sets the closure and the source of the next round. The mole fractions are relaxed in their logarithm, and the loop ends when they, the ions and the beam's or the growth rate change by less than 2×10^{-3} between two rounds and the closure no longer changes. A growth rate's change is measured against the larger of itself and 10^6 s^{-1}, one e-fold in the pulse, below which a growth rate multiplies nothing and its relative change says nothing. At the boundary between the closures the two can alternate from round to round, with the beam's rate alternating between zero and a finite value and the excited atoms differing by up to 1 % between the two states, which is one and the same plasma within that: a point whose excited atoms and ions have settled within ten tolerances and whose closure has alternated for three rounds is accepted and flagged as a boundary point. It took 3 to 72 rounds per point, the most at the one boundary cell of Section 5.6. Each frequency starts from

Table 5.3: The best frequency f_{best} in THz at $L = 10$ cm and 300 kW/cm^3 with the excited atoms inside the electrons' balances. Marked a where the field multiplies the electrons more than e-fold in the pulse of 1 μs, and b where the beam's electrons bring more than a third of the absorbed power.

fill	p [atm]	$n_e = 10^{14}$	3×10^{14}	10^{15}	3×10^{15}	10^{16} cm^{-3}
Ne	0.1	0.64^a	1^a	1.7^a	2.9^a	5.7^a
Ne	0.2	0.86^a	1.4^a	2.3^a	3.8^a	7.2
Ne	0.5	1.3^a	2.1^a	3.4^a	5.7	10^b
Ne	1	1.7^a	2.8^a	4.7	7.7^b	14^b
Ne	2	2.4^a	3.9	6.6	11^b	19^b
Ar	0.1	1^a	1.7^a	2.9^a	4.6^a	8.2
Ar	0.2	1.3^a	2.2^a	3.8^a	6	10^b
Ar	0.5	1.9^a	3	5.3	8.6^b	15^b
Ar	1	2.6	4.4	7.5^b	11^b	17^b
Ar	2	3.5	6.3	11^b	16^b	23^b
Kr	0.1	1^a	1.8^a	3.1^a	5^a	8.8^b
Kr	0.2	1.4^a	2.3^a	4	6.6	11^b
Kr	0.5	–	3.4	6	9.4^b	16^b
Kr	1	2.7	4.9	8.6^b	13^b	19^b
Kr	2	3.8	7.1^b	13^b	19^b	27^b

the state of the one below it. The electrons and the excited atoms book the same energy for every collision, which Appendix 5.A tests row by row as an identity.

5.5 Results

The table. Table 5.3 and Figure 5.1 give the best frequency at the 75 points with the excited atoms inside the electrons' balances and the measured 5s to 5p cross section, and Table 5.11 of Appendix 5.C every point with its electron temperature, its marks, the mole fraction of Kr*, the net energy the excited atoms take from the electrons, and the bands of Section 5.6. One cell of the krypton fill, 0.5 atm and 10^{14} cm^{-3}, is a marginal plasma whose two closures alternate at most of its frequencies, with the excited atoms moving by more than the boundary rule allows: 9 of its 13 frequencies closed, 4 of them at the flipping boundary, and the cell is reported incomplete, blank in the table and the figure and left out of the bands.

The krypton fill against the argon fill. At the same pressure and density the krypton fill's best frequency lies 1.04 to 1.20 times the argon fill's, the least at 0.2 atm and 3×10^{14} cm^{-3} and the most at 2 atm and 3×10^{15} cm^{-3}. The reason is the collision frequency. Above the collision frequency α falls as ν_m/ω^2, so the frequency at which $\alpha L = 1$ scales as the square root of the effective momentum-transfer frequency $\nu_m = \omega\mu_r/\mu_i$, the ratio of the real to the imaginary part of the mobility times ω. At the best frequencies ν_m of the krypton fill is 1.08 to 1.52 times the argon fill's, and its square root gives the ratio of the best frequencies within 3.5 %. Krypton's momentum-transfer cross section rises from 0.54 of argon's at 1 eV to 1.55 times at 3 eV [2], and the argon fill is 40 % krypton. The electrons of the krypton fill sit at 0.94 to 2.44 eV, those of the argon fill at 0.87 to 2.65 eV and those of the neon fill at 1.40 to 4.69 eV.

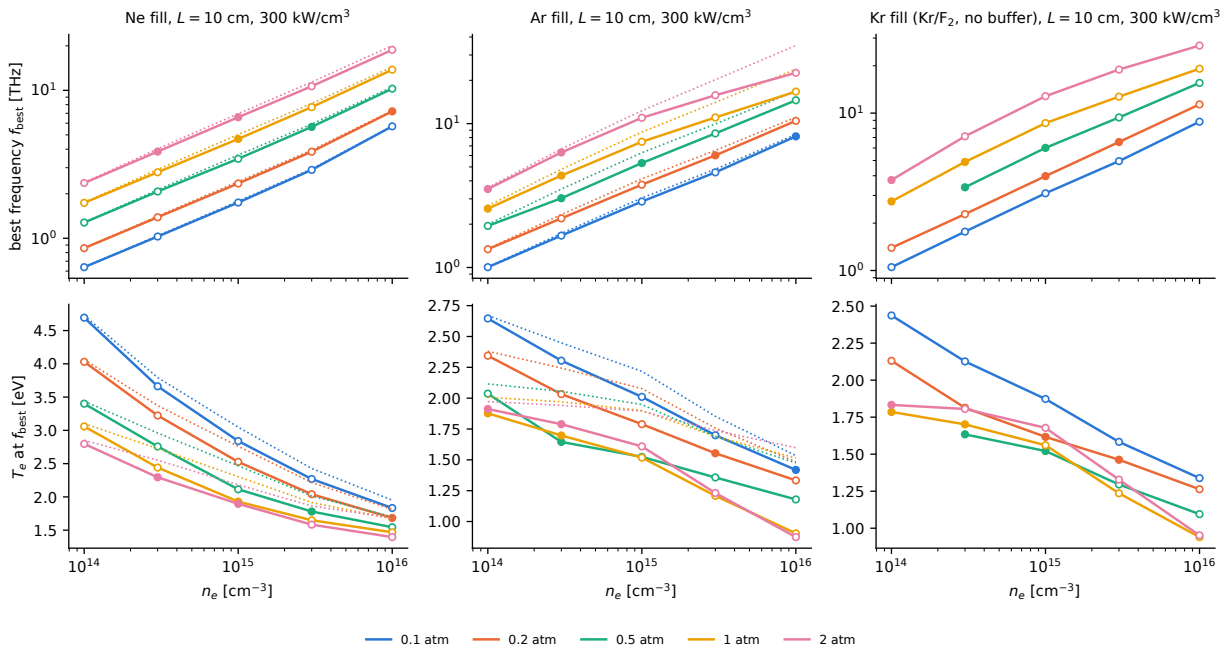


Figure 5.1: The best frequency at $L = 10$ cm and 300 kW/cm^3 against n_e, top, and the electron temperature $2\langle\epsilon\rangle/3$ there, bottom, with the excited atoms as solid lines and, for the two buffered fills, without them dotted. An open marker is a point marked a or b in Table 5.3. The krypton fill's cell at 0.5 atm and 10^{14} cm^{-3} is left blank.

How the density is held. The field multiplies the electrons at 15 points of the neon fill, 8 of the argon fill and 6 of the krypton fill, marked a, and the beam brings more than a third of the absorbed power at 5, 9 and 11, marked b. The excited atoms take from the electrons, net of what they give back in superelastic collisions, 0.007 to 2.9 times the absorbed power in the neon fill, 0.012 to 15.0 times in the argon fill and 0.017 to 23.5 times in the krypton fill, the most at 2 atm and 10^{16} cm^{-3} in each. Where they take more than the field gives, the beam holds the density and its slow electrons bring the difference. With the measured cross section the best frequency of the neon fill lies at 0.933 to 0.997 of its value without the excited atoms and that of the argon fill at 0.646 to 0.995, falling at every point, by the most at 2 atm and 10^{16} cm^{-3} in the argon fill, from 34.9 to 22.5 THz. The electrons there are at 0.55 to 0.99 of their temperature without the excited atoms in the argon fill and 0.83 to 0.99 in the neon fill. The krypton fill has no counterpart without excited atoms.

5.6 What limits the answer

Three bands. Three more scans of the 75 points bound what the rates of the 5p rung can do, Table 5.11. With Kannari's fit in place of the measured 5s to 5p cross section the best frequency rises to 1.000 to 1.049 times the reference, up to 1.010 in the neon fill, 1.047 in the argon fill and 1.049 in the krypton fill, the most at 2 atm and 10^{16} cm^{-3}, where the krypton fill moves from 26.9 to 28.2 THz, with the electrons hotter by at most 6.7 %. With every electron collision of the 5p rung left out, so that collisions and radiation alone return Kr** to 5s, it rises to 1.00 to 1.46 times the reference, up to 1.054 in the neon fill, 1.460 in the argon fill and 1.436 in the krypton fill, the

most at 2 atm and 10^{16} cm^{-3} in the argon fill, where it returns to 32.9 THz: the 5s to 5p cycle carries the fall. With the return of 5p to 5s by collisions raised from Kannari's estimate to the measured total loss rate of Gazeli et al., 1.7×10^{-10} cm^3 s^{-1}, it falls to 0.868 to 1.000 times the reference, to 0.999 in the neon fill, 0.951 in the argon fill and 0.868 in the krypton fill, the most at 0.5 atm and 10^{16} cm^{-3} in the krypton fill, where it moves from 15.6 to 13.5 THz with the electrons at 0.81 of their reference temperature. A faster return empties the 5p manifold before the electrons can ionize it or take their energy back, so more of the energy of the cycle ends in the gas and the electrons are colder. The return rate by collisions is thus the largest uncertainty of the answer at the dense krypton points, 13 % in the best frequency and 19 % in the electron temperature, and a level-resolved measurement in krypton [30] would replace it.

What the model leaves out. Kr** lumps 5p and the levels above at the lowest 5p level, and argon and neon have no second rung. The four argon levels give back up to 0.28 eV more in superelastic collisions than the lumped Ar* holds. The growing closure leaves out a share of order $g/\nu_m < 3 \times 10^{-3}$ in the anisotropic part of the distribution, one point of the argon fill lies on the boundary between the closures and one cell of the krypton fill is incomplete. The recombination of Kr_2^+ puts the sum of the two branches that Kannari et al. print at 300 K on a $T_e^{-1/2}$ law [7]. The excited atoms settle within 220 ns, but the gas is held at its fill composition, while attachment alone would take the F_2 down e-fold in 24 to 56 ns at 10^{16} cm^{-3} and in 0.44 to 0.97 μs at 10^{15} cm^{-3}. What that means for the table is the subject of the next section.

5.7 The pulse

The march. The steady state of the table is marched in time at three of its points. The kinetic solution of the electrons is renewed every few nanoseconds: whenever the slowest of the electron density, the F_2 and the excited atoms would move by 2 %, at least every 10 ns, 5 ns in the third pulse, until a twentieth of the time elapsed exceeds that, and every twentieth of the time elapsed after, and between renewals the excited species, the cations, F^- and F_2 are integrated with the renewed rate coefficients held, the electron density following from quasineutrality. The F_2 burns by attachment, by the harpoon reactions of the excited atoms and by its own ionization. At each renewal the field is solved again so that the wave deposits P, the solve carries the electrons' sources, the beam's secondaries and the Penning electrons at their release energies, and the instantaneous net rate g of the density enters as the uniform loss of Eq. (5.6), given rather than solved, so that the distribution's shape is that of a plasma changing at that rate. The beam is a constant current, at the pair rate that held the density at the point's steady state at $t = 0$. Held still are the gas temperature, 300 K, the atomic fluorine beyond the consumption of F_2, the excimers, which are sinks, and the buffer and krypton densities. At chosen times the 13-frequency scan is solved at the frozen state and the best frequency of that state found by the definitions of Section 5.2.

What the pulses say. Table 5.4 follows the krypton fill and the argon fill at 1 atm and 10^{15} cm^{-3} at the table's best frequencies, 10 and 7.7 THz, and the krypton fill at 1 atm and 10^{16} cm^{-3} at its own, 19.2 THz, and Table 5.5 the best frequency of the frozen state at the checkpoints. The attachment that holds the density burns the F_2 at 3×10^8 to 6×10^8 molecules per electron per second. At 10^{15} cm^{-3} half of it is gone at 134 ns and all but 1 % by 365 ns in the krypton fill and 383 ns in the argon fill. The density then runs up, 3.75 times in the krypton fill and 3.95 times in the argon fill, until the beam's pairs balance recombination, which they do by 0.44 to 0.47 μs, the same power spread over four times the electrons cools them from 1.56 and 1.52 eV to 0.535 and

Table 5.4: Three pulses at 300 kW/cm^3, the beam holding the density at $t = 0$: the electron density in cm^{-3}, the electron temperature in eV, the mole fraction of Kr* in units of 10^{-4}, the F_2 over its fill value, the net rate g of the density in 10^6 s^{-1}, the rms field in kV/cm and the absorption coefficient in m^{-1} at the wave's frequency.

t [ns]	n_e	T_e	x_{Kr^*}	F_2	g	E	α
Kr fill, 1 atm, 10^{15} *cm*$^{-3}$*, 10 THz*							
0	1.00×10^{15}	1.560	0.74	1.000	0.00	38.9	7.48
51	1.11×10^{15}	1.476	0.85	0.798	1.65	38.3	7.71
100	1.25×10^{15}	1.370	1.02	0.613	1.50	38.2	7.76
198	1.61×10^{15}	1.081	1.71	0.288	1.56	40.5	6.91
300	2.36×10^{15}	0.728	3.85	0.060	4.16	46.9	5.14
489	3.74×10^{15}	0.541	7.62	0.000	0.21	47.3	5.05
1000	3.75×10^{15}	0.535	7.70	0.000	0.00	47.6	4.99
Ar fill, 1 atm, 10^{15} *cm*$^{-3}$*, 7.7 THz*							
0	1.00×10^{15}	1.519	0.72	1.000	0.00	34.6	9.44
49	1.12×10^{15}	1.442	0.80	0.797	2.11	33.9	9.82
100	1.28×10^{15}	1.336	0.94	0.601	2.34	33.6	10.03
198	1.68×10^{15}	1.074	1.44	0.281	2.19	34.5	9.50
300	2.41×10^{15}	0.763	2.80	0.066	3.95	37.6	8.00
489	3.88×10^{15}	0.554	5.60	0.000	0.54	36.8	8.35
1000	3.95×10^{15}	0.542	5.81	0.000	0.00	37.0	8.29
Kr fill, 1 atm, 10^{16} *cm*$^{-3}$*, 19.2 THz*							
0	1.00×10^{16}	0.943	5.09	1.000	0.00	33.6	10.04
10	1.32×10^{16}	0.838	6.42	0.536	29.93	31.8	11.20
20	1.97×10^{16}	0.691	10.73	0.193	41.35	28.7	13.77
30	3.22×10^{16}	0.599	17.89	0.028	48.70	21.9	23.64
49	5.58×10^{16}	0.554	28.97	0.000	14.87	14.4	55.17
100	6.86×10^{16}	0.537	33.90	0.000	0.35	12.0	78.62
300	6.84×10^{16}	0.533	34.12	0.000	-0.01	12.0	78.54
1000	6.83×10^{16}	0.532	34.13	0.000	0.00	12.1	78.38

Table 5.5: The best frequency in THz of the frozen state at the checkpoints of the three pulses, with f_R in parentheses.

pulse	0 ns	30 ns	100 ns	300 ns	1000 ns
Kr fill, 1 atm, 10^{15} cm^{-3}, 10 THz	8.64 (0.23)	–	8.80 (0.31)	7.18 (0.55)	7.09 (0.72)
Ar fill, 1 atm, 10^{15} cm^{-3}, 7.7 THz	7.48 (0.27)	–	7.71 (0.34)	6.89 (0.56)	7.03 (0.75)
Kr fill, 1 atm, 10^{16} cm^{-3}, 19.2 THz	19.19 (1.16)	29.11 (2.18)	52.86 (3.17)	52.83 (3.17)	52.77 (3.16)

0.542 eV, Kr* accumulates 10 and 8 times once its harpoon partner is gone, and the field for the same deposited power rises from 38.9 to 47.6 kV/cm in the krypton fill. The best frequency of the plasma rises at first, by 2 and 3 % at 100 ns, then falls, to 0.82 and 0.94 of its start at 1 μs. At 100 ns the density is up by 25 and 28 %, the electrons are at 1.37 and 1.34 eV and the F_2 at 0.61 and 0.60 of its fill, so the state of the table holds, within 3 % in the best frequency, for the first 100 ns at this density. At 10^{16} cm^{-3} half of the F_2 is gone at 11 ns and all but 1 % at 35 ns, the density runs up 6.8 times by 92 ns, the electrons cool from 0.94 to 0.53 eV, the field for the same power falls from 33.6 to 12.1 kV/cm as α at 19 THz climbs from 10.0 to 78 m^{-1}, so that the wave then stops within 1.3 cm of the 10-cm layer, and the best frequency of the new plasma is 1.52 times the table's at 30 ns and 2.75 times, 53 THz, from 100 ns on. At 10 ns the density is up by 32 % and the F_2 at 0.54 of its fill: at 10^{16} cm^{-3} the table's answer holds for the first 10 ns. A shorter pulse, a lower beam current or more F_2 keeps the table's answer longer, and the march gives the time it holds at any point.

5.8 The kinetic solver against a second solver

The same equation twice. The two-term Boltzmann equation of Section 5.2, with its wave term, its electron–electron term, its growth closure and the electron–ion absorption of Turnbull et al., was solved a second time by a solver written independently for a time-domain model of the KrF laser, with its own implementation of the Turnbull construction from the same two papers [3, 4] and the same cross sections [2]. Table 5.6 compares the two on the krypton fill at seven states, from 10^{12} to 10^{16} cm^{-3}, 0.1 to 1 atm and 95 GHz to 10 THz, with the ions carrying from none to a quarter of the absorption: a given rms field and frequency, no excited atoms, no sources, the net frequency of ionization and attachment self-consistent in both, the second solver started from the first's converged distribution. The mean energies agree within 0.2 %, the wave's absorbed power within 0.2 % and the net frequency within 0.6 % at every state. The share of the absorption the two book to the ions differs by definition at 95 GHz, 0.19 against 0.17 at 0.1 atm and 10^{16} cm^{-3}, while the total power agrees to 0.02 %: the first books the ions' part by their share of the momentum transfer in the field term, the second re-evaluates the field term with the atoms' momentum transfer alone, and below $\omega \sim \nu_m$ that is not the complement. The optics of Eq. (5.1) computed on the second solver's distributions, from the same momentum-transfer sum its wave term uses, give α within 0.2 % and R within 0.4 % of the first's at the seven states, over α from 2.3 to 3.4×10^4 m^{-1} and R from 3×10^{-8} to 0.91, and the power the mobility implies equals the second solver's own absorbed power to 10^{-4}. The absorption that defines the best frequency is therefore the same in both.

The two full models from the same cold start. The second solver belongs to a time-domain model of the KrF laser that evolves the gas temperature, the excimers, the photons and the walls as well as the species followed here. The two full models were started from one and the same state, the krypton fill at 1 atm with a seed of 10^{15} electrons/cm^3 and as many Kr^+, no excited atoms, no F^-, under a wave of 38.9 kV/cm rms at 10 THz, the field the table needs for 300 kW/cm^3 at this point, and no beam, and run for 100 ns, Table 5.7. Without a beam, attachment outruns the wave's own ionization at this fill and both models decay the plasma: the second model at 2.17×10^8 s^{-1} between 10 and 100 ns, the march of Section 5.7 at 2.15×10^8 s^{-1} from 40 ns on, 1 % apart, with the electrons of the decaying plasma at 1.747 and 1.742 eV, 0.3 % apart. The two differ only in the first 10 ns, where the second model heats its seed from the gas temperature and loses three decades of density to the attachment of the still-cold electrons on the way, while the march starts from the

Table 5.6: The two kinetic solvers at one and the same state of the krypton fill: the rms field in kV/cm, the reduced field in Td, the mean energy in eV, first solver / second solver, the ratio of the wave's absorbed power per electron, second over first, the share of the absorption each books to the ions, and the ratio of the net frequency of ionization and attachment, second over first.

p [atm]	n_e [cm^{-3}]	E	f	E/N	$\langle\epsilon\rangle$	P_2/P_1	ions' share	ν_2/ν_1
1	10^{12}	1.5	95 GHz	6.1	3.8061 / 3.8083	0.9994	0.0000 / 0.0000	0.9999
1	10^{14}	1.5	95 GHz	6.1	3.5558 / 3.5580	0.9990	0.0009 / 0.0005	1.0032
1	10^{13}	3	1 THz	12.3	3.1950 / 3.1995	1.0014	0.0000 / 0.0000	0.9984
1	10^{15}	3	1 THz	12.3	2.5933 / 2.5964	1.0011	0.0024 / 0.0021	1.0045
1	10^{15}	30	10 THz	122.6	2.5825 / 2.5854	1.0016	0.0013 / 0.0013	1.0047
0.1	10^{16}	2	1 THz	81.8	1.8584 / 1.8614	1.0016	0.2496 / 0.2483	1.0046
0.1	10^{16}	0.6	95 GHz	24.5	2.3229 / 2.3242	1.0002	0.1928 / 0.1719	1.0058

Table 5.7: The two full models from the same cold start, the krypton fill at 1 atm under 38.9 kV/cm at 10 THz with no beam: the electron density in cm^{-3}, the electron temperature in eV and the Kr* density in cm^{-3}, each second model / march, the F_2 over its fill value in both, and the second model's gas temperature in K.

t [ns]	n_e, second	march	T_e, second	march	Kr*, second	march	F_2, second	march	T_g
0	1.00×10^{15}	1.00×10^{15}	0.026	1.795	0	2.4×10^{7}	1.0000	1.0000	300.0
10	5.18×10^{11}	7.55×10^{13}	1.744	1.661	7.9×10^{12}	3.7×10^{14}	0.9915	0.9899	302.6
20	5.69×10^{10}	6.03×10^{12}	1.747	1.696	2.8×10^{12}	1.5×10^{14}	0.9914	0.9871	302.8
50	8.44×10^{7}	7.13×10^{9}	1.746	1.740	9.8×10^{10}	5.1×10^{12}	0.9914	0.9858	303.2
100	1.72×10^{3}	1.73×10^{5}	1.747	1.742	3.5×10^{8}	2.0×10^{10}	0.9914	0.9856	303.4

distribution solved hot at $t = 0$. The second model burns 0.9 % of the F_2, the march 1.4 %, the difference again in the first 10 ns, and the gas warms by 3.4 K in the second model while the march holds it at 300 K.

5.9 Where the calculation stands

The best frequency of Table 5.3 rests on the measured 5s to 5p cross section, on the measured lifetimes of the 5p levels, on an estimate for the return of 5p to 5s by collisions that one measured total loss rate bounds, and on a calculated stepwise ionization that the answer does not rest on. Its largest uncertainty is the return rate, 13 % in the best frequency and 19 % in the electron temperature at the dense krypton points, and a level-resolved measurement would remove it. The table is the state of the first 100 ns of a beam-held pulse at 10^{15} cm^{-3} and of the first 10 ns at 10^{16} cm^{-3}, after which the F_2 is gone and the best frequency of the plasma that remains is the march's, not the table's. The kinetic solver behind both agrees with an independent one to 0.2 %. What the calculation still leaves out, the gas temperature, the excimers' own kinetics and the photons, the atomic fluorine and the walls, belongs to the time-domain model the second solver was written for, and the next step is to carry the absorption of Eq. (5.1) and the excited atoms' electron collisions of Table 5.2 into it.

5.A The checks

Every number of this note and every step of the calculation behind it is tested by the checks listed in Table 5.8, which end with all of them passing. Section K tests the electron collisions of the excited atoms: each fitted process against its source's printed rate on Maxwellians, the measured cross section against its tables and its own integral, each de-excitation against detailed balance, every energy against the ladder, the 5p lifetime against the ten level lifetimes, and the identity that the energy the excited atoms take from the electrons equals the sum, over the collisions the Boltzmann equation carries, of their events times the energy each takes. Section C checks every point of the four scans: each must close the same way for the electrons and the ions, the two energy books must agree, the residuals must be small, and one point is solved again in a fresh process. Section B recomputes a best frequency from the scan. Section L checks the comparison with the second solver and section P the pulses, their quasineutrality, the monotone burn of the F_2 and their checkpoints. Section N computes every number the text prints from the data, section E ties each printed number and claim to them, and section S checks the style and that no file or unpublished note is named.

Table 5.8: The checks of this note.

	check	what	value	
1	`K.rows_added`	the added rows are on: the 5p manifold's radiative decay among the rows, the return band	V4 True, PRAD rows [('DZIERZEGA', 'PRAD')], KPS_Kr band {'default': 1e-16, 'lo': 3e-17, 'hi': 1.7e-16, 'unit': 'm3/s', 'marks': ((1.7e-16, "Gazeli 2021, the ...	pass
2	`K.fit_maxwell`	each fitted process on Maxwellians (T 1, 2, 4 eV) = the source's printed rate x exp(-(onset - E)/T), onset max(E, gap), the electron losing the ledger's gap (three fills, Kannari's stepwise fit)	10 processes, largest 2.5e-05 at ('Ar', 'KSTEP', 1.0), wrong energies []	pass
3	`K.jung_tables`	the measured 5s → 5p cross sections: the tables read back at the tabulated energies, the lowest threshold against the ledger's gap, the fit's pseudo cross section returning the fit	Table II read back at the tabulated energies: [np. float64(27.0), np. float64(47.0), np. float64(52.0), np. float64(52.0), np. float64(51.0), np. float64(49.0)]; T...	pass
4	`K.jung_lumped`	the solver's K76 at 3 eV = the tables' sums weighted 8:4 by hand (1e-16 cm^2), a measured cross section above the ledger's gap	99.19 vs 99.19, gap 1.385	pass
5	`K.jung_rate`	the solver's rate of the measured K76 on Maxwellians (1, 2, 4 eV) = the module's own integral	largest relative difference 2.4e-05	pass
6	`K.jung_reverse`	its reverse by detailed balance at 2 eV: k_down/k_up = g_5s/g_5p exp(gap/T)	0.66624 vs 0.66624	pass

	check	what	value	
7	K.down_db	every de-excitation (Kr* and the four Ar levels to the ground, Kr** to Kr*) is its excitation's detailed balance on Maxwellians, at the energy the excitation takes	30 values, largest relative difference 1.8e-06	pass
8	K.const	ArKr* + e → Ar + Kr + e as sigma = k/(gamma sqrt(eps)) gives the printed constant on Maxwellians	k 1.000e-13 mˆ3/s, largest relative difference 4.4e-16	pass
9	K.ledger	every process the solver carries takes the ledger's energy of its row (the Ar superelastic rows give each level's own, above the lumped Ar*), and every lumped excited atom stands at or below the lowest field excitation booked into it	off the ledger [], Ar levels above Ar* by [-0.0, 0.075, 0.175, 0.28] eV, level vs lowest excitation {'Ar*': (11.548, 11.548), 'Kr*': (9.915, 9.915), 'Kr**': ...	pass
10	K.prad	Kr** → Kr* + light: a one-body row at 1/29.0 ns, the 2J+1-weighted mean decay rate of the ten semi-empirical 5p lifetimes of Dzierzega et al. (their Table 2, 23.1 to 38.1 ns; the row's energy on the ledger leaves as light)	row 3.4483e+07 /s = 1/29.00 ns, the ten give 29.04 ns	pass
11	K.kps_band	the 5p → 5s return in collisions with Kr: Kannari's estimate 1.0e-10 cmˆ3/s as the default, Gazeli et al.'s measured total loss 1.7e-10 as the band's top, and the four scans' settings (the reference with the measured K76, the fit, the 5p rows out, the return at the top)	default 1.00e-16, at the top 1.70e-16 mˆ3/s; {'jung': ('kannari', {'k76': 'jung'}), 'kannari': ('kannari', {}), 'jung_no5p': ('kannari', {'k76': 'jung', 'no5...	pass
12	K.kr_fill	the Kr fill: Kr 0.995, F2 0.005, the cations Kr+ and Kr2+, the beam's pairs shared by the stopping powers, 0.28 Kr* per pair made in krypton, its energy per pair	[('Kr', 0.995), ('F2', 0.005)], cations ('Kr+', 'Kr2+'), pairs {'Kr+': 0.9978230972792193, 'F2+': 0.0021769027207806492}, Kr* per pair {'Kr*': 0.279390467238...	pass
13	K.energy_rows_ Kr2_1e+16	the ladder's net energy from the electrons = the sum over the solver's processes of the ladder's events x the energy each takes, and every electron row on an excited species is a process of the solver (the Kr fill, the state without excited atoms, kinetic rates, the measured K76)	p_net/P 34.5379 vs 34.5379, rows not carried []	pass

	check	what	value	
14	C.jung	the scan 'jung': 75 cells x 13 frequencies (errors only at the Kr fill's 0.5 atm 1e14 cell), every point closed the same way for the electrons and the ions (residual against the point's largest particle rate below the loop's tolerance), the solver's net power to the excited gases against the ladder's, the residuals, the rounds, the boundary and floor points	75 cells, 971 points, 4 errors at ['Kr 0.5 atm 1e14'], closure off at 0 [], closure residual 2.2e-04 held / 1.2e-04 growing, energy 8.2e-03, solver 3.6e-05, ...	pass
15	C.kannari	the scan 'kannari': 75 cells x 13 frequencies (errors only at the Kr fill's 0.5 atm 1e14 cell), every point closed the same way for the electrons and the ions (residual against the point's largest particle rate below the loop's tolerance), the solver's net power to the excited gases against the ladder's, the residuals, the rounds, the boundary and floor points	75 cells, 967 points, 8 errors at ['Kr 0.5 atm 1e14'], closure off at 0 [], closure residual 2.5e-04 held / 1.3e-04 growing, energy 5.9e-03, solver 3.6e-05, ...	pass
16	C.jung_no5p	the scan 'jung_no5p': 75 cells x 13 frequencies (errors only at the Kr fill's 0.5 atm 1e14 cell), every point closed the same way for the electrons and the ions (residual against the point's largest particle rate below the loop's tolerance), the solver's net power to the excited gases against the ladder's, the residuals, the rounds, the boundary and floor points	75 cells, 975 points, 0 errors at [], closure off at 0 [], closure residual 2.6e-04 held / 1.6e-04 growing, energy 4.7e-03, solver 3.4e-05, ladder 6.6e-16, |...	pass
17	C.jung_kpshi	the scan 'jung_kpshi': 75 cells x 13 frequencies (errors only at the Kr fill's 0.5 atm 1e14 cell), every point closed the same way for the electrons and the ions (residual against the point's largest particle rate below the loop's tolerance), the solver's net power to the excited gases against the ladder's, the residuals, the rounds, the boundary and floor points	75 cells, 974 points, 1 errors at ['Kr 0.5 atm 1e14'], closure off at 0 [], closure residual 2.5e-04 held / 1.3e-04 growing, energy 6.3e-03, solver 3.6e-05, ...	pass
18	C.recompute	the Kr fill, 1 atm, 1e15, first frequency, solved again in a fresh process (the measured K76)	<eps> 2.842079513 vs 2.842079513, x_ Kr* 5.129746938e-05 vs 5.129746938e-05	pass
19	C.same_rates	there, the solver's rate of every excited process = the same integral over its distribution, and the excited atoms' rates (the round before) within the loop's tolerance	solver vs the integral 2.2e-16, excited atoms vs solver 4.1e-03	pass
20	B.complete	a best frequency at every cell but the incomplete one	75 cells, incomplete ['Kr 0.5 atm 1e14']	pass
21	B.recompute	the Kr fill, 1 atm, 1e15 analysed again from the scan (f_best, T_e, the marks)	8.64033e+12 vs 8.64033e+12 Hz, T_e 1.5604	pass

	check	what	value	
22	`L.solvers`	the two kinetic solvers at the seven stored states: mean energy, the wave's absorbed power and the net frequency, second over first, and the ions' share at 0.1 atm, 1e16, 95 GHz	7 states, |mean - 1| 1.6e-03, |power - 1| 1.6e-03, |nu - 1| 5.8e-03, shares 0.1928 / 0.1719, power there 1.00019, quant_IB residual 1.3e-04	pass
23	`L.recompute`	quant_IB's side at 1 atm, 1e15, 3 kV/cm, 1 THz solved again in this process (the scans' electron-ion model)	<eps> 2.5932833 vs stored 2.5932833	pass
24	`L.optics`	the optics on the second solver's distributions at the seven states against quant_IB's (alpha, R), the power its mobility implies against its own, and quant_IB's alpha at one state recomputed here	|alpha - 1| 1.6e-03, |R - 1| 3.4e-03, |P - 1| 1.3e-04; alpha 551.245 vs stored 551.245	pass
25	`L.cold`	the two full models from the same cold start (Kr fill, 1 atm, 1e15 seed, 38.9 kV/cm at 10 THz, no beam): the decay rate of n_e (the second solver 10 to 100 ns, the march 40 to 100 ns), T_e at 100 ns, the F2 burnt	decay 2.1695e+08 / 2.1496e+08 /s, T_e 1.7469 / 1.7416 eV, F2 burnt 0.0086 / 0.0144, T_g 303.4 K	pass
26	`P.Kr15`	the pulse Kr15: quasineutral (n_e = cations - F-), F2 never rising, time increasing to 1 us, the renewal rule (at most dt or a twentieth of the time elapsed), the wave's absorbed power at the target at every renewal, the closure, the checkpoints' 13-frequency scans	86 renewals, quasineutral to 0.0e+00, F2 monotone True, t monotone True, rule True, |P/P* - 1| 1.0e-04, closures ['source+loss'], checkpoints [0, 100, 300, 1...	pass
27	`P.Ar15`	the pulse Ar15: quasineutral (n_e = cations - F-), F2 never rising, time increasing to 1 us, the renewal rule (at most dt or a twentieth of the time elapsed), the wave's absorbed power at the target at every renewal, the closure, the checkpoints' 13-frequency scans	87 renewals, quasineutral to 0.0e+00, F2 monotone True, t monotone True, rule True, |P/P* - 1| 9.7e-05, closures ['source+loss'], checkpoints [0, 100, 300, 1...	pass
28	`P.Kr16`	the pulse Kr16: quasineutral (n_e = cations - F-), F2 never rising, time increasing to 1 us, the renewal rule (at most dt or a twentieth of the time elapsed), the wave's absorbed power at the target at every renewal, the closure, the checkpoints' 13-frequency scans	103 renewals, quasineutral to 0.0e+00, F2 monotone True, t monotone True, rule True, |P/P* - 1| 9.6e-05, closures ['source+loss'], checkpoints [0, 30, 100, 3...	pass

	check	what	value	
29	N.cells	the reference's cells: three fills x 5 pressures x 5 densities, those complete, those not	75 cells, 74 complete, incomplete ['Kr 0.5 atm 1e14']	pass
30	N.fills	per fill: T_e at f_best, the marks, the net energy the excited atoms take over P	{'Ne': ('T_e 1.396-4.694', {'': 5, 'a': 15, 'b': 5, 'ab': 0}, 'p_net/P 0.007-2.9'), 'Ar': ('T_e 0.874-2.646', {'': 8, 'a': 8, 'b': 9, 'ab': 0}, 'p_net/P 0.01...	pass
31	N.kr_over_ar	the Kr fill's best frequency over the Ar fill's at the same p and n_e	24 cells, 1.0394 (at Kr 0.2 atm 3e14) to 1.2013 (at Kr 2 atm 3e15)	pass
32	N.kr_over_ar_mech	the effective momentum-transfer frequency (omega mu_r/mu_i at f_best) of the Kr fill over the Ar fill's at the same cell, and how far its square root lies from the ratio of the best frequencies	nu ratio 1.081 to 1.521, sqrt vs f ratio within 0.035	pass
33	N.sigma_m	the momentum-transfer cross section of krypton over argon's at 1 to 3 eV	0.54 to 1.55	pass
34	N.band_kannari	the best frequency with the case 'kannari' against the reference, by fill (the cells complete in both)	{'Ne': (25, '0.9999-1.0099', 'Ne 1 atm 3e14'), 'Ar': (25, '0.9999-1.0467', 'Ar 2 atm 1e16'), 'Kr': (24, '0.9999-1.0491', 'Kr 2 atm 1e16')}	pass
35	N.band_jung_no5p	the best frequency with the case 'jung_no5p' against the reference, by fill (the cells complete in both)	{'Ne': (25, '0.9995-1.0541', 'Ne 1 atm 1e15'), 'Ar': (25, '1.0035-1.4596', 'Ar 2 atm 1e16'), 'Kr': (24, '1.0056-1.4362', 'Kr 2 atm 1e16')}	pass
36	N.band_jung_kpshi	the best frequency with the case 'jung_kpshi' against the reference, by fill (the cells complete in both)	{'Ne': (25, '0.9990-1.0000', 'Ne 2 atm 1e16'), 'Ar': (25, '0.9509-0.9999', 'Ar 1 atm 1e16'), 'Kr': (24, '0.8685-0.9998', 'Kr 0.5 atm 1e16')}	pass
37	N.step_band	stepwise ionization at the template (2.1 to 2.3 x Kannari's fit) or at none against Kannari's fit, in the calculation with his 5s → 5p fit (Ne and Ar, 50 cells each): the largest move of f_best	0.0144	pass
38	N.jung	Jung et al.'s lumped 5s → 5p cross section over Kannari's fit on Maxwellians at the reference's lowest and highest T_e	1.358 at 0.874 eV, 0.920 at 4.694 eV	pass

	check	what	value	
39	N.nist	the atomic levels and ionization limits of the ledger against the NIST database	{'Kr*': 0.0002, 'Kr**': 0.0035, 'Kr+': 0.0036, 'Ar*': 0.0004, 'Ar+': 0.0004, 'Ne*': 0.0001, 'Ne+': 0.0045}	pass
40	N.prad	the 5p manifold's lifetime: the 2J+1-weighted mean decay rate of the ten level lifetimes against the row's 29.0 ns	29.04 ns from 23.1 to 38.1 ns, the row 29.0 ns	pass
41	N.loop	the reference scan's points, the loop's rounds, the growth rate over the momentum-transfer frequency at the growing points, the points accepted at a flipping boundary and at the pulse floor	971 points, 3 to 72 rounds, g/nu up to 2.83e-03 (394 growing), boundary 5, floor 0	pass
42	N.return_5p	how the 5p manifold returns to 5s at the Kr fill, 1 atm, 1e15, 10 THz (the events)	{'collisions': '0.838', 'superelastic': '0.150', 'radiation': '0.012'}	pass
43	N.f2_burn	the time attachment alone takes the F2 down e-fold at f_best, and the slowest excited species' relaxation time	F2: 24.4 to 56.2 ns at 1e16, 436 to 974 ns at 1e15; excited atoms within 212 ns	pass
44	N.pulse_Kr15	the pulse Kr15: F2 below 1 % of its fill, n_e steady (|g| < 1e6/s), the rise of n_e, the fall of T_e, the field, the best frequency of the frozen state at the checkpoints	F2 < 1 % at 365 ns (half at 134.0), g ~ 0 at 443 ns, n_e x3.75, T_e 1.560 → 0.535, E 38.9 → 47.6 kV/cm, f_best 0 ns 8.64, 100 ns 8.80, 300 ns 7.18, 1000 ns...	pass
45	N.pulse_Ar15	the pulse Ar15: F2 below 1 % of its fill, n_e steady (|g| < 1e6/s), the rise of n_e, the fall of T_e, the field, the best frequency of the frozen state at the checkpoints	F2 < 1 % at 383 ns (half at 134.0), g ~ 0 at 465 ns, n_e x3.95, T_e 1.519 → 0.542, E 34.6 → 37.0 kV/cm, f_best 0 ns 7.48, 100 ns 7.71, 300 ns 6.89, 1000 ns...	pass
46	N.pulse_Kr16	the pulse Kr16: F2 below 1 % of its fill, n_e steady (|g| < 1e6/s), the rise of n_e, the fall of T_e, the field, the best frequency of the frozen state at the checkpoints	F2 < 1 % at 35 ns (half at 11.0), g ~ 0 at 92 ns, n_e x6.83, T_e 0.943 → 0.532, E 33.6 → 12.1 kV/cm, f_best 0 ns 19.19, 30 ns 29.11, 100 ns 52.86, 300 ns 5...	pass
47	N.recorded	the note's numbers computed from the scans, the best frequencies and the pulses	13 of 13 groups recorded	pass
48	N.inc_cell	the incomplete cell's frequencies closed in the reference and those accepted at the flipping boundary, and the cell with the most rounds	{'records': 9, 'boundary': 4}, most rounds at Kr 0.5 atm 1e14 (72)	pass

	check	what	value	
49	E.third	claim: the binding of Kr* over Kr's ionization energy is under a third (0.292)		pass
50	E.table75	claim: three fills, 75 points, the krypton fill with 0.5 % F2 ([('Kr', 0.995), ('F2', 0.005)])		pass
51	E.ans_kr_lo	printed 1.04: the Kr fill's f_best over the Ar fill's, smallest	1.03944	pass
52	E.ans_kr_hi	printed 1.2: the same, largest	1.20134	pass
53	E.ans_f01_lo	printed 1.05: the Kr fill's f_best at 0.1 atm, lowest [THz]	1.04983	pass
54	E.ans_f01_hi	printed 8.8: the same, highest	8.81888	pass
55	E.ans_f2_lo	printed 3.8: at 2 atm, lowest	3.75663	pass
56	E.ans_f2_hi	printed 26.9: at 2 atm, highest	26.8752	pass
57	E.ans_te_lo	printed 0.94: the Kr fill's T_e, lowest	0.940115	pass
58	E.ans_te_hi	printed 2.44: the same, highest	2.43674	pass
59	E.ans_b11	claim: the beam holds 11 of the Kr fill's cells (b marks {'': 7, 'a': 6, 'b': 11, 'ab': 0})		pass
60	E.ans_ne_lo	printed 0.933: the Ne fill over the calculation without excited atoms, smallest	0.932873	pass
61	E.ans_ne_hi	printed 0.997: the same, largest	0.996997	pass
62	E.ans_ar_lo	printed 0.646: the Ar fill, smallest	0.646198	pass
63	E.ans_ar_hi	printed 0.995: the Ar fill, largest	0.995163	pass
64	E.ans_fit	printed 5: Kannari's fit in place of the measurement raises f_best by up to (%)	4.91307	pass
65	E.ans_no5p	printed 1.46: the 5p rung's electron rows left out, up to	1.45957	pass
66	E.ans_kps	printed 13: the return at the measured bound lowers f_best by up to (%)	13.1509	pass
67	E.ans_kps_te	printed 19: and T_e there by (%)	19.0497	pass
68	E.ans_kps_where	claim: the most at Kr 0.5 atm 1e16 (a dense krypton point)		pass
69	E.ans_hold15	claim: at 1e15 the frozen state's f_best at 100 ns within 3 % of the start (1.0189, 1.0306)		pass
70	E.ans_f2_15lo	printed 0.36: F2 below 1 % of its fill at 1e15, earliest [us]	0.364652	pass
71	E.ans_f2_15hi	printed 0.38: the same, latest	0.382884	pass
72	E.ans_f2_16	printed 35: at 1e16 [ns]	34.7287	pass
73	E.ans_ne_rise_lo	printed 3.7: n_e runs up, least	3.74961	pass
74	E.ans_ne_rise_hi	printed 6.8: the same, most	6.83088	pass
75	E.ans_te_cool_lo	printed 0.53: the electrons cool to, lowest [eV]	0.531929	pass
76	E.ans_te_cool_hi	printed 0.54: the same, highest	0.541683	pass
77	E.ans_fmove_kr	printed -18: f_best at 1 us over the start, the Kr fill at 1e15 (%)	-17.9178	pass
78	E.ans_fmove_ar	printed -6: the Ar fill (%)	-6.09407	pass
79	E.ans_fmove_16	printed 2.8: the Kr fill at 1e16, times	2.75057	pass
80	E.ans_seven	claim: seven states from 1e12 to 1e16 (7, (1000000000000.0, 1e+16))		pass
81	E.ans_02	claim: within 0.2 % in mean energy (1.65e-03) and power (1.64e-03)		pass

	check	what	value	
82	E.ans_cold	claim: the two full models' decay rates 1 % apart (2.1695e+08 vs 2.1496e+08) and T_e 0.2 % apart (1.7469 vs 1.7416)		pass
83	E.fills	claim: the fills' mole fractions {'Ne': [('Ne', 0.948), ('Kr', 0.05), ('F2', 0.002)], 'Ar': [('Ar', 0.597), ('Kr', 0.4), ('F2', 0.003)], 'Kr': [('Kr', 0.995), ('F2', 0.005)]}		pass
84	E.grid	claim: pressures (0.1, 0.2, 0.5, 1.0, 2.0), densities (100000000000000.0, 300000000000000.0, 1000000000000000.0, 3000000000000000.0, 1e+16)		pass
85	E.freqs	claim: 13 frequencies, five per decade, 0.1 to 25.12 THz		pass
86	E.rmax	claim: R_max 0.05		pass
87	E.cations	claim: the cations {'Ne': ('Ne+', 'Ne2+', 'NeKr+', 'Kr+', 'Kr2+'), 'Ar': ('Ar+', 'Ar2+', 'ArKr+', 'Kr+', 'Kr2+'), 'Kr': ('Kr+', 'Kr2+')}		pass
88	E.epssec	printed 4.506: the mean energy of the beam's slow secondaries	4.506	pass
89	E.tau	claim: the pulse 1e-06 s		pass
90	E.kr915	printed 9.915: Kr* on the ledger	9.915	pass
91	E.kr1130	printed 11.3: Kr** on the ledger	11.3	pass
92	E.ar11548	printed 11.548: Ar* on the ledger	11.548	pass
93	E.ne16619	printed 16.619: Ne* on the ledger	16.619	pass
94	E.kr_species	claim: the Kr fill carries Kr*, Kr** and Kr2*		pass
95	E.nist	claim: the atoms' levels within 0.005 eV of NIST (largest 0.0045 eV)		pass
96	E.bethe	claim: the stopping share at 600 keV (6.0e5 eV in the code)		pass
97	E.beam	claim: the beam's energies per pair and excited atoms per pair as the calculation uses them		pass
98	E.harp	printed 8.47e-10: Kr* + F2, all channels, cm3/s	8.47e-10	pass
99	E.yield	printed 0.9: the KrF* yield of Kr* + F2	0.9	pass
100	E.jung_meas	claim: Jung's tables: 2 to 300 eV, +-35 % (1s5), +-40 % (1s3), the ten 5p levels		pass
101	E.jung_weights	claim: the 5s manifold lumped 5:3:1:3 (weights {'1s5': 0.6666666666666666, '1s3': 0.3333333333333333})		pass
102	E.gap	printed 1.385: the ledger's gap Kr* → Kr** [eV]	1.385	pass
103	E.jung_lo	printed 1.36: the measured lumped rate over Kannari's fit at the table's coldest T_e	1.35791	pass
104	E.jung_tlo	printed 0.87: that T_e [eV]	0.873805	pass
105	E.jung_hi	printed 0.92: the same at the hottest	0.919652	pass
106	E.jung_thi	printed 4.69: that T_e [eV]	4.69366	pass
107	E.jung_cal	claim: both ratios within the measurement's 35 to 40 %		pass

	check	what	value	
108	E.tau5p_lo	printed 23.1: the ten 5p lifetimes, shortest [ns]	23.1	pass
109	E.tau5p_hi	printed 38.1: the same, longest	38.1	pass
110	E.tau5p	printed 29.0: the 2J+1 mean [ns]	29.0377	pass
111	E.gap_light	printed 1.385: the energy that leaves as light	1.385	pass
112	E.kps	printed 1e-10: the return by collisions, the default [cm3/s]	1e-10	pass
113	E.gazeli	claim: Gazeli's measured total loss, the band's top: 1.70e-10 cm3/s		pass
114	E.ret_coll	printed 84: the 5p manifold's return by collisions at Kr 1 atm 1e15 10 THz (%)	83.7806	pass
115	E.ret_super	printed 15: superelastically (%)	15.0325	pass
116	E.ret_rad	printed 1.2: by radiation (%)	1.18689	pass
117	E.kstepA	printed 7e-08: Kannari's stepwise fit, A [cm3/s]	7e-08	pass
118	E.kstepB	printed 0.68: its exponent B	0.68	pass
119	E.kstepE	printed 4.04: its activation E [eV]	4.04	pass
120	E.jh49	claim: Johnson & Hunter's row 49, 4.8e-8 from Vriens		pass
121	E.tmpl_lo	printed 2.1: the template over Kannari's fit at the table's temperatures, smallest	2.09613	pass
122	E.tmpl_hi	printed 2.3: the same, largest	2.30965	pass
123	E.step_band	claim: the stepwise band moved f_best by less than 1.5 % (largest 0.0144)		pass
124	E.box	printed 0.2: the width of a Penning electron's line	0.2	pass
125	E.pen_lo	printed 2.1: the lowest Penning release energy	2.104	pass
126	E.pen_hi	printed 12.0: the highest	12	pass
127	E.gnu	claim: g/nu_m below 3e-3 at every growing point (largest 2.830e-03)		pass
128	E.tol	claim: the loop's tolerance 0.002		pass
129	E.floor	claim: the pulse floor 1000000.0 /s		pass
130	E.bnd_rule	claim: the boundary rule: within 10.0 tolerances, three flips		pass
131	E.rounds	claim: the loop took 3 to 72 rounds, the most at Kr 0.5 atm 1e14		pass
132	E.inc_cell	claim: the incomplete cell: {'records': 9, 'boundary': 4}		pass
133	E.kr_lo	printed 1.04: the Kr fill over the Ar fill, smallest	1.03944	pass
134	E.kr_hi	printed 1.2: the same, largest	1.20134	pass
135	E.kr_where	claim: the least at Kr 0.2 atm 3e14, the most at Kr 2 atm 3e15		pass
136	E.nu_lo	printed 1.08: nu_m of the Kr fill over the Ar fill's at f_best, smallest	1.08116	pass
137	E.nu_hi	printed 1.52: the same, largest	1.52106	pass
138	E.nu_sqrt	printed 3.5: sqrt of the nu ratio against the f ratio, largest difference (%)	3.46553	pass
139	E.sm_lo	printed 0.54: sigma_m(Kr)/sigma_m(Ar) at 1 eV	0.536335	pass

	check	what	value	
140	E.sm_hi	printed 1.55: the same at 3 eV	1.54808	pass
141	E.ar40	claim: the argon fill is 40 % krypton		pass
142	E.te_kr_lo	printed 0.94: the Kr fill's T_e, lowest	0.940115	pass
143	E.te_kr_hi	printed 2.44: highest	2.43674	pass
144	E.te_ar_lo	printed 0.87: the Ar fill's T_e, lowest	0.873805	pass
145	E.te_ar_hi	printed 2.65: highest	2.64608	pass
146	E.te_ne_lo	printed 1.4: the Ne fill's T_e, lowest	1.39604	pass
147	E.te_ne_hi	printed 4.69: highest	4.69366	pass
148	E.flags_a	claim: a marks [15, 8, 6]		pass
149	E.flags_b	claim: b marks [5, 9, 11]		pass
150	E.pnet_ne_lo	printed 0.007: p_net/P in the Ne fill, smallest	0.0070093	pass
151	E.pnet_ne_hi	printed 2.9: largest	2.89352	pass
152	E.pnet_ar_lo	printed 0.012: the Ar fill, smallest	0.0119025	pass
153	E.pnet_ar_hi	printed 15.0: largest	15.0481	pass
154	E.pnet_kr_lo	printed 0.017: the Kr fill, smallest	0.0173269	pass
155	E.pnet_kr_hi	printed 23.5: largest	23.4638	pass
156	E.pnet_where	claim: the most at 2 atm 1e16 in each: ['Ne 2 atm 1e16', 'Ar 2 atm 1e16', 'Kr 2 atm 1e16']		pass
157	E.vs_ne_lo	printed 0.933: the Ne fill over the calculation without excited atoms, smallest	0.932873	pass
158	E.vs_ne_hi	printed 0.997: largest	0.996997	pass
159	E.vs_ar_lo	printed 0.646: the Ar fill, smallest	0.646198	pass
160	E.vs_ar_hi	printed 0.995: largest	0.995163	pass
161	E.falls	claim: falling at every point (25 + 25)		pass
162	E.vs_where	claim: the most at Ar 2 atm 1e16		pass
163	E.f0	printed 34.9: f_best there without the excited atoms [THz]	34.8906	pass
164	E.f1	printed 22.5: and with them	22.5463	pass
165	E.tev_ar_lo	printed 0.55: T_e over without, the Ar fill, smallest	0.546861	pass
166	E.tev_ar_hi	printed 0.99: largest	0.990902	pass
167	E.tev_ne_lo	printed 0.83: the Ne fill, smallest	0.831089	pass
168	E.tev_ne_hi	printed 0.99: largest	0.992994	pass
169	E.bcells	claim: every band at 74 cells		pass
170	E.bk_lo	printed 1.0: Kannari's fit over the reference, smallest	0.999887	pass
171	E.bk_hi	printed 1.049: largest	1.04913	pass
172	E.bk_ne	printed 1.01: the Ne fill, largest	1.00989	pass
173	E.bk_ar	printed 1.047: the Ar fill, largest	1.04674	pass
174	E.bk_kr	printed 1.049: the Kr fill, largest	1.04913	pass
175	E.bk_where	claim: the most at Kr 2 atm 1e16		pass
176	E.bk_f0	printed 26.9: f_best there, the reference [THz]	26.8752	pass
177	E.bk_f1	printed 28.2: with the fit	28.1956	pass
178	E.bk_te	printed 6.7: the electrons hotter by at most (%)	6.67175	pass
179	E.b5p_lo	printed 1.0: the 5p rows out over the reference, smallest	0.999536	pass
180	E.b5p_hi	printed 1.46: largest	1.45957	pass
181	E.b5p_ne	printed 1.054: the Ne fill	1.05409	pass

	check	what	value	
182	E.b5p_ar	printed 1.46: the Ar fill	1.45957	pass
183	E.b5p_kr	printed 1.436: the Kr fill	1.43624	pass
184	E.b5p_where	claim: the most at Ar 2 atm 1e16		pass
185	E.b5p_f	printed 32.9: f_best there with the 5p rows out [THz]	32.9078	pass
186	E.br_lo	printed 0.868: the return at the measured bound over the reference, smallest	0.868491	pass
187	E.br_hi	printed 1.0: largest	0.999997	pass
188	E.br_ne	printed 0.999: the Ne fill, smallest	0.999023	pass
189	E.br_ar	printed 0.951: the Ar fill, smallest	0.950916	pass
190	E.br_kr	printed 0.868: the Kr fill, smallest	0.868491	pass
191	E.br_where	claim: the most at Kr 0.5 atm 1e16		pass
192	E.br_f0	printed 15.6: f_best there, the reference [THz]	15.5601	pass
193	E.br_f1	printed 13.5: at the bound	13.5138	pass
194	E.br_te	printed 0.81: T_e there over the reference	0.809503	pass
195	E.br_13	printed 13: the return rate's uncertainty in f_best (%)	13.1509	pass
196	E.br_19	printed 19: in T_e (%)	19.0497	pass
197	E.arspread	printed 0.28: the Ar levels above the lumped Ar*, largest (eV)	0.28	pass
198	E.gnu2	claim: the same g/nu_m bound		pass
199	E.nbnd	claim: one boundary point of the Ar fill ([('Ar 0.5 atm 3e14', 630957000000.0)])		pass
200	E.settle	claim: the excited atoms settle within 220 ns (slowest 211.5 ns)		pass
201	E.tf2_16lo	printed 24: attachment's F2 e-folding at 1e16, shortest [ns]	24.4416	pass
202	E.tf2_16hi	printed 56: longest	56.1545	pass
203	E.tf2_15lo	printed 0.44: at 1e15, shortest [us]	0.435913	pass
204	E.tf2_15hi	printed 0.97: longest	0.973668	pass
205	E.march_rule	claim: the drift 0.02, the renewal floor in the code		pass
206	E.march_held	claim: T_g 300 K held, the excimers sinks		pass
207	E.pulse_points	claim: the three pulses' points and frequencies		pass
208	E.pulse_f	claim: the wave's frequencies [10000000000000.0, 7700000000000.0, 19150000000000.0] and the points		pass
209	E.burn_lo	printed 300000000.0: F2 molecules burnt per electron per second at t = 0, least	3.18072e+08	pass
210	E.burn_hi	printed 600000000.0: most	6.43812e+08	pass
211	E.half15	printed 134: half the F2 gone at 1e15 [ns] (both fills)	134.01	pass
212	E.half15_ar	claim: the same in the Ar fill		pass
213	E.gone_kr	printed 365: F2 below 1 %, the Kr fill [ns]	364.652	pass
214	E.gone_ar	printed 383: the Ar fill [ns]	382.884	pass
215	E.rise_kr	printed 3.75: n_e runs up, the Kr fill	3.74961	pass
216	E.rise_ar	printed 3.95: the Ar fill	3.95174	pass
217	E.gzero_lo	printed 0.44: n_e steady (|g| < 1e6/s) by, earliest [us]	0.443237	pass
218	E.gzero_hi	printed 0.47: latest	0.465398	pass

	check	what	value	
219	E.te0_kr	printed 1.56: T_e at t = 0, the Kr fill	1.5596	pass
220	E.te0_ar	printed 1.52: the Ar fill	1.51863	pass
221	E.te1_kr	printed 0.535: at 1 us, the Kr fill	0.53494	pass
222	E.te1_ar	printed 0.542: the Ar fill	0.541683	pass
223	E.xkr_kr	printed 10: Kr* accumulates, the Kr fill (times)	10.3634	pass
224	E.xkr_ar	printed 8: the Ar fill	8.08276	pass
225	E.E0_kr	printed 38.9: the field at t = 0, the Kr fill [kV/cm]	38.8871	pass
226	E.E1_kr	printed 47.6: at 1 us	47.6354	pass
227	E.f100_kr	printed 2: f_best at 100 ns over the start, the Kr fill (%)	1.88791	pass
228	E.f100_ar	printed 3: the Ar fill (%)	3.05967	pass
229	E.fend_kr	printed 0.82: at 1 us over the start, the Kr fill	0.820822	pass
230	E.fend_ar	printed 0.94: the Ar fill	0.939059	pass
231	E.ne100_kr	printed 25: n_e at 100 ns over the start, the Kr fill (%)	25.0987	pass
232	E.ne100_ar	printed 28: the Ar fill (%)	27.7966	pass
233	E.te100_kr	printed 1.37: T_e at 100 ns, the Kr fill	1.37008	pass
234	E.te100_ar	printed 1.34: the Ar fill	1.33598	pass
235	E.f2100_kr	printed 0.61: F2 at 100 ns over its fill, the Kr fill	0.612692	pass
236	E.f2100_ar	printed 0.6: the Ar fill	0.600919	pass
237	E.hold3	claim: within 3 % in f_best at 100 ns (1.0189, 1.0306)		pass
238	E.half16	printed 11: half the F2 gone at 1e16 [ns]	11	pass
239	E.gone16	printed 35: below 1 % [ns]	34.7287	pass
240	E.rise16	printed 6.8: n_e runs up (times)	6.83088	pass
241	E.gzero16	printed 92: n_e steady by [ns]	92.1457	pass
242	E.te0_16	printed 0.94: T_e at t = 0	0.942549	pass
243	E.te1_16	printed 0.53: at 1 us	0.531929	pass
244	E.E0_16	printed 33.6: the field at t = 0 [kV/cm]	33.5685	pass
245	E.E1_16	printed 12.1: at 1 us	12.0529	pass
246	E.a0_16	printed 10.0: alpha at t = 0 [1/m]	10.0409	pass
247	E.a1_16	printed 78: at 1 us	78.3845	pass
248	E.depth16	printed 1.3: 1/alpha at 1 us [cm]	1.27576	pass
249	E.f30_16	printed 1.52: f_best at 30 ns over the start	1.51723	pass
250	E.fend_16	printed 2.75: at 1 us over the start	2.75057	pass
251	E.fend_16_thz	printed 53: f_best at 1 us [THz]	52.7725	pass
252	E.f100_16	claim: from 100 ns on: 100, 300 and 1000 ns within 0.3 % of each other ({'0': 19186025949982.11, '30': 29109668959022.47, '100': 52856943530711.71, '300': 52825798062356.71, '1000': 52772522820037.086})		pass
253	E.ne10_16	printed 32: n_e at 10 ns over the start (%)	31.6008	pass
254	E.f210_16	printed 0.54: F2 at 10 ns over its fill	0.53612	pass

	check	what	value	
255	E.sv_range	claim: seven states: n_e (1000000000000.0, 1e+16), p (0.1, 1.0), f (95000000000.0, 10000000000000.0), ions' share up to 0.250		pass
256	E.sv_agree	claim: mean energy within 0.2 % (1.65e-03), power within 0.2 % (1.64e-03), nu within 0.6 % (5.77e-03)		pass
257	E.sv_share1	printed 0.19: the ions' share at 0.1 atm 1e16 95 GHz, first solver	0.192773	pass
258	E.sv_share2	printed 0.17: second solver	0.171885	pass
259	E.sv_p95	printed 0.02: the total power there, |ratio - 1| (%)	0.0191881	pass
260	E.opt_agree	claim: alpha within 0.2 % (1.59e-03), R within 0.3 % (3.37e-03), the mobility's power within 1e-4 of the second solver's own (1.35e-04)		pass
261	E.opt_a_lo	printed 2.3: alpha over the seven states, smallest [1/m]	2.26871	pass
262	E.opt_a_hi	printed 34000.0: largest	33681.1	pass
263	E.opt_r_lo	printed 3e-08: R, smallest	2.50963e-08	pass
264	E.opt_r_hi	printed 0.91: largest	0.907518	pass
265	E.opt_p	claim: the mobility's power = the second solver's own to 1e-4 (1.35e-04)		pass
266	E.cold_state	claim: the cold start: the march's field 38900.0 V/cm, its closure ['growth'], both run 100 ns, no beam		pass
267	E.cold_g2	printed 217000000.0: the second model's decay rate, 10 to 100 ns [1/s]	2.16952e+08	pass
268	E.cold_g1	printed 215000000.0: the march's, 40 to 100 ns	2.14962e+08	pass
269	E.cold_g_apart	printed 1: the two apart (%)	0.925893	pass
270	E.cold_te2	printed 1.747: T_e at 100 ns, the second model	1.74692	pass
271	E.cold_te1	printed 1.742: the march	1.74164	pass
272	E.cold_te_apart	printed 0.3: apart (%)	0.302988	pass
273	E.cold_decades	claim: the second model loses three decades of n_e in its first 10 ns (3.29)		pass
274	E.cold_f2_2	printed 0.9: F2 burnt in 100 ns, the second model (%)	0.861654	pass
275	E.cold_f2_1	printed 1.4: the march (%)	1.43963	pass
276	E.cold_tg	printed 3.4: the gas warms by [K]	3.41622	pass
277	E.st_13	printed 13: the return rate's uncertainty in f_best (%)	13.1509	pass
278	E.st_19	printed 19: in T_e (%)	19.0497	pass
279	E.st_hold	claim: the first 100 ns at 1e15 and 10 ns at 1e16		pass
280	E.st_02	claim: the solvers to 0.2 %		pass
281	S.semicolon	no semicolon in the prose	0 found	pass
282	S.emdash	no em-dash in the prose	0 found	pass
283	S.placeholders	no unfilled placeholder	0 found []	pass

	check	what	value	
284	`S.self_contained`	no local file, path, model or unpublished note named in the printed source, the generated tables or the PDF's text	source [], PDF []	pass

5.B The rows

The heavy-particle reactions of the excited atoms, Table 5.9, at 300 K, and the reactions of the cations, Table 5.10, in SI units as the ion balance takes them, $\mathrm{m^3\,s^{-1}}$ for two bodies and $\mathrm{m^6\,s^{-1}}$ for three, with T_e in eV. F(a, s) is $aN/(N^2+1.7)\,s$ and L(A, b, c) is $AN/(N^2+bN+c)$, both in $\mathrm{cm^3\,s^{-1}}$ with N in amagat. A rate run as a band stands at its default.

Table 5.9: The heavy-particle reactions of the excited atoms.

reaction	rate	standing	source
$2\ \mathrm{Kr^*} \rightarrow \mathrm{Kr^+} + \mathrm{Kr} + e$	$5.00 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	printed	5.0×10^{-10} of the second table, run as a band, [7]
$\mathrm{Kr^*} + \mathrm{F_2} \rightarrow \mathrm{KrF(B)} + \mathrm{F}$	$4.73 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	measured	the mean of three measurements, $\mathrm{KrF^*}$ yield 0.90, B to C 62 to 38, [13, 14, 15]
$\mathrm{Kr^*} + \mathrm{F_2} \rightarrow \mathrm{KrF(C)} + \mathrm{F}$	$2.90 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	measured	as the row above, the C share, [13, 14, 15]
$\mathrm{Kr^*} + \mathrm{F_2} \rightarrow \mathrm{Kr} + 2\ \mathrm{F}$	$8.47 \times 10^{-11}\ \mathrm{cm^3\,s^{-1}}$	measured	the 10 % without $\mathrm{KrF^*}$, [13]
$\mathrm{Kr^*} + 2\ \mathrm{Kr} \rightarrow \mathrm{Kr_2^*} + \mathrm{Kr}$	$1.70 \times 10^{-32}\ \mathrm{cm^6\,s^{-1}}$, $\times(T_g/300)^{-0.64}$	measured	the temperature law of the second, [14, 16]
$\mathrm{Kr_2^*} \rightarrow 2\ \mathrm{Kr}$	$3.70 \times 10^{6}\ \mathrm{s^{-1}}$	measured	270 ns, [17]
$\mathrm{Kr_2^*} + \mathrm{F_2} \rightarrow \mathrm{KrF(B)} + \mathrm{Kr} + \mathrm{F}$	$1.05 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	measured	the total measured, split evenly, [18]
$\mathrm{Kr_2^*} + \mathrm{F_2} \rightarrow \mathrm{Kr_2F^*} + \mathrm{F}$	$1.05 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	measured	the total measured, split evenly, [18]
$\mathrm{Kr_2^*} + \mathrm{F} \rightarrow \mathrm{KrF(B)} + \mathrm{Kr}$	$0\ \mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$2\ \mathrm{Kr_2^*} \rightarrow \mathrm{Kr_2^+} + 2\ \mathrm{Kr} + e$	$0\ \mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$2\ \mathrm{Ar^*} \rightarrow \mathrm{Ar^+} + \mathrm{Ar} + e$	$0\ \mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$\mathrm{Ar^*} + \mathrm{F_2} \rightarrow \mathrm{ArF(B)} + \mathrm{F}$	$4.05 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	measured	the mean of three groups, $\mathrm{ArF^*}$ yield 0.5, [13, 21, 14, 22]
$\mathrm{Ar^*} + \mathrm{F_2} \rightarrow \mathrm{Ar} + 2\ \mathrm{F}$	$4.05 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	measured	the half without $\mathrm{ArF^*}$, [22]
$\mathrm{Ar^*} + 2\ \mathrm{Ar} \rightarrow \mathrm{Ar_2^*} + \mathrm{Ar}$	$6.00 \times 10^{-33}\ \mathrm{cm^6\,s^{-1}}$	measured	[14]
$\mathrm{Ar_2^*} \rightarrow 2\ \mathrm{Ar}$	$3.12 \times 10^{5}\ \mathrm{s^{-1}}$	measured	the triplet, 3.2 μs, [20]
$\mathrm{Ar_2^*} + \mathrm{F_2} \rightarrow \mathrm{ArF(B)} + \mathrm{Ar} + \mathrm{F}$	$4.73 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	measured	$\mathrm{ArF^*}$ share 0.91, [21, 22]
$\mathrm{Ar_2^*} + \mathrm{F_2} \rightarrow \mathrm{Ar_2F^*} + \mathrm{F}$	$4.68 \times 10^{-11}\ \mathrm{cm^3\,s^{-1}}$	measured	the rest, $\mathrm{Ar_2F^*}$, [21, 22]
$\mathrm{Ar_2^*} + \mathrm{F} \rightarrow \mathrm{ArF(B)} + \mathrm{Ar}$	$0\ \mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$2\ \mathrm{Ar_2^*} \rightarrow \mathrm{Ar_2^+} + 2\ \mathrm{Ar} + e$	$0\ \mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$2\ \mathrm{Ne^*} \rightarrow \mathrm{Ne^+} + \mathrm{Ne} + e$	$5.00 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	printed	[25]
$\mathrm{Ne^*} + 2\ \mathrm{Ne} \rightarrow \mathrm{Ne_2^*} + \mathrm{Ne}$	$4.00 \times 10^{-34}\ \mathrm{cm^6\,s^{-1}}$, $\times(T_g/300)^{-0.75}$	printed	[24, 23]
$\mathrm{Ne^*} + \mathrm{Kr} \rightarrow \mathrm{Kr^+} + \mathrm{Ne} + e$	$6.50 \times 10^{-11}\ \mathrm{cm^3\,s^{-1}}$	analog	of $\mathrm{Ne^*}$ + Ar, [23]
$\mathrm{Ne^*} + \mathrm{Kr} \rightarrow \mathrm{NeKr^+} + e$	$2.30 \times 10^{-11}\ \mathrm{cm^3\,s^{-1}}$	analog	of $\mathrm{Ne^*}$ + Ar, associative, [23]
$\mathrm{Ne^*} + \mathrm{F_2} \rightarrow \mathrm{NeF^*} + \mathrm{F}$	$5.30 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	printed	[23]
$\mathrm{Kr^*} + \mathrm{Kr} + \mathrm{Ne} \rightarrow \mathrm{Kr_2^*} + \mathrm{Ne}$	$0\ \mathrm{cm^6\,s^{-1}}$, $\times(T_g/300)^{-0.75}$	none, 0	analog of $\mathrm{Ar^*}$ + Ar + Ne, a band from 0
$\mathrm{Ne_2^*} \rightarrow 2\ \mathrm{Ne}$	$7.50 \times 10^{7}\ \mathrm{s^{-1}}$	printed	[24]
$\mathrm{Ne_2^*} + \mathrm{Kr} \rightarrow \mathrm{Kr^+} + 2\ \mathrm{Ne} + e$	$6.50 \times 10^{-11}\ \mathrm{cm^3\,s^{-1}}$	analog	of $\mathrm{Ne_2^*}$ + Ar, [23, 26]
$2\ \mathrm{Ne_2^*} \rightarrow \mathrm{Ne_2^+} + 2\ \mathrm{Ne} + e$	$5.00 \times 10^{-10}\ \mathrm{cm^3\,s^{-1}}$	printed	[26]
$\mathrm{Ar^*} + \mathrm{Kr} \rightarrow \mathrm{Kr^*} + \mathrm{Ar}$	$6.20 \times 10^{-12}\ \mathrm{cm^3\,s^{-1}}$	printed	[8]
$\mathrm{Kr^*} + \mathrm{Kr} + \mathrm{Ar} \rightarrow \mathrm{Kr_2^*} + \mathrm{Ar}$	$2.70 \times 10^{-32}\ \mathrm{cm^6\,s^{-1}}$, $\times(T_g/300)^{-0.75}$	printed	[7]
$\mathrm{Kr^*} + 2\ \mathrm{Ar} \rightarrow \mathrm{ArKr^*} + \mathrm{Ar}$	$1.00 \times 10^{-32}\ \mathrm{cm^6\,s^{-1}}$	printed	[7, 8]

reaction	rate	standing	source
$Ar^* + Kr + Ar \rightarrow ArKr^* + Ar$	1.00×10^{-32} cm^6 s^{-1}	printed	[7, 8]
$Ar_2^* + Kr \rightarrow Kr^* + 2\ Ar$	4.00×10^{-10} cm^3 s^{-1}	printed	[7, 8]
$ArKr^* + F_2 \rightarrow KrF(B) + Ar + F$	3.00×10^{-10} cm^3 s^{-1}	printed	split evenly to B and C, [7]
$ArKr^* + F_2 \rightarrow KrF(C) + Ar + F$	3.00×10^{-10} cm^3 s^{-1}	printed	split evenly to B and C, [7]
$ArKr^* + F_2 \rightarrow ArKrF^* + F$	3.00×10^{-10} cm^3 s^{-1}	printed	[7]
$ArKr^* + Kr \rightarrow Kr_2^* + Ar$	1.00×10^{-10} cm^3 s^{-1}	printed	[7, 8]
$ArKr^* \rightarrow Ar + Kr$	8.00×10^{7} s^{-1}	estimated	estimated there, [7]
$Kr^{**} + Ne \rightarrow Kr^* + Ne$	1.00×10^{-10} cm^3 s^{-1}	estimated	of $Kr^{**} + Ar$, estimated there, [7]
$Kr^{**} + Ar \rightarrow Kr^* + Ar$	1.00×10^{-10} cm^3 s^{-1}	estimated	estimated there, [7]
$Kr^{**} + Kr \rightarrow Kr^* + Kr$	1.00×10^{-10} cm^3 s^{-1}	estimated	of $Kr^{**} + Ar$, estimated there, run as a band to the measured total loss of one $5p'$ level, [7, 28]
$Kr^{**} + F_2 \rightarrow KrF(B) + F$	4.05×10^{-10} cm^3 s^{-1}	printed	split evenly to B and C, [8]
$Kr^{**} + F_2 \rightarrow KrF(C) + F$	4.05×10^{-10} cm^3 s^{-1}	printed	split evenly to B and C, [8]
$Kr^{**} \rightarrow Kr^*$	3.45×10^{7} s^{-1}	measured	the $2J + 1$ mean of the ten level lifetimes, 29.0 ns, [29]

Table 5.10: The reactions of the cations.

reaction	rate	standing	source
$Ne^+ + Ne + Ne \rightarrow Ne_2^+ + Ne$	4.4×10^{-44}	printed	[25]
$Ne^+ + Kr \rightarrow Kr^+ + Ne$	1×10^{-17}	analog	of $Ne^+ + Ar$ and $Ne^+ + Xe$, [26]
$Ne^+ + Kr + Ne \rightarrow NeKr^+ + Ne$	1×10^{-43}	printed	assumed there, [27]
$Ne_2^+ + Kr \rightarrow Kr^+ + Ne + Ne$	5×10^{-19}	printed	the published upper bound, [27]
$e + Ne_2^+ \rightarrow Ne + Ne^*$	$3.7 \times 10^{-14}\, T_e^{-0.43}$	printed	exponent of the second, [25, 26]
$NeKr^+ + Kr \rightarrow Kr^+ + Ne + Kr$	1.1×10^{-15}	analog	of $NeAr^+ + Ar$, [23]
$NeKr^+ + Kr \rightarrow Kr_2^+ + Ne$	1.2×10^{-17}	analog	of $NeAr^+ + Ar$, [23]
$e + NeKr^+ \rightarrow Kr^{**} + Ne$	$1 \times 10^{-13}\, T_e^{-1/2}$	printed	[27]
$Kr^+ + Kr + Ne \rightarrow Kr_2^+ + Ne$	2.5×10^{-43}	estimated	within the measured analogs
$Kr^+ + Kr + Kr \rightarrow Kr_2^+ + Kr$	2.5×10^{-43}	printed	[8, 7, 27]
$e + Kr_2^+ \rightarrow Kr^* + Kr$	$2.24 \times 10^{-13}\, T_e^{-1/2}$	printed	both branches on $T_e^{-1/2}$, [7]
$Ar^+ + Ar + Ar \rightarrow Ar_2^+ + Ar$	2.5×10^{-43}	printed	[7]
$Ar^+ + Kr \rightarrow Kr^+ + Ar$	3×10^{-17}	printed	[7]
$Ar^+ + Kr + Ar \rightarrow ArKr^+ + Ar$	1×10^{-43}	printed	[7, 8]
$Ar_2^+ + Kr \rightarrow Kr^+ + Ar + Ar$	7.5×10^{-16}	printed	[7, 8]
$e + Ar_2^+ \rightarrow Ar^* + Ar$	$1.14 \times 10^{-13}\, T_e^{-1/2}$	printed	both branches on $T_e^{-1/2}$, [7]
$Kr^+ + Ar + Ar \rightarrow ArKr^+ + Ar$	1×10^{-43}	printed	[7, 8]
$ArKr^+ + Kr \rightarrow Kr_2^+ + Ar$	3.2×10^{-16}	printed	[7, 8]
$e + ArKr^+ \rightarrow Kr^{**} + Ar$	$1.93 \times 10^{-13}\, T_e^{-1/2}$	printed	on $T_e^{-1/2}$, [7]
$Kr^+ + Kr + Ar \rightarrow Kr_2^+ + Ar$	2.5×10^{-43}	printed	[7, 8]
$Kr^+ + F^- \rightarrow KrF^*$	F(2.4, 3.417×10^{-6})	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]
$Kr_2^+ + F^- \rightarrow KrF^* + Kr$	F(2.65, 3.096×10^{-6})	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]
$Ne^+ + F^- \rightarrow NeF^*$	L(2.8×10^{-5}, 1.44, 10.9)	printed	[23]
$Ne_2^+ + F^- \rightarrow NeF^* + Ne$	L(3.66×10^{-5}, 1.74, 17.3)	printed	[23]
$NeKr^+ + F^- \rightarrow KrF^* + Ne$	L(5.27×10^{-5}, 6.81, 20.3)	analog	of $NeAr^+ + F^-$, [23]
$Ar^+ + F^- \rightarrow ArF^*$	F(2.4, 3.417×10^{-6})	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]
$Ar_2^+ + F^- \rightarrow ArF^* + Ar$	F(2.65, 3.096×10^{-6})	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]

reaction	rate	standing	source
$ArKr^+ + F^- \to KrF^* + Ar$	$F(2.65,\ 3.096 \times 10^{-6})$	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]

5.C Every point

Table 5.11: The 75 points at $L = 10$ cm and 300 kW/cm^3 with the excited atoms inside the electrons' balances and the measured 5s to 5p cross section: the best frequency in THz, the electron temperature at it in eV, the marks of Table 5.3, the mole fraction of Kr^* in units of 10^{-4}, the net energy the excited atoms take from the electrons over the absorbed power, then the best frequency of each band over this one, with Kannari's fit in place of the measured cross section (k), with the 5p rung's electron collisions left out ($5p$) and with the collisional return of 5p at the measured bound (r), and the best frequency over the calculation without excited atoms where that exists (f/f_0).

fill	p	n_e	f_{best}	T_e		x_{Kr^*}	p_{net}/P	k	$5p$	r	f/f_0
Ne	0.1	10^{14}	0.64	4.69	a	0.25	0.01	1.000	1.001	1.000	0.997
Ne	0.1	3×10^{14}	1.03	3.66	a	0.74	0.05	1.001	1.007	1.000	0.989
Ne	0.1	10^{15}	1.74	2.84	a	0.88	0.13	1.001	1.010	1.000	0.979
Ne	0.1	3×10^{15}	2.90	2.27	a	0.63	0.17	1.000	1.003	1.000	0.984
Ne	0.1	10^{16}	5.71	1.83	a	0.36	0.18	1.000	1.000	1.000	0.997
Ne	0.2	10^{14}	0.86	4.03	a	0.20	0.01	1.000	1.002	1.000	0.997
Ne	0.2	3×10^{14}	1.39	3.22	a	0.62	0.07	1.002	1.011	1.000	0.984
Ne	0.2	10^{15}	2.35	2.53	a	0.70	0.19	1.002	1.017	1.000	0.970
Ne	0.2	3×10^{15}	3.85	2.04	a	0.47	0.23	1.000	1.009	1.000	0.975
Ne	0.2	10^{16}	7.21	1.68	–	0.28	0.25	1.000	1.001	1.000	0.989
Ne	0.5	10^{14}	1.28	3.40	a	0.15	0.02	1.001	1.003	1.000	0.996
Ne	0.5	3×10^{14}	2.07	2.76	a	0.49	0.13	1.005	1.022	1.000	0.974
Ne	0.5	10^{15}	3.45	2.11	a	0.60	0.36	1.006	1.043	0.999	0.943
Ne	0.5	3×10^{15}	5.66	1.78	–	0.35	0.41	1.001	1.023	1.000	0.957
Ne	0.5	10^{16}	10.27	1.54	b	0.24	0.52	1.000	1.009	1.000	0.974
Ne	1	10^{14}	1.74	3.06	a	0.12	0.02	1.001	1.004	1.000	0.995
Ne	1	3×10^{14}	2.79	2.44	a	0.44	0.21	1.010	1.038	1.000	0.958
Ne	1	10^{15}	4.69	1.93	–	0.40	0.47	1.008	1.054	1.000	0.933
Ne	1	3×10^{15}	7.70	1.65	b	0.27	0.65	1.003	1.039	0.999	0.945
Ne	1	10^{16}	13.79	1.47	b	0.24	1.15	1.001	1.022	1.000	0.958
Ne	2	10^{14}	2.36	2.80	a	0.10	0.04	1.002	1.007	1.000	0.993
Ne	2	3×10^{14}	3.86	2.29	–	0.24	0.22	1.009	1.036	1.000	0.958
Ne	2	10^{15}	6.58	1.90	–	0.19	0.46	1.008	1.046	1.000	0.945
Ne	2	3×10^{15}	10.66	1.58	b	0.19	1.00	1.006	1.049	0.999	0.938
Ne	2	10^{16}	18.78	1.40	b	0.26	2.89	1.003	1.045	0.999	0.935
Ar	0.1	10^{14}	1.00	2.65	a	0.64	0.01	1.001	1.003	1.000	0.995
Ar	0.1	3×10^{14}	1.67	2.30	a	1.93	0.08	1.005	1.024	0.997	0.966
Ar	0.1	10^{15}	2.87	2.01	a	1.96	0.19	1.003	1.031	0.996	0.944
Ar	0.1	3×10^{15}	4.58	1.70	a	1.43	0.24	1.000	1.017	0.998	0.952
Ar	0.1	10^{16}	8.16	1.42	–	0.97	0.29	1.000	1.006	0.999	0.969
Ar	0.2	10^{14}	1.34	2.34	a	0.62	0.02	1.002	1.008	0.999	0.991
Ar	0.2	3×10^{14}	2.19	2.03	a	1.80	0.13	1.011	1.050	0.995	0.942
Ar	0.2	10^{15}	3.76	1.79	a	1.71	0.30	1.008	1.065	0.989	0.910
Ar	0.2	3×10^{15}	6.02	1.55	–	1.18	0.36	1.001	1.036	0.995	0.925
Ar	0.2	10^{16}	10.46	1.33	b	0.95	0.51	1.000	1.019	0.998	0.944
Ar	0.5	10^{14}	1.95	2.04	a	0.71	0.05	1.007	1.023	0.999	0.976
Ar	0.5	3×10^{14}	3.03	1.65	–	1.94	0.29	1.031	1.138	0.996	0.864

fill	p	n_e	f_{best}	T_e		x_{Kr^*}	p_{net}/P	k	$5p$	r	f/f_0
Ar	0.5	10^{15}	5.32	1.52	–	1.34	0.53	1.023	1.140	0.988	0.847
Ar	0.5	3×10^{15}	8.55	1.36	*b*	1.13	0.86	1.010	1.116	0.980	0.860
Ar	0.5	10^{16}	14.53	1.18	*b*	1.46	1.81	1.004	1.097	0.974	0.865
Ar	1	10^{14}	2.56	1.88	–	0.71	0.09	1.013	1.043	1.000	0.956
Ar	1	3×10^{14}	4.35	1.70	–	0.72	0.23	1.024	1.093	0.998	0.905
Ar	1	10^{15}	7.48	1.52	*b*	0.72	0.62	1.027	1.140	0.993	0.857
Ar	1	3×10^{15}	11.03	1.21	*b*	1.19	1.79	1.035	1.226	0.974	0.786
Ar	1	10^{16}	16.70	0.90	*b*	3.57	6.48	1.036	1.349	0.951	0.699
Ar	2	10^{14}	3.51	1.91	–	0.20	0.05	1.007	1.020	1.000	0.979
Ar	2	3×10^{14}	6.31	1.79	–	0.23	0.16	1.015	1.052	0.999	0.945
Ar	2	10^{15}	10.98	1.61	*b*	0.33	0.65	1.022	1.105	0.997	0.893
Ar	2	3×10^{15}	15.75	1.23	*b*	0.88	3.08	1.043	1.245	0.986	0.780
Ar	2	10^{16}	22.55	0.87	*b*	3.37	15.05	1.047	1.460	0.971	0.646
Kr	0.1	10^{14}	1.05	2.44	*a*	1.26	0.02	1.002	1.006	0.998	–
Kr	0.1	3×10^{14}	1.77	2.13	*a*	3.44	0.12	1.008	1.039	0.989	–
Kr	0.1	10^{15}	3.10	1.87	*a*	3.06	0.24	1.004	1.044	0.987	–
Kr	0.1	3×10^{15}	4.95	1.58	*a*	2.31	0.29	1.001	1.025	0.992	–
Kr	0.1	10^{16}	8.82	1.34	*b*	2.00	0.42	1.000	1.013	0.997	–
Kr	0.2	10^{14}	1.39	2.13	*a*	1.25	0.03	1.004	1.014	0.998	–
Kr	0.2	3×10^{14}	2.28	1.81	*a*	3.42	0.21	1.022	1.089	0.975	–
Kr	0.2	10^{15}	3.98	1.62	–	2.88	0.41	1.012	1.103	0.972	–
Kr	0.2	3×10^{15}	6.55	1.46	–	1.89	0.48	1.002	1.056	0.981	–
Kr	0.2	10^{16}	11.36	1.26	*b*	2.10	0.87	1.000	1.039	0.988	–
Kr	0.5	10^{14}	–	–	–	–	–	–	–	–	–
Kr	0.5	3×10^{14}	3.38	1.63	–	1.78	0.26	1.028	1.112	0.993	–
Kr	0.5	10^{15}	6.01	1.52	–	1.45	0.57	1.022	1.131	0.978	–
Kr	0.5	3×10^{15}	9.38	1.30	*b*	1.81	1.25	1.016	1.151	0.940	–
Kr	0.5	10^{16}	15.56	1.10	*b*	3.43	3.48	1.010	1.165	0.868	–
Kr	1	10^{14}	2.75	1.79	–	0.59	0.07	1.010	1.031	0.999	–
Kr	1	3×10^{14}	4.89	1.70	–	0.62	0.20	1.019	1.069	0.997	–
Kr	1	10^{15}	8.64	1.56	*b*	0.74	0.66	1.024	1.118	0.989	–
Kr	1	3×10^{15}	12.72	1.24	*b*	1.59	2.47	1.035	1.222	0.951	–
Kr	1	10^{16}	19.15	0.94	*b*	5.13	10.28	1.036	1.364	0.907	–
Kr	2	10^{14}	3.76	1.83	–	0.16	0.04	1.005	1.014	1.000	–
Kr	2	3×10^{14}	7.12	1.81	*b*	0.19	0.14	1.011	1.036	0.999	–
Kr	2	10^{15}	12.81	1.68	*b*	0.33	0.69	1.018	1.080	0.996	–
Kr	2	3×10^{15}	18.92	1.33	*b*	1.01	4.04	1.037	1.203	0.975	–
Kr	2	10^{16}	26.88	0.95	*b*	4.30	23.46	1.049	1.436	0.936	–

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Chapter 6

Nonlinear inverse bremsstrahlung and the efficiency of gas lasers

The note of October 20, 2026.

Inverse bremsstrahlung absorption is nonlinear: a strong enough field reshapes the electron distribution that absorbs it, as Langdon showed, as Milder et al. observed directly and as Turnbull et al. have now measured in absorption [1, 5, 3, 4]. This note asks whether that nonlinearity can be used to make gas lasers more efficient. It locates the lasers and the available sources against the thresholds of the three nonlinearities, builds a kinetic equation for Langdon's and the gas's nonlinearity (the field, electron–atom and electron–ion collisions, and the electron–electron collisions that oppose the field, in the classical small-angle form of the collision integral whose quantum form is derived in [6]), checks it against Langdon's absorption factor, and applies it to the KrF laser gas, with growing and with beam-sustained electrons, and to the target that excimer drivers illuminate. Silin's regime is reached nowhere here. Every number computed here is checked numerically, as Appendix 6.B describes. The results are in silico.

SI units are used, with electron energies ϵ and temperatures T in eV. The isotropic electron distribution $F(\epsilon)$ is normalized as $\int_0^\infty \epsilon^{1/2} F \,\mathrm{d}\epsilon = 1$, $\gamma = (2e/m)^{1/2}$, $v = \gamma\epsilon^{1/2}$, and 1 Td $= 10^{-21}$ V m^2. A field $E_0 \cos\omega t$ has the rms value $E = E_0/\sqrt{2}$, the quiver amplitude $v_0 = eE_0/m\omega$ and, for linear polarization, the intensity $I = c\epsilon_0 E_0^2/2$. Two fills of the KrF laser gas are used: the Ne fill, Ne/Kr/F_2 = 0.948/0.05/0.002 at 2 atm, and the Ar fill, Ar/Kr/F_2 = 0.597/0.40/0.003 at 1 atm, both at 300 K, with the LXCat cross sections BSR for neon, Biagi-v7.1 for krypton, IST-Lisbon for argon and SIGLO for F_2 [27]. The ions (cations and F^-) are taken to number $1.15\,n_e$ at every n_e, the mean over the pump window that a kinetic model of an electron-beam-pumped laser gives for Kr/F_2 at 665 Torr with 2.8 Torr of F_2, taken for both fills. A revised form of that model gives 1.35, and with it krypton's share at beam-on density moves by at most 0.02 points.

6.1 The question and the answer

(1) A gas laser converts the energy of its electrons into excitation. The efficiency of that step depends on the shape of the electron distribution, because the excitation thresholds sit in its tail. Inverse bremsstrahlung (IB) is the way a field of frequency ω heats electrons, and its nonlinearity is that the heating reshapes the distribution. The question is whether a gas laser can be run in the nonlinear regime, by its own field or by an auxiliary one, so that the reshaped distribution puts more of the electrons' energy where the laser wants it.

(2) The answer has four parts.

- The lasing field of an excimer laser is linear. At a few saturation intensities (10 MW cm^{-2} for KrF, 30 for ArF) its intensity is $10^{3.8}$ to $10^{4.2}$ times below the intensity at which IB alone would supply the electrons' energy budget at $T_e = 1.5$ eV (at least 1.4×10^3 times for T_e from 1.08 to 2.04 eV), and further still below Langdon's and Silin's thresholds (Sec. 6.5). Of the lasers examined, the CO_2 laser is the exception: by an order-of-magnitude estimate its own field supplies about a tenth of its discharge electrons' budget at 100 MW cm^{-2}.

- An auxiliary field (microwave to far infrared) reaches only the nonlinearity of a weakly ionized gas, in which the field sets the electron temperature. It does so long before Langdon's flattening or Silin's saturation, which need tens and hundreds of times more intensity (33 to 80 and 380 to 920 times at the temperatures of Table 6.2).

- In that regime the frequency is a real lever on the distribution while electron–electron collisions are rare. A field far above the collision frequency roughly doubles krypton's share of the absorbed power at low power (×2.0 to ×2.1 in the Ne fill at the power of 2 Td and ×2.1 to ×2.4 in the Ar fill at 6 Td, with growing or with beam-sustained electrons at a fixed electron density), and multiplies it by 12 to 32 in the Ar fill at 3 Td, where the DC share is only 1 to 5 %. Those collisions remove most of the lever at the densities of a pumped laser. Where the lever is large they do not spoil the high-frequency distribution: they pull both distributions towards the Maxwellian, and the DC one catches up. The advantage halves at a generalized Langdon parameter $\alpha_g = 0.41$ to 0.55. At the beam-on density of a pumped laser, 2.72×10^{21} m^{-3}, with the growth model or with beam-sustained electrons that recombine as Kr_2^+, it is 2 to 7 % where the field's power exceeds the beam's own and 10 to 14 % at a fixed beam current where the beam's is larger (up to 27 % at fixed n_e). Without recombination it is 1 to 18 %, and 3 to 7 % at a fixed beam current. With one electron per ion pair the beam-sustained model holds that density only with 13 times the beam power that gives it in a full kinetic model of the laser, and at the five times lower density it holds with that power the advantage is up to ×1.24 at a fixed beam current and ×1.355 at fixed n_e (Sec. 6.6). The krypton share of a self-sustained discharge differs from its DC value by at most 1.5 percentage points in the Ne fill, and falls by up to ten points at high frequency in the Ar fill (Secs. 6.6 and 6.7).

- At the target the nonlinearity favours the short wavelengths of the gas lasers. At the quarter-critical conditions that NRL computed for 351, 248 and 193 nm, the Langdon factor of a homogeneous plasma is 0.94, 0.96 and 0.98 with the velocity-dependent logarithm. That factor alone raises the local absorption rate of KrF and ArF light by 2.3 and 3.8 % over 351 nm, on top of the much larger linear gain of the short wavelengths. Speckle and nonlocal transport, left out here, may change it (Sec. 6.9).

(3) The nonlinearity of IB is therefore not, by itself, a lever on the efficiency of a gas laser. What the computation does show is nearly frequency-blind and favourable. Field pumping of the beam-ionized Ne fill at beam-on density puts 0.30 to 0.35 of the absorbed energy into KrF^* where its power exceeds the beam's own, counted at the photon energy, depending on the power and on the electron-source model. The e-beam itself reaches 0.20 in the same accounting once its sub-excitation electrons are credited, so field pumping forms 1.5 to 1.8 times as much there, and the frequency moves that by −4 to +6 % (Sec. 6.8). The accounting leaves out electron collisions with Kr^* and KrF^*, which compete with the formation at beam-on density. Whether field pumping pays is a question of source efficiency, stability and chemistry, which this note does not address. The

most useful application of the nonlinear absorption is an instrument: a THz pulse that sets T_e on demand and a transmission that reads it back.

6.2 Three nonlinearities of inverse bremsstrahlung

(1) *Langdon's nonlinearity: the distribution flattens.* In a fully ionized plasma with $\omega \gg \nu_{ei}$ the field heats electrons through their collisions with ions with the velocity-space diffusion coefficient $(v_0^2/6)\nu_{ei}(v)$ [1], and $\nu_{ei} \propto v^{-3}$ heats the slow electrons most. Electron–electron collisions restore the Maxwellian. Their competition is measured by Langdon's parameter

$$\alpha_L = \frac{Z_{\text{eff}} v_0^2}{v_t^2}, \qquad v_t^2 = \frac{eT}{m}, \qquad Z_{\text{eff}} = \frac{\langle Z^2 \rangle}{\langle Z \rangle}, \tag{6.1}$$

As α_L grows the distribution becomes flat-topped, which lowers the absorption by the factor f_L. Langdon found the absorption 12 % lower at $\alpha_L = 0.05$, 67 % of its Maxwellian value at 0.5 and 45 % in the limit [1]. This note takes the onset at $\alpha_L = 0.1$, where the self-similar solution of Sec. 6.4 gives $f_L = 0.83$ (Langdon's fit 0.82). Turnbull et al. measured the absorption of a 527 nm probe in gas-jet plasmas and found that it needs this factor [3]. The path-integrated f_L averaged 0.76 over all their data, and even over the eight pure-hydrogen shots it averaged 0.81 and fell as low as 0.74.

(2) *Silin's nonlinearity: the quiver outruns the electron.* When $v_0 \gtrsim v_t$ the collision samples the oscillating relative velocity and the absorption falls as $(v_t/v_0)^3$ up to a logarithm [16].

(3) *The nonlinearity of a weakly ionized gas: the field sets the temperature.* In a laser gas, n_e/N is 10^{-7} to 10^{-4} and electrons lose energy mainly to atoms and molecules. Elastic collisions remove only the fraction $2m/M$ of the energy per collision, 1.3×10^{-5} (Kr) to 5.4×10^{-5} (Ne), and the inelastic channels raise the effective fraction δ_{eff} to 4×10^{-4} to 8×10^{-4} in the excimer fills (Sec. 6.5) and to about 10^{-2} in a CO_2 laser discharge. A field heats an electron at the rate $P = \sigma_{\text{abs}} I$, where, in the Lorentz two-term approximation,

$$\sigma_{\text{abs}} = \frac{e \,\text{Re}\, \mu(\omega)}{c\epsilon_0} \xrightarrow{\omega \gg \nu} \frac{e^2}{mc\epsilon_0} \frac{\nu_{\text{eff}}}{\omega^2}, \tag{6.2}$$

μ being the complex mobility. The field takes over the temperature once P exceeds the collisional loss $P_{\text{loss}}(T_e)$ that holds T_e, at

$$I_{\text{heat}} = \frac{P_{\text{loss}}(T_e)}{\sigma_{\text{abs}}(\omega)}, \qquad I_L(\alpha_L) = \frac{c\epsilon_0}{2}\left(\frac{m\omega}{e}\right)^2 \frac{\alpha_L \, eT}{mZ_{\text{eff}}}, \qquad I_S = \frac{c\epsilon_0}{2}\left(\frac{m\omega}{e}\right)^2 \frac{eT}{m}, \tag{6.3}$$

the last two being the intensities of a given α_L and of $v_0 = v_t$. For $\omega \gg \nu$, Eqs. (6.2) and (6.3) give

$$\frac{I_L}{I_{\text{heat}}} = \frac{\alpha_L \, eT \, \nu_{\text{eff}}}{2 Z_{\text{eff}} P_{\text{loss}}} = \frac{\alpha_L}{3 Z_{\text{eff}} \, \delta_{\text{eff}}}, \qquad P_{\text{loss}} = \tfrac{3}{2} \, eT \, \delta_{\text{eff}} \, \nu_{\text{eff}}. \tag{6.4}$$

A gas's electrons thus answer to an intensity $\alpha_L / 3 Z_{\text{eff}} \delta_{\text{eff}}$ times lower than the one that gives a plasma's distribution a given α_L, 33 to 80 times lower for $\alpha_L = 0.1$ at the temperatures of Table 6.2. I_{heat} is the threshold that Ginzburg and Gurevich called the plasma field: below it a wave crosses a weakly ionized plasma without changing it appreciably [29, 30]. In the gas, the nonlinearity is that the distribution, and with it ν_{eff} and the absorption, follows the field. Its shape depends on ω: a DC field heats each electron in proportion to $1/\nu_m(\epsilon)$, a field with $\omega \gg \nu_m$ in proportion to $\nu_m(\epsilon)$ (Sec. 6.3).

(4) *The generalized Langdon parameter.* In every case the departure from the Maxwellian is decided by the field's heating against the Maxwellization by electron–electron collisions, while in a gas the energy dependence of ν_m and the inelastic losses also set its form. Define

$$\alpha_g = \frac{P_{\rm abs}}{eT_e\,\nu_{ee}}, \qquad \nu_{ee} = 2.91\times10^{-12}\,n_e\,\ln\Lambda\,T_e^{-3/2}\ \mathrm{s}^{-1} \quad (n_e \text{ in m}^{-3},\ T_e \text{ in eV}), \tag{6.5}$$

with $P_{\rm abs}$ the absorbed power per electron and ν_{ee} the electron collision rate of the NRL formulary [28]. For the Maxwellian of a fully ionized plasma the IB heating is $P = (e^2E^2/m\omega^2)\,\nu_T L$ with $\nu_T = (4\sqrt{2\pi}/3)\,n_i Z^2 e^4/[(4\pi\epsilon_0)^2 m^{1/2}(eT)^{3/2}]$ [7]. In practical units $\nu_T = 2.906\times10^{-12}\,n_i Z^2 T^{-3/2}$, the coefficient of ν_{ee}, so

$$\alpha_g = \frac{\alpha_L}{2}\,\frac{L}{\ln\Lambda}, \tag{6.6}$$

and Langdon's α_L is $2\alpha_g$ when the two logarithms are equal. The same α_g applies to a gas, where $P_{\rm abs}$ comes mostly from collisions with atoms.

6.3 The kinetic equation

(1) The isotropic part of the distribution obeys, in the two-term approximation and per electron,

$$\epsilon^{1/2}\frac{\partial F}{\partial t} = -\frac{\partial G}{\partial \epsilon} + S[F], \qquad G = G_{\rm field} + G_{\rm el} + G_{ee}, \tag{6.7}$$

with S the inelastic collisions (and the source or growth term of item (6)). The field flux is $G_{\rm field} = -D_{\rm field}F'$ with

$$D_{\rm field} = \frac{\gamma^2}{3}\,E^2\epsilon^{3/2}\,\frac{\nu}{\nu^2+\omega^2}, \qquad \nu = \nu_a + \nu_c, \tag{6.8}$$

where $\nu_a = N\gamma\epsilon^{1/2}\sigma_m$ is the momentum-transfer frequency on atoms and ν_c that on ions. At $\omega = 0$ this is the DC term of Hagelaar and Pitchford [9], $D = (\gamma/3)(E/N)^2\epsilon N/\sigma_m$, and for $\omega \gg \nu$ it is $(\gamma^2/3)(E/\omega)^2\epsilon^{3/2}\nu$, which is Langdon's operator with the velocity-space diffusion coefficient $(v_0^2/6)\nu$ [1]. The two-term expansion needs a small anisotropy, so it cannot describe Silin's regime. At every scan point of Secs. 6.6 and 6.7 the electrons' oscillation velocity $eE_0/m(\omega^2+\nu_m^2)^{1/2}$, with the Maxwellian $\langle\nu_m\rangle$ on atoms, is at most $0.13\,v_t$. The elastic flux on atoms is $G_{\rm el} = -\gamma\epsilon^2 N\sigma_\epsilon(F + T_g F')$ with $\sigma_\epsilon = \sum_j x_j(2m/M_j)\sigma_{m,j}$.

(2) *Ions.* A singly charged ion has the Coulomb momentum-transfer cross section

$$\sigma_c(\epsilon) = \pi\Big(\frac{e}{4\pi\epsilon_0\,\epsilon}\Big)^2 L(\epsilon), \qquad L = \tfrac{1}{2}\ln\Big[1+\Big(\frac{b_{\max}}{b_{\min}}\Big)^2\Big], \tag{6.9}$$

with $b_{\max} = v/(\omega^2+\omega_p^2)^{1/2}$ and $b_{\min}$ the larger of $e^2/4\pi\epsilon_0 mv^2$ and $\hbar/2mv$. For $\omega \gg \omega_p$, $b_{\max}$ is the IB cutoff v/ω, set by the laser frequency rather than the plasma frequency as Turnbull et al. found the absorption to need [3]. Below ω_p it becomes the screening length v/ω_p. Then $\nu_c = \pi n_i Z^2\gamma(e/4\pi\epsilon_0)^2 L\,\epsilon^{-3/2}$, and for $\omega \gg \nu_c$ Eq. (6.8) becomes, with $E^2/\omega^2 = (v_0^2/v_t^2)\,T/\gamma^2$ and $n_i Z = n_e$,

$$D_{\rm IB} = \frac{\pi\gamma^3}{3}\,n_i Z^2\Big(\frac{e}{4\pi\epsilon_0}\Big)^2 L\,\frac{E^2}{\omega^2} = \frac{\pi\gamma}{3}\,\alpha_L\,T\,n_e\Big(\frac{e}{4\pi\epsilon_0}\Big)^2 L. \tag{6.10}$$

For a constant logarithm $D_{\rm IB}$ does not depend on ϵ: in energy, Langdon's operator is pure diffusion with a constant coefficient. The absorbed power is $P = \int D_{\rm IB}(-F')\,\mathrm{d}\epsilon = D_{\rm IB}F(0)$, proportional to the number of slow electrons, so $f_L = F(0)/F_M(0)$ at equal T, where F_M is the Maxwellian.

(3) *Langdon's v^5 distribution, exactly.* With heating alone the distribution obeys $\epsilon^{1/2}\partial_t F = D_{\rm IB}\partial_\epsilon^2 F$. Try $F = A(t)\exp[-\epsilon^{5/2}/b(t)]$. Then $\epsilon^{1/2}\partial_t F = \epsilon^{1/2}F(\dot A/A + \epsilon^{5/2}\dot b/b^2)$ and $\partial_\epsilon^2 F = F[(25/4)\epsilon^3/b^2 - (15/4)\epsilon^{1/2}/b]$. The powers $\epsilon^{1/2}$ and ϵ^3 match separately when

$$\frac{\mathrm{d}b}{\mathrm{d}t} = \frac{25}{4}\,D_{\rm IB}, \qquad \frac{\mathrm{d}\ln A}{\mathrm{d}t} = -\frac{15}{4}\frac{D_{\rm IB}}{b}. \tag{6.11}$$

The distribution keeps the shape $\exp(-v^5)$ while it heats, Langdon's limit for $\alpha_L \to \infty$, and its absorption factor, that of the super-Gaussian $\exp[-(v/v_m)^M]$ with $M = 5$ at the same density and energy, is $F(0)/F_M(0) = M(\pi/2)^{1/2}\Gamma(5/M)^{3/2}/[3^{3/2}\Gamma(3/M)^{5/2}] = 0.4456$.
(4) *Electron–electron collisions.* The isotropic Rosenbluth operator [10] is the classical small-angle (Landau) form of the electron–electron collision integral, whose quantum form is derived in [6]. In energy it reads (Appendix 6.A)

$$G_{ee} = -2a\,n_e\int_0^\infty \min(\epsilon,\epsilon')^{3/2}\big[F(\epsilon')F'(\epsilon) - F(\epsilon)F'(\epsilon')\big]\,\mathrm{d}\epsilon', \qquad a = \frac{e^2\gamma\ln\Lambda}{24\pi\epsilon_0^2}, \tag{6.12}$$

which is the form $-a\,n_e[3A_1F + 2(A_2 + \epsilon^{3/2}A_3)F']$ of Hagelaar and Pitchford [9], $A_1 = \int_0^\epsilon u^{1/2}F$, $A_2 = \int_0^\epsilon u^{3/2}F$, $A_3 = \int_\epsilon^\infty F$. The kernel is symmetric and the bracket antisymmetric in (ϵ,ϵ'), so

$$\int_0^\infty G_{ee}\,\mathrm{d}\epsilon = 0, \tag{6.13}$$

which, since $\int \epsilon\,\partial_\epsilon G\,\mathrm{d}\epsilon = -\int G\,\mathrm{d}\epsilon$, is the conservation of energy. Any $F \propto e^{-\epsilon/T}$ makes the bracket vanish. Far above the bulk, $\min = \epsilon'$, the friction is $2an_e\int \epsilon'^{3/2}F'\,\mathrm{d}\epsilon' = -3an_e$ and the diffusion is constant, so a fast electron loses energy at

$$\frac{\mathrm{d}(e\epsilon)}{\mathrm{d}t} = -\frac{3a\,n_e\,e}{\epsilon^{1/2}} = -\frac{e^4\ln\Lambda\,n_e}{4\pi\epsilon_0^2\,m\,v}, \tag{6.14}$$

the classical slowing-down rate of a fast electron by the field electrons, $7.7\times10^{-12}\,n_e\ln\Lambda\,\epsilon^{-3/2}$ s^{-1} in the units of Eq. (6.5) [28]. Discretized on the cell faces with face averages and differences of F, Eq. (6.13) holds to round-off and a sampled Maxwellian is an exact zero for every grid step.
(5) *The heated plasma at fixed α_L.* For a fully ionized plasma heated by IB, write $x = \epsilon/T$ and $F = T^{-3/2}\Phi(x)$ with T two thirds of the mean energy. When the shape is stationary while T rises, the flux in the frame that moves with T vanishes everywhere. Dividing by $2an_eT^{-3/2}$,

$$-\beta\,g(x)\Phi' - \big[\Phi' P(x) - \Phi\,Q(x)\big] = \lambda\,x^{3/2}\Phi, \qquad \beta = \frac{\alpha_L L}{4\ln\Lambda}, \tag{6.15}$$

with $P = \int \min(x,x')^{3/2}\Phi(x')\,\mathrm{d}x'$, $Q = \int \min(x,x')^{3/2}\Phi'(x')\,\mathrm{d}x'$, $g = L(xT)/L$ for a velocity-dependent logarithm (otherwise 1), and λ the heating rate $\dot T/T$ in e-e units. The ratio β follows from Eq. (6.10) and $2an_e = (4\pi/3)\gamma\ln\Lambda\,n_e(e/4\pi\epsilon_0)^2$. Integrating Eq. (6.15) over x, the e-e term drops out by Eq. (6.13) and $\int x^{3/2}\Phi = 3/2$ gives $\lambda = (2/3)\beta\int g(-\Phi')\,\mathrm{d}x$. For $\beta \to \infty$, Eq. (6.15) gives $\Phi \propto \exp(-cx^{5/2})$, the distribution of Eq. (6.11). For $\beta \to 0$ it gives the Maxwellian. By Eq. (6.6), $\beta = \alpha_g/2$ for the Maxwellian.
(6) *Method.* Equation (6.7) is solved on an energy grid with the two-term discretization of Hagelaar and Pitchford [9], extended by Eqs. (6.9) and (6.12). Three choices matter, and the first was forced by checks of Appendix 6.B.

- A scheme whose inelastic gains are not exactly number-conserving, and whose LU solve replaces the top row by the normalization, satisfies the rows of the tail only to the accuracy of the largest F. Against an elimination that satisfies every row to round-off, such an LU solution is off by 2 % where $F/F_{\max}$ is 10^{-6} to 10^{-3} and by order one below. The solver therefore distributes each source cell's inelastic loss over its landing cells (number-conserving to round-off) and solves by Gaussian elimination from the top in ratio form, $c_i = F_i/F_{i-1}$, which satisfies every row but the lowest. That row is then exactly the particle balance, and the growth frequency is found as the root of it. Krypton's share of the power changes by 3×10^{-5} between the two schemes.

- The e-e coefficients are lagged and iterated with Anderson mixing [11] of $\ln F$, so that the tail converges in relative terms.

- A beam-sustained mode replaces the growth term by electrons injected with the sub-excitation spectrum of the beam's secondary electrons that a kinetic model of the laser uses (linear on 0 to 9.915 eV, mean 4.506 eV), lost by attachment and by recombination with a cross section $\propto 1/\epsilon$ (Sec. 6.6).

6.4 Checks of the kinetic model

(1) *Langdon.* Table 6.1 and Fig. 6.1 compare the solutions of Eq. (6.15) with equal logarithms ($\beta = \alpha_L/4$) to Langdon's fit $f_L = 1 - 0.553/[1 + (0.27/\alpha_L)^{0.75}]$, which he gives as good to ± 0.005 [1] and Turnbull et al. use [3]. They agree within 1.8 % from $\alpha_L = 0.01$ to 100, the largest difference being at $\alpha_L = 0.5$. The kinetic solution is therefore the problem Langdon solved. The fitted super-Gaussian exponent depends on the fit, because the distribution is not exactly super-Gaussian. Matte's $m(\alpha_L) = 2 + 3/(1 + 1.66\alpha_L^{-0.724})$ [2, 4] lies between the exponents of two fits for $\alpha_L \le 5$ and within 0.09 of the three-parameter fit above. The energy relation of item (5) of Sec. 6.3 holds to 3×10^{-6}, and $\beta \to \infty$ reproduces the v^5 distribution's factor 0.4456 to 4×10^{-6}.

Table 6.1: The heated plasma at fixed α_L, Eq. (6.15) with $\beta = \alpha_L/4$, against Langdon's absorption fit and Matte's exponent. m is fitted in two ways: to $\ln\ln(\Phi(0)/\Phi)$ against $\ln x$ on $0.1 < x < 3$, and with three free parameters on $x < 3$.

α_L	f_L	Langdon's fit	ratio	m, $\ln\ln$	m, three-parameter	Matte's m
0.01	0.9564	0.9569	0.9994	2.12	2.03	2.06
0.03	0.9125	0.9108	1.0019	2.25	2.08	2.14
0.1	0.8292	0.8220	1.0088	2.51	2.21	2.31
0.3	0.7242	0.7126	1.0163	2.91	2.46	2.60
1	0.6065	0.5977	1.0147	3.53	2.97	3.13
3	0.5256	0.5250	1.0011	4.13	3.61	3.71
10	0.4761	0.4815	0.9888	4.62	4.31	4.28
30	0.4568	0.4627	0.9873	4.85	4.71	4.63
100	0.4491	0.4535	0.9904	4.95	4.91	4.83

(2) *The gas.* With $n_e = 0$, no ions and the discretization of the first scheme of Sec. 6.3, the solver reproduces an independent two-term solver to 7×10^{-11}. A constant cross section in a DC field gives the Druyvesteyn distribution [12] to 1.2×10^{-7} in $\ln F$, a field with $\omega \gg \nu$ and elastic losses the Maxwellian of Margenau [13], with or without e-e collisions. The e-e flux conserves energy to

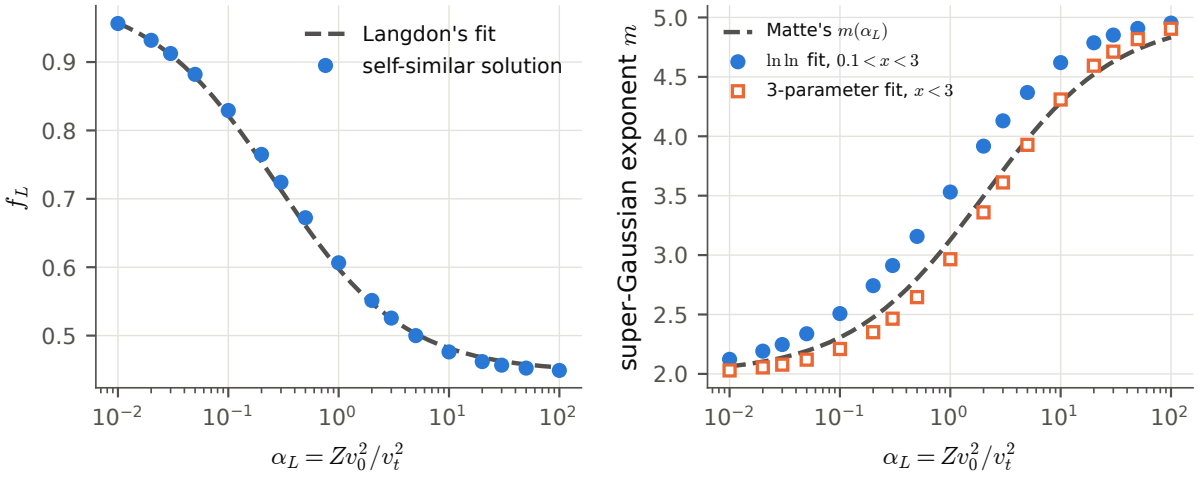


Figure 6.1: Left: the absorption factor of the heated plasma against Langdon's fit. Right: fitted super-Gaussian exponents against Matte's.

1.4×10^{-17}, its drift on a fast electron is Spitzer's Eq. (6.14) to 2×10^{-5}, and two mixed Maxwellians relax to one. The Coulomb heating of a 3 eV Maxwellian is $(e^2E^2/m\omega^2)\nu_T L$ to 9×10^{-5}, once the half cell next to $\epsilon = 0$ is accounted for. In a DC field e-e collisions turn the Druyvesteyn distribution, whose $(2/3)\langle\epsilon\rangle/T_{\rm slope}$ is 1.50 with $T_{\rm slope}$ fitted to $\ln F$ on $0.1\langle\epsilon\rangle < \epsilon < 4\langle\epsilon\rangle$, into a Maxwellian: the ratio is 1.0005 at $n_e/N = 10^{-2}$.

6.5 Where gas lasers sit

(1) Table 6.2 evaluates Eq. (6.3) for the fields of three lasers and for auxiliary sources heating the Ne fill, with Maxwellian electrons, the losses of the LXCat sets, and the ions of Eq. (6.9). Every beam-on row has $T_e = 1.5$ eV, between the temperature of a Maxwellian power balance at beam-on density, 1.08 eV, and that of the beam-sustained electrons of Sec. 6.6 at beam-on density, 1.88 (Ar fill) and 2.04 eV (Ne fill), and $n_e = 2.72\times10^{21}$ m^{-3}. The afterglow rows have 0.63 eV and 10^{20} m^{-3}. The ArF fill is Electra's, Ar/F$_2$ = 0.997/0.003 at 0.82 atm [14]. The CO$_2$ row uses a CO$_2$:N$_2$:He = 1 : 1 : 8 mixture at 1 atm, $T_e = 1$ eV and $n_e = 10^{19}$ m^{-3} with the IST-Lisbon cross sections [27], and singly charged cations only ($Z_{\rm eff} = 1$). Langdon's and Silin's intensities are left out where $\nu_m > \omega$, where the electrons do not quiver freely. Figure 6.2 shows the thresholds against frequency.

(2) *The lasers' own fields.* An amplifier extracts at a few saturation intensities. For KrF, $I_{\rm sat} = h\nu(A+Q)/\sigma$ is 1.0 and 1.2 MW cm^{-2} in the two fills, with the cross section 2.6×10^{-16} cm^2 and radiative rate of Kannari et al. [31] and the quench rates of a KrF kinetic network at each fill's composition. Electra's ArF amplifier saturates at 10 MW cm^{-2} [14]. At 10 MW cm^{-2} the KrF field is 7.1×10^3 times below $I_{\rm heat}$ in the Ne fill and 1.5×10^4 times in the Ar fill, and at 30 MW cm^{-2} the ArF field is 5.8×10^3 times below it. These margins depend on T_e through the inelastic losses: for T_e from 1.08 to 2.04 eV they are $10^{3.2}$ to $10^{5.0}$. They also rest on the classical kernel at $u = \hbar\omega/eT_e = 3.3$ and 4.3, where it is not validated (the quantum absorption on these atoms, from exact phase shifts of model potentials, is 10 to 28 % below it at $u = 4.5$, for 300 K electrons at 28.3 THz, and its size for 5.0 and 6.4 eV photons is not known). The excimer lasers' own fields

Table 6.2: Thresholds of the three nonlinearities, Eq. (6.3), in $\mathrm{W\,cm^{-2}}$. $u = \hbar\omega/eT_e$; ν_m/ω with the Maxwellian $\langle\nu_m\rangle$; P_{loss} in $\mathrm{eV\,s^{-1}}$ per electron; Langdon at $\alpha_L = 0.1$ with $Z_{\mathrm{eff}} = 1.15$ (1 for the CO_2 discharge); Langdon and Silin left out where $\nu_m > \omega$.

field, gas	T_e	n_e ($\mathrm{m^{-3}}$)	u	ν_m/ω	P_{loss}	I_{heat}	I_L	I_S
KrF, 248 nm, Ne fill	1.5	2.7×10^{21}	3.3	2.0×10^{-4}	1.1×10^{9}	7.1×10^{10}	5.7×10^{12}	6.5×10^{13}
ArF, 193 nm, ArF fill	1.5	2.7×10^{21}	4.3	1.4×10^{-4}	1.6×10^{9}	1.7×10^{11}	9.3×10^{12}	1.1×10^{14}
CO_2, 10.6 μm, discharge	1	1.0×10^{19}	0.12	0.0097	3.5×10^{10}	9.6×10^{8}	2.4×10^{9}	2.4×10^{10}
2.45 GHz, afterglow	0.63	1.0×10^{20}	1.6×10^{-5}	32	3.6×10^{8}	2.8×10^{2}	–	–
170 GHz, afterglow	0.63	1.0×10^{20}	0.0011	0.46	3.6×10^{8}	1.4×10^{3}	4.7×10^{4}	5.4×10^{5}
0.67 THz, afterglow	0.63	1.0×10^{20}	0.0044	0.12	3.6×10^{8}	1.5×10^{4}	7.3×10^{5}	8.4×10^{6}
1 THz, beam-on	1.5	2.7×10^{21}	0.0028	0.24	1.1×10^{9}	5.2×10^{4}	3.9×10^{6}	4.5×10^{7}
28.3 THz, beam-on	1.5	2.7×10^{21}	0.078	0.0086	1.1×10^{9}	3.9×10^{7}	3.1×10^{9}	3.6×10^{10}

are linear by every measure. The CO_2 laser's field supplies a tenth (0.10) of the energy budget of its discharge electrons at 100 $\mathrm{MW\,cm^{-2}}$, and all of it at $I_{\mathrm{heat}} = 9.6 \times 10^{8}$ $\mathrm{W\,cm^{-2}}$. That is an order-of-magnitude estimate. It assumes Maxwellian electrons at 1 eV, which at $n_e/N = 4 \times 10^{-7}$ nothing Maxwellizes, and discharge electrons present while the pulse passes, which depends on the timing of the amplifier. Of the lasers examined it is the one in which the lasing field can reach the gas nonlinearity. Lasing is known to perturb CO_2-laser discharges, the optogalvanic effect, but through the vibrational kinetics of the laser levels [26]. No study was found of the field's IB heating of the discharge electrons as an efficiency or stability term.

(3) *Auxiliary fields.* Heating the Ne fill's electrons takes 1.4×10^{3} $\mathrm{W\,cm^{-2}}$ at 170 GHz and 1.5×10^{4} $\mathrm{W\,cm^{-2}}$ at 0.67 THz in the afterglow, and 5.2×10^{4} $\mathrm{W\,cm^{-2}}$ at 1 THz beam-on, within reach of gyrotron beams [32, 8]. Such a beam heats only its absorption length, 2.5 mm at 170 GHz and 2.6 cm at 0.67 THz in the afterglow and 1.1 mm at 1 THz beam-on. Where $\nu_m < \omega/2$, Langdon's $\alpha_L = 0.1$ needs 62 to 80 times I_{heat} in the Ne fill beam-on, 33 to 51 times in its afterglow, and 37 and 54 times at the laser frequencies of the Ar and ArF fills. Silin's $v_0 = v_t$ needs a further $Z_{\mathrm{eff}}/\alpha_L = 11.5$ times. For $\omega \gg \nu_m$ the ratio is Eq. (6.4), $\alpha_L/3Z_{\mathrm{eff}}\delta_{\mathrm{eff}}$, with $\delta_{\mathrm{eff}} = 4 \times 10^{-4}$ to 8×10^{-4}. In the excimer fills a field strong enough to matter is in the gas nonlinearity, and only there. The CO_2 discharge, whose vibrational losses make δ_{eff} large, has Langdon's intensity only 2.5 times its I_{heat}.

6.6 Frequency as a lever on the distribution

(1) Without e-e collisions a field above the collision frequency puts more of its power into krypton than a DC field of the same absorbed power: at the lowest power about twice as much in the Ne fill and up to 32 times as much in the Ar fill, where the DC share is 1.3 %. This section asks whether e-e collisions erase that. The mechanism is item (3) of Sec. 6.2. A DC field heats each electron in proportion to $1/\nu_m$, so in a gas whose ν_m rises with energy it heats the slow electrons most and gives the Druyvesteyn-like distribution, with a depleted tail. A field with $\omega \gg \nu_m$ heats in proportion to ν_m, which with elastic losses gives Margenau's Maxwellian [13]. Its tail reaches the 9.9 eV krypton levels. That the energy dependence of ν_m decides how DC and microwave distributions differ is known for discharges in general [17], high-frequency argon distributions have been computed from $\omega \ll \nu$ to $\omega \gg \nu$ [18], and e-e collisions were added for argon discharges [19, 20]. What is new here is the KrF laser gas and its formation chain. A scan repeats the comparison at the absorbed powers per electron of DC fields of 2, 5 and 10 Td in the Ne fill and 3, 6 and 15 Td in the Ar fill, with e-e collisions and ions at each electron density, and with the temporal-growth model of two-term

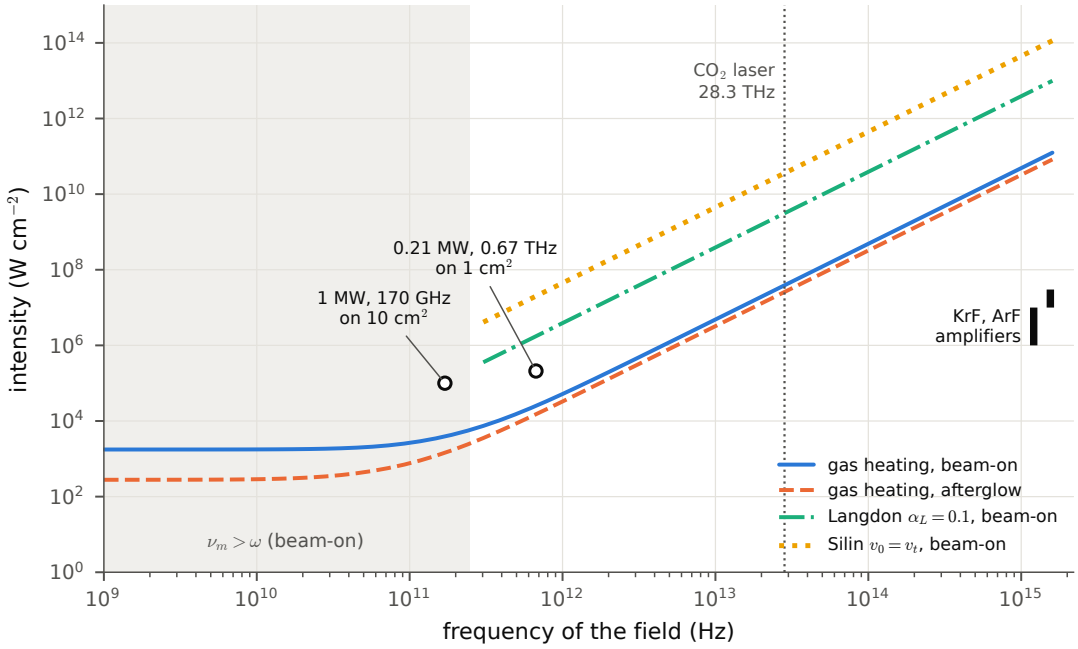


Figure 6.2: The three thresholds against frequency in the Ne fill: gas heating (I_{heat}) beam-on ($T_e = 1.5$ eV, $n_e = 2.72 \times 10^{21}$ m^{-3}) and in the afterglow (0.63 eV, 10^{20} m^{-3}), Langdon at $\alpha_L = 0.1$ and Silin, both beam-on and drawn only where $\nu_m < \omega$. Bars: KrF amplifiers at 1 to 10 and ArF amplifiers at 10 to 30 MW cm^{-2} (Sec. 6.5); dotted line: the CO_2-laser frequency 28.3 THz; circles: a 1 MW gyrotron at 170 GHz [32] and a 0.21 MW, 0.67 THz gyrotron [8], each on an assumed spot.

solvers: the electrons grow or decay with the shape of F. Item (3) repeats it with electrons born from the beam. The scan treats the field as given inside the gas. At beam-on density the plasma frequency is 0.47 THz, so a 0.3 THz wave would not enter the gas there. How a field reaches the gas is a separate question of coupling.

(2) Table 6.3 and Fig. 6.3 give the result. Electron–electron collisions raise krypton's share in the DC field, from 27.5 to 80.1 % at the power of 2 Td in the Ne fill, as n_e rises from 0 to the beam-on density 2.72×10^{21} m^{-3}. The 28.3 THz field goes from 54.2 to 84.5 %. Collisions bring both towards the Maxwellian, which is where the high frequency was already taking the distribution. At the 9.9 eV krypton threshold, relative to a Maxwellian of the same mean energy, the DC distribution holds 0.556 of the population and the 28.3 THz one 0.742 without e-e collisions, and 0.900 and 0.969 at beam-on density. Over the 30 series of the scan that ratio moves towards 1 between $n_e = 0$ and beam-on density in 29, and stays at 0.94 in the Ar fill at 6 Td and 3 THz. Where the advantage is large, at 2 Td in the Ne fill and at 3 and 6 Td in the Ar fill, the 28.3 THz ratio rises and the DC one rises more. The advantage of the frequency therefore falls from ×1.97 to ×1.055. Normalized by its value without e-e collisions, the advantage falls along nearly one curve of α_g for the three power levels in Fig. 6.3. It halves at $\alpha_g = 0.41$, 0.46 and 0.55, found by root finding in n_e between the scan's grid points. At beam-on density α_g is 0.023 to 0.14 for these three levels and the frequency is worth 2.0 to 5.5 %. In the Ne fill at the highest power, 10 Td, the high frequency is worse than DC. The fatter tail excites neon itself (its share rises from 1.9 to 7.2 %), and krypton's share falls by 3.0 points. In the Ar fill at 15 Td the high frequency is still 2.9 points better without e-e collisions,

and 0.6 points worse at beam-on density.

Table 6.3: Krypton's share of the absorbed power (%) and the formation efficiency η_{form} of Sec. 6.8, DC against 28.3 THz at the same absorbed power per electron (that of a DC field of the given E/N without e-e collisions). α_g of the DC point, Eq. (6.5).

fill, power	n_e (m^{-3})	α_g	Kr, DC	Kr, 28.3 THz	ratio	η_{form}, DC	η_{form}, 28.3 THz
Ne, 2 Td	0	–	27.5	54.2	1.973	0.114	0.224
Ne, 2 Td	10^{19}	4.85	29.9	57.6	1.928	0.123	0.237
Ne, 2 Td	10^{20}	0.538	42.7	66.4	1.556	0.176	0.274
Ne, 2 Td	10^{21}	0.0591	70.5	79.4	1.126	0.290	0.325
Ne, 2 Td	2.72×10^{21}	0.0228	80.1	84.5	1.055	0.327	0.344
Ne, 5 Td	0	–	78.5	86.8	1.105	0.322	0.351
Ne, 5 Td	10^{19}	28.7	78.6	86.8	1.104	0.323	0.351
Ne, 5 Td	10^{20}	3.21	79.5	86.9	1.093	0.326	0.351
Ne, 5 Td	10^{21}	0.356	83.2	87.0	1.045	0.339	0.350
Ne, 5 Td	2.72×10^{21}	0.137	85.2	86.9	1.020	0.345	0.349
Ar, 6 Td	0	–	38.4	81.3	2.117	0.149	0.325
Ar, 6 Td	10^{19}	5.74	40.9	83.7	2.047	0.158	0.335
Ar, 6 Td	10^{20}	0.63	53.4	85.3	1.596	0.208	0.342
Ar, 6 Td	10^{21}	0.0676	77.6	87.2	1.124	0.311	0.356
Ar, 6 Td	2.72×10^{21}	0.0259	83.9	87.9	1.048	0.344	0.365

(3) *Beam-sustained electrons.* The growth model lets new electrons appear with the shape of F, and at the lowest power its term carries up to 0.66 of the absorbed power at the points kept, those with $P_{\text{field}} > 1.5\,|P_{\text{growth}}|$. In an e-beam pumped gas the electrons are born from the beam and lost by attachment and by recombination. A second scan therefore repeats the comparison at the absorbed powers of Table 6.3 with electrons injected with the beam's sub-excitation spectrum (item (6) of Sec. 6.3) and lost by attachment and by dissociative recombination on $1.075\,n_e$ cations, taken as Kr_2^+ with the rate $2.24 \times 10^{-7}\,T_e^{-1/2}$ $\text{cm}^3\,\text{s}^{-1}$, built on the dissociative recombination rates of Kannari et al. [31]. The loss has a cross section $\propto 1/\epsilon$, which gives that rate for a Maxwellian, so it takes the slow electrons first. At beam-on density recombination removes electrons 3.1 (Ne fill) and 3.9 (Ar fill) times faster than attachment, and with the DC field the beam's own injected power is then 2.4 to 8.3 times the field's, except at 5 Td in the Ne fill (0.425). Table 6.4 gives the field's marginal share to krypton at fixed n_e, the change of krypton's power per unit field power. It charges the field with the change of the beam's injected power, which follows the attachment and recombination of the heated distribution. Crediting the beam with its own partition at that power gives the second ratio of the table. Neither ratio holds the beam current fixed. Holding it means finding the electron density at which the beam's injected power per unit volume, and so its current, keeps its value without the field, and the field's absorbed power per unit volume keeps its value at fixed n_e. At beam-on density the advantage at a fixed beam current is within 0.01 of the second ratio. At 10^{18} m^{-3} the field changes the density by a factor $x = 0.36$ to 1.44, and the advantage at a fixed current, $\times 1.23$ to $\times 13.5$, exceeds both ratios. That excess is the field's power per electron, which a fixed absorbed power per unit volume changes by $1/x$: with the field's amplitude held instead, the advantage at a fixed current is the fixed-n_e one within 1 %. The ranges below leave out the points where, with the DC field, the beam injects more power than the field

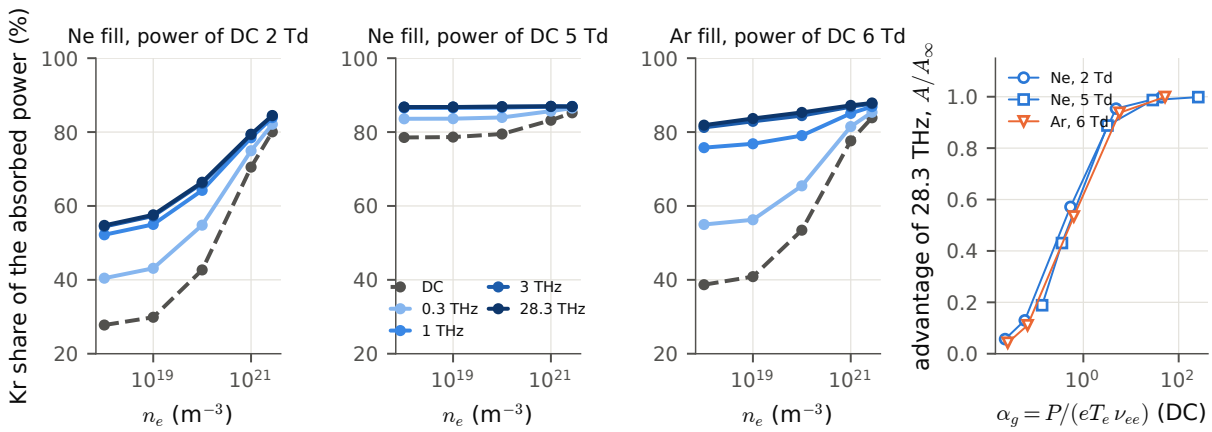


Figure 6.3: Krypton's share of the absorbed power against n_e, DC (dashed) and 0.3, 1, 3 and 28.3 THz, and (right) the advantage of 28.3 THz over DC, A = ratio -1, normalized by its value without e-e collisions, against α_g. Every point drawn has $P_{\text{field}} > 1.5\,|P_{\text{growth}}|$. The Ar fill at 3 Td is not shown: from 10^{19} m^{-3} up its points at 1 to 28.3 THz fall below that.

absorbs. At 10^{18} m^{-3} and fixed n_e the frequency raises krypton's share $\times$2.06 in the Ne fill at 2 Td and $\times$2.40 in the Ar fill at 6 Td, against $\times$1.97 and $\times$2.12 with the growth model, and $\times$11.7 in the Ar fill at 3 Td, against $\times$27 with the growth model at the same density. At beam-on density, where only the Ne fill at 5 Td is kept, the advantage is $\times$1.05 to $\times$1.07, against $\times$1.020 with the growth model at the same power, and the field sends 75 to 85 % of its power to krypton, against 85 to 87 %. Where the beam outweighs the field the advantage is $\times$1.10 to $\times$1.14 at a fixed beam current and up to $\times$1.27 at fixed n_e. Counted per change of the total input power, the beam's and the field's together, it is $\times$1.03 to $\times$1.07 at every beam-on point. Where the frequency's advantage is large it collapses in both models, and its size at beam-on density depends on how the electrons are created and lost and on how the field's share is counted. Without a field, the e-e collisions at beam-on density send 38.9 % of the injected sub-excitation energy into krypton in the Ne fill and 44.7 % in the Ar fill, against 0.0 % at 10^{18} m^{-3}, and recombination takes 34 to 35 % of it. That is an e-beam effect independent of any field. A loss that takes electrons of every energy alike at the same Maxwellian rate gives 28.5 and 34.2 %, with recombination taking 47 to 49 %, and nearly the same advantage at 5 Td in the Ne fill, $\times$1.06 to $\times$1.07. The beam's own numbers depend on the form of the recombination, and the frequency's advantage much less: at beam-on density each of its four counts moves by at most 0.013 at 5 Td in the Ne fill and 0.044 where the beam outweighs the field.

The beam-on density used throughout, 2.72×10^{21} m^{-3}, is the peak that a full kinetic model of the Ne-fill laser gives under 0.6 MW cm^{-3} of beam. This note's model holds it only with 1.6×10^{30} ion pairs per m^3 and second, 7.9 MW cm^{-3} at the chain's 30.4 eV per pair, which is 13 times that beam, or 2.5 MW cm^{-3} without recombination. Taking every cation as Kr_2^+ overstates the recombination of both fills: Kr^+ forms Kr_2^+ by three-body association in 33 ns in the Ne fill and 17 ns in the Ar fill at that model's 2.5×10^{-31} cm^6 s^{-1}, while Kr_2^+ recombines in 2 ns at beam-on density. That model by default extrapolates its attachment below the floor of its measured tables, and with the measured tables its peak falls to 0.56 of that density. With its beam this note's model holds 5.2×10^{20} m^{-3} in the Ne fill, and there the advantage is $\times$1.157 to $\times$1.355 at 2 Td and $\times$1.051 to $\times$1.107 at 5 Td in the four counts. That density is a lower bound. Besides

overstating recombination, the model injects one electron per ion pair, where the chain of Sec. 6.8 frees 1.56 through the Penning ionization of krypton by Ne*, and with those it holds 6.2×10^{20} to 7.1×10^{20} m^{-3} at 36.6 to 30.4 eV per pair. Without recombination the advantage at beam-on density is ×1.01 to ×1.18. The density of the beam-pumped gas, and with it what is left of the frequency's advantage, is uncertain by a factor of about five.

Table 6.4: The beam-sustained model: electrons injected with the beam's sub-excitation spectrum and lost by attachment and recombination, at the absorbed powers of Table 6.3. Krypton's marginal share $(P_{\mathrm{Kr}} - P_{\mathrm{Kr,beam}})/P_{\mathrm{field}}$ (%) at fixed n_e, the ratio of 28.3 THz over DC with it, with the beam credited with its own partition at its injected power (fixed beam), and at a fixed beam current (see the text), and the formation efficiency of the field's marginal power at fixed n_e. $P_{\mathrm{beam}}/P_{\mathrm{field}}$ is the beam's injected power over the field's, with the DC field; in the rows marked † the beam outweighs the field.

fill, power	n_e (m^{-3})	$P_{\mathrm{beam}}/P_{\mathrm{field}}$	Kr, DC	Kr, 28.3 THz	ratio, fixed n_e	ratio, fixed beam	ratio, fixed current	η_{form}, DC	η_{form}, 28.3 THz
Ne, 2 Td	10^{18}	0.34	30.7	63.4	2.063	2.063	2.647	0.127	0.261
Ne, 2 Td	10^{20}	0.51	42.4	72.4	1.709	1.630	1.633	0.175	0.298
Ne, 2 Td†	2.72×10^{21}	2.85	65.3	75.6	1.157	1.102	1.098	0.265	0.305
Ne, 5 Td	10^{18}	0.05	78.2	88.0	1.125	1.125	1.234	0.321	0.356
Ne, 5 Td	10^{20}	0.06	78.1	87.2	1.117	1.111	1.215	0.320	0.352
Ne, 5 Td	2.72×10^{21}	0.42	75.4	80.8	1.071	1.048	1.052	0.304	0.323
Ar, 3 Td	10^{18}	0.75	5.5	64.5	11.737	11.737	13.515	0.021	0.251
Ar, 3 Td†	10^{20}	1.30	26.0	87.3	3.351	2.597	2.201	0.101	0.343
Ar, 3 Td†	2.72×10^{21}	8.29	60.0	76.2	1.271	1.138	1.135	0.250	0.324
Ar, 6 Td	10^{18}	0.19	39.8	95.5	2.401	2.401	2.777	0.154	0.381
Ar, 6 Td	10^{20}	0.30	49.2	93.4	1.899	1.745	1.769	0.192	0.375
Ar, 6 Td†	2.72×10^{21}	2.45	62.6	76.9	1.227	1.116	1.115	0.262	0.329

6.7 The self-sustained discharge

(1) A discharge-pumped laser needs ionization to balance attachment. At $\nu_{\mathrm{net}} = 0$ the growth model drops out, which makes this the cleanest comparison of field shapes. A third scan finds the rms field of self-sustainment from DC to $\omega/N = 10^{-11}$ m^3 s^{-1} (78 THz in the Ne fill) for $n_e/N = 0$ to 10^{-4} (Table 6.5, Fig. 6.4).
(2) In the Ne fill the discharge sustains itself at 8.33 Td DC with 87.2 % of its power in krypton. On the grid of the scan the share peaks at 88.1 % at $\omega/N = 3 \times 10^{-14}$ m^3 s^{-1}, a third of the Maxwellian mean $\langle \nu_m \rangle/N = 9.2 \times 10^{-14}$ m^3 s^{-1} at the discharge's mean energy: the optimum sits where ω and ν_m are of the same order. A Canon patent on microwave-excited excimer lasers states that at 35 GHz "the plasma excitation efficiency based on identical electric power is highest" at the pressure, 160 Torr, at which the collision frequency of electrons with neon, the main body of the Kr/Ne/F$_2$ gas, equals the microwave frequency [15]. The optimum exists but is worth 0.9 points. Far above the collision frequency the share falls to 86.4 %, the fatter tail exciting neon. In the Ar fill the discharge needs 35.9 Td DC and gives 80.5 % to krypton, falling to 70.2 % at high frequency as argon's 11.5 eV levels take the tail. The formation efficiency moves less, from 0.351 to 0.337, because argon's excitation partly reaches KrF*. Raising n_e/N to 10^{-4} changes these shares by at most 2.6 points.

Table 6.5: The self-sustained discharge: DC, the frequency of the largest krypton share, and $\omega/N = 10^{-11}$ m^3 s^{-1}. P in eV s^{-1} per electron; Kr and buffer shares in %.

fill	n_e/N	ω/N (m^3 s^{-1})	E/N (Td)	$\langle\epsilon\rangle$ (eV)	P	Kr	buffer	η_{form}
Ne	0	DC	8.33	5.31	1.24×10^{10}	87.2	1.0	0.353
Ne	0	3×10^{-14}	10.6	4.94	1.34×10^{10}	88.1	1.8	0.354
Ne	0	10^{-11}	1855	4.06	1.27×10^{10}	86.4	5.2	0.344
Ne	10^{-4}	DC	7.13	4.43	1.13×10^{10}	86.4	3.3	0.345
Ne	10^{-4}	10^{-14}	7.58	4.40	1.14×10^{10}	86.5	3.4	0.345
Ne	10^{-4}	10^{-11}	1769	3.92	1.13×10^{10}	84.9	6.2	0.338
Ar	0	DC	35.9	5.32	2.88×10^{10}	80.5	15.3	0.351
Ar	0	10^{-11}	2507	2.84	3.02×10^{10}	70.2	27.8	0.337
Ar	10^{-4}	DC	30.4	4.86	2.56×10^{10}	78.9	17.0	0.348
Ar	10^{-4}	10^{-11}	2009	3.34	2.39×10^{10}	72.8	24.5	0.340

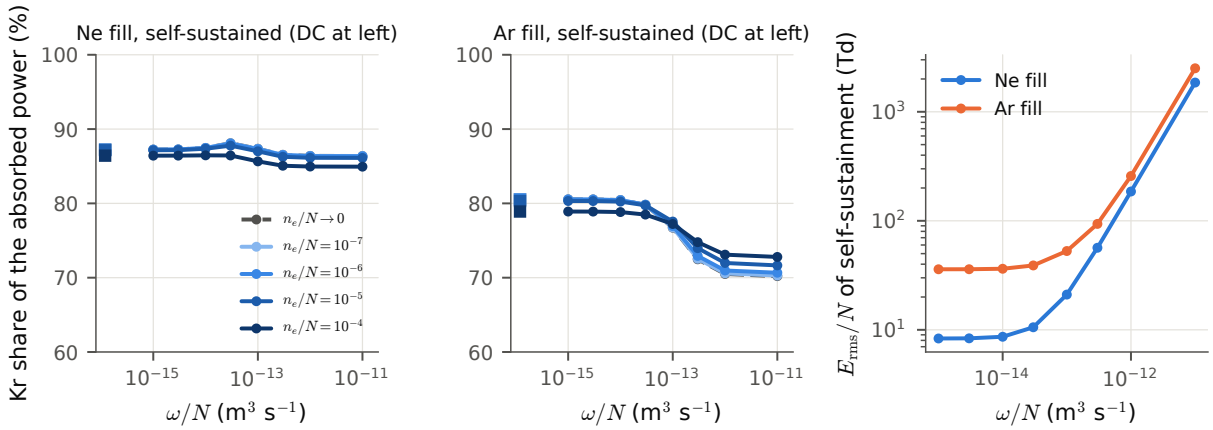


Figure 6.4: Self-sustained discharges against ω/N: krypton's share in the two fills (DC at the left, squares) and the rms E/N of self-sustainment.

6.8 What field pumping buys

(1) A formation efficiency turns a power partition into KrF* per absorbed energy,

$$\eta_{\text{form}} = \sum_c f_c \frac{h\nu}{u_c} b_c, \tag{6.16}$$

with f_c the shares, u_c the energy per event (Kr excitation 10.5 eV, ionization 14.0, Ne 17.5, Ar 12.5), $h\nu = 4.997$ eV, and b_c the branchings of a KrF kinetic network for each fill (Kr* harpoon 0.87 and 0.81, ion channel 0.68). The same accounting gives e-beam pumping of the Ne fill 0.175, from 30.4 eV per ion pair and 0.7 excitations per ion with this fill's Ne* $\to$Kr$^+$ branching 0.80. That accounting gives the beam's sub-excitation electrons nothing. At beam-on density their e-e collisions put 0.159 of their energy into KrF* (Sec. 6.6, where recombination takes about a third of it), and one such electron of 4.506 eV per 30.4 eV ion pair adds 0.024: the like-for-like e-beam figure there is 0.198. A full kinetic model of the Ne-fill laser, whose Spencer–Fano partition has

36.6 eV per ion pair where the chain counts 30.4, also forms 0.194 of its deposited energy into KrF^* at the photon energy, its 1.42 KrF^* per ion pair against 1.21 here making up for the larger energy per pair. The chain has no e-beam accounting for the Ar fill.
(2) Field pumping of the beam-ionized gas at beam-on density has $\eta_{\text{form}} = 0.327$ to 0.365 where the field dominates in the growth model (Table 6.3), 0.327 to 0.349 in the Ne fill, and 0.30 to 0.34 for the field's marginal power with beam-sustained electrons in the Ne fill at 5 Td, in both attributions and at a fixed beam current (Table 6.4 gives the fixed-n_e values). The self-sustained discharge gives 0.337 to 0.354. In the Ne fill at beam-on density, where the field's power exceeds the beam's own, field pumping thus forms 1.5 to 1.8 times as much as the e-beam's 0.198, and the frequency moves it by −4 to +6 %. At 2 Td, where the beam outweighs the field, the field's marginal power gives 0.30 to 0.33 at a fixed beam current, 1.5 to 1.7 times the e-beam's, and the frequency raises it by 9 % (from 0.265 and by 15 % counted at fixed n_e). This is the rationale of the e-beam-controlled discharge, demonstrated for KrF by Mangano and Jacob [33], and a KrF laser pumped by the discharge that an IR laser beam drives in the gas has also been modelled [21]. Field pumping beats beam pumping at the wall plug when $\eta_{\text{src}}\,\eta_{\text{form,f}} > \eta_{\text{beam}}\,\eta_{\text{form,b}}$, i.e. when the source efficiency exceeds 0.57 to 0.65 of the efficiency with which the pulsed power puts energy into the gas through the foil. The chain's branchings do not depend on n_e. At beam-on density the stepwise ionization $e + \text{Kr}^* \to \text{Kr}^+ + 2e$ runs at 0.39 to 1.63 of the harpoon rate in the Ne fill, rising with the power (Kannari's rate at the scan's T_e [31]). It moves Kr^* from the harpoon channel (0.87) to the ion channel (0.68), which lowers the KrF^* yield of a Kr^* by 4.8 to 11.8 %, and so lowers field pumping, which goes mostly through Kr^*, more than beam pumping, which goes through ions. In a full kinetic model of microwave co-pumping, the sign of the laser energy that a co-pumping field gains rests on that one rate, and the electron quench of KrF^* is a candidate second such rate. The first depends on T_e, and the second may. Those rates, not the frequency, decide whether field pumping pays.

6.9 At the target

(1) In a direct-drive corona Langdon's $\alpha_L \propto Z_{\text{eff}} I\lambda^2/T_e$ favours short wavelengths. Obenschain et al. computed the quarter-critical temperature and intensity of a 2.6 mm CH sphere driven at 351, 248 and 193 nm (their Table 1, averaged over 2 to 3 ns) [14]. With fully ionized CH, $Z_{\text{eff}} = 37/7$, Table 6.6 gives α_L and four estimates of f_L: Langdon's fit, Matte's super-Gaussian, Eq. (6.15) with a constant logarithm, and Eq. (6.15) with the quantum logarithm $L = \ln(2mv^2/\hbar\omega)$ of one electron in both the operator and the absorption. The last is the kinetic counterpart of the improved factor of Turnbull et al. [4], which weights a super-Gaussian with a velocity-dependent Gaunt factor and which Sherlock et al. treat more thoroughly [34]. The e-e logarithm is the NRL formulary's electron–electron value at $n_c/4$ [28], $\ln\Lambda_{ee} = 7.7$, 7.2 and 6.8 at 351, 248 and 193 nm. All of it is for a homogeneous plasma at the time-averaged conditions, with the self-similar distribution at fixed α_L. Nonlocal transport near $n_c/4$ and the structure of the beam are left out.

Table 6.6: The Langdon factor at the quarter-critical conditions of [14], Table 1. I in 10^{14} W cm^{-2}.

λ (nm)	T_e (keV)	I	α_L	fit	Matte	kinetic, const. L	kinetic, $L(v)$
351	3.88	5.19	0.0325	0.906	0.905	0.908	0.940
248	3.32	5.11	0.0187	0.934	0.933	0.935	0.962
193	2.87	4.44	0.0114	0.953	0.951	0.953	0.975

(2) The kinetic solution with a constant logarithm reproduces the fit within 0.22 %. The velocity-dependent logarithm cuts the penalty $1 - f_L$ to 0.64, 0.58 and 0.52 of the fit's, in line with the finding of Turnbull et al. that the velocity dependence of the Gaunt factor "somewhat mitigates the Langdon effect" [4]. The nonlinear factor alone makes the local absorption rate at quarter critical higher than at 351 nm by ×1.031 (KrF) and ×1.052 (ArF) with Langdon's fit, and by ×1.023 and ×1.038 with the velocity-dependent logarithm. The linear coefficient itself, proportional to $n_c T_e^{-3/2} \propto \lambda^{-2} T_e^{-3/2}$ at quarter critical apart from the logarithm, is ×2.53 and ×5.20 that of 351 nm at the same conditions. The nonlinearity thus adds a few percent to a much larger linear advantage of the excimer drivers. The Langdon effect has been reviewed for laser plasmas [22], computed for short KrF pulses on selenium [23], and quantified against $I\lambda^2$ for the plasmas of extreme-ultraviolet sources, where it lowers the absorption by 10 to 20 % [35]. No published quantification of this wavelength dependence for excimer drivers was found, though not every candidate could be read in full. The bandwidths of Obenschain et al.'s Table 1, 1, 3 and 10 THz, give speckle lifetimes $1/\Delta\nu$ of 1.0, 0.33 and 0.10 ps, shorter than the e-e time $1/\nu_{ee}$ at quarter critical (4.8, 2.0 and 1.0 ps). Whether that changes the Langdon factor of a speckled beam is open. Shaffer et al. found that the Langdon operator predicts the heating well in the simple cases they considered. For a two-colour field it gave slightly less heating than their full calculation, as it did without bandwidth, and a weaker, phase-shifted beat modulation, which they attribute to its steady-state assumption breaking down for broadband light [24]. The scattering of electrons on speckles changes both absorption and transport [25].

6.10 Verdict and next steps

(1) The nonlinear behavior of IB does not by itself improve a gas laser. The lasing field of an excimer laser stays more than 10^3 times below it. An auxiliary field reaches only the gas nonlinearity, whose effect on the distribution the electron–electron collisions of a pumped laser largely supply: what is left of the frequency's advantage at the beam-on density is 2 to 7 % where the field's power exceeds the beam's own and 10 to 14 % at a fixed beam current where the beam's is larger (1 to 18 % without recombination), and up to ×1.24 at a fixed beam current (×1.355 at fixed n_e) at the five times lower density that this note's electron source and losses give with the beam of the full kinetic model. At the target the nonlinearity is worth a few percent of absorption rate to KrF and ArF. The ranked next steps, cheapest decisive first:

1. *The instrument.* Fit THz transmission of the laser gas with this solver, which carries the field, the ions and e-e, to read n_e and T_e, and use a THz T_e jump to measure $e + \mathrm{Kr}^* \to \mathrm{Kr}^+ + 2e$ and the electron quench of KrF^*. The first decides the sign of the co-pump in a full kinetic model, the second is the candidate for a second such rate.

2. *Field co-pumping at beam-class power*, frequency-blind: run a full kinetic model of the laser in pairs, with and without the field, with this solver's partition (Tables 6.3 and 6.4) and the uncertain rates bracketed, with the electron collisions on Kr^* and KrF^* that the formation chain leaves out.

3. *The e-beam's own sub-excitation electrons.* Item (3) of Sec. 6.6 sends 39 % of their energy to krypton at beam-on density through e-e collisions in the Ne fill (45 % in the Ar fill), which raises the e-beam's formation efficiency from 0.175 to 0.198. Check whether the energy accounting of full kinetic models of the laser carries it.

4. *The electron source.* The residual advantage at beam-on density depends on how the electrons are created and lost and on how the field's share is counted (1 to 28 %), and on the density itself, which this note's electron source and losses and a full kinetic model put about five times apart. Held at a fixed beam current in the steady state it is within 0.01 of the fixed-beam ratio (Sec. 6.6). A run in time with the beam's source, attachment, recombination on the fill's own mix of cations and the field together would settle it.

5. *CO_2 lasers.* Of the lasers examined, the one whose own field heats its electrons at the 10 % level, by an order-of-magnitude estimate. Compute it with the discharge's own distribution and timing, and its effect on the vibrational partition and on discharge stability for high-intensity amplifiers.

6. *The target.* Use the velocity-dependent Langdon factor in the wavelength studies of excimer drivers, and test whether THz bandwidth and speckle change it.

6.A The electron–electron operator in energy

(1) The isotropic Rosenbluth operator for electrons of distribution $f(v)$, $\int 4\pi v^2 f\,\mathrm{d}v = n_e$, is [10]

$$\frac{\partial f}{\partial t} = \frac{4\pi\Gamma}{v^2}\frac{\partial}{\partial v}\Big[f\,I_2 + \frac{I_4 + v^3 J_1}{3v}\frac{\partial f}{\partial v}\Big], \qquad \Gamma = \frac{e^4 \ln\Lambda}{4\pi\epsilon_0^2 m^2}, \tag{6.17}$$

with $I_2 = \int_0^v u^2 f$, $I_4 = \int_0^v u^4 f$, $J_1 = \int_v^\infty u f$. For a Maxwellian $J_1 = v_t^2 f$ and $I_4 = 3v_t^2 I_2 - v^3 v_t^2 f$, so the bracket vanishes.
(2) Write $h(v,w) = \min(v,w)^3/3$. Since $\partial_w h = w^2$ for $w < v$ and 0 above, $-\int f'(w) h\,\mathrm{d}w = I_2$. And $(I_4 + v^3 J_1)/3v = \int f(w)(w/v) h\,\mathrm{d}w$. The bracket is therefore

$$\int_0^\infty w\,h(v,w)\Big[f(w)\frac{f'(v)}{v} - f(v)\frac{f'(w)}{w}\Big]\mathrm{d}w, \tag{6.18}$$

so v times the bracket has the symmetric weight $v\,w\,h$ and an antisymmetric bracket, which is the Landau form. In energy, $f'(v)/v = (m/e)\,\partial_\epsilon f$ and $w\,\mathrm{d}w = (e/m)\mathrm{d}\epsilon'$, and with $f = n_e(2\pi)^{-1}(m/2e)^{3/2}F$ the upward number flux per electron becomes Eq. (6.12).
(3) Integrating by parts in ϵ', $\int \min^{3/2}[F(\epsilon')F' - FF'(\epsilon')]\mathrm{d}\epsilon' = F'(A_2 + \epsilon^{3/2}A_3) + \frac{3}{2}A_1 F$, which is half the bracket of Hagelaar and Pitchford, so their a [9] equals $(2\Gamma/3)(m/2e)^{3/2} = e^2\gamma\ln\Lambda/24\pi\epsilon_0^2$, as written in Eq. (6.12).

6.B The checks

A check script computes every result, or reloads it from the stored output of the scan that made it and recomputes a sample of that output, and tests it. It recomputes the computed numbers of the prose to their printed digits, each inside a sentence that occurs once in the text, compares the inputs the text states with the code that uses them, and tests the worded claims against the data. It re-derives the displayed formulas numerically and requires them in the text as printed. The numbers quoted from sources were checked against the sources by hand. Table 6.7 counts the checks by the section they support.

Table 6.7: The checks of these notes, by the section whose statements, equations, tables and figures they support: all checks, the computed numbers of the prose recomputed to their printed digits, and the worded claims tested against the data.

supports	checks	printed numbers	claims
Sec. 6.1	58	40	16
Sec. 6.2	18	8	1
Sec. 6.3	46	12	5
Sec. 6.4	20	8	3
Sec. 6.5	45	20	19
Sec. 6.6	150	95	30
Sec. 6.7	23	14	5
Sec. 6.8	61	47	4
Sec. 6.9	29	18	4
Sec. 6.10	17	14	3
Appendix 6.A	1		
Appendix 6.B	2		
the references and the text as a whole	65		
Total	535	276	90

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Chapter 7

A ladder of models for inverse bremsstrahlung absorption

The note of October 12, 2026 (version 3).

Abstract. The interatomic potential was built as a ladder: Coulomb, then Yukawa, which screens it, then Lennard-Jones, which adds a repulsive core, each rung keeping the one below as a limit and adding one physical length. Inverse bremsstrahlung (IB) absorption is organized the same way here, from the Drude model with the Spitzer logarithm through the quantum logarithm, screening by the electrons, the correlations of the ions, strong collisions and their cross term with the medium, the non-Maxwellian distribution, speckles, strong fields and the anisotropy; for neutral atoms through exact phase shifts, polarization bremsstrahlung and multiple collisions. The correlations of the ions are computed in the hypernetted-chain approximation for the ion mixture; for hydrogen they equal Debye–Hückel to 2×10^{-4}, and with them the medium lowers the absorption of the nitrogen–hydrogen mixtures by 2 to 7 %. The anisotropy of the heated distribution is computed with two logarithms and electron–electron collisions and lowers the absorption by 0.8 to 1.3 % for hydrogen. The heaters' own strong-field factor and the exact factor at $v_{\rm os} \geq v_t$ are computed. In argon the single-collision photon exchange beyond the formula of Kroll and Watson is computed exactly and equals double scattering in the target at 2 to 7×10^{13} cm^{-3}. Applied to the fourteen shots of Turnbull et al., the ladder, with no fitted parameter, gives 0.96 ± 0.04 of the measured absorption for the ten nitrogen–hydrogen shots and 1.19 ± 0.06 for the four hydrogen shots. The residual is confined to hydrogen, where every rung of the theory is small.

7.1 The idea

(1) Every IB model in use is the Drude absorption coefficient with a collision frequency [1],

$$K = \frac{\nu_{\rm eff}}{c}\,\frac{\omega_p^2}{\omega^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right)^{-1/2}, \tag{7.1}$$

and the models differ only in $\nu_{\rm eff}$. For electrons on ions of charge Ze,

$$\nu_{\rm eff} = \nu_T\,\Lambda, \qquad \nu_T = \frac{4\sqrt{2\pi}}{3}\,\frac{n_i Z^2 e^4}{m^{1/2}(k_B T_e)^{3/2}}, \tag{7.2}$$

and everything sits in the dimensionless "logarithm" Λ and in factors multiplying it. For electrons on neutral atoms $\nu_{\rm eff}$ is built from the cross sections of the atom.

(2) Each rung below has four parts: the formula, the effect it adds, the dimensionless parameter that says when the effect matters, and its size at the reference conditions. Each rung reduces to the rung below when its parameter goes to zero. Gaussian units are used.
(3) *The plasma reference.* Turnbull et al. measured the absorption of a 527-nm probe beam in gas-jet plasmas heated by thirteen 351-nm beams, with the plasma conditions from Thomson scattering [2, 3]. Of their fifteen analysed shots, fourteen have published profiles: four in hydrogen and ten in a mixture of 43 % nitrogen and 57 % hydrogen nuclei. All are read at the density peak of their profiles (Tables 7.1 and 7.2). The probe carried 1 to 4 J in 100 ps over a 165-μm spot, about 10^{14} W cm^{-2}, so that at the late shots the Langdon parameter is the probe's own; at the co-timed shots the probe carries the share w_p of the overlapped intensity. The Langdon parameter α is computed from the 351-nm-equivalent intensity plotted by Turnbull et al.; it agrees with the one that Matte's formula gives from their plotted exponent m to 5.2 % at every shot. Their composite, the exact (Sommerfeld) logarithm and the Langdon factor with the velocity-dependent Gaunt factor, without their screening factor, exceeds the measured absorption by a factor 1.29 ± 0.06 for the hydrogen shots and 1.07 ± 0.04 for the mixture shots, the averages and spreads they quote [3]. The *hydrogen point*, $n_e/n_c = 0.020$, $T_e = 0.36$ keV, $T_i = 0.05$ keV, serves for the rungs that do not depend on the field. *The gas reference* is argon, the main constituent of one fill of an electron-beam-pumped KrF laser, at electron temperatures of 300 K and 1.08 eV and at 1, 4 and 28.3 THz.

Table 7.1: The four hydrogen shots of Turnbull et al. [3] at the density peak: probe time, n_e/n_c at 527 nm, T_e, the Langdon parameter α, the probe's share w_p of the intensity, and $u = \hbar\omega/k_BT_e$. The ion temperature is 0.05 keV.

shot	t_{probe} (ns)	n_e/n_c	T_e (keV)	α	w_p	u
102727	0.6	0.0177	0.37	0.044	1	0.00636
104031	0.3	0.0229	0.34	0.119	0.27	0.00692
106310	0.3	0.0100	0.26	0.180	0.29	0.00905
106311	0.3	0.0047	0.20	0.232	0.29	0.01177

7.2 The electron–ion ladder

7.2.1 Drude–Spitzer

The classical logarithm $\Lambda = \ln(b_{\max}/b_{\min})$ with the Debye length $b_{\max} = \lambda_{De} = v_t/\omega_p$ and $b_{\min}$ the larger of the 90° impact parameter $Ze^2/3k_BT_e$ and the de Broglie length $\hbar/2mv$, both at $mv^2 = 3k_BT_e$ [1]; $v_t = (k_BT_e/m)^{1/2}$. This is the "Coulomb potential" of the ladder: right in form, with cutoffs put in by hand. At the hydrogen point $\lambda_{De} = 15.73$ nm, the de Broglie length $0.2887\,\hbar/mv_t = 4.20 \times 10^{-3}$ nm exceeds $Ze^2/3k_BT_e = 1.33 \times 10^{-3}$ nm, and $\Lambda_0 = 8.23$.

7.2.2 The quantum logarithm

Second order in the ion potential, with Volkov (field-dressed) electron states and a Maxwell distribution, gives the collision frequency of Eq. (7.2) with

$$\Lambda_1(u) = \frac{1 - e^{-u}}{u}\, e^{u/2} K_0\Big(\frac{u}{2}\Big), \qquad u = \frac{\hbar\omega}{k_BT_e}, \tag{7.3}$$

Table 7.2: The ten nitrogen–hydrogen shots of Turnbull et al. [3] at the density peak: probe time, n_e/n_c, T_e, T_i, the nitrogen charge Z_N used in their calculations, $Z_{\text{eff}} = \langle Z^2\rangle/\langle Z\rangle$, the coupling Γ_{NN} of the nitrogen ions, α and w_p.

shot	t_{probe} (ns)	n_e/n_c	T_e (keV)	T_i (keV)	Z_N	Z_{eff}	Γ_{NN}	α (w_p)
102729	0.6	0.0149	0.65	0.070	6.5	5.57	0.276	0.086 (1.00)
102737	0.3	0.0179	0.42	0.050	5	4.16	0.261	0.189 (0.40)
102738	0.3	0.0189	0.70	0.070	6	5.10	0.260	0.547 (0.15)
104035	0.3	0.0179	0.62	0.070	5	4.16	0.186	0.299 (0.23)
104037	0.6	0.0179	0.62	0.100	6	5.10	0.179	0.067 (1.00)
104041	0.6	0.0194	0.43	0.090	6	5.10	0.204	0.108 (1.00)
106315	0.6	0.0184	0.63	0.090	6	5.10	0.201	0.147 (1.00)
106316	0.6	0.0080	0.45	0.045	5.5	4.63	0.261	0.150 (1.00)
106317	0.6	0.0039	0.30	0.030	5	4.16	0.261	0.196 (1.00)
106319	0.6	0.0279	0.80	0.120	7	6.04	0.225	0.086 (1.00)

absorption net of stimulated emission, and no cutoff. This is Kramers' result with the Born Gaunt factor, $\Lambda_1 = (\pi/\sqrt{3})\langle g_B\rangle(1-e^{-u})/u$ with $\langle g_B\rangle = (\sqrt{3}/\pi)e^{u/2}K_0(u/2)$ [4, 5]. For $u \ll 1$, $K_0(x) = -\ln(x/2) - \gamma + O(x^2\ln x)$ gives

$$\Lambda_1 \simeq \ln\frac{4k_BT_e}{e^\gamma\hbar\omega} = \ln\frac{b_{\max}}{b_{\min}}, \qquad b_{\max} = \frac{v_t}{\omega}, \qquad b_{\min} = \frac{e^\gamma}{4}\,\frac{\hbar}{mv_t} = 0.445\,\frac{\hbar}{mv_t}, \tag{7.4}$$

with γ Euler's constant. The theory fixes both cutoffs. The upper one is the distance an electron travels in a radian of the laser period, the dynamic cutoff of Dawson and Oberman [6], not the Debye length. This is the "Yukawa" step: the long range is cut by a physical length. Parameter: ω/ω_p for the upper cutoff, u for the photon energy. At the hydrogen point $b_{\max} = 2.225$ nm and $\Lambda_1 = 5.84$. The logarithm Λ_0 of Sec. 7.2.1 exceeds it by $\ln(\omega/\omega_p) + \ln(0.445/0.2887) = 1.96 + 0.43$, or 40.9 %, in line with the finding of Turnbull et al. that transport logarithms with a Debye length miss their data by more than 50 % [2].

7.2.3 Dynamic screening by the electrons

With the electrons' own response, second order in the ion potential gives the high-frequency collision frequency of Dawson and Oberman [6],

$$\nu_{\text{eff}} = -\frac{2Ze^2}{3\pi m\omega}\int_0^\infty k^2 S(k)\,\operatorname{Im}\frac{1}{\varepsilon(k,\omega)}\,\mathrm{d}k, \tag{7.5}$$

with ε the longitudinal dielectric function of the electrons and S the static structure factor of the ions. For bare ions ($S = 1$) and the quantum (Lindhard) dielectric function of Maxwell electrons, with $x = kv_t/\omega$ and $r = \omega_p/\omega$,

$$\Lambda_2 = -\frac{1}{r^2}\Big(\frac{2}{\pi}\Big)^{1/2}\int_0^\infty x^2\,\operatorname{Im}\frac{1}{\varepsilon(x)}\,\mathrm{d}x, \qquad \varepsilon(x) = 1 - \frac{r^2}{\sqrt{2}\,ux^3}\big[Z(\zeta_+) - Z(\zeta_-)\big], \quad \zeta_\pm = \frac{1}{\sqrt{2}}\Big(\frac{1}{x} \pm \frac{ux}{2}\Big), \tag{7.6}$$

where Z is the plasma dispersion function. Replacing $\operatorname{Im}\varepsilon^{-1}$ by $-\operatorname{Im}\varepsilon$ gives back Eq. (7.3). Parameter: $r = \omega_p/\omega$. At the hydrogen point the electrons lower the logarithm by 0.0076, or 0.13 %.

7.2.4 Correlations of the ions

The ions enter Eq. (7.5) only through $S(k)$. Their mean spacing does not appear by itself; uncorrelated ions add whatever their spacing. In a weakly coupled plasma S has the Debye–Hückel form

$$S(k) = \frac{k^2 + k_{De}^2}{k^2 + k_{De}^2 + k_{Di}^2}, \qquad k_{Di}^2 = a\,k_{De}^2, \qquad a = \frac{Z T_e}{T_i}. \tag{7.7}$$

With static (Debye) electrons and soft photons the ion term has a closed form. The one-speed logarithm changes by $\int_{\omega/v}^{\infty}[S(q)-1]\,\mathrm{d}q/q = -(k_{Di}^2/2k_t^2)\ln(1+k_t^2v^2/\omega^2)$, $k_t^2 = k_{De}^2 + k_{Di}^2$, and the Maxwell average, with $\int_0^\infty e^{-s}\ln(1+\kappa s)\,\mathrm{d}s = e^{1/\kappa}E_1(1/\kappa)$, gives

$$\Delta\Lambda_3 \simeq \bar{I} = -\frac{a}{2(1+a)}\,e^{1/\kappa}E_1(1/\kappa), \qquad \kappa = 2(1+a)\,\frac{n_e}{n_c}. \tag{7.8}$$

Parameters: $k_{Di}/k_{De} = a^{1/2}$ for the size, and the ion coupling $\Gamma_{ii} = Z^2e^2/a_ik_BT_i$ for the validity of Eq. (7.7), with $a_i = (3/4\pi n_i)^{1/3}$. At the hydrogen point $a = 7.2$, $\kappa = 0.328$ and $\Gamma_{ii} = 0.020$. Equation (7.5) with Eq. (7.7) and the Lindhard ε lowers the logarithm by a further 0.114, or 1.96 %; Eq. (7.8) gives −0.1136, within 0.44 % of it. With the electron screening of Sec. 7.2.3, the medium lowers Λ_1 by 2.08 %. At the four shots the medium factor $\Lambda_{2,3}/\Lambda_1$ is 0.981, 0.978, 0.991 and 0.996 (Table 7.6); it would be 0.968 at the first shot with $T_i = 0.025$ keV and 0.989 with 0.1 keV.

Against the fitted factor. Turnbull et al. multiply the absorption by $f_{sc} = \{\ln[(2k_BT_e/\hbar\omega)\,y/(1+y^2)^{1/2}] - 1/[2(1+y^2)]\}/\ln(2k_BT_e/\hbar\omega)$, $y = l_{sc}\omega/v_t$, the Born logarithm of a Yukawa ion of screening length l_{sc} [2]. With $l_{sc} = \lambda_{D,ei} = 5.49$ nm, the Debye length of electrons and ions, it lowers the absorption by 2.56 %, close to the medium of Secs. 7.2.3 and 7.2.4. With the interionic distance $l_{sc} = a_i = 1.437$ nm, chosen to match their data, it lowers it by 16.8 %, about eight times what the medium gives.

Beyond Debye–Hückel. Equation (7.7) is the weak-coupling limit. The structure factor of an ion mixture follows at any coupling from the Ornstein–Zernike relation with the hypernetted-chain closure [26]. With ions i of charge Z_ie and density n_i at the temperature T_i, interacting through the electron-screened potential $\phi_{ij} = Z_iZ_je^2e^{-k_{De}r}/r$,

$$h_{ij}(r) = \exp\Big[-\frac{\phi_{ij}(r)}{k_BT_i} + h_{ij}(r) - c_{ij}(r)\Big] - 1, \qquad \tilde{h}_{ij}(k) = \tilde{c}_{ij}(k) + \sum_l \tilde{c}_{il}(k)\,n_l\,\tilde{h}_{lj}(k), \tag{7.9}$$

and the absorption takes the charge structure factor

$$S_{ZZ}(k) = \frac{\sum_i n_iZ_i^2 + \sum_{ij} n_in_jZ_iZ_j\tilde{h}_{ij}(k)}{\sum_i n_iZ_i^2} \tag{7.10}$$

in place of S in Eq. (7.5). The equations are solved by iteration on a radial grid, with the long-range parts of c_{ij} and $h_{ij}-c_{ij}$ subtracted and transformed analytically, so that the Debye–Hückel solution is the exact starting point. For the one-component plasma the excess energy is −0.5704 at $\Gamma = 1$, against −0.5726 from the Monte Carlo fit of Slattery, Doolen and DeWitt [27], and −0.02569 at $\Gamma = 0.1$, against −0.02590 from Abe's expansion [28] and −0.02739 from Debye–Hückel. At the hydrogen point the hypernetted-chain S differs from Eq. (7.7) by at most 6.2×10^{-4}, and at the four hydrogen shots the medium factor changes by at most 1.9×10^{-4}: for hydrogen the correlations beyond Debye–Hückel are nil. The mixtures are another matter. There $k_{Di}^2/k_{De}^2 = Z_{\mathrm{eff}}T_e/T_i$ is 24 to 52 and $\Gamma_{NN} = 0.18$ to 0.28, the medium factor is 0.930 to 0.982, and the hypernetted chain raises it by 0.0037 to 0.0064 against Debye–Hückel. The medium lowers the absorption by 2 to 7 %

in the mixtures and by 0.4 to 2.2 % in hydrogen. Halving or doubling T_i, the least certain reading, moves the mixture medium factor between 0.904 and 0.989. The fitted factor of Turnbull et al., with the interionic distance of the nitrogen ions for the mixtures as they use it and of the protons for hydrogen, is 0.879 to 0.969 for the mixtures and 0.827 to 0.927 for hydrogen; at the density peaks it reproduces their reported changes $1.29 \to 1.13$ and $1.07 \to 1.00$ to 0.8 and 2.3 %.

7.2.5 Strong collisions

The Born approximation needs $\eta = Ze^2/\hbar v \ll 1$. The exact Coulomb (Sommerfeld) free–free Gaunt factor removes that condition [7, 5]:

$$\Lambda_4 = \frac{\pi}{\sqrt{3}}\,\langle g\rangle(\gamma^2, u)\,\frac{1 - e^{-u}}{u}, \qquad \gamma^2 = \frac{Z^2\,\mathrm{Ry}}{k_B T_e} = \frac{\eta_t^2}{2}, \quad \eta_t = \frac{Ze^2}{\hbar v_t}, \tag{7.11}$$

with the thermally averaged exact Gaunt factor $\langle g\rangle$ tabulated by van Hoof et al. [8]. As $\gamma^2 \to 0$ it tends to $\langle g_B\rangle$ and Λ_4 to Λ_1. In the soft-photon limit the exact logarithm of one electron is the Born one plus Bloch's constant $\Delta(\eta) = -\gamma - \mathrm{Re}\,\psi(1 + i\eta)$ [9], which is $-\zeta(3)\eta^2$ for $\eta \ll 1$ and $-\gamma - \ln\eta$ for $\eta \gg 1$, where it turns the quantum lower cutoff into the classical one. Parameter: η_t. At the hydrogen point $\eta_t = 0.275$ and $\langle g\rangle/\langle g_B\rangle = 0.99315$, a correction of -0.685 %; at the four shots it is -0.67 to -0.88 %. For fully stripped nitrogen ($Z = 7$) at the same T_e and u, $\eta_t = 1.92$ and the correction is -15.4 %.

The cross term with the medium. The exact correction is added to the logarithm of Sec. 7.2.4, $\Lambda_{2,3} + (\Lambda_4 - \Lambda_1)$. Whether the medium changes the strong-collision correction itself is decided by the partial waves of an ion screened to $V = -(Ze^2/r)e^{-r/\lambda}$. In the soft-photon limit the logarithm of one electron is the transport sum $S = \sum_l (l+1)\sin^2(\delta_{l+1} - \delta_l)$ of its phase shifts, cut at the partial wave $J = kb_{\max} = k^2/\omega$ of the dynamic cutoff, in units of η^2; with the first Born phases, $S^B = \sum_l (l+1)(\delta^B_{l+1} - \delta^B_l)^2$. For the bare Coulomb field $\delta_l = \mathrm{ph}\,\Gamma(l + 1 - i\eta)$, $\delta^B_l = -\eta\psi(l+1)$, and $(S - S^B)/\eta^2 = \Delta(\eta) + O(\eta^2/J^2)$ exactly. For the screened field the exact phases come from the radial equation, integrated numerically, and the Born phases are $\delta^B_l = \eta\,Q_l(1 + 1/2k^2\lambda^2)$ with the Legendre function of the second kind. The cross term

$$C = \frac{S^Y - S^{Y,B}}{\eta^2} - \frac{S^C - S^{C,B}}{\eta^2} \tag{7.12}$$

is the part of the exact-minus-Born correction that the screening changes; the Born part of the screening, $(S^{Y,B} - S^{C,B})/\eta^2$, is the second-order medium of Sec. 7.2.4 and is not in C. For the Coulomb field only odd powers of η enter the phases, so there is no second-Born term in their differences; for the screened field there is, and C is first order in $b_{\max}/\lambda$, not in b_{90}/λ as the size of the strong-collision region would suggest. Averaged over the Maxwellian with the soft-photon weight e^{-s}, $s = E/k_B T_e$, at $T_e = 0.36$ keV: for hydrogen $\langle C\rangle = +0.0098$ with $\lambda = \lambda_{D,ei} = 5.49$ nm and $+0.0035$ with $\lambda = \lambda_{De} = 15.73$ nm, against $\langle\Delta\rangle = -0.134$ for the bare ion, that is 7.4 % and 2.6 % of the strong-collision correction and 1.7×10^{-3} and 6.0×10^{-4} of Λ_1. For $Z = 7$, $\langle C\rangle = +0.0359$ and $+0.0128$ against $\langle\Delta\rangle = -1.235$, that is 2.9 % and 1.0 % of the correction and 6.1×10^{-3} and 2.2×10^{-3} of Λ_1. The sign is positive: the medium weakens the strong-collision correction. Doubling the cutoff J changes C by 9 % at $s = 0.25$ and by 2 to 3 % at $s \geq 1.2$. The soft-photon average $\langle\Delta\rangle$ overstates the exact thermal correction of Eq. (7.11) by a factor 3.3 for hydrogen, since the slow electrons, for which the photon is not soft, are focused by the ion; C shares that weighting, so these are upper values. In the absorption the cross term is below 0.2 % for hydrogen and about 0.6 % for the nitrogen ions of the mixed shots. At the conditions of the mixture shots, $Z = 6$, $T_e = 0.62$ keV and the total screening length $\lambda_t = 3.58$ nm of shot 106315, $\langle C\rangle = +0.0383$ against $\langle\Delta\rangle = -0.870$, 6.0×10^{-3} of Λ_1. This closes the item.

7.2.6 The non-Maxwellian distribution (Langdon)

IB heats slow electrons fastest. When it outpaces the electron–electron collisions the distribution flattens toward $F \propto \exp[-(v/v_m)^M]$ with $M > 2$ [10]. The parameter is $\alpha = Zv_{\mathrm{os}}^2/v_t^2$, with v_{os} the peak quiver velocity, and Matte et al. fit the exponent as $m(\alpha) = 2 + 3/(1 + 1.66/\alpha^{0.724})$ [11]. For an isotropic distribution the heating rate is $P \propto -\int_0^\infty L(v)\,\partial_v F\,\mathrm{d}v$, with the one-speed Born logarithm $L(v) = \ln(2mv^2/\hbar\omega)$. At fixed L this is $LF(0)$, and the factor is $F_M(0)/F_2(0)$ at fixed density and energy; Langdon's fit $f_L = 1 - 0.553/[1 + (0.27/\alpha)^{0.75}]$ [10] equals it at $M = m(\alpha)$ to within 0.27 % for $\alpha = 10^{-3}$ to 10^3. With the velocity-dependent L, the substitution $s = (v/v_m)^M$ and $\int_0^\infty e^{-s}\ln s\,\mathrm{d}s = -\gamma$ give

$$\frac{P_M}{P_2} = \frac{F_M(0)}{F_2(0)}\,\frac{\ln\left(2v_m^2/uv_t^2\right) - 2\gamma/M}{\ln(4/u) - \gamma}, \qquad \frac{v_m^2}{v_t^2} = \frac{3\Gamma(3/M)}{\Gamma(5/M)}. \tag{7.13}$$

The second factor, one at $M = 2$, raises the absorption: a flattened distribution moves electrons to speeds where the logarithm is larger. At the four shots $m = 2.18$, 2.34, 2.45 and 2.52, $f_L = 0.887$, 0.806, 0.765 and 0.739, the logarithm factor C_L is 1.031, 1.056, 1.072 and 1.085, and the product $f_L C_L$ is 0.914, 0.851, 0.820 and 0.802. At the mixture shots $\alpha = 0.067$ to 0.547.

7.2.7 Speckles

A real beam has an intensity pattern. If the distribution adjusts locally and T_e is uniform, the absorbed power is that of the mean intensity times an average of f_L over the pattern. The probe's own pattern has the exponential distribution of a fully developed speckle field, $x = I/\langle I\rangle \sim e^{-x}$, and the probe is absorbed where its own intensity is, with the weight x; the thirteen heater beams carry independent patterns, whose sum y has the gamma distribution of shape 13 and mean one. With $\alpha_p = w_p\alpha$ and $\alpha_h = (1 - w_p)\alpha$,

$$f_{sp} = \int_0^\infty \mathrm{d}x\, x\, e^{-x} \int_0^\infty \mathrm{d}y\, \frac{13^{13} y^{12} e^{-13y}}{12!}\, f_L(\alpha_h y + \alpha_p x). \tag{7.14}$$

Since f_L falls with α, the bright speckles absorb less per unit intensity. Parameter: α, and the ratio of the speckle size to the electron relaxation length (Eq. (7.14) is the local limit; in the opposite limit $f_{sp} = f_L$). At the four shots $f_{sp}/f_L = 0.955$, 0.977, 0.973 and 0.972: the late shot, where the probe is the only beam, is the most affected.

7.2.8 Strong fields: the probe's own field and the heaters' quiver

When the quiver velocity is not small against the thermal velocity the field cannot be treated to lowest order. Second order in the ion potential but exact in the field gives the heating rate as a sum over photon numbers ℓ, each weighted by $J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)$ with $\mathbf{a}_0$ the quiver amplitude, inside the dielectric form of Secs. 7.2.3 and 7.2.4 [12, 13, 14]. Parameter: $\beta = v_{\mathrm{os}}/v_t = (\alpha/Z)^{1/2}$. At the four shots the probe has $\beta_p = 0.21$, 0.18, 0.23 and 0.26, and the heaters $\beta_h = 0$, 0.29, 0.36 and 0.40. Two effects follow, computed here with the free-electron response χ_0 of Maxwell electrons and bare ions, in the variables $x = kv_t/\omega_p$, $u = \hbar\omega_p/k_B T_e$ and

$$D(x;\nu) = e^{-\frac{1}{2}(\nu/x - ux/2)^2} - e^{-\frac{1}{2}(\nu/x + ux/2)^2}, \qquad \int_0^\infty D(x;1)\,\frac{\mathrm{d}x}{x} = u\Lambda_1(u), \tag{7.15}$$

D being proportional to $x\,\mathrm{Im}\,\chi_0(k, \nu\omega_p)$, absorption net of emission.

The probe alone. With $B_\ell(z)=\int_{-1}^{1}J_\ell^2(z\mu)\,\mathrm{d}\mu$ the exact ratio of the probe's collision frequency to its weak-field value is

$$R_{\mathrm{self}}(\beta_p,u)=\frac{3}{u\beta_p^2\Lambda_1}\sum_{\ell=-\infty}^{\infty}\ell\int_0^\infty\frac{\mathrm{d}x}{x^3}\,D(x;\ell)\,B_\ell(x\beta_p),\tag{7.16}$$

which tends to one as $\beta_p\to0$. Its quasi-static limit at a fixed logarithm is Silin's $1-\frac{9}{40}\beta_p^2+O(\beta_p^4)$; the exact coefficient at $u=0.0064$ is 0.195, smaller because the largest momentum transfer grows with the field. At the four shots $R_{\mathrm{self}}=0.991$, 0.994, 0.990 and 0.987.

The probe inside the heaters' quiver. With two fields the quiver is $\mathbf{a}(t)=\mathbf{a}_h\sin\omega_h t+\mathbf{a}_p\sin\omega_p t$, and the Volkov phase $e^{i\mathbf{k}\cdot\mathbf{a}(t)}$ has sidebands at $\ell\omega_h+m\omega_p$ with the weights $J_\ell^2(\mathbf{k}\cdot\mathbf{a}_h)J_m^2(\mathbf{k}\cdot\mathbf{a}_p)$; for $\omega_h/\omega_p=3/2$ no two sidebands with $|m|\le1$ coincide. The probe's photons are counted by m, and to second order in the probe field only $m=\pm1$ enter, so that

$$R_{pp}(\psi)=\frac{\displaystyle\sum_{\ell=-\infty}^{\infty}\int_0^\infty\frac{\mathrm{d}x}{x}\,D\Big(x;1+\ell\frac{\omega_h}{\omega_p}\Big)\,W_\ell^{\psi}\Big(\frac{x\beta_h\omega_p}{\omega_h}\Big)}{\dfrac{2}{3}\displaystyle\int_0^\infty\frac{\mathrm{d}x}{x}\,D(x;1)},\tag{7.17}$$

$$W_\ell^{\psi}(z)=\int_{-1}^{1}J_\ell^2(z\mu)\Big[\frac{\sin^2\psi}{2}(1-\mu^2)+\cos^2\psi\,\mu^2\Big]\mathrm{d}\mu,\tag{7.18}$$

with ψ the angle between the two polarizations; averaged over ψ, $W_\ell=B_\ell/3$, and $W_\ell^{\psi}(0)=\frac{2}{3}\delta_{\ell0}$ returns the weak field. The terms with $\ell<0$ have $1+\ell\omega_h/\omega_p<0$ and $D<0$: an electron that absorbs a probe photon while emitting a heater photon loses energy, so the reverse process, a heater photon absorbed and a probe photon emitted, is favoured, and the heater feeds the probe through the ions. At large k the sidebands cancel, since $\sum_\ell J_\ell^2=1$ and $\sum_\ell\ell J_\ell^2=0$ and D is linear in ν there; the effect is a logarithmic integral from the thermal cutoff $x\sim1$ upward, of relative size β_h^2. Two independent checks: the small-field series of Eq. (7.17), from $B_0=2-z^2/3$ and $B_{\pm1}=z^2/6$, gives $R_{pp}=1-c\beta_h^2$ with $c=0.212$ averaged over ψ at $u=0.0064$, which the full sum reproduces to 6×10^{-5} at $\beta_h=0.2$; and the classical picture of ions drifting through the electron gas with the velocity $-(\mathbf{u}_h+\mathbf{u}_p)(t)$, a drag $-h(s)\mathbf{s}$ with $s=u/\sqrt{2}v_t$ and $h=[\sqrt{\pi}\,\mathrm{erf}(s)/s-2e^{-s^2}]/s^2$, gives for the probe's power $\langle F\cdot u_p\rangle$ to second order $R_{pp}^{qs}(\psi)=\frac{3}{4}\langle h(s_h)+s_hh'(s_h)\cos^2\psi\rangle$, which at small β_h is $1-\frac{9}{20}\beta_h^2$ for parallel polarizations, $1-\frac{3}{20}\beta_h^2$ for perpendicular ones and $1-\frac{1}{4}\beta_h^2$ averaged. A pump is therefore twice as effective on a probe as a beam on itself, $\frac{9}{40}\to\frac{9}{20}$, because the two phases are independent. At $\beta_h=0.2$ the exact $1-R_{pp}$ is 0.0151, 0.0051 and 0.0084 for the three cases against 0.0178, 0.0060 and 0.0099 quasi-static, the same ratio 0.847 in all three, as for R_{self} against Silin. The thirteen heaters surround the plasma, so the average over ψ is the physical case. At the co-timed shots $R_{pp}=0.982$, 0.974 and 0.967 averaged, 0.968, 0.953 and 0.941 for parallel and 0.989, 0.984 and 0.980 for perpendicular polarizations. The product $R_{\mathrm{self}}R_{pp}$ is 0.991, 0.976, 0.964 and 0.954 at the four shots. For the heater beams themselves the single-beam factor at their own $\beta_h=0.29$, 0.36 and 0.40 is $R_{\mathrm{self}}=0.983$, 0.976 and 0.969 at the co-timed shots: the heating rate is 2 to 3 % below its weak-field value. At $\beta\ge1$ Eq. (7.16) at $u=0.0064$ gives 0.953, 0.828 and 0.507 for $\beta=0.5$, 1 and 2, against Silin's 0.946, 0.805 and 0.460: the exact rate exceeds the quasi-static one by 1, 3 and 10 %, because the largest momentum transfer grows with the quiver velocity. This closes the item.

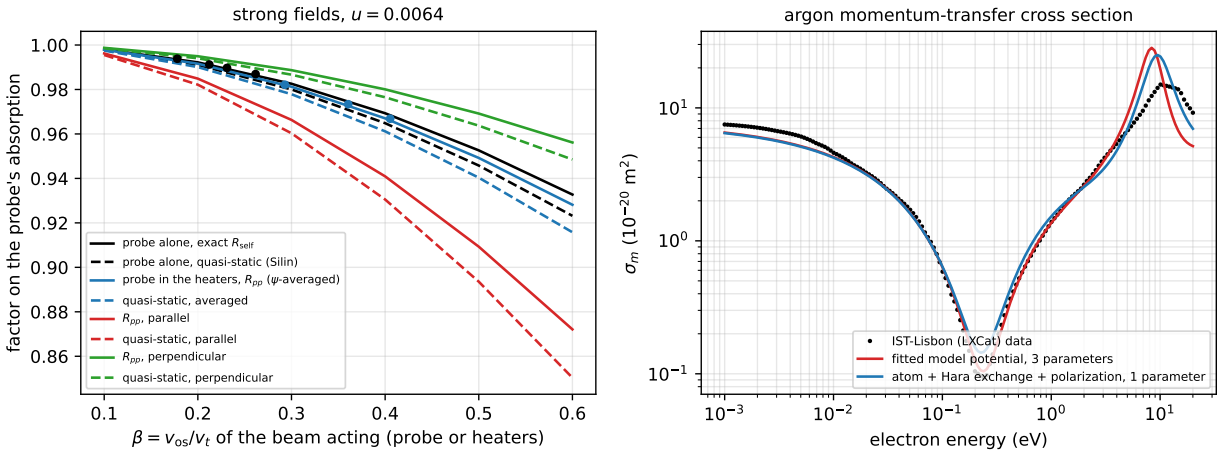


Figure 7.1: Left: the strong-field factors on a probe's absorption at $u = 0.0064$ against $\beta = v_{os}/v_t$ of the beam that acts. Black: the probe's own field, exact (Eq. (7.16)) and quasi-static (Silin). Colours: the probe inside the heaters' quiver, Eq. (7.17), averaged over the polarizations (blue), parallel (red) and perpendicular (green), exact (solid) and in the quasi-static drag picture (dashed). Dots: the four hydrogen shots. Right: the momentum-transfer cross section of argon from the IST-Lisbon data, from the three-parameter model potential of Eq. (7.22), and from the Hartree–Fock atom with Hara's exchange and one fitted cutoff.

7.2.9 The anisotropy

A linearly polarized field heats along its polarization, and the distribution acquires a part of degree 2 in the angle, $f = f_0 + f_2(v)P_2(\hat{\mathbf{v}}\cdot\boldsymbol{\epsilon})$. Its direct effect on the absorption of the same beam is

$$\frac{\delta P}{P} = -\frac{2}{5}\frac{v_{os}^2}{v_t^2}\,Q, \qquad Q = \int_0^\infty \frac{f_2(v)}{(v_{os}^2/3v_t^2)\,f_0(0)}\,\frac{\mathrm{d}v}{v}, \tag{7.19}$$

with v_{os} the probe's own quiver velocity. In the Lorentz limit, $Z \gg 1$, the field drives f_2 through the ion collisions and the same collisions relax it, $f_2 = S_2/3\nu_{ei}$, and for the self-similar heated distribution Q is a number of order one, 1.32, 1.03, 0.93 and 0.87 at the α of the four shots. Two corrections enter. The drive is inverse bremsstrahlung and carries the logarithm Λ_1 with the dynamic cutoff, while the relaxation of the quasi-static f_2 is pitch-angle scattering and carries the transport logarithm $\ln(\lambda_{De}/b_{\min})$ of Sec. 7.2.1, 8.10 to 8.36 at the shots: f_2 is smaller by $\Lambda_1/\Lambda_{tr} = 0.63$ to 0.71. And at $Z = 1$ the electron–electron collisions relax f_2 as well. With the test-particle collision operator on a Maxwellian background [29], in units of $4\pi n_e e^4\Lambda_{tr}/m^2v_t^3$ and with $w = v/v_t$, $x = w^2/2$ and $\psi(x) = \operatorname{erf}\sqrt{x} - 2(x/\pi)^{1/2}e^{-x}$, f_2 solves

$$3\Big(\frac{Z}{w^3} + \nu_D\Big)f_2 - \frac{1}{w^2}\frac{\mathrm{d}}{\mathrm{d}w}\Big[w^2\frac{\psi}{w^3}\Big(\frac{\mathrm{d}f_2}{\mathrm{d}w} + wf_2\Big)\Big] = \frac{3Z}{w^3}\frac{\Lambda_1}{\Lambda_{tr}}\,S_2, \qquad \nu_D = \frac{(1-1/2x)\psi + \psi'}{w^3}, \tag{7.20}$$

with the drive S_2 of the Lorentz limit. The operator keeps the Maxwellian, and for fast electrons at $Z = 1$ the deflection by electrons equals that by ions, as it must [30]. For a Maxwellian drive the electron–electron collisions lower Q by a factor 0.77 to 0.80 at the hydrogen shots and 0.94 to 0.96 at the mixture shots, where $Z_{\text{eff}} \approx 5$. Applied to the self-similar Q, the direct effect is 1.30, 0.76, 1.02 and 1.15 % at the four hydrogen shots, against 2.3, 1.3, 2.0 and 2.4 % in the Lorentz limit with one logarithm, and 0.32 to 1.07 % at the mixture shots. The feedback of f_2 on the isotropic part and the field-particle part of the electron–electron operator are not computed; the direct part is.

7.2.10 The plasma ladder at the hydrogen point

Table 7.3: The electron–ion ladder at the hydrogen point ($n_e/n_c = 0.020$, $T_e = 0.36$ keV, $T_i = 0.05$ keV, 526.7 nm). The logarithm after each rung and its change against Λ_1.

Sec.	Adds	Parameter	Λ	against Λ_1 (%)
7.2.1	Drude–Spitzer, cutoffs by hand	—	8.228	+40.9
7.2.2	quantum logarithm, dynamic cutoff	u, ω/ω_p	5.839	0
7.2.3	screening by the electrons	ω_p/ω	5.832	−0.13
7.2.4	correlations of the ions	$a^{1/2}$, Γ_{ii}	5.717	−2.08
7.2.5	strong collisions	η_t	5.678	−2.77
7.2.5	cross term with the medium	$\eta_t\, b_{\max}/\lambda$	5.681 to 5.687	−2.71 to −2.60

The rungs of Secs. 7.2.6 to 7.2.9 multiply the absorption rather than the logarithm (Sec. 7.4).

7.3 The electron–atom ladder

In a weakly ionized gas the electrons absorb mostly while they collide with atoms. For ω much larger than the momentum-transfer collision frequency $\nu_m(v) = n_a \sigma_m(v) v$ the Lorentz-gas conductivity gives, for any isotropic F normalized to one, $\nu_{\mathrm{eff}} = -(4\pi/3)\int_0^\infty v^3 \nu_m(v)\, \partial_v F \, \mathrm{d}v$.

7.3.1 Drude with a mean collision frequency

$\nu_{\mathrm{eff}} = \langle \nu_m \rangle = n_a \langle \sigma_m v \rangle$ over the Maxwellian. This is the usual input of a Drude model.

7.3.2 The energy-dependent kernel

For a Maxwellian, $\partial_v F = -(v/v_t^2)F$ and

$$\nu_{\mathrm{eff}} = \frac{\langle \nu_m v^2 \rangle}{3 v_t^2} = \left\langle \nu_m \, \frac{2\varepsilon}{3 k_B T_e} \right\rangle, \tag{7.21}$$

with $\varepsilon = m v^2/2$, which reduces to $\langle \nu_m \rangle$ of Sec. 7.3.1 only for a constant ν_m. The ratio to $\langle \nu_m \rangle$ is 4/3 for a constant σ_m. The IST-Lisbon momentum-transfer cross section of argon [15, 16, 17] has its Ramsauer minimum near 0.25 eV, and with it the ratio is 0.836 at 300 K and 2.02 at 1.08 eV. A Drude model with a mean collision frequency is wrong by up to a factor of two here. Parameter: the logarithmic slope of $\sigma_m(\varepsilon)$. By the low-energy theorem of Low [18] and the formula of Kroll and Watson [19], this kernel is exact for soft photons, $\hbar\omega \ll \varepsilon$, and the atom enters only through σ_m.

7.3.3 Exact phase shifts

When $\hbar\omega$ is comparable with the electron energy the free–free cross section depends on more than σ_m. It is computed from the acceleration-form dipole matrix elements between exact partial waves of a model potential, here

$$V(r) = -\frac{Z}{r}\,\frac{1}{1+H(e^{r/d}-1)} - \frac{\alpha_d}{2(r^2+r_c^2)^2} \tag{7.22}$$

in atomic units, with the screening function of Green, Sellin and Zachor [20], $Z = 18$ and the polarizability $\alpha_d = 11.083$ [21]. Its parameters, $H = 4.321$, $d = 0.7869$ and $r_c = 0.8266$, minimize the rms of $\ln\sigma_m$ against the IST-Lisbon set over 36 energies from 0.003 to 4 eV (rms 0.081), and it binds the three s and two p orbitals of argon. With $G(\varepsilon)$ the ratio of the exact cross section to its soft-photon limit, the ratio of the quantum to the classical absorption coefficient is

$$R = \frac{1-e^{-u}}{u}\,\frac{\langle\sigma_m G\rangle_w}{\langle\sigma_m\rangle_w}, \tag{7.23}$$

with the weight $\varepsilon^2 e^{-\varepsilon/k_BT_e}$ of Eq. (7.21). Parameter: u. For argon $R = 0.9984$ at 1 THz and 0.9749 at 4 THz for 300 K electrons, and 0.591 at 28.3 THz, the CO_2 laser; at 1.08 eV it stays within 10^{-3} of one at all three frequencies.

From the atom. Equation (7.22) is fitted to σ_m with three parameters, so the mid-infrared value rests on the fit. A potential built from the atom replaces it: the static field of the closed-shell Hartree–Fock atom, $V_{st} = -Z/r + V_H$, whose total energy is -526.831 against the Hartree–Fock limit -526.818 hartree, plus a local exchange, plus the same polarization tail with the single parameter r_c. The semiclassical exchange of Furness and McCarthy, $V_{ex} = \frac{1}{2}(E-V_{st}) - \frac{1}{2}[(E-V_{st})^2 + 4\pi\rho]^{1/2}$ with ρ the electron density [22], is far too attractive at these energies: no r_c brings its σ_m within a factor of 4.5 of the data. The free-electron-gas exchange of Hara [23],

$$V_{ex} = -\frac{k_F}{\pi}\left[1 + \frac{1-\eta^2}{2\eta}\ln\left|\frac{1+\eta}{1-\eta}\right|\right], \qquad k_F = (3\pi^2\rho)^{1/3}, \quad \eta = \frac{(k^2+2I+k_F^2)^{1/2}}{k_F}, \tag{7.24}$$

with I the ionization energy, does reproduce the Ramsauer region: with r_c =1.548 the rms of $\ln\sigma_m$ over the same 36 energies is 0.172, against 0.081 for the three-parameter fit, and the minimum lies at 0.226 eV against about 0.25 eV in the data; over 0.003 to 20 eV the rms is 0.215. With this potential R at 300 K is 0.9986 at 1 THz, 0.9744 at 4 THz and 0.566 at 28.3 THz, against 0.9984, 0.9749 and 0.591 for Eq. (7.22); at 1.08 eV it is 1.0009 at 28.3 THz. The exchange depends on the energy, and the matrix elements between k and k' use the potential at the mean energy. The mid-infrared reduction is therefore a property of the phase shifts that any potential reproducing σ_m shares, not of the fit.

7.3.4 Polarization bremsstrahlung

The field of the electron induces a dipole $\alpha_d\hat{\mathbf{r}}/r^2$ (atomic units) in the atom, which radiates together with the electron. In the acceleration form, where the electron's dipole matrix element is that of $V'(r)/\omega^2$, the radial force $V'(r)$ becomes $V'(r) - \omega^2\alpha_d/(r^2+r_c^2)$, with the cutoff r_c of the polarization tail. Parameter: $\alpha_d\omega^2$ against the atomic force. For argon with Eq. (7.22) it changes R by $-0.03\,\%$ at 1 THz and $-0.43\,\%$ at 4 THz for 300 K electrons, and by $-13.0\,\%$ at 28.3 THz, where R becomes 0.514; at 1.08 eV the change is below $0.1\,\%$. With the potential from the atom, R at 28.3 THz and 300 K becomes 0.491.

7.3.5 Multiple collisions

An electron can scatter on two atoms in a row. The kinetic equation counts every collision with its own photon-exchange cross section whatever collision follows, so sequences of collisions add nothing to the absorption of a gas. They do add a signal proportional to the square of the density in laser-assisted scattering experiments, which may explain why photon exchange in argon in CO_2-laser fields was measured far above the Kroll–Watson prediction [24, 25]. That anomaly is a question about single-collision cross sections, and measurements at several target densities would settle it.

Table 7.4: The electron–atom ladder for argon. $\nu_{\mathrm{eff}}/\langle\nu_m\rangle$ is the ratio of the collision frequency of Eq. (7.21) to that of Sec. 7.3.1; R is Eq. (7.23) without and with polarization bremsstrahlung (Secs. 7.3.3 and 7.3.4), for the fitted potential of Eq. (7.22) and for the potential from the atom with Hara's exchange.

			fitted potential		atom + Hara exchange	
T_e	$\nu_{\mathrm{eff}}/\langle\nu_m\rangle$	f (THz)	R (Sec. 7.3.3)	R (Sec. 7.3.4)	R (Sec. 7.3.3)	R (Sec. 7.3.4)
300 K	0.836	1	0.9984	0.9981	0.9986	0.9983
		4	0.9749	0.9707	0.9744	0.9702
		28.3	0.591	0.514	0.566	0.491
1.08 eV	2.02	1	1.0000	1.0000	1.0000	1.0000
		4	1.0000	1.0000	1.0000	1.0001
		28.3	1.0008	1.0018	1.0009	1.0019

7.3.6 One collision against two

The anomaly of Sec. 7.3.5 needs the size of the single-collision photon exchange where Kroll and Watson give zero, $\boldsymbol{\epsilon}\cdot\mathbf{Q}=0$. To first order in the field and exactly in the atomic potential, the one-photon amplitude between the scattering states of momenta $\mathbf{k}_i$ and $\mathbf{k}_f$ is, in the acceleration form and atomic units,

$$M=\sum_{l,\,l'=l\pm1} i^{\,l-l'}e^{i(\delta^f_{l'}+\delta^i_l)}R_{l'l}\sum_{mm'}Y_{l'm'}(\hat{\mathbf{k}}_f)\,\langle l'm'|\boldsymbol{\epsilon}\cdot\hat{\mathbf{r}}|lm\rangle\,Y^*_{lm}(\hat{\mathbf{k}}_i),\qquad R_{l'l}=\frac{1}{\omega^2}\int_0^\infty u_{k_f l'}V' u_{k_i l}\,\mathrm{d}r, \tag{7.25}$$

with radial waves of unit amplitude, and the ratio of the one-photon signal to the elastic one in a field of amplitude E_0 is

$$\frac{\mathrm{d}\sigma_{+1}}{\mathrm{d}\sigma_{\mathrm{el}}}=\frac{16\pi^2E_0^2|M|^2}{k_i^3k_f\,|f_{\mathrm{el}}|^2}, \tag{7.26}$$

with f_{el} the elastic amplitude. Three properties check it. Averaged over the polarization and integrated over $\hat{\mathbf{k}}_f$, $|M|^2$ gives back the integrated free–free cross section of Sec. 7.3.3, to 5×10^{-14}. At 0.3 THz, where the photon is soft, Eq. (7.26) with $\boldsymbol{\epsilon}\parallel\mathbf{Q}$ equals the Kroll–Watson value $(k_f/k_i)(\boldsymbol{\epsilon}\cdot\mathbf{Q})^2a_0^2/4$ to 6.4×10^{-4}. In the geometry with $\boldsymbol{\epsilon}$ normal to the scattering plane it vanishes to 10^{-31}, as the reflection symmetry of Sec. 7.3.5 requires. The partial waves are taken to $l=22$, with the polarization tail of the elastic amplitude summed in the Born approximation, $\delta_l=\pi\alpha_d k^2/[(2l-1)(2l+1)(2l+3)]$, beyond [31]; going to $l=26$ changes the signal by 4.8×10^{-3}

and halving the radial step by 3.5×10^{-5}. In the expansion in the field the next term carries $a_0 k\,\hbar\omega/E \approx 0.03$ at 10 eV.

In the in-plane geometry, $\boldsymbol{\epsilon}$ in the scattering plane and perpendicular to $\mathbf{Q}$, argon at 10 eV in a CO_2 laser at 10^8 W cm^{-2} gives a one-photon signal of 2.25×10^{-3} of the elastic one at 90° with the fitted potential of Eq. (7.22) and 1.23×10^{-3} with the potential from the atom; 1.72×10^{-3} and 1.00×10^{-3} at 8 eV, 3.80×10^{-4} and 3.37×10^{-4} at 20 eV. This is the dipole amplitude f_D of the in-plane geometry, 37 and 20 times the nondipole term, which with the laser beam along $\mathbf{Q}$ gives $(k_f/k_i)J_1^2(x_1) = 6.09 \times 10^{-5}$ with $x_1 = a_0(\hat{\mathbf{k}} \cdot \mathbf{Q})(\boldsymbol{\epsilon} \cdot \bar{\mathbf{v}})/c = 0.0156$, $\bar{\mathbf{v}}$ the mean of the two velocities. To first order in v/c the in-plane amplitude is $f_D + f_N \cos\varphi$, with φ the angle between the laser direction $\hat{\mathbf{k}}$ and $\mathbf{Q}$ (the reflection through the scattering plane removes a term in $\sin\varphi$), so the interference $2\mathrm{Re}(f_D^* f_N)\cos\varphi$ can move the in-plane signal by up to $2|f_N/f_D|$, 33 and 44 % of $|f_D|^2$, as the laser direction turns. The polarization bremsstrahlung of Sec. 7.3.4 changes the in-plane signal by at most 0.6 %, since its first order vanishes with $\boldsymbol{\epsilon} \cdot \mathbf{Q}$. In absolute terms the single-collision cross section at 10 eV stays below 1.3 % of the elastic one at 90° at every angle; its ratio to the elastic signal rises only where the elastic cross section has a minimum, to 1.19×10^{-2} and 0.18 at 120°, and there the two potentials disagree.

Double scattering in the target is computed with the same potentials, for the gas cylinder of radius 1 mm and length 2 mm about the beam: each collision follows Kroll and Watson, the second collision returns the photon count to one, and the signal per incident electron is $n^2 D_1$ with

$$D_1 = \int \mathrm{d}\Omega_1\, G(\hat{\mathbf{u}}_1)\, \frac{\mathrm{d}\sigma_{\mathrm{el}}}{\mathrm{d}\Omega}(\hat{\mathbf{z}} \cdot \hat{\mathbf{u}}_1)\, \frac{\mathrm{d}\sigma_{\mathrm{el}}}{\mathrm{d}\Omega}(\hat{\mathbf{u}}_1 \cdot \hat{\mathbf{u}}_d) \sum_\ell J_\ell^2(\xi) J_{1-\ell}^2(\xi), \qquad \xi = k a_0\, \boldsymbol{\epsilon} \cdot (\hat{\mathbf{u}}_1 - \hat{\mathbf{z}}), \tag{7.27}$$

where G is the path length in the gas integrated along the beam. Relative to the single-collision elastic signal it is $n d_1$, with $d_1 = D_1/(L\,\mathrm{d}\sigma_{\mathrm{el}}/\mathrm{d}\Omega)$. At 90°, $d_1 = 3.27 \times 10^{-17}$ and 2.77×10^{-17} cm^3 in the in-plane geometry and 4.50×10^{-17} and 4.26×10^{-17} cm^3 in the normal geometry. The single collision and the double scattering are therefore equal at $n^* = 6.9 \times 10^{13}$ and 4.5×10^{13} cm^{-3} at 90°, at 5.3 and 2.8×10^{13} cm^{-3} at 60° and at 2.4 and 2.9×10^{13} cm^{-3} at 120°: at the densities of the targets used, the two mechanisms are of the same size in the in-plane geometry. In the normal geometry the single collision is absent. Both rise toward the minima of the elastic cross section, both stay one to two orders of magnitude below the measured 12 % away from them, and a density scan in the two geometries separates them: an intercept in the in-plane geometry only, a slope in both. The two potentials give σ_m(10 eV) 33 and 63 % above the data, so the numbers at 10 eV carry that uncertainty; their crossover densities n^* agree within a factor 2.

Table 7.5: Argon at 28.3 THz and 10^8 W cm^{-2}: the exact single-collision one-photon signal relative to the elastic one in the in-plane geometry with $\boldsymbol{\epsilon} \perp \mathbf{Q}$, double scattering d_1 in the same geometry, and the density n^* at which the two are equal, for the fitted potential and the potential from the atom, at 10 eV.

	fitted potential			atom + Hara exchange		
angle	single	d_1 (cm^3)	n^* (cm^{-3})	single	d_1 (cm^3)	n^* (cm^{-3})
60°	2.93×10^{-3}	5.51×10^{-17}	5.32×10^{13}	1.61×10^{-3}	5.79×10^{-17}	2.79×10^{13}
90°	2.25×10^{-3}	3.27×10^{-17}	6.89×10^{13}	1.23×10^{-3}	2.77×10^{-17}	4.45×10^{13}
120°	1.19×10^{-2}	5.05×10^{-16}	2.36×10^{13}	1.81×10^{-1}	6.36×10^{-15}	2.85×10^{13}

7.4 Putting the rungs together

7.4.1 The two kinds of scatterer add

Electron–ion and electron–atom collisions are separate binary events, and electron–electron collisions radiate no dipole radiation, since two electrons have the same charge-to-mass ratio. So $\nu_{\text{eff}} = \nu_T\Lambda + \nu_{ea}$. In argon at 1 atm and 300 K, at 1 THz, the two are equal at $n_e = 1.0\times10^{13}$ cm^{-3} for 300 K electrons, the Ramsauer minimum making the atoms weak scatterers, and at 1.0×10^{16} cm^{-3} for 1.08 eV electrons, close to the critical density 1.24×10^{16} cm^{-3} at that frequency.

7.4.2 The standard model

Collecting the rungs, the absorption coefficient of a probe is Eq. (7.1) with

$$\boxed{\begin{aligned}\nu_{\text{eff}} = \nu_T\Big[\Lambda_{2,3}(r,u,\{Z_i\}) + \Delta\Lambda_4(\eta_t,u) + C\Big]\, f_L(\alpha)\, C_L\big(m(\alpha),u\big) \\ \times\, \frac{f_{sp}}{f_L}\, R_{\text{self}}(\beta_p,u)\, R_{pp}(\beta_h,u)\left(1+\frac{\delta P}{P}\right) \\ +\; n_a\Big\langle \sigma_m v\, \frac{2\varepsilon}{3k_BT_e}\Big\rangle R(u),\end{aligned}} \tag{7.28}$$

where $\Lambda_{2,3}$ is Eq. (7.5) in the form of Eq. (7.6) with the charge structure factor of the ion mixture from Eq. (7.9), $\Delta\Lambda_4 = \Lambda_4 - \Lambda_1$, C is the cross term of Eq. (7.12), C_L is the second factor of Eq. (7.13), f_{sp} is Eq. (7.14), R_{self} and R_{pp} are Eqs. (7.16) and (7.17), $\delta P/P$ is Eq. (7.19) with f_2 from Eq. (7.20), and the last term is Eq. (7.21) times R of Secs. 7.3.3 and 7.3.4. Products of factors are a first-order approximation in each correction.

7.4.3 The budgets of the fourteen shots

The ladder shares the rungs of Secs. 7.2.2, 7.2.5 and 7.2.6 with the composite of Turnbull et al. What it adds is the product

$$X = \frac{\Lambda_{2,3}+\Delta\Lambda_4+C}{\Lambda_1+\Delta\Lambda_4}\,\frac{f_{sp}}{f_L}\,R_{\text{self}}\,R_{pp}\left(1+\frac{\delta P}{P}\right), \tag{7.29}$$

listed shot by shot in Tables 7.6 and 7.7. The exact-Gaunt correction $\Delta\Lambda_4$ and the cross term C are weighted by $n_iZ_i^2$ over the ion species: $\Delta\Lambda_4$ is -0.67 to -0.88 % of Λ_1 for hydrogen and -8.3 to -12.1 % for the mixtures, where the nitrogen ions carry it, and $C = +0.0098$ for the protons and $+0.0383$ for the nitrogen ions. For hydrogen X is 0.916 to 0.926, with a mean of 0.921; for the mixtures it is 0.871 to 0.911, with a mean of 0.895.

7.4.4 Rules against counting an effect twice

1. *One upper cutoff.* The dynamic cutoff v_t/ω of Sec. 7.2.2 and the Debye length of Sec. 7.2.1 are two models of the same long-range cut. Electron screening enters through ε in Sec. 7.2.3, never as a second cutoff.

2. *One account of the ions.* The rung of Sec. 7.2.4 and a fitted screening factor both describe how the ions modify the long-range part of the logarithm. Applying f_{sc} on top of that rung counts the ion correlations twice.

Table 7.6: The budget of the four hydrogen shots: the logarithm factor of Eq. (7.29), the speckle factor, the strong-field factors, the anisotropy, their product X of Eq. (7.29) and the ladder against the measurements, $1.29X$. The fitted factor f_{sc} of Turnbull et al. is listed for comparison.

shot	logarithm	f_{sp}/f_L	R_{self}	R_{pp}	$-\delta P/P$	X	ladder/data	f_{sc}
102727	0.983	0.955	0.991	1	0.0130	0.918	1.185	0.837
104031	0.979	0.977	0.994	0.982	0.0076	0.926	1.195	0.827
106310	0.993	0.973	0.990	0.974	0.0102	0.922	1.189	0.884
106311	0.998	0.972	0.987	0.967	0.0115	0.916	1.181	0.927

Table 7.7: The budget of the ten nitrogen–hydrogen shots, as Table 7.6; the ladder against the measurements is $1.07X$.

shot	logarithm	f_{sp}/f_L	R_{self}	R_{pp}	$-\delta P/P$	X	ladder/data	f_{sc}
102729	0.944	0.946	0.997	1	0.0049	0.886	0.948	0.911
102737	0.949	0.965	0.996	0.994	0.0047	0.903	0.966	0.915
102738	0.936	0.986	0.997	0.981	0.0032	0.899	0.962	0.896
104035	0.949	0.977	0.997	0.988	0.0037	0.910	0.973	0.898
104037	0.953	0.949	0.997	1	0.0045	0.898	0.961	0.904
104041	0.957	0.944	0.996	1	0.0064	0.894	0.956	0.917
106315	0.949	0.942	0.994	1	0.0080	0.881	0.943	0.902
106316	0.968	0.942	0.994	1	0.0084	0.898	0.961	0.943
106317	0.987	0.942	0.991	1	0.0107	0.911	0.975	0.969
106319	0.928	0.946	0.997	1	0.0047	0.871	0.932	0.879

3. *Strong collisions once.* The exact Gaunt factor of Sec. 7.2.5 already contains the Born term; it is added as $\Lambda_4 - \Lambda_1$ to the medium logarithm, not multiplied by it, and the cross term of Eq. (7.12) is the only further piece.

4. *One logarithm throughout.* The Langdon factor of Sec. 7.2.6 must use the same velocity-dependent logarithm as Sec. 7.2.2 or 7.2.5; Langdon's fit alone assumes a fixed one.

5. *The field once per beam.* R_{self} holds the probe's own quiver, R_{pp} the heaters'; the Langdon parameter α of Sec. 7.2.6 takes the overlapped intensity because it describes the time-averaged distribution, which every beam shapes. Neither replaces the other.

6. *No separate term for collision sequences.* Double scattering is already in the kinetic equation (Sec. 7.3.5).

7. *Electron–electron collisions shape, they do not absorb.* They enter through α in Sec. 7.2.6 and through the relaxation of f_2 in Sec. 7.2.9, not as a third scatterer.

8. *Two logarithms in the anisotropy.* The field drives f_2 with the inverse-bremsstrahlung logarithm, and collisions relax it with the transport logarithm; using one for both overstates the anisotropy by $\Lambda_{\text{tr}}/\Lambda_1 \approx 1.5$.

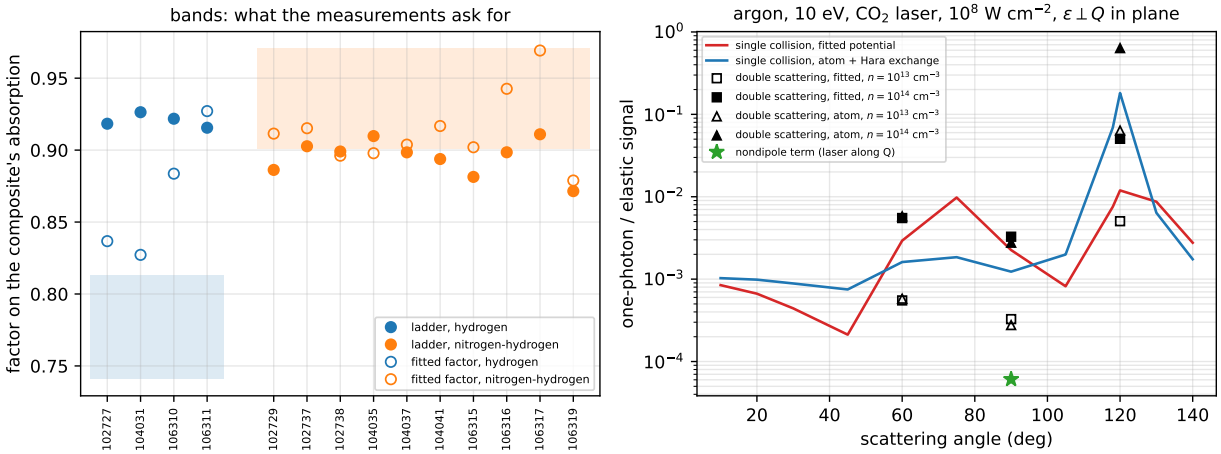


Figure 7.2: Left: per shot, the factor X of Eq. (7.29) that the ladder applies beyond the composite of Turnbull et al. (filled) and their fitted screening factor f_{sc} (open), against the factor the measurements ask for, $1/(1.29\pm0.06)$ for hydrogen and $1/(1.07\pm0.04)$ for the mixtures (bands). Right: argon at 10 eV in a CO_2 laser at 10^8 W cm^{-2}, in the in-plane geometry with $\boldsymbol{\epsilon} \perp \mathbf{Q}$: the exact single-collision one-photon signal relative to the elastic one for the two potentials (lines), double scattering in the target at 10^{13} and 10^{14} cm^{-3} (markers), and the nondipole term (star).

7.5 Have the models improved?

Plasma. Against the composite of Turnbull et al. without their screening factor, which is 1.29 ± 0.06 times the measured absorption of the hydrogen shots and 1.07 ± 0.04 times that of the mixture shots, the ladder is

$$\frac{K_{\text{ladder}}}{K_{\text{measured}}} = \begin{cases} 1.07\,\bar{X}_{\text{N}_2/\text{H}_2} = 1.07 \times 0.895 = 0.96 \pm 0.04 & \text{(ten nitrogen–hydrogen shots)}, \\ 1.29\,\bar{X}_{\text{H}_2} = 1.29 \times 0.921 = 1.19 \pm 0.06 & \text{(four hydrogen shots)}. \end{cases} \tag{7.30}$$

For the mixtures the ladder, with no fitted parameter, agrees with the measurements as well as the fitted composite of Turnbull et al., 1.00 ± 0.04 [3]. In these plasmas the fitted factor and the computed correlations of the ions do the same thing, 0.879 to 0.969 against a medium factor of 0.930 to 0.982, because the heavy ions make the correlations strong. For hydrogen they part: the fitted factor lowers the absorption by 7 to 17 %, the medium by 0.4 to 2.2 %, and the hypernetted chain confirms Debye–Hückel to 2×10^{-4} at $\Gamma_{ii} = 0.02$. *The ladder explains the mixture shots and leaves the hydrogen shots 19 % too absorbing.* Every rung that could act on hydrogen alone is now computed and small: the anisotropy, which scales as $1/Z$, 0.8 to 1.3 %; the strong fields 1 to 5 %; the speckles 2.3 to 4.5 %. The residual is the size of a 12 % error in T_e or a 9 % error in n_e of the hydrogen shots, whose measured absorption, 0.15 to 1.48 %, is the smallest of the data set (0.47 to 5.05 % for the mixtures).

The discriminating measurement. Helium against hydrogen at the conditions of the four hydrogen shots, each normalized by the unscreened model of its own charge: the ladder predicts 0.987, 0.985, 0.993 and 0.997, the fitted factor 1.042, 1.044, 1.034 and 1.024. The two differ by 2.7 to 5.8 % and move in opposite directions.

Gas. Against a Drude model with a mean collision frequency,

$$\boxed{\frac{K_{\mathrm{ladder}}}{K_{\mathrm{Drude}}} = \frac{\langle \nu_m \, 2\varepsilon/3k_B T_e \rangle}{\langle \nu_m \rangle} \; R = 0.836 \text{ (300 K) to } 2.02 \text{ (1.08 eV) at 1 to 4 THz,}} \tag{7.31}$$

where R of Secs. 7.3.3 and 7.3.4 is 0.97 to 1.00 at 1 to 4 THz, and at 28.3 THz R lowers it further, to 0.514 of the classical kernel for 300 K electrons with the fitted potential and 0.491 with the potential from the atom (Table 7.4). The single collisions beyond Kroll and Watson are computed (Table 7.5); they equal double scattering at the densities of the targets used.

7.6 What remains open

1. *The hydrogen residual.* The ladder accounts for the mixture shots and leaves 19 % for hydrogen, with every hydrogen-specific rung computed. The theory therefore places the residual in the hydrogen measurements or their plasma conditions, not in the absorption physics; the helium measurement of Sec. 7.5 decides between this and a fitted screening length.

2. *The anisotropy beyond its direct part.* The feedback of f_2 on the isotropic distribution and the field-particle part of the electron–electron operator are not computed. They change the 0.8 to 1.3 % of Sec. 7.2.9 by a factor of order one, not the residual.

3. *Argon at the elastic minima.* The single-collision and double-scattering signals are computed; near the minima of the elastic cross section, where the measured photon exchange is largest, both depend on the depth and the position of the minimum, which the two potentials do not agree on. A density scan in the in-plane and normal geometries measures both.

Closed here: the ion correlations beyond Debye–Hückel (Sec. 7.2.4), the strong-collision cross term at the mixtures' conditions (Sec. 7.2.5), the heaters' own strong-field factor and the exact factor at $\beta \geq 1$ (Sec. 7.2.8), the anisotropy with two logarithms and electron–electron collisions (Sec. 7.2.9), and the single-collision photon exchange beyond Kroll and Watson in argon (Sec. 7.3.6).

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Chapter 8

Detailed derivation of the quantum theory of inverse bremsstrahlung heating of the free electrons of a plasma

The note of October 5, 2026.

These notes derive, with every intermediate result written out, the rate at which a laser field heats the free electrons of a fully ionized plasma by inverse bremsstrahlung. The starting point is the Schrödinger equation of an electron in the field of the laser and of the ions. The electron in the laser field is described exactly by the Volkov states [2], the ion potential is treated in Born approximation as in Bunkin and Fedorov [3], and the occupation numbers obey a kinetic equation that is derived with the operator method of the quantum Boltzmann equation paper [4]. The classical treatment of Atzeni and Meyer-ter-Vehn [1] is the baseline. Every boxed result is checked numerically by a script, which Appendix 8.C describes.

For notation, Gaussian units are used throughout, as in [1]. The electron has charge $-e$ and mass m. The ions have charge Ze, are infinitely heavy and sit at fixed positions $\mathbf{R}_j$ with density n_i, and the plasma is neutral, $n_e = Zn_i$. The temperature T is in kelvin and k_B is written explicitly. The laser field is uniform and linearly polarized along the unit vector $\hat{\mathbf{e}}$. The electrons are in a box of volume $\mathcal{V}$ with periodic boundary conditions, a momentum transfer is written $\hbar\mathbf{k}$, and the integer ℓ counts the photons absorbed in one collision, with $\ell < 0$ for emission. The mean occupation of the electron state of momentum $\mathbf{p}$, per spin state, is $f_{\mathbf{p}}$. It is the $n_{\mathbf{p}}$ of the quantum Boltzmann equation paper. The Fourier transform of a function of position is $\tilde{v}(\mathbf{k}) = \int v(\mathbf{r})\, e^{-i\mathbf{k}\cdot\mathbf{r}}\, \mathrm{d}^3 r$. A term that is struck through vanishes, for the reason given at that step. A boxed equation is a result that is used later. An equation of another document is cited by its own number followed by the reference, as in Eq. (11.12) [1], and an equation number without a reference is one of these notes.

8.1 Why a quantum theory of inverse bremsstrahlung?

(1) Inverse bremsstrahlung is the absorption of laser photons by free electrons while they collide with ions. In laser-driven inertial fusion it is the preferred absorption mechanism, because the absorbed energy goes into the thermal electrons and stays where it was absorbed [1, Sec. 11.2].

(2) A free electron alone cannot absorb a photon. Let an electron of momentum $\mathbf{p}$ absorb a photon of energy $\hbar\omega$ and momentum $\hbar\mathbf{k}_L$. Conservation of energy and momentum requires

$$\frac{|\mathbf{p}+\hbar\mathbf{k}_L|^2}{2m} - \frac{p^2}{2m} = \hbar\omega \quad \Longrightarrow \quad \frac{\omega}{k_L} = \frac{\mathbf{p}\cdot\hat{\mathbf{k}}_L}{m} + \frac{\hbar k_L}{2m}. \tag{8.1}$$

The left side is the phase velocity of the transverse wave. In a plasma it is c/n_r with the refractive index $n_r = (1-\omega_p^2/\omega^2)^{1/2} < 1$, so it exceeds c. The right side is the velocity of a nonrelativistic electron plus $\hbar k_L/2m = \hbar\omega n_r/(2mc)$, which is of order $10^{-5}c$ for optical photons. The two sides cannot be equal, and a third body has to take up the momentum. In a fully ionized plasma that body is an ion.
(3) In the classical theory [1, Sec. 11.1.1] the electrons oscillate in the laser field and a friction force $-\nu_e m\mathbf{v}$ represents their collisions with the ions. The absorption coefficient follows from the imaginary part of the dielectric function, and ν_e is an input. Its Coulomb logarithm needs two cutoffs. For $k_BT_e \geq 10$ eV the lower one is the de Broglie wavelength of the electron [1, Eq. (10.136)], which is a quantum statement put in by hand.
(4) The photon energy enters the classical theory twice, through the factor $1-e^{-\hbar\omega/k_BT}$ for stimulated emission [1, Eq. (11.34)] and through the Gaunt factor of the free–free cross section [1, Sec. 10.6]. Table 8.1 shows that $u = \hbar\omega/k_BT$ ranges from 0.05 at 100 eV to 0.001 at 3 keV for ultraviolet lasers. It is small, but it sets the Coulomb logarithm, as Sec. 8.8 shows.
(5) At the intensities of inertial fusion the quiver velocity $v_0 = eE_0/m\omega$ is a sizable fraction of the thermal velocity $v_t = (k_BT/m)^{1/2}$, and at 100 eV it is close to it, as Table 8.1 shows. The absorption is then not linear in the intensity, and several photons can be exchanged in one collision.
(6) A quantum theory removes all of these inputs. The electron in the laser field is described exactly by the Volkov states, the ion potential is treated in Born approximation, and the result holds for any field strength, any photon energy and any degeneracy of the electrons. The Born approximation needs $\eta = Ze^2/\hbar v \ll 1$ [3, Eq. (5)], which Table 8.1 shows to hold for hydrogen above about 100 eV and for carbon–hydrogen plastic above about 1 keV.

Table 8.1: Scales of inverse bremsstrahlung for the third harmonic of the Nd:glass laser and for the KrF laser, at the density $n_e = n_c/4$ and the vacuum intensity 10^{15} W cm^{-2}. Here $u = \hbar\omega/k_BT$, η/Z is the Born parameter at the thermal velocity, $\Lambda(u)$ is the unscreened quantum Coulomb logarithm of Eq. (8.87), $\Lambda_{\rm RPA}$ the dynamically screened one of Eq. (8.125), and $\ln\Lambda_e$ is Eq. (10.136) [1].

λ_L (μm)	k_BT (keV)	u	η/Z	v_0/v_t	$\Lambda(u)$	$\Lambda_{\rm RPA}$	$\ln\Lambda_e$
0.351	0.1	0.0353	0.52	0.68	4.15	4.05	4.39
0.351	1	0.0035	0.16	0.21	6.45	6.36	6.69
0.351	3	0.0012	0.10	0.12	7.55	7.45	7.79
0.248	0.1	0.0500	0.52	0.48	3.81	3.71	4.04
0.248	1	0.0050	0.16	0.15	6.11	6.01	6.34
0.248	3	0.0017	0.10	0.09	7.21	7.11	7.44

The plan of these notes. Section 8.2 sets out the classical theory that the quantum theory must reproduce. Section 8.3 writes the Hamiltonian and the quiver motion. Section 8.4 derives the Volkov states. Section 8.5 derives the multiphoton transition rates in Born approximation and the cross section of Bunkin and Fedorov. Section 8.6 derives the kinetic equation for the occupation numbers with the operator method of the quantum Boltzmann equation paper, and from it the heating rate. Section 8.7 writes the heating rate through the Lindhard function. Section 8.8

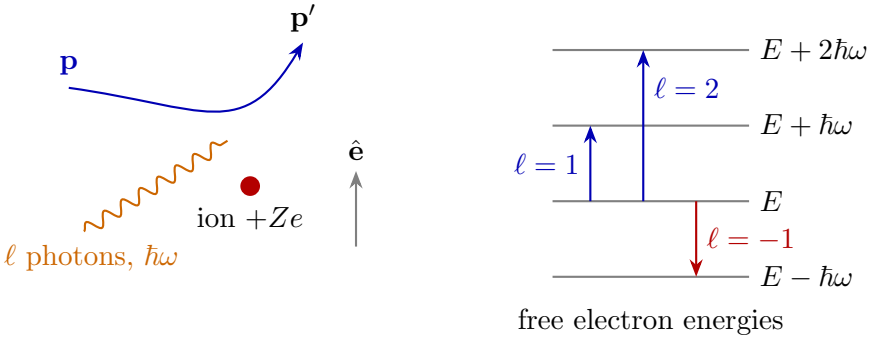


Figure 8.1: Inverse bremsstrahlung. Left, an electron of momentum $\mathbf{p}$ scatters on an ion into $\mathbf{p}'$ while absorbing ℓ photons from the laser, which is polarized along $\hat{\mathbf{e}}$. The ion takes up the momentum $\mathbf{p}-\mathbf{p}'$. Right, the energy of the free electron changes by $\ell\hbar\omega$, with $\ell<0$ for stimulated emission.

evaluates it in closed form for weak fields, and Section 8.9 connects it to the bremsstrahlung cross section, the Gaunt factor and Kirchhoff's law. Section 8.10 derives the same rate for the interacting electron gas and adds dynamic screening. Section 8.11 treats strong fields, and Section 8.12 derives the kinetic equation of an isotropic distribution and its Fokker–Planck limit. Section 8.13 collects the results. Appendix 8.A reduces the results to the classical treatment of Atzeni and Meyer-ter-Vehn equation by equation, Appendix 8.B proves the mathematical identities, and Appendix 8.C lists the numerical checks.

8.2 The classical baseline

8.2.1 The Drude model

This subsection follows [1, Sec. 11.1.1] and uses its convention, in which a perturbation varies as $e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ and the physical quantity is the real part.
(1) The electron velocity obeys Newton's equation with a friction force,

$$-i\omega m\mathbf{v} = -e\mathbf{E} - \nu m\mathbf{v}, \tag{8.2}$$

which is Eq. (11.7) [1]. Collect the terms in $\mathbf{v}$,

$$m(\nu-i\omega)\,\mathbf{v} = -e\mathbf{E} \quad\longrightarrow\quad \mathbf{v} = -\frac{e\mathbf{E}}{m(\nu-i\omega)} = -\frac{ie\mathbf{E}}{m(\omega+i\nu)}, \tag{8.3}$$

where the last step uses $\nu - i\omega = -i(\omega+i\nu)$ and $1/i=-i$.
(2) The current density and the conductivity are

$$\mathbf{j} = -en_e\mathbf{v} = \frac{ie^2n_e}{m(\omega+i\nu)}\,\mathbf{E}, \qquad \sigma = \frac{ie^2n_e}{m(\omega+i\nu)} = \frac{e^2n_e}{m(\nu-i\omega)}. \tag{8.4}$$

Eq. (11.8) [1] prints $\mathbf{j}_1 = e^2n_0\mathbf{E}_1/[im_e(\omega+i\nu_e)]$, which is the negative of Eq. (8.4). The sign of Eq. (8.4) is the one that Eq. (11.9) [1] requires, as the next step shows.
(3) With the dielectric tensor $\varepsilon = 1 + (4\pi i/\omega)\sigma$ of Eq. (11.4) [1],

$$\boxed{\varepsilon_T(\omega) = 1 + \frac{4\pi i}{\omega}\,\frac{ie^2n_e}{m(\omega+i\nu)} = 1 - \frac{\omega_p^2}{\omega(\omega+i\nu)}, \qquad \omega_p^2 = \frac{4\pi e^2n_e}{m},} \tag{8.5}$$

which is Eq. (11.9) [1]. Its imaginary part is

$$\operatorname{Im}\varepsilon_T = \frac{\omega_p^2}{\omega}\operatorname{Im}\frac{-1}{\omega+i\nu} = \frac{\omega_p^2}{\omega}\frac{\nu}{\omega^2+\nu^2} = \frac{\omega_p^2\nu}{\omega^3}\Big[1+O\Big(\frac{\nu^2}{\omega^2}\Big)\Big]. \tag{8.6}$$

(4) The power absorbed per unit volume is the cycle average of $\mathbf{j}\cdot\mathbf{E}$. For the field $\operatorname{Re}[\mathbf{E}_1 e^{-i\omega t}]$ with the real amplitude $E_0 = |\mathbf{E}_1|$,

$$\langle \mathbf{j}\cdot\mathbf{E}\rangle_t = \tfrac{1}{2}\operatorname{Re}\sigma\, E_0^2 = \frac{\omega}{8\pi}\operatorname{Im}\varepsilon_T\, E_0^2 = \frac{e^2 n_e}{2m}\frac{\nu}{\nu^2+\omega^2}\,E_0^2. \tag{8.7}$$

With the quiver velocity $v_0 = eE_0/m\omega$ that Section 8.3.1 derives,

$$\boxed{P = \frac{\omega}{8\pi}\operatorname{Im}\varepsilon_T\, E_0^2 = n_e\,\nu\,\frac{mv_0^2}{2}\Big[1+O\Big(\frac{\nu^2}{\omega^2}\Big)\Big].} \tag{8.8}$$

The power is the collision rate times the kinetic energy $mv_0^2/2$ of the quiver amplitude. Every quantum result of these notes is written in this form, which defines the effective collision frequency $\nu_{\text{eff}} \equiv P/(n_e m v_0^2/2)$.

(5) The wave number of a transverse wave obeys $k^2 = (\omega^2/c^2)\varepsilon_T$, Eq. (11.5) [1]. To first order in $\operatorname{Im}\varepsilon_T$,

$$k = \frac{\omega}{c}\Big[(\operatorname{Re}\varepsilon_T)^{1/2} + \frac{i\operatorname{Im}\varepsilon_T}{2(\operatorname{Re}\varepsilon_T)^{1/2}}\Big], \tag{8.9}$$

and the intensity, which varies as $|E|^2 \propto e^{-2\operatorname{Im}k\,x}$, decays with the coefficient

$$\boxed{K = 2\operatorname{Im}k = \frac{\omega}{c}\frac{\operatorname{Im}\varepsilon_T}{(\operatorname{Re}\varepsilon_T)^{1/2}} = \frac{\nu_{\text{eff}}}{c}\frac{\omega_p^2}{\omega^2}\frac{1}{n_r}, \qquad n_r = \Big(1-\frac{\omega_p^2}{\omega^2}\Big)^{1/2},} \tag{8.10}$$

which is Eq. (11.12) [1]. The same coefficient follows as the ratio of P to the energy flux $I = cn_r E_0^2/8\pi$ of the wave in the plasma.

8.2.2 The collision frequency and the Coulomb logarithm

(1) The electron collision time of Eq. (10.133) [1] is

$$\frac{1}{\tau_e} = \frac{4\sqrt{2\pi}}{3}\frac{n_i Z^2 e^4}{m^{1/2}(k_B T)^{3/2}}\ln\Lambda_e \equiv \nu_T \ln\Lambda_e, \qquad \boxed{\nu_T = \frac{4\sqrt{2\pi}}{3}\frac{n_i Z^2 e^4}{m^{1/2}(k_B T)^{3/2}}.} \tag{8.11}$$

These notes write every collision frequency as ν_T times a dimensionless logarithm Λ, so that Λ can be compared with $\ln\Lambda_e$ directly.

(2) The Coulomb logarithm of Eq. (10.136) [1] is $\ln\Lambda_e = 7.1 - 0.5\ln n_e + \ln T_e$ with n_e in 10^{21} cm^{-3} and T_e in keV. It is the logarithm of the ratio of the Debye length to the de Broglie wavelength $\hbar/2mv$ at the velocity $mv^2/2 = 3k_BT/2$. The lower cutoff is quantum mechanical, and the theory does not derive it.

(3) Two further points are not settled by the classical theory. The collision time τ_e belongs to the slow relaxation of the electrons, while the absorption of light at $\omega \gg \nu$ involves the collisions within one period of the field. Dawson and Oberman [6] and Johnston and Dawson [7] showed that the upper cutoff is then v_t/ω rather than the Debye length. The quantum theory produces both cutoffs, Eq. (8.88).

8.2.3 The free–free cross section

(1) Kramers' cross section for the emission of a photon of frequency ν by an electron of velocity v in the field of an ion is [8]

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\nu} = \frac{32\pi^2}{3\sqrt{3}}\,\frac{Z^2e^6}{m^2c^3v^2h\nu}, \tag{8.12}$$

Eq. (10.89) [1]. Quantum corrections are written as a Gaunt factor g that multiplies it.
(2) Averaged over a Maxwellian, Eq. (8.12) gives the emission coefficient of Eq. (10.91) [1], and Kirchhoff's law gives the free–free opacity of Eq. (10.92) [1]. Corrected for stimulated emission and refraction, the absorption coefficient of the laser light is Eq. (11.34) [1],

$$K = \rho\,\kappa_\nu^{\mathrm{ff}}\big(1 - e^{-h\nu/k_BT_e}\big)\Big/\Big(1 - \frac{\omega_p^2}{\omega_L^2}\Big)^{1/2}. \tag{8.13}$$

Atzeni and Meyer-ter-Vehn state that this agrees with Eq. (8.10) in its dependence on density, temperature and wavelength, and in its numerical factor within a few percent. The dependences agree, but with $g = 1$ in Eq. (10.92) [1] and $\ln\Lambda_e = 8$ the numerical factors differ by a factor 4.4. Section 8.9 derives the Gaunt factor that removes the difference, and Appendix 8.A.5 gives the numbers.

In Summary: The classical theory takes three inputs. The collision frequency enters the Drude model as a parameter, its Coulomb logarithm needs a lower cutoff that is quantum mechanical, and the free–free route needs a Gaunt factor. The quantum theory of the following sections supplies all three, and its result is written as $P = n_e\nu_{\mathrm{eff}}\,mv_0^2/2$ with $\nu_{\mathrm{eff}} = \nu_T\Lambda$.

8.3 The Hamiltonian

8.3.1 The laser field and the quiver motion

(1) The laser field is uniform over the region of a collision. Its electric field and a vector potential that produces it, with zero scalar potential, are

$$\mathbf{E}(t) = E_0\,\hat{\mathbf{e}}\cos\omega t, \qquad \mathbf{A}(t) = -\frac{cE_0}{\omega}\,\hat{\mathbf{e}}\sin\omega t, \tag{8.14}$$

since $-c^{-1}\partial_t\mathbf{A} = E_0\hat{\mathbf{e}}\cos\omega t$. The validity of the uniform field is the subject of Section 8.3.4.
(2) A classical electron obeys $m\dot{\mathbf{v}} = -eE_0\hat{\mathbf{e}}\cos\omega t$. Integrating once and twice,

$$\mathbf{v}(t) = \mathbf{v}_d - v_0\,\hat{\mathbf{e}}\sin\omega t, \qquad \mathbf{r}(t) = \mathbf{r}_d + \mathbf{v}_dt + a_0\,\hat{\mathbf{e}}\cos\omega t, \tag{8.15}$$

where $\mathbf{v}_d$ and $\mathbf{r}_d$ are constants. The oscillating parts define the quiver displacement and the quiver velocity,

$$\boxed{\begin{aligned}&\boldsymbol{\alpha}(t) = \mathbf{a}_0\cos\omega t, \qquad \mathbf{v}_{\mathrm{osc}}(t) = \dot{\boldsymbol{\alpha}}(t) = -v_0\,\hat{\mathbf{e}}\sin\omega t, \qquad \mathbf{a}_0 = a_0\hat{\mathbf{e}},\\ &v_0 = \frac{eE_0}{m\omega}, \qquad a_0 = \frac{v_0}{\omega} = \frac{eE_0}{m\omega^2}, \qquad U_{\mathrm{p}} = \Big\langle\frac{m\mathbf{v}_{\mathrm{osc}}^2}{2}\Big\rangle_t = \frac{mv_0^2}{4} = \frac{e^2E_0^2}{4m\omega^2}.\end{aligned}} \tag{8.16}$$

Here $\langle\cdots\rangle_t$ is the average over one period, and U_{p} is the ponderomotive energy. The second derivative is $\ddot{\boldsymbol{\alpha}} = -\omega^2\boldsymbol{\alpha} = -(e/m)\mathbf{E}(t)$, and the quiver velocity is also

$$\dot{\boldsymbol{\alpha}}(t) = \frac{e}{mc}\mathbf{A}(t). \tag{8.17}$$

(3) For a charge $-e$ the canonical momentum is $\mathbf{p} = m\mathbf{v} - (e/c)\mathbf{A}$. With Eq. (8.14), $(e/c)\mathbf{A} = -mv_0\hat{\mathbf{e}}\sin\omega t$ and $\mathbf{p} = m\mathbf{v} + mv_0\hat{\mathbf{e}}\sin\omega t = m\mathbf{v}_d$ is constant. The canonical momentum is the drift momentum, and the kinetic momentum $m\mathbf{v} = \mathbf{p} + (e/c)\mathbf{A} = \mathbf{p} + m\mathbf{v}_{\mathrm{osc}}$ carries the quiver motion.

8.3.2 The Hamiltonian of one electron

(1) The Hamiltonian of one electron in the laser field and in the field of the ions is

$$\boxed{H(t) = \frac{1}{2m}\Big[\hat{\mathbf{p}} + \frac{e}{c}\mathbf{A}(t)\Big]^2 + V(\hat{\mathbf{r}}), \qquad V(\mathbf{r}) = \sum_{j=1}^{N_i} v(\mathbf{r} - \mathbf{R}_j),} \tag{8.18}$$

where $v(\mathbf{r})$ is the potential energy of the electron in the field of one ion and $N_i = n_i\mathcal{V}$.
(2) For a bare Coulomb ion $v(\mathbf{r}) = -Ze^2/r$, and for a statically screened one $v(\mathbf{r}) = -Ze^2e^{-\kappa r}/r$. The Fourier transform follows from the angular integral $\int e^{-i\mathbf{k}\cdot\mathbf{r}}\,\mathrm{d}\Omega_r = 4\pi\sin(kr)/kr$,

$$\int \frac{e^{-\kappa r}}{r}\, e^{-i\mathbf{k}\cdot\mathbf{r}}\,\mathrm{d}^3r = \frac{4\pi}{k}\int_0^\infty \sin(kr)\, e^{-\kappa r}\,\mathrm{d}r = \frac{4\pi}{k}\,\frac{k}{k^2+\kappa^2}, \tag{8.19}$$

so that

$$\boxed{\tilde{v}(\mathbf{k}) = -\frac{4\pi Ze^2}{k^2+\kappa^2} \xrightarrow{\kappa\to 0} -\frac{4\pi Ze^2}{k^2}.} \tag{8.20}$$

These notes use the bare Coulomb form unless stated otherwise. Screening is treated dynamically in Section 8.10.
(3) With the Fourier series $v(\mathbf{r}) = \mathcal{V}^{-1}\sum_{\mathbf{k}} \tilde{v}(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{r}}$ the potential of all the ions is

$$V(\mathbf{r}) = \frac{1}{\mathcal{V}}\sum_{\mathbf{k}} \tilde{v}(\mathbf{k})\,\rho_i(\mathbf{k})\,e^{i\mathbf{k}\cdot\mathbf{r}}, \qquad \rho_i(\mathbf{k}) = \sum_j e^{-i\mathbf{k}\cdot\mathbf{R}_j}, \qquad \tilde{V}(\mathbf{k}) \equiv \int V(\mathbf{r})e^{-i\mathbf{k}\cdot\mathbf{r}}\mathrm{d}^3r = \tilde{v}(\mathbf{k})\rho_i(\mathbf{k}). \tag{8.21}$$

8.3.3 The oscillating frame of Kramers and Henneberger

The quiver motion can be removed from the kinetic energy by going into the frame that oscillates with the electron [12]. This frame gives the clearest picture of inverse bremsstrahlung, and Section 8.10 uses it for many electrons.
(1) Let $\hat{T}(t) = \exp[-(i/\hbar)\boldsymbol{\alpha}(t)\cdot\hat{\mathbf{p}}]$, which translates a wave function by $\boldsymbol{\alpha}$, $(\hat{T}\phi)(\mathbf{r}) = \phi(\mathbf{r}-\boldsymbol{\alpha})$. Then $\hat{T}^\dagger V(\hat{\mathbf{r}})\hat{T} = V(\hat{\mathbf{r}}+\boldsymbol{\alpha})$, $\hat{T}$ commutes with $\hat{\mathbf{p}}$, and $i\hbar\,\partial_t\hat{T} = (\dot{\boldsymbol{\alpha}}\cdot\hat{\mathbf{p}})\hat{T}$, because $\boldsymbol{\alpha}\cdot\hat{\mathbf{p}}$ and $\dot{\boldsymbol{\alpha}}\cdot\hat{\mathbf{p}}$ commute.
(2) Write the wave function as $\psi = e^{-iS(t)/\hbar}\hat{T}\phi$ with a real function $S(t)$. Insert it into $i\hbar\partial_t\psi = H\psi$,

$$e^{-iS/\hbar}\hat{T}\big[\dot{S} + \dot{\boldsymbol{\alpha}}\cdot\hat{\mathbf{p}} + i\hbar\partial_t\big]\phi = e^{-iS/\hbar}\hat{T}\big[\hat{T}^\dagger H\hat{T}\big]\phi, \tag{8.22}$$

and multiply by $e^{iS/\hbar}\hat{T}^\dagger$,

$$i\hbar\partial_t\phi = \Big[\frac{\hat{p}^2}{2m} + \frac{e}{mc}\mathbf{A}\cdot\hat{\mathbf{p}} + \frac{e^2A^2}{2mc^2} + V(\hat{\mathbf{r}}+\boldsymbol{\alpha}) - \dot{\boldsymbol{\alpha}}\cdot\hat{\mathbf{p}} - \dot{S}\Big]\phi. \tag{8.23}$$

(3) By Eq. (8.17) the second and fifth terms cancel. Choosing $\dot{S} = e^2A^2/2mc^2$ removes the third, which is independent of position. What remains is

$$\boxed{i\hbar\partial_t\phi = \Big[\frac{\hat{p}^2}{2m} + V\big(\hat{\mathbf{r}} + \boldsymbol{\alpha}(t)\big)\Big]\phi.} \tag{8.24}$$

In this frame the electron is free and the ions oscillate. The ion j sits at $\mathbf{R}_j - \boldsymbol{\alpha}(t)$. A collision is elastic in the rest frame of the ion, which is the laboratory, and the energy that the electron gains in the oscillating frame comes from the motion of the ion relative to that frame. Step (3) of Section 8.5.6 makes this precise.

8.3.4 Validity of the uniform field and of nonrelativistic motion

The field of Eq. (8.14) neglects the variation of the laser field over the region of a collision and the magnetic force. Bunkin and Fedorov state the conditions as $v/c \ll 1$ and $\omega x_0/c \ll 1$, where x_0 is the larger of the distance v/ω traveled in one period and the quiver amplitude a_0 [3, Eq. (4)]. Since $\omega a_0 = v_0$, both conditions are $v \ll c$ and $v_0 \ll c$, the nonrelativistic motion of the electron. At 10^{15} W cm^{-2} and $\lambda_L = 0.351$ μm, $v_0/c = 0.0095$, and at 1 keV, $v_t/c = 0.044$.

8.4 The Volkov states

8.4.1 The solution

(1) Without the ions the Schrödinger equation is

$$i\hbar\,\partial_t\psi = \frac{1}{2m}\Big[-i\hbar\nabla + \frac{e}{c}\mathbf{A}(t)\Big]^2\psi. \tag{8.25}$$

Since $\mathbf{A}$ does not depend on position, the plane wave $e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}$ is an eigenfunction of the operator on the right at every time. Try $\psi = \mathcal{V}^{-1/2}e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}e^{-i\Phi_\mathbf{p}(t)}$, which gives

$$\hbar\dot\Phi_\mathbf{p} = \frac{1}{2m}\Big[\mathbf{p} + \frac{e}{c}\mathbf{A}(t)\Big]^2. \tag{8.26}$$

(2) Expand the square with Eq. (8.14), where $(e/c)\mathbf{A} = -mv_0\hat{\mathbf{e}}\sin\omega t$,

$$\frac{1}{2m}\Big[\mathbf{p}+\frac{e}{c}\mathbf{A}\Big]^2 = \frac{p^2}{2m} - v_0(\mathbf{p}\cdot\hat{\mathbf{e}})\sin\omega t + \frac{mv_0^2}{2}\sin^2\omega t = E_\mathbf{p} - v_0(\mathbf{p}\cdot\hat{\mathbf{e}})\sin\omega t + U_\mathrm{p}\,(1-\cos 2\omega t), \tag{8.27}$$

with $E_\mathbf{p} = p^2/2m$ and $U_\mathrm{p} = mv_0^2/4$. Integrate from 0 to t with $\int_0^t \sin\omega t'\mathrm{d}t' = (1-\cos\omega t)/\omega$ and $\int_0^t \cos 2\omega t'\mathrm{d}t' = \sin(2\omega t)/2\omega$,

$$\hbar\Phi_\mathbf{p}(t) = (E_\mathbf{p} + U_\mathrm{p})\,t - \mathbf{p}\cdot\mathbf{a}_0\,(1-\cos\omega t) - \frac{U_\mathrm{p}}{2\omega}\sin 2\omega t. \tag{8.28}$$

The Volkov state is therefore [2]

$$\boxed{\psi_\mathbf{p}(\mathbf{r},t) = \frac{1}{\mathcal{V}^{1/2}}\exp\Big\{\frac{i}{\hbar}\Big[\mathbf{p}\cdot\mathbf{r} - (E_\mathbf{p}+U_\mathrm{p})\,t + \mathbf{p}\cdot\mathbf{a}_0\,(1-\cos\omega t) + \frac{U_\mathrm{p}}{2\omega}\sin 2\omega t\Big]\Big\},} \tag{8.29}$$

which is Eq. (1) [3] with the integral in its phase carried out for the field of Eq. (8.14), apart from the normalization.

(3) The phase $\mathbf{p}\cdot\mathbf{r} + \mathbf{p}\cdot\mathbf{a}_0(1-\cos\omega t) = \mathbf{p}\cdot[\mathbf{r} - \boldsymbol{\alpha}(t) + \boldsymbol{\alpha}(0)]$ shows that the Volkov wave is the plane wave of the oscillating frame of Section 8.3.3, carried along with the quiver motion. The mean energy is $E_\mathbf{p} + U_\mathrm{p}$, and U_p is the same for all states, so it drops out of every energy difference.

8.4.2 Orthonormality and completeness

At equal times the time-dependent phases are the same for the two factors of a product with equal momenta, so

$$\int \psi^*_{\mathbf{p}'}(\mathbf{r},t)\psi_\mathbf{p}(\mathbf{r},t)\mathrm{d}^3r = \delta_{\mathbf{p}\mathbf{p}'}, \qquad \sum_\mathbf{p}\psi_\mathbf{p}(\mathbf{r},t)\psi^*_\mathbf{p}(\mathbf{r}',t) = \frac{1}{\mathcal{V}}\sum_\mathbf{p} e^{i\mathbf{p}\cdot(\mathbf{r}-\mathbf{r}')/\hbar} = \delta(\mathbf{r}-\mathbf{r}'). \tag{8.30}$$

The Volkov states at time t are a complete orthonormal set, and they can be used as the basis of the many-electron field operator in Section 8.6.

8.4.3 The Jacobi–Anger expansion and three sum rules

(1) The generating function of the Bessel functions is $e^{\frac{1}{2}z(\tau-\tau^{-1})}=\sum_\ell \tau^\ell J_\ell(z)$, Eq. (10.12.1) [11]. With $\tau=ie^{i\theta}$, $\tau-\tau^{-1}=ie^{i\theta}+ie^{-i\theta}=2i\cos\theta$, and

$$\boxed{e^{iz\cos\theta}=\sum_{\ell=-\infty}^{\infty} i^\ell J_\ell(z)\,e^{i\ell\theta}.} \tag{8.31}$$

(2) Parseval's theorem for the Fourier series in θ and for the series of its derivative $\partial_\theta e^{iz\cos\theta}=-iz\sin\theta\,e^{iz\cos\theta}=\sum_\ell i^\ell(i\ell)J_\ell e^{i\ell\theta}$ gives

$$\sum_\ell J_\ell^2(z)=\frac{1}{2\pi}\int_0^{2\pi}\left|e^{iz\cos\theta}\right|^2\mathrm{d}\theta=1,\qquad \sum_\ell \ell^2 J_\ell^2(z)=\frac{1}{2\pi}\int_0^{2\pi} z^2\sin^2\theta\,\mathrm{d}\theta=\frac{z^2}{2}. \tag{8.32}$$

Since $J_{-\ell}=(-1)^\ell J_\ell$, the squares are even in ℓ and the first moment vanishes. Altogether,

$$\boxed{\sum_\ell J_\ell^2(z)=1,\qquad \sum_\ell \ell\,J_\ell^2(z)=0,\qquad \sum_\ell \ell^2 J_\ell^2(z)=\frac{z^2}{2},\qquad J_{-\ell}^2=J_\ell^2.} \tag{8.33}$$

(3) For small arguments $J_\ell(z)=(\pm z/2)^{|\ell|}/|\ell|!+O(z^{|\ell|+2})$, so that

$$J_0^2(z)=1-\frac{z^2}{2}+O(z^4),\qquad J_{\pm1}^2(z)=\frac{z^2}{4}+O(z^4),\qquad J_{\pm\ell}^2(z)=O(z^{2|\ell|}). \tag{8.34}$$

(4) Multiplying Eq. (8.31) by the complex conjugate of the same expansion at $\theta+s$ and averaging over θ gives the generating function of the squares. With $\cos\theta-\cos(\theta+s)=2\sin(s/2)\sin(\theta+s/2)$ and $(2\pi)^{-1}\int_0^{2\pi}e^{ix\sin\varphi}\mathrm{d}\varphi=J_0(x)$,

$$\boxed{\sum_\ell J_\ell^2(z)\,e^{-i\ell s}=\frac{1}{2\pi}\int_0^{2\pi} e^{iz[\cos\theta-\cos(\theta+s)]}\mathrm{d}\theta=J_0\!\left(2z\sin\frac{s}{2}\right).} \tag{8.35}$$

This identity gives the classical limit in Section 8.5.6.

8.5 Multiphoton transition rates in Born approximation

8.5.1 The transition amplitude

(1) Expand the state of one electron in the Volkov states, $\psi(t)=\sum_{\mathbf{p}} c_{\mathbf{p}}(t)\psi_{\mathbf{p}}(t)$. Since each $\psi_{\mathbf{p}}$ solves the equation without ions, the Schrödinger equation with H of Eq. (8.18) reduces to

$$i\hbar\,\dot c_{\mathbf{p}'}=\sum_{\mathbf{p}} V_{\mathbf{p}'\mathbf{p}}(t)\,c_{\mathbf{p}},\qquad V_{\mathbf{p}'\mathbf{p}}(t)=\int \psi^*_{\mathbf{p}'}(\mathbf{r},t)V(\mathbf{r})\psi_{\mathbf{p}}(\mathbf{r},t)\mathrm{d}^3r. \tag{8.36}$$

To first order in V, starting from $c_{\mathbf{p}}(0)=\delta_{\mathbf{p}\mathbf{p}_i}$,

$$c_{\mathbf{p}'}(t)=-\frac{i}{\hbar}\int_0^t V_{\mathbf{p}'\mathbf{p}_i}(t')\mathrm{d}t'. \tag{8.37}$$

(2) In the matrix element the factor $\mathcal{V}^{-1}e^{-i(\mathbf{p}'-\mathbf{p})\cdot\mathbf{r}/\hbar}$ multiplies $V(\mathbf{r})$, and with $\hbar\mathbf{k}=\mathbf{p}'-\mathbf{p}$ the spatial integral is $\tilde V(\mathbf{k})/\mathcal{V}$ of Eq. (8.21). The time-dependent phases of Eq. (8.29) combine to

$$\begin{aligned}\hbar(\Phi_{\mathbf{p}'}-\Phi_{\mathbf{p}}) &= (E_{\mathbf{p}'}-E_{\mathbf{p}})\,t + \cancel{(U_{\mathrm{p}}-U_{\mathrm{p}})\,t} - (\mathbf{p}'-\mathbf{p})\cdot\mathbf{a}_0\,(1-\cos\omega t) - \cancel{\Big(\frac{U_{\mathrm{p}}}{2\omega}-\frac{U_{\mathrm{p}}}{2\omega}\Big)\sin 2\omega t}\\ &= (E_{\mathbf{p}'}-E_{\mathbf{p}})\,t - \hbar\mathbf{k}\cdot\mathbf{a}_0\,(1-\cos\omega t),\end{aligned} \tag{8.38}$$

in which the terms with U_{p} cancel because the ponderomotive energy is the same for every state. With $z\equiv\mathbf{k}\cdot\mathbf{a}_0$ and Eq. (8.31),

$$\boxed{V_{\mathbf{p}'\mathbf{p}}(t)=\frac{\tilde V(\mathbf{k})}{\mathcal{V}}\,e^{-iz}\sum_{\ell=-\infty}^{\infty} i^{\ell}J_\ell(z)\,\exp\Big[\frac{i}{\hbar}\big(E_{\mathbf{p}'}-E_{\mathbf{p}}+\ell\hbar\omega\big)t\Big],\qquad \hbar\mathbf{k}=\mathbf{p}'-\mathbf{p},\quad z=\mathbf{k}\cdot\mathbf{a}_0.} \tag{8.39}$$

The matrix element is a sum of harmonics of the laser frequency, and the ℓth harmonic carries the weight $J_\ell(z)$.

8.5.2 The long-time limit

(1) Insert Eq. (8.39) into Eq. (8.37),

$$c_{\mathbf{p}'}(t)=-\frac{i}{\hbar}\,\frac{\tilde V(\mathbf{k})}{\mathcal{V}}\,e^{-iz}\sum_{\ell} i^{\ell}J_\ell(z)\,I_\ell(t),\qquad I_\ell(t)=\int_0^t e^{i\Omega_\ell t'}\,\mathrm{d}t'=e^{i\Omega_\ell t/2}\,\frac{\sin(\Omega_\ell t/2)}{\Omega_\ell/2}, \tag{8.40}$$

with $\hbar\Omega_\ell=E_{\mathbf{p}'}-E_{\mathbf{p}}+\ell\hbar\omega$. The probability is

$$|c_{\mathbf{p}'}(t)|^2=\frac{|\tilde V(\mathbf{k})|^2}{\hbar^2\mathcal{V}^2}\Big[\sum_{\ell}J_\ell^2(z)\,|I_\ell(t)|^2+\sum_{\ell\neq\ell'}i^{\ell-\ell'}J_\ell(z)J_{\ell'}(z)\,I_\ell(t)I_{\ell'}^*(t)\Big]. \tag{8.41}$$

(2) The diagonal terms grow linearly in time. With $x=\Omega t/2$, $\int\sin^2(\Omega t/2)/(\Omega/2)^2\,\mathrm{d}\Omega = 2t\int\sin^2x/x^2\,\mathrm{d}x=2\pi t$, and the function concentrates at $\Omega=0$ as t grows, so that

$$|I_\ell(t)|^2=\frac{\sin^2(\Omega_\ell t/2)}{(\Omega_\ell/2)^2}\;\longrightarrow\;2\pi t\,\delta(\Omega_\ell)=2\pi\hbar t\,\delta(\hbar\Omega_\ell). \tag{8.42}$$

(3) The cross terms do not grow. The peaks of $|I_\ell|$ and $|I_{\ell'}|$ sit at energies that differ by $(\ell-\ell')\hbar\omega$, so an integral of $I_\ell I_{\ell'}^*$ over the final energies stays bounded as $t\to\infty$. They describe the modulation of the scattering within a period of the field, which produces harmonics of the current [9] and averages out of the absorbed power.
(4) The transition rate $W=\mathrm{d}|c_{\mathbf{p}'}|^2/\mathrm{d}t$ is then

$$W_{\mathbf{p}\to\mathbf{p}'}=\frac{2\pi}{\hbar}\,\frac{|\tilde V(\mathbf{k})|^2}{\mathcal{V}^2}\sum_{\ell}J_\ell^2(z)\,\delta\big(E_{\mathbf{p}'}-E_{\mathbf{p}}+\ell\hbar\omega\big). \tag{8.43}$$

Replacing ℓ by $-\ell$, which leaves J_ℓ^2 unchanged by Eq. (8.33), puts it in the form in which $\ell>0$ counts absorbed photons,

$$\boxed{W_{\mathbf{p}\to\mathbf{p}'}=\frac{2\pi}{\hbar}\,\frac{|\tilde V(\mathbf{k})|^2}{\mathcal{V}^2}\sum_{\ell=-\infty}^{\infty}J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\,\delta\big(E_{\mathbf{p}'}-E_{\mathbf{p}}-\ell\hbar\omega\big),\qquad \hbar\mathbf{k}=\mathbf{p}'-\mathbf{p}.} \tag{8.44}$$

The rate is symmetric, $W_{\mathbf{p}\to\mathbf{p}'}=W_{\mathbf{p}'\to\mathbf{p}}$, because the term ℓ of one equals the term $-\ell$ of the other and $|\tilde V(-\mathbf{k})|=|\tilde V(\mathbf{k})|$.

8.5.3 The average over the ion positions

(1) By Eq. (8.21), $|\tilde V(\mathbf{k})|^2 = |\tilde v(\mathbf{k})|^2 |\rho_i(\mathbf{k})|^2$ with

$$|\rho_i(\mathbf{k})|^2 = \sum_{j,j'} e^{-i\mathbf{k}\cdot(\mathbf{R}_j - \mathbf{R}_{j'})} = N_i + \sum_{j\neq j'} e^{-i\mathbf{k}\cdot(\mathbf{R}_j - \mathbf{R}_{j'})}. \tag{8.45}$$

(2) For ions placed independently and uniformly in the box, $\langle e^{-i\mathbf{k}\cdot(\mathbf{R}_j-\mathbf{R}_{j'})}\rangle = |\langle e^{-i\mathbf{k}\cdot\mathbf{R}}\rangle|^2 = \delta_{\mathbf{k}0}$ for $j \neq j'$, so $\langle|\rho_i(\mathbf{k})|^2\rangle = N_i$ for $\mathbf{k}\neq 0$. For correlated ions the average defines the static structure factor, $\langle|\rho_i(\mathbf{k})|^2\rangle = N_i S(\mathbf{k})$, and $S = 1$ for independent ions.
(3) The averaged rate is

$$\boxed{W_{\mathbf{p}\to\mathbf{p}'} = \frac{2\pi}{\hbar}\,\frac{n_i}{\mathcal{V}}\,|\tilde v(\mathbf{k})|^2 S(\mathbf{k}) \sum_\ell J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\,\delta\big(E_{\mathbf{p}'} - E_{\mathbf{p}} - \ell\hbar\omega\big).} \tag{8.46}$$

8.5.4 The cross section of Bunkin and Fedorov

(1) The rate of transitions of one electron into the final directions $\mathrm{d}\Omega'$ is the sum of Eq. (8.46) over the final states in that cone. With $\sum_{\mathbf{p}'} \to \mathcal{V}\int p'^2 \mathrm{d}p'\,\mathrm{d}\Omega'/(2\pi\hbar)^3$ and $\mathrm{d}E_{\mathbf{p}'} = p'\mathrm{d}p'/m$, the energy delta function sets $p' = p'_\ell \equiv (p^2 + 2m\ell\hbar\omega)^{1/2}$ and contributes mp'_ℓ,

$$\mathrm{d}\Gamma_\ell = \frac{2\pi}{\hbar}\, n_i |\tilde v(\mathbf{k})|^2 S(\mathbf{k}) J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\,\frac{mp'_\ell}{(2\pi\hbar)^3}\,\mathrm{d}\Omega'. \tag{8.47}$$

(2) Dividing by the flux $n_i v = n_i p/m$ of ions seen by the electron gives the cross section for the absorption of ℓ photons,

$$\boxed{\begin{aligned} \frac{\mathrm{d}\sigma_\ell}{\mathrm{d}\Omega'} &= \frac{m^2}{4\pi^2\hbar^4}\,\frac{p'_\ell}{p}\,|\tilde v(\mathbf{k})|^2 S(\mathbf{k})\, J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0) = \frac{p'_\ell}{p}\, J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\,\frac{\mathrm{d}\sigma_B}{\mathrm{d}\Omega}(\mathbf{k}), \\ \frac{\mathrm{d}\sigma_B}{\mathrm{d}\Omega} &= \Big(\frac{m}{2\pi\hbar^2}\Big)^2 |\tilde v(\mathbf{k})|^2 S(\mathbf{k}). \end{aligned}} \tag{8.48}$$

Here $\mathrm{d}\sigma_B/\mathrm{d}\Omega$ is the field-free Born cross section at the momentum transfer $\hbar\mathbf{k}$. Equation (8.48) is Eq. (3) [3], which normalizes the cross section by the photon flux $I/\hbar\omega$ with $I = cE_0^2/8\pi$ and writes the argument of the Bessel function as $\gamma[\cos\theta_0(1-\lambda\cos\theta) - \lambda\sin\theta_0\sin\theta\cos\varphi]$, which is $-\mathbf{k}\cdot\mathbf{a}_0$ in the present notation.
(3) For a bare Coulomb ion and $S = 1$, with $\hbar k = 2p\sin(\theta/2)$ for elastic scattering,

$$\frac{\mathrm{d}\sigma_B}{\mathrm{d}\Omega} = \frac{4m^2Z^2e^4}{\hbar^4k^4} = \Big(\frac{Ze^2}{2mv^2}\Big)^2 \frac{1}{\sin^4(\theta/2)}, \tag{8.49}$$

the Rutherford cross section. At $E_0 = 0$ only $\ell = 0$ survives, with $J_0(0) = 1$, and Eq. (8.48) reduces to it.
(4) When $\ell\hbar\omega \ll E_{\mathbf{p}}$ the factor p'_ℓ/p is close to one, and the sum rule $\sum_\ell J_\ell^2 = 1$ shows that the field does not change the total scattering. It distributes each scattering event over the energies $E_{\mathbf{p}} + \ell\hbar\omega$.

8.5.5 Weak fields and one photon

(1) For $|\mathbf{k}\cdot\mathbf{a}_0| \ll 1$, Eq. (8.34) keeps $\ell = 0, \pm 1$. With Eq. (8.20) at $\kappa = 0$ and $S = 1$, the one-photon cross sections are

$$\frac{\mathrm{d}\sigma_{\pm 1}}{\mathrm{d}\Omega'} = \frac{m^2}{4\pi^2\hbar^4}\,\frac{p'_\pm}{p}\,\frac{16\pi^2 Z^2 e^4}{k^4}\,\frac{(\mathbf{k}\cdot\mathbf{a}_0)^2}{4} = \frac{m^2 Z^2 e^4 a_0^2}{\hbar^2}\,\frac{p'_\pm}{p}\,\frac{[(\mathbf{p}'-\mathbf{p})\cdot\hat{\mathbf{e}}]^2}{|\mathbf{p}'-\mathbf{p}|^4}, \tag{8.50}$$

with $p'_\pm = (p^2 \pm 2m\hbar\omega)^{1/2}$. The upper sign is absorption and the lower one stimulated emission, which exists only for $E_\mathbf{p} > \hbar\omega$.

(2) Let the incident momentum make the angle θ_0 with $\hat{\mathbf{e}}$. Put $\mathbf{p}$ along the z axis, $\hat{\mathbf{e}} = (\sin\theta_0, 0, \cos\theta_0)$ and $\mathbf{p}' = p'(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$. The average over φ is

$$\big\langle[(\mathbf{p}'-\mathbf{p})\cdot\hat{\mathbf{e}}]^2\big\rangle_\varphi = \tfrac{1}{2}p'^2\sin^2\theta\,\sin^2\theta_0 + (p'\cos\theta - p)^2\cos^2\theta_0. \tag{8.51}$$

With $D = p^2 + p'^2 - 2pp'c$, $c = \cos\theta$, the remaining integrals are elementary. Using $b = 2pp'$, $a = p^2 + p'^2$, $\Delta = p^2 - p'^2$ and $L = \ln[(p+p')/|p-p'|]$,

$$\int_{-1}^{1}\frac{\mathrm{d}c}{D^2} = \frac{2}{\Delta^2}, \qquad \int_{-1}^{1}\frac{c\,\mathrm{d}c}{D^2} = \frac{2a}{b\Delta^2} - \frac{2L}{b^2}, \qquad \int_{-1}^{1}\frac{c^2\mathrm{d}c}{D^2} = \frac{2a^2}{b^2\Delta^2} - \frac{4aL}{b^3} + \frac{2}{b^2}, \tag{8.52}$$

from which, using $b^2 - a^2 = -\Delta^2$ and $(a - 2p^2)^2 = \Delta^2$,

$$\int_{-1}^{1}\frac{(1-c^2)\,\mathrm{d}c}{D^2} = \frac{(p^2+p'^2)L}{2p^3p'^3} - \frac{1}{p^2p'^2}, \qquad \int_{-1}^{1}\frac{(p'c-p)^2\mathrm{d}c}{D^2} = \frac{1}{p^2} + \frac{(p^2-p'^2)L}{2p^3p'}. \tag{8.53}$$

(3) Collecting,

$$\boxed{\sigma_{\pm1}(\theta_0) = \frac{2\pi m^2 Z^2 e^4 a_0^2}{\hbar^2 p^2}\,\frac{p'_\pm}{p}\Big\{\sin^2\theta_0\Big[\frac{(p^2+p'^2_\pm)L_\pm}{4pp'_\pm} - \frac{1}{2}\Big] + \cos^2\theta_0\Big[1 + \frac{(p^2-p'^2_\pm)L_\pm}{2pp'_\pm}\Big]\Big\},} \tag{8.54}$$

with $L_\pm = \ln[(p+p'_\pm)/|p-p'_\pm|]$.

(4) The average over the orientation of $\mathbf{p}$ uses $\langle\sin^2\theta_0\rangle = 2/3$ and $\langle\cos^2\theta_0\rangle = 1/3$. The terms without L cancel and

$$\boxed{\bar\sigma_{\pm1}(p) = \frac{2\pi}{3}\,\frac{m^2Z^2e^4a_0^2}{\hbar^2p^2}\,L_\pm, \qquad L_\pm = \ln\frac{p+p'_\pm}{|p-p'_\pm|}.} \tag{8.55}$$

(5) Bunkin and Fedorov expand Eq. (8.54) for $\xi = \hbar\omega/mv^2 \ll 1$, in their normalization per photon flux. Multiplying by $n_i v\hbar\omega/I$ with $I = cE_0^2/8\pi$ and $a_0^2 = e^2E_0^2/m^2\omega^4$ turns the prefactor of Eq. (8.54) into $16\pi^2 n_i Z^2 e^6/(m^2 vc\hbar\omega^3)$. With $p'_\pm = p(1\pm2\xi)^{1/2}$, $L_\pm = \ln(2/\xi) \pm \xi + O(\xi^2)$ and the curly bracket to first order in ξ, the result is

$$\sigma^{\mathrm{BF}}_{a,e} = \frac{8\pi^2Z^2e^6n_i}{m^2vc\hbar\omega^3}\Big[(3\cos^2\theta_0 - 1) + (1-\cos^2\theta_0)\ln\frac{2}{\xi} \mp (3\cos^2\theta_0 - 1)\,\xi\ln\frac{2}{\xi} \pm 2\xi\cos^2\theta_0\Big], \tag{8.56}$$

which is Eq. (10) [3] term by term, with the upper sign for absorption. The net emission $\sigma_e - \sigma_a$ is proportional to $(3\cos^2\theta_0 - 1)\ln(2/\xi) - 2\cos^2\theta_0$. It is positive for electrons moving along the field, which then amplify it, as Marcuse first found [13]. Averaged over directions it is $-2/3$ times the prefactor, and the plasma absorbs.

8.5.6 The classical limit of the photon-number distribution

(1) Let $\hbar\omega \to 0$ at a fixed quiver velocity, so that $z = \mathbf{k}\cdot\mathbf{a}_0 \to \infty$ while $Z_E \equiv z\hbar\omega = \hbar\mathbf{k}\cdot\hat{\mathbf{e}}\,v_0$ stays fixed. Take a smooth test function $\phi(\epsilon)$ of the energy transfer with Fourier representation $\phi(\epsilon) = \int \hat{\phi}(\tau)e^{-i\epsilon\tau}\mathrm{d}\tau$. By Eq. (8.35) with $s = \hbar\omega\tau$,

$$\sum_\ell J_\ell^2(z)\,\phi(\ell\hbar\omega) = \int \hat{\phi}(\tau)\,J_0\Big(2z\sin\frac{\hbar\omega\tau}{2}\Big)\mathrm{d}\tau \;\longrightarrow\; \int \hat{\phi}(\tau)\,J_0(Z_E\tau)\,\mathrm{d}\tau. \tag{8.57}$$

(2) With $J_0(x) = (2\pi)^{-1}\int_0^{2\pi} e^{-ix\cos\theta}\mathrm{d}\theta$ the limit is $(2\pi)^{-1}\int_0^{2\pi}\phi(Z_E\cos\theta)\,\mathrm{d}\theta$. In distributional form,

$$\boxed{\sum_\ell J_\ell^2(z)\,\delta(\epsilon-\ell\hbar\omega) \;\longrightarrow\; \frac{1}{2\pi}\int_0^{2\pi}\delta(\epsilon - Z_E\cos\theta)\,\mathrm{d}\theta = \frac{1}{\pi}\,\frac{\Theta(Z_E^2-\epsilon^2)}{(Z_E^2-\epsilon^2)^{1/2}}, \qquad Z_E = \hbar|\mathbf{k}\cdot\hat{\mathbf{e}}|\,v_0.} \tag{8.58}$$

(3) This is the classical picture. The ion is at rest in the laboratory, so a collision that is short compared with the period conserves the kinetic energy $m|\mathbf{v}|^2/2$ of the electron while its momentum changes by $\hbar\mathbf{k}$. The kinetic momentum is $m\mathbf{v} = \mathbf{p} + m\mathbf{v}_{\rm osc}$, so the drift energy $E_\mathbf{p} = |m\mathbf{v} - m\mathbf{v}_{\rm osc}|^2/2m$ changes by $\Delta\epsilon = -\hbar\mathbf{k}\cdot\mathbf{v}_{\rm osc}(t_c) = \hbar(\mathbf{k}\cdot\hat{\mathbf{e}})v_0\sin\omega t_c$, where t_c is the time of the collision. A uniformly distributed phase ωt_c gives the arcsine density of Eq. (8.58).

Remarks:
(1) The Born approximation for a Coulomb ion needs $\eta = Ze^2/\hbar v \ll 1$, Eq. (5) [3]. Since $\eta = r_0/\lambda\!\!\!^{-}$ with $r_0 = Ze^2/mv^2$ and $\lambda\!\!\!^{-} = \hbar/mv$, it is the condition that the de Broglie wavelength exceeds the distance of closest approach. It is the same condition that selects the quantum lower cutoff of the Coulomb logarithm in [1, Eq. (10.136)] and the Bethe rather than the Bohr stopping number in [1, Eq. (11.74)]. At the thermal velocity $\eta = Z(2\,\mathrm{Ry}/k_BT)^{1/2}$, with $2\,\mathrm{Ry} = 27.2$ eV.
(2) At first order in E_0 the amplitude of Eq. (8.39) with $\ell = 1$ is that of second-order perturbation theory, with one photon vertex $(e/mc)\mathbf{A}\cdot\hat{\mathbf{p}}$ and one ion vertex in both orders. The photon vertex is diagonal in momentum, so the two orders have the intermediate states $\mathbf{p}$ and $\mathbf{p}'$, the energy denominators $\hbar\omega$ and $-\hbar\omega$, and the amplitude $(e/mc\hbar\omega)\mathbf{A}_1\cdot(\mathbf{p}-\mathbf{p}')\tilde{V}(\mathbf{k})/\mathcal{V}$, where $\mathbf{A}_1$ is the amplitude of $e^{-i\omega t}$ in $\mathbf{A}$, of magnitude $cE_0/2\omega$. Its magnitude is $\frac{1}{2}|\mathbf{k}\cdot\mathbf{a}_0||\tilde{V}|/\mathcal{V}$, which is $J_1(\mathbf{k}\cdot\mathbf{a}_0)|\tilde{V}|/\mathcal{V}$ to this order. The term $e^2A^2/2mc^2$ only shifts all energies by U_p and does not contribute.
(3) The weak-field cross sections of Eq. (8.55) are what the kinetic equation of Section 8.12 needs for an isotropic distribution.

8.6 The kinetic equation

This section derives the equation of motion of the occupation numbers with the operator method that Section 2 of the quantum Boltzmann equation paper [4] uses for a quartic interaction. The electrons are taken as independent here. Their interaction enters in Section 8.10.

8.6.1 The field operator in the Volkov basis

(1) Expand the electron field operator of spin σ in the Volkov states, which are complete at every time by Eq. (8.30),

$$\hat{\Psi}_\sigma(\mathbf{r},t) = \sum_\mathbf{p}\psi_\mathbf{p}(\mathbf{r},t)\,\hat{b}_{\mathbf{p}\sigma}, \qquad \hat{b}_{\mathbf{p}\sigma}\hat{b}^\dagger_{\mathbf{p}'\sigma'} - \eta_s\,\hat{b}^\dagger_{\mathbf{p}'\sigma'}\hat{b}_{\mathbf{p}\sigma} = \delta_{\mathbf{p}\mathbf{p}'}\delta_{\sigma\sigma'}, \tag{8.59}$$

with $\eta_s = -1$ for electrons. The value $\eta_s = +1$ for bosons is kept to show that the result does not depend on it.
(2) In the interaction picture with respect to the motion in the laser field, the time dependence of the free motion is carried by the mode functions $\psi_{\mathbf{p}}$, the operators $\hat{b}_{\mathbf{p}\sigma}$ are constant in the absence of ions, and the ion potential is the one-body operator

$$\hat{H}_V(t) = \sum_\sigma \int \hat{\Psi}^\dagger_\sigma V \hat{\Psi}_\sigma \, \mathrm{d}^3 r = \sum_\sigma \sum_{\mathbf{p},\mathbf{p}'} V_{\mathbf{p}'\mathbf{p}}(t)\, \hat{b}^\dagger_{\mathbf{p}'\sigma} \hat{b}_{\mathbf{p}\sigma}, \tag{8.60}$$

with the matrix elements of Eq. (8.39). The potential does not act on the spin, and the spin label is dropped from here on.

8.6.2 The exact equation of motion of the one-body density matrix

(1) The commutator of two bilinears is the same for both statistics. For fermions,

$$\hat{b}^\dagger_a \hat{b}_b \hat{b}^\dagger_c \hat{b}_d = \delta_{bc} \hat{b}^\dagger_a \hat{b}_d - \hat{b}^\dagger_a \hat{b}^\dagger_c \hat{b}_b \hat{b}_d, \qquad \hat{b}^\dagger_c \hat{b}_d \hat{b}^\dagger_a \hat{b}_b = \delta_{ad} \hat{b}^\dagger_c \hat{b}_b - \hat{b}^\dagger_c \hat{b}^\dagger_a \hat{b}_d \hat{b}_b, \tag{8.61}$$

and the quartic terms are equal because $\hat{b}^\dagger_c \hat{b}^\dagger_a \hat{b}_d \hat{b}_b = (-\hat{b}^\dagger_a \hat{b}^\dagger_c)(-\hat{b}_b \hat{b}_d)$. For bosons both signs change and the quartic terms are again equal. Hence

$$\big[\hat{b}^\dagger_a \hat{b}_b, \hat{b}^\dagger_c \hat{b}_d\big] = \delta_{bc}\, \hat{b}^\dagger_a \hat{b}_d - \delta_{ad}\, \hat{b}^\dagger_c \hat{b}_b. \tag{8.62}$$

(2) The Heisenberg equation for $\hat{b}^\dagger_k \hat{b}_{k'}$, with k, k' short for momenta, is then

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(\hat{b}^\dagger_k \hat{b}_{k'}\big) = \frac{i}{\hbar}\big[\hat{H}_V, \hat{b}^\dagger_k \hat{b}_{k'}\big] = \frac{i}{\hbar} \sum_{c,d} V_{cd}\big(\delta_{dk} \hat{b}^\dagger_c \hat{b}_{k'} - \delta_{ck'} \hat{b}^\dagger_k \hat{b}_d\big) = \frac{i}{\hbar} \sum_c \big(V_{ck}\, \hat{b}^\dagger_c \hat{b}_{k'} - V_{k'c}\, \hat{b}^\dagger_k \hat{b}_c\big). \tag{8.63}$$

With the one-body density matrix $\rho_{ab} = \langle \hat{b}^\dagger_a \hat{b}_b \rangle$,

$$\boxed{\frac{\mathrm{d}\rho_{kk'}}{\mathrm{d}t} = \frac{i}{\hbar} \sum_c \big[V_{ck}(t)\, \rho_{ck'} - V_{k'c}(t)\, \rho_{kc}\big].} \tag{8.64}$$

(3) Equation (8.64) is exact and closed. In the quantum Boltzmann equation paper the quartic interaction makes the right side of Eq. (4) [4] a product of four operators, whose average at second order needs the six-field correlator and Wick's theorem. Here the ion potential is a one-body operator, the right side is again bilinear, and no factorization is needed.

8.6.3 Second order and the Markov limit

(1) Let the initial state be homogeneous, $\rho_{ab}(0) = \delta_{ab} f_a$. To first order in V, Eq. (8.64) has $\sum_c V_{ca}\rho_{cb} = V_{ba} f_b$ and $\sum_c V_{bc}\rho_{ac} = V_{ba} f_a$, so

$$\rho^{(1)}_{ab}(t) = \frac{i}{\hbar}\,(f_b - f_a) \int_0^t V_{ba}(t')\, \mathrm{d}t'. \tag{8.65}$$

The diagonal elements do not change at first order, because $f_k - f_k = 0$. This is the vanishing of the mean-field term that the quantum Boltzmann equation paper finds for $\mathbf{k} = \mathbf{k}'$ in its Eq. (5) [4].

(2) For the diagonal element at second order, insert Eq. (8.65) into Eq. (8.64) with $k' = k$. Since $\rho^{(1)}_{kc} = \rho^{(1)*}_{ck}$ and $V_{kc} = V^*_{ck}$, the two terms are $\zeta_c - \zeta_c^* = 2i\,\mathrm{Im}\,\zeta_c$ with $\zeta_c = V_{ck}(t)\rho^{(1)}_{ck}(t) = (i/\hbar)(f_k - f_c)V_{ck}(t)\int_0^t V_{kc}(t')\mathrm{d}t'$, and

$$\frac{\mathrm{d}f_k}{\mathrm{d}t} = -\frac{2}{\hbar}\sum_c \mathrm{Im}\,\zeta_c = \frac{2}{\hbar^2}\sum_c (f_c - f_k)\;\mathrm{Re}\left[V^*_{kc}(t)\int_0^t V_{kc}(t')\,\mathrm{d}t'\right]. \tag{8.66}$$

Since $\mathrm{Re}[V^*(t)\int_0^t V\mathrm{d}t'] = \frac{1}{2}\mathrm{d}|\int_0^t V\mathrm{d}t'|^2/\mathrm{d}t$, the bracket is $\hbar^2/2$ times the rate of growth of the transition probability of Eq. (8.37).
(3) Insert Eq. (8.39). The phases e^{-iz} cancel between V^*_{kc} and V_{kc}, and with $\hbar\Omega_\ell = E_k - E_c + \ell\hbar\omega$ and $\tau = t - t'$,

$$\mathrm{Re}\left[V^*_{kc}(t)\int_0^t V_{kc}(t')\mathrm{d}t'\right] = \frac{|\tilde{V}|^2}{\mathcal{V}^2}\sum_{\ell,\ell'}\mathrm{Re}\left[i^{\ell'-\ell}J_\ell J_{\ell'}\,e^{i(\ell'-\ell)\omega t}\int_0^t e^{-i\Omega_{\ell'}\tau}\mathrm{d}\tau\right]. \tag{8.67}$$

The terms with $\ell' \neq \ell$ oscillate at multiples of ω and average to zero over a period. The terms with $\ell' = \ell$ contain

$$\mathrm{Re}\int_0^t e^{-i\Omega\tau}\mathrm{d}\tau = \frac{\sin\Omega t}{\Omega} \;\longrightarrow\; \pi\,\delta(\Omega) \qquad (t\to\infty). \tag{8.68}$$

The one-sided time integral gives $\pi\delta$, and the factor 2 of Eq. (8.66), which comes from the two Hermitian-conjugate terms, completes the 2π of the golden rule. This is the step at which the published Eqs. (5)–(7) [4] carry twice the coefficient, as the erratum [5] corrects.
(4) With $\delta(\Omega) = \hbar\,\delta(\hbar\Omega)$ the result is $\mathrm{d}f_k/\mathrm{d}t = \sum_c W_{c\to k}(f_c - f_k)$ with the rate of Eq. (8.44), and after the average over the ion positions with the rate of Eq. (8.46),

$$\boxed{\frac{\partial f_{\mathbf{p}}}{\partial t} = \sum_{\mathbf{p}'} W_{\mathbf{p}'\to\mathbf{p}}\,(f_{\mathbf{p}'} - f_{\mathbf{p}}), \qquad W_{\mathbf{p}'\to\mathbf{p}} = W_{\mathbf{p}\to\mathbf{p}'}.} \tag{8.69}$$

This is the kinetic equation of inverse bremsstrahlung, a master equation in which each pair of states is connected by the multiphoton rates of Bunkin and Fedorov.

8.6.4 Why there is no Pauli blocking

The Uehling–Uhlenbeck form of the collision integral, Eq. (7) [4], carries the factors $1 + \eta_s n$ of the final states. For scattering on a static potential the gain and the loss term of one pair of states combine as

$$f_{\mathbf{p}'}(1 + \eta_s f_{\mathbf{p}}) - f_{\mathbf{p}}(1 + \eta_s f_{\mathbf{p}'}) = f_{\mathbf{p}'} - f_{\mathbf{p}}, \tag{8.70}$$

because the quadratic terms are equal. The exact one-body equation of motion never produced them. Pauli blocking does not suppress inverse bremsstrahlung on static ions, but the Fermi–Dirac distribution enters through f itself, Section 8.8.2.

8.6.5 Conservation, equilibrium and the heating rate

(1) Summing Eq. (8.69) over $\mathbf{p}$ and using the symmetry of W shows that the number of electrons is conserved. Momentum is not conserved, because the ions absorb it.
(2) Without the field only $\ell = 0$ survives, W contains $\delta(E_{\mathbf{p}'} - E_{\mathbf{p}})$, and every distribution that depends on the energy alone is stationary. The field drives the electrons through the terms with $\ell \neq 0$.

(3) The kinetic energy of the drift motion per unit volume, counting both spin states, is $u_e = (2/\mathcal{V})\sum_{\mathbf{p}} E_{\mathbf{p}} f_{\mathbf{p}}$. The quiver energy $U_{\rm p}$ per electron is constant and absorbs nothing on average. With Eq. (8.69), and then with the labels $\mathbf{p}, \mathbf{p}'$ exchanged and the two forms averaged,

$$P = \frac{\mathrm{d}u_e}{\mathrm{d}t} = \frac{2}{\mathcal{V}} \sum_{\mathbf{p},\mathbf{p}'} E_{\mathbf{p}} W_{\mathbf{p}'\to\mathbf{p}} (f_{\mathbf{p}'} - f_{\mathbf{p}}) = \frac{1}{\mathcal{V}} \sum_{\mathbf{p},\mathbf{p}'} \big(E_{\mathbf{p}} - E_{\mathbf{p}'}\big) W_{\mathbf{p}'\to\mathbf{p}} (f_{\mathbf{p}'} - f_{\mathbf{p}}). \tag{8.71}$$

(4) The term ℓ of the rate fixes $E_{\mathbf{p}} - E_{\mathbf{p}'} = \ell\hbar\omega$. Pairing the term $-\ell$ with the term ℓ after exchanging $\mathbf{p}$ and $\mathbf{p}'$ in it, which uses $J_{-\ell}^2 = J_\ell^2$, gives

$$\boxed{\begin{aligned} P &= \frac{2}{\mathcal{V}} \sum_{\ell=1}^{\infty} \ell\hbar\omega \sum_{\mathbf{p},\mathbf{p}'} w_\ell(\mathbf{p}' \to \mathbf{p}) \, (f_{\mathbf{p}'} - f_{\mathbf{p}}), \\ w_\ell(\mathbf{p}' \to \mathbf{p}) &= \frac{2\pi}{\hbar} \frac{n_i}{\mathcal{V}} |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0) \, \delta(E_{\mathbf{p}} - E_{\mathbf{p}'} - \ell\hbar\omega), \end{aligned}} \tag{8.72}$$

with $\hbar\mathbf{k} = \mathbf{p} - \mathbf{p}'$. Each term is a transition from the lower energy $E_{\mathbf{p}'}$ to the higher one $E_{\mathbf{p}} = E_{\mathbf{p}'} + \ell\hbar\omega$, absorption minus stimulated emission.
(5) If f is a decreasing function of the energy alone, every difference $f_{\mathbf{p}'} - f_{\mathbf{p}}$ is positive and $P \geq 0$. Free–free gain needs either an inverted energy distribution or an anisotropic one, as for the electrons moving along the field in Eq. (8.56).

In Summary: The occupation numbers obey the master equation (8.69), derived exactly from the operator equation of motion to second order in the ion potential. No Pauli blocking appears, and the heating rate is Eq. (8.72).

8.7 The heating rate and the Lindhard function

8.7.1 The heating rate through the Lindhard function

(1) In Eq. (8.72) write the initial momentum as $\mathbf{q}$ and the final one as $\mathbf{q} + \hbar\mathbf{k}$, with $\sum_{\mathbf{q}} \to \mathcal{V}\int \mathrm{d}^3q/(2\pi\hbar)^3$ and $\sum_{\mathbf{k}} \to \mathcal{V}\int \mathrm{d}^3k/(2\pi)^3$,

$$P = \sum_{\ell=1}^{\infty} \ell\hbar\omega \, \frac{2\pi}{\hbar} \, n_i \int \frac{\mathrm{d}^3k}{(2\pi)^3} \, |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0) \Big\{ 2 \int \frac{\mathrm{d}^3q}{(2\pi\hbar)^3} \, \delta\big(E_{\mathbf{q}+\hbar\mathbf{k}} - E_{\mathbf{q}} - \ell\hbar\omega\big) \big(f_{\mathbf{q}} - f_{\mathbf{q}+\hbar\mathbf{k}}\big) \Big\}. \tag{8.73}$$

(2) The density response function of free electrons, the Lindhard function [10], is

$$\boxed{\chi_0(\mathbf{k},\Omega) = 2 \int \frac{\mathrm{d}^3q}{(2\pi\hbar)^3} \, \frac{f_{\mathbf{q}} - f_{\mathbf{q}+\hbar\mathbf{k}}}{\hbar\Omega + E_{\mathbf{q}} - E_{\mathbf{q}+\hbar\mathbf{k}} + i0^+},} \tag{8.74}$$

the factor 2 counting the spin. With $\mathrm{Im}(x + i0^+)^{-1} = -\pi\delta(x)$ the curly bracket is $-\,\mathrm{Im}\,\chi_0(\mathbf{k}, \ell\omega)/\pi$, and

$$P = -2n_i \sum_{\ell=1}^{\infty} \ell\omega \int \frac{\mathrm{d}^3k}{(2\pi)^3} \, |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0) \; \mathrm{Im}\,\chi_0(\mathbf{k}, \ell\omega). \tag{8.75}$$

(3) Changing $\mathbf{q} \to \mathbf{q} + \hbar\mathbf{k}$ in Eq. (8.74) shows $\chi_0(-\mathbf{k}, -\Omega)^* = \chi_0(\mathbf{k}, \Omega)$, so $\mathrm{Im}\,\chi_0(-\mathbf{k}, -\Omega) = -\,\mathrm{Im}\,\chi_0(\mathbf{k}, \Omega)$. Since the rest of the integrand is even in $\mathbf{k}$, the term $-\ell$ equals the term ℓ, and the

sum can run over all ℓ,

$$\boxed{P = -n_i \sum_{\ell=-\infty}^{\infty} \ell\omega \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k})\, J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\ \mathrm{Im}\,\chi_0(\mathbf{k},\ell\omega).} \tag{8.76}$$

This form holds for any distribution f, isotropic or not, and any field strength. Section 8.10 shows that the interacting electron gas gives the same formula with χ_0 replaced by its density response.

8.7.2 The imaginary part of the Lindhard function

(1) For an isotropic distribution $f_\mathbf{q} = f(E_\mathbf{q})$, write $\mathrm{Im}\,\chi_0 = -2\pi \int \mathrm{d}^3q\,(2\pi\hbar)^{-3}[f(E_\mathbf{q}) - f(E_{\mathbf{q}+\hbar\mathbf{k}})]\,\delta(\hbar\Omega + E_\mathbf{q} - E_{\mathbf{q}+\hbar\mathbf{k}})$. In the first term $E_{\mathbf{q}+\hbar\mathbf{k}} - E_\mathbf{q} = \hbar k q_\parallel/m + \hbar^2k^2/2m$ with $q_\parallel = \mathbf{q}\cdot\hat{\mathbf{k}}$, so the delta function is $(m/\hbar k)\,\delta(q_\parallel - mc_-)$ with $c_- = \Omega/k - \hbar k/2m$.
(2) The integral over the transverse momentum is $\int f\,\mathrm{d}^2q_\perp = \pi\int f\,\mathrm{d}(q_\perp^2) = 2\pi m\int_{E_-}^\infty f(E)\,\mathrm{d}E$, where $E_- = mc_-^2/2$ is the energy at $q_\perp = 0$.
(3) In the second term the shift $\mathbf{q}\to\mathbf{q}-\hbar\mathbf{k}$ turns the condition into $\hbar k q_\parallel/m - \hbar^2k^2/2m = \hbar\Omega$, that is $q_\parallel = mc_+$ with $c_+ = \Omega/k + \hbar k/2m$ and $E_+ = mc_+^2/2$.
(4) Subtracting the two and collecting the constants $2\pi\cdot 2\pi m\cdot m/[(2\pi\hbar)^3\hbar k] = m^2/2\pi\hbar^4 k$,

$$\boxed{\mathrm{Im}\,\chi_0(\mathbf{k},\Omega) = -\frac{m^2}{2\pi\hbar^4 k}\int_{E_-}^{E_+} f(E)\,\mathrm{d}E, \qquad E_\pm = \frac{m}{2}\Big(\frac{\Omega}{k} \pm \frac{\hbar k}{2m}\Big)^2.} \tag{8.77}$$

For $\Omega > 0$, $E_+ > E_-$ and $\mathrm{Im}\,\chi_0 < 0$.
(5) For the Fermi–Dirac distribution $f = [e^{(E-\mu)/k_BT}+1]^{-1}$, with the antiderivative $-k_BT\ln(1+e^{(\mu-E)/k_BT})$,

$$\boxed{\mathrm{Im}\,\chi_0(\mathbf{k},\Omega) = -\frac{m^2 k_BT}{2\pi\hbar^4 k}\,\ln\frac{1+e^{(\mu-E_-)/k_BT}}{1+e^{(\mu-E_+)/k_BT}}.} \tag{8.78}$$

(6) For nondegenerate electrons $f = e^{(\mu-E)/k_BT}$ with $e^{\mu/k_BT} = \frac{1}{2}n_e(2\pi\hbar^2/mk_BT)^{3/2}$, and the constants combine into $(\pi/2)^{1/2}/\hbar k v_t$,

$$\boxed{\mathrm{Im}\,\chi_0(\mathbf{k},\Omega) = -n_e\Big(\frac{\pi}{2}\Big)^{1/2}\frac{1}{\hbar k v_t}\Big[e^{-E_-/k_BT} - e^{-E_+/k_BT}\Big], \qquad \frac{E_\pm}{k_BT} = \frac{1}{2}\Big(\frac{\Omega}{kv_t} \pm \frac{\hbar k v_t}{2k_BT}\Big)^2.} \tag{8.79}$$

(7) In the classical limit $\hbar\to 0$ the bracket is $(\hbar\Omega/k_BT)e^{-m\Omega^2/2k^2k_BT}$, and with $\zeta = \Omega/\sqrt{2}kv_t$,

$$\mathrm{Im}\,\chi_0 \longrightarrow -\frac{n_e}{k_BT}\sqrt{\pi}\,\zeta\,e^{-\zeta^2}, \tag{8.80}$$

the Landau damping of the classical dielectric function, Eq. (11.19) [1]. The quantum form differs from it only for $\hbar k \gtrsim mv_t$, and that difference cuts off the Coulomb logarithm.

8.7.3 Weak fields

(1) For $|\mathbf{k}\cdot\mathbf{a}_0| \ll 1$ only $\ell = \pm 1$ contribute to Eq. (8.76), each with $J_1^2 = (\mathbf{k}\cdot\mathbf{a}_0)^2/4$,

$$P = -\frac{n_i\omega}{2}\int\frac{\mathrm{d}^3k}{(2\pi)^3}\,|\tilde{v}(\mathbf{k})|^2 S(\mathbf{k})\,(\mathbf{k}\cdot\mathbf{a}_0)^2\ \mathrm{Im}\,\chi_0(\mathbf{k},\omega). \tag{8.81}$$

(2) For an isotropic distribution and isotropic ions the rest of the integrand depends on k only, and the angular average gives $(\mathbf{k}\cdot\mathbf{a}_0)^2 \to k^2a_0^2/3$. With $\int \mathrm{d}^3k/(2\pi)^3 = (2\pi^2)^{-1}\int k^2\mathrm{d}k$,

$$P = -\frac{n_i\omega a_0^2}{12\pi^2}\int_0^\infty k^4|\tilde v(k)|^2 S(k)\ \mathrm{Im}\,\chi_0(k,\omega)\,\mathrm{d}k. \tag{8.82}$$

(3) Dividing by $n_e m\omega^2 a_0^2/2 = n_e m v_0^2/2$ and, for a Coulomb ion, using $k^4|\tilde v|^2 = 16\pi^2Z^2e^4$ and $n_iZ^2/n_e = Z$,

$$\boxed{\nu_{\rm eff}(\omega) = -\frac{n_i}{6\pi^2 n_e m\omega}\int_0^\infty k^4|\tilde v(k)|^2 S(k)\ \mathrm{Im}\,\chi_0(k,\omega)\,\mathrm{d}k = -\frac{8Ze^4}{3m\omega}\int_0^\infty S(k)\ \mathrm{Im}\,\chi_0(k,\omega)\,\mathrm{d}k.} \tag{8.83}$$

By Eq. (8.8) this collision frequency gives the transverse dielectric function $\mathrm{Im}\,\varepsilon_T = \omega_p^2\nu_{\rm eff}/\omega^3$ and the absorption coefficient of Eq. (8.10). The quantum theory has replaced the input ν of the Drude model by a formula.

8.8 Closed forms for weak fields

Throughout this section the ions are bare Coulomb charges, $S = 1$, and the field is weak, so that Eq. (8.83) applies.

8.8.1 Maxwell electrons

(1) Insert Eq. (8.79) at $\Omega = \omega$ into Eq. (8.83),

$$\nu_{\rm eff} = \frac{8Ze^4}{3m\omega}\,n_e\Big(\frac{\pi}{2}\Big)^{1/2}\frac{1}{\hbar v_t}\int_0^\infty\frac{\mathrm{d}k}{k}\Big[e^{-E_-/k_BT} - e^{-E_+/k_BT}\Big]. \tag{8.84}$$

(2) Expand the exponents of Eq. (8.79). With $mv_t^2 = k_BT$ and $u = \hbar\omega/k_BT$,

$$\frac{E_\pm}{k_BT} = \frac{\omega^2}{2k^2v_t^2} \pm \frac{u}{2} + \frac{\hbar^2k^2}{8mk_BT}, \qquad e^{-E_-/k_BT} - e^{-E_+/k_BT} = 2\sinh\frac{u}{2}\ e^{-a/k^2-bk^2}, \tag{8.85}$$

with $a = \omega^2/2v_t^2$ and $b = \hbar^2/8mk_BT$.
(3) The integral representation $K_0(z) = \frac12\int_0^\infty \exp(-t - z^2/4t)\,t^{-1}\mathrm{d}t$, Eq. (10.32.10) [11] at $\nu = 0$, with $t = bk^2$ and $\mathrm{d}t/t = 2\mathrm{d}k/k$ gives

$$\int_0^\infty\frac{\mathrm{d}k}{k}\,e^{-a/k^2-bk^2} = K_0\big(2\sqrt{ab}\big), \qquad 2\sqrt{ab} = 2\Big(\frac{\omega^2\hbar^2}{16(k_BT)^2}\Big)^{1/2} = \frac{u}{2}. \tag{8.86}$$

(4) Collect. Since $m\omega\hbar v_t = u\,m^{1/2}(k_BT)^{3/2}$, $\frac83(\pi/2)^{1/2} = \frac43(2\pi)^{1/2}$ and $Zn_e = Z^2n_i$,

$$\boxed{\nu_{\rm eff} = \nu_T\,\Lambda(u), \qquad \Lambda(u) = \frac{2\sinh(u/2)}{u}\,K_0\Big(\frac{u}{2}\Big) = \frac{1-e^{-u}}{u}\,e^{u/2}K_0\Big(\frac{u}{2}\Big), \qquad u = \frac{\hbar\omega}{k_BT},} \tag{8.87}$$

with ν_T of Eq. (8.11). The quantum theory gives the collision frequency of the Drude model with the logarithm $\ln\Lambda_e$ of Eq. (10.133) [1] replaced by the function $\Lambda(u)$, and with no cutoff.
(5) The limits follow from $K_0(x) = -\ln(x/2) - \gamma + O(x^2\ln x)$, Eq. (10.31.2) [11], and $K_0(x) \sim (\pi/2x)^{1/2}e^{-x}$, Eq. (10.40.2) [11], where $\gamma = 0.5772$ is Euler's constant,

$$\boxed{\Lambda(u) = \ln\frac{4k_BT}{e^\gamma\hbar\omega} + O(u^2\ln u) \quad (u \ll 1), \qquad \Lambda(u) \simeq \sqrt{\pi}\,u^{-3/2} \quad (u \gg 1).} \tag{8.88}$$

(6) For $u \ll 1$ the logarithm is $\ln(b_{\max}/b_{\min})$ with

$$b_{\max} = \frac{v_t}{\omega}, \qquad b_{\min} = \frac{e^\gamma}{4}\,\frac{\hbar}{mv_t} = 0.445\,\frac{\hbar}{mv_t}. \tag{8.89}$$

The upper cutoff is the distance an electron travels in a radian of the period, the dynamic cutoff of Dawson and Oberman, not the Debye length. At $\omega = \omega_p$ the two coincide, $v_t/\omega_p = \lambda_D$. The lower cutoff is fixed by the theory. The de Broglie length $\hbar/2mv$ at $mv^2 = 3k_BT$ of Eq. (10.136) [1] is $0.289\,\hbar/mv_t$, smaller by a factor 0.65, so the classical prescription overestimates the logarithm by $\ln(0.445/0.289) = 0.43$ at the critical density. Table 8.2 and Fig. 8.2 give $\Lambda(u)$ and its asymptotes.

Table 8.2: The quantum Coulomb logarithm of inverse bremsstrahlung for Maxwell electrons, Eq. (8.87), its two asymptotes of Eq. (8.88), and the thermally averaged Born Gaunt factor of Eq. (8.107).

u	$\Lambda(u)$	$\ln(4/e^\gamma u)$	$\sqrt{\pi}\,u^{-3/2}$	$\langle g\rangle$
0.001	7.717	7.717	56050	4.257
0.003	6.618	6.618	10787	3.654
0.01	5.414	5.414	1772	3.000
0.03	4.316	4.316	341.1	2.415
0.1	3.116	3.112	56.05	1.805
0.3	2.038	2.013	10.79	1.300
1	0.963	0.809	1.772	0.840
3	0.304	−0.290	0.341	0.528
10	0.0548	−1.494	0.0560	0.302

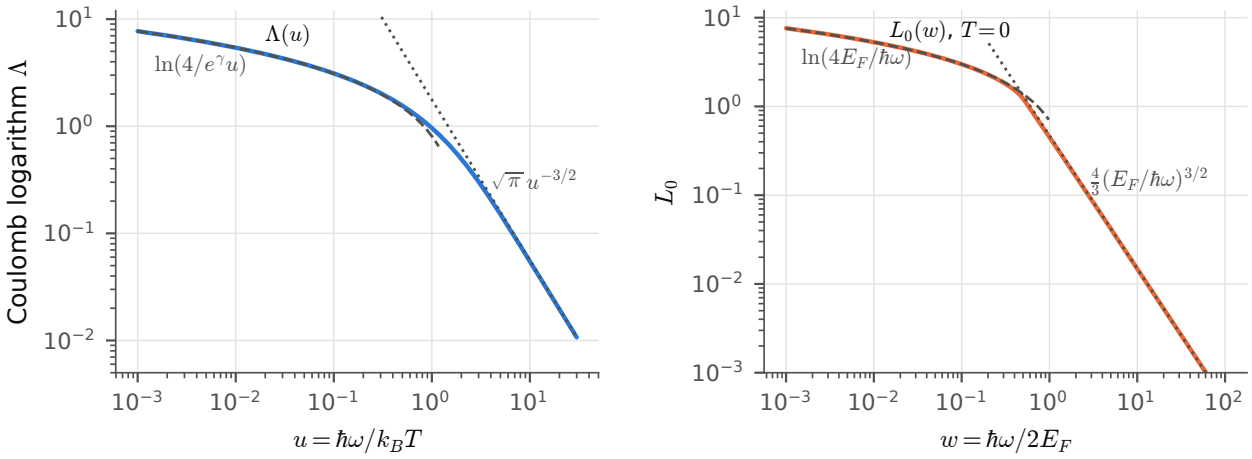


Figure 8.2: Left, the Coulomb logarithm $\Lambda(u)$ of Maxwell electrons, Eq. (8.87), with its asymptotes of Eq. (8.88). Right, the function $L_0(w)$ of the degenerate gas at $T = 0$, Eq. (8.94), with its asymptotes of Eq. (8.95), against $w = \hbar\omega/2E_F$.

8.8.2 Arbitrary degeneracy

Insert Eq. (8.78) into Eq. (8.83),

$$\boxed{\nu_{\rm eff} = \frac{4Ze^4 m k_B T}{3\pi\hbar^4\omega}\int_0^\infty \frac{{\rm d}k}{k}\,\ln\frac{1+e^{(\mu-E_-)/k_BT}}{1+e^{(\mu-E_+)/k_BT}}, \qquad E_\pm = \frac{m}{2}\Big(\frac{\omega}{k}\pm\frac{\hbar k}{2m}\Big)^2.} \tag{8.90}$$

For $e^{\mu/k_BT}\ll 1$ the logarithm is $e^{\mu/k_BT}(e^{-E_-/k_BT}-e^{-E_+/k_BT})$ and Eq. (8.87) follows. The chemical potential is fixed by $n_e = 2\int {\rm d}^3q\,(2\pi\hbar)^{-3}f$. Pauli blocking does not appear, by Section 8.6.4, but the occupation of the states does.

8.8.3 The degenerate electron gas at zero temperature

(1) At $T=0$, $k_BT\ln[(1+e^{(\mu-E_-)/k_BT})/(1+e^{(\mu-E_+)/k_BT})] \to (E_F-E_-)_+ - (E_F-E_+)_+$, where $(y)_+ = \max(y,0)$ and $\mu = E_F = mv_F^2/2$. With $x = k/k_F$, $k_F = mv_F/\hbar$ and $w = \hbar\omega/2E_F = \omega/k_Fv_F$, $E_\pm/E_F = (w/x \pm x/2)^2$ and

$$\nu_{\rm eff} = \frac{4Ze^4 m E_F}{3\pi\hbar^4\omega}\,I_0(w), \qquad I_0(w) = \int_0^\infty \frac{{\rm d}x}{x}\Big\{\Big[1-\Big(\frac{w}{x}-\frac{x}{2}\Big)^2\Big]_+ - \Big[1-\Big(\frac{w}{x}+\frac{x}{2}\Big)^2\Big]_+\Big\}. \tag{8.91}$$

(2) The first bracket is positive for $|w/x - x/2|<1$, that is for $x_1<x<x_2$ with $x_{1,2} = \sqrt{1+2w}\mp 1$. There the integrand is $(1+w)/x - w^2/x^3 - x/4$, with the antiderivative $(1+w)\ln x + w^2/2x^2 - x^2/8$. At the end points $w/x - x/2 = \pm 1$, and $\frac12(w^2/x^2 - x^2/4) = \frac12(w/x - x/2)(w/x + x/2)$ equals $(x_1+1)/2$ at x_1 and $-(x_2-1)/2$ at x_2. Hence

$$I_-(w) = (1+w)\ln\frac{\sqrt{1+2w}+1}{\sqrt{1+2w}-1} - \sqrt{1+2w}. \tag{8.92}$$

(3) The second bracket is positive for $w/x + x/2 < 1$, which needs $w<1/2$ and then $x_3<x<x_4$ with $x_{3,4} = 1\mp\sqrt{1-2w}$. The same steps, with the integrand $(1-w)/x - w^2/x^3 - x/4$, give

$$I_+(w) = (1-w)\ln\frac{1+\sqrt{1-2w}}{1-\sqrt{1-2w}} - \sqrt{1-2w} \quad (w<\tfrac12), \qquad I_+ = 0 \quad (w\ge\tfrac12). \tag{8.93}$$

(4) With $4Ze^4mE_F/(3\pi\hbar^4\omega) = (4Ze^4m/3\pi\hbar^3)/(2w)$,

$$\boxed{\nu_{\rm eff} = \frac{4Ze^4m}{3\pi\hbar^3}\,L_0(w), \qquad L_0(w) = \frac{I_-(w)-I_+(w)}{2w}, \qquad w = \frac{\hbar\omega}{2E_F}.} \tag{8.94}$$

(5) For $w\ll 1$, $I_-(w) - I_+(w) = 2w\ln(2/w) + O(w^2)$, and for $w\gg 1$, $I_- = \frac43(2w)^{-1/2} + O(w^{-3/2})$, so

$$\boxed{L_0(w) \simeq \ln\frac{4E_F}{\hbar\omega} \quad (\hbar\omega\ll E_F), \qquad L_0(w)\simeq\frac43\Big(\frac{E_F}{\hbar\omega}\Big)^{3/2} \quad (\hbar\omega\gg E_F).} \tag{8.95}$$

(6) The low-frequency form is the resistivity formula of Ziman. For an electron at the Fermi surface the Born cross section $(m/2\pi\hbar^2)^2|\tilde v(q)|^2$ with $q = 2k_F\sin(\theta/2)$, $1-\cos\theta = q^2/2k_F^2$ and ${\rm d}\Omega = 2\pi q\,{\rm d}q/k_F^2$ gives the momentum relaxation rate

$$\frac{1}{\tau} = n_i v_F\int(1-\cos\theta)\frac{{\rm d}\sigma}{{\rm d}\Omega}{\rm d}\Omega = \frac{n_i m}{4\pi\hbar^3k_F^3}\int_{k_{\rm min}}^{2k_F} q^3|\tilde v(q)|^2\,{\rm d}q = \frac{4Ze^4m}{3\pi\hbar^3}\ln\frac{2k_F}{k_{\rm min}}, \tag{8.96}$$

where the last step uses the Coulomb form and $k_F^3 = 3\pi^2 n_e = 3\pi^2 Z n_i$. With the minimum momentum transfer $k_{\rm min} = \omega/v_F$ that energy conservation imposes on an electron at the Fermi surface, $\ln(2k_F/k_{\rm min}) = \ln(4E_F/\hbar\omega)$, and the two agree.

8.8.4 The universal high-frequency tail

(1) When $\hbar\omega$ exceeds every electron energy that is occupied, Eq. (8.77) has $E_+ = E_- + \hbar\omega$ far above the occupied energies, and E_- is small only near $k_\omega = (2m\omega/\hbar)^{1/2}$, where $\omega/k = \hbar k/2m$. There $c_- = \omega/k - \hbar k/2m \simeq -(\hbar/m)(k - k_\omega)$.
(2) Hence

$$\int_0^\infty \frac{\mathrm{d}k}{k} \int_{E_-}^{E_+} f \,\mathrm{d}E \simeq \frac{m}{\hbar k_\omega} \int_{-\infty}^{\infty} \mathrm{d}c \int_{mc^2/2}^{\infty} f(E)\,\mathrm{d}E = \frac{m}{\hbar k_\omega}\, 2 \int_0^\infty f(E) \Big(\frac{2E}{m}\Big)^{1/2} \mathrm{d}E = \frac{m}{\hbar k_\omega}\, \frac{n_e (2\pi\hbar)^3}{4\pi m^2}, \tag{8.97}$$

where the last step uses $n_e = 8\pi m(2\pi\hbar)^{-3} \int (2mE)^{1/2} f \,\mathrm{d}E$.
(3) With Eq. (8.77), $\int_0^\infty \mathrm{Im}\,\chi_0 \,\mathrm{d}k = -\pi m n_e/\hbar^2 k_\omega$ for every distribution, and Eq. (8.83) gives

$$\boxed{\nu_{\mathrm{eff}} \;\longrightarrow\; \frac{4\sqrt{2}\,\pi}{3}\, \frac{Z n_e e^4}{m^{1/2} (\hbar\omega)^{3/2}}.} \tag{8.98}$$

Both the Maxwell form, $\nu_T \sqrt{\pi} u^{-3/2}$, and the degenerate form, $(4Ze^4 m/3\pi\hbar^3)\frac{4}{3}(E_F/\hbar\omega)^{3/2}$ with $E_F^{3/2} = 3\pi^2\hbar^3 n_e/(2m)^{3/2}$, reduce to it. When the photon energy exceeds the kinetic energies the electrons absorb as if they were at rest.

Remarks:
(1) For hydrogen at $k_B T = 1$ keV and $\lambda_L = 0.351\ \mu$m, $u = 0.0035$ and $\Lambda = 6.45$, while the logarithm of the cutoffs stated with Eq. (10.136) [1] at $n_e = n_c/4$ is 7.58, and the printed formula gives 6.69. The difference between the printed formula and its stated cutoffs is taken up in Appendix 8.A.2.
(2) Screening changes the lower limit of the k integral, where $\mathrm{Im}\,\chi_0$ is already exponentially small below $k \sim \omega/v_t$. Its effect is computed in Section 8.10.5 and is a few tenths in Λ near the critical density.

8.9 Spontaneous bremsstrahlung, the Gaunt factor and Kirchhoff's law

8.9.1 From stimulated to spontaneous emission

(1) Quantize one mode of the radiation field in the box, with wave vector $\mathbf{k}_L$, polarization $\hat{\mathbf{e}}_\lambda$ and frequency $\omega = ck_L$,

$$\hat{\mathbf{E}} = i\Big(\frac{2\pi\hbar\omega}{\mathcal{V}}\Big)^{1/2} \hat{\mathbf{e}}_\lambda \big(\hat{a}\, e^{i(\mathbf{k}_L\cdot\mathbf{r} - \omega t)} - \hat{a}^\dagger e^{-i(\mathbf{k}_L\cdot\mathbf{r} - \omega t)}\big). \tag{8.99}$$

In a state with N photons the absorption matrix element carries $\langle N-1|\hat{a}|N\rangle = \sqrt{N}$ and the emission one $\langle N+1|\hat{a}^\dagger|N\rangle = \sqrt{N+1}$.
(2) The classical field $E_0 \hat{\mathbf{e}} \cos\omega t$ has the amplitude $E_0/2$ in front of $e^{-i\omega t}$. For $N \gg 1$ it corresponds to $E_0/2 = (2\pi\hbar\omega N/\mathcal{V})^{1/2}$, that is $E_0^2/8\pi = N\hbar\omega/\mathcal{V}$, the energy density of N photons. The weak-field rates of Section 8.5.5 are therefore N times the rates for one photon, and the spontaneous emission into the mode is the stimulated rate with $N \to 1$,

$$\boxed{W^{\mathrm{sp}}_{\mathbf{p}\to\mathbf{p}'}(\mathbf{k}_L, \lambda) = W^{(\ell=-1)}_{\mathbf{p}\to\mathbf{p}'}\Big|_{E_0^2 = 8\pi\hbar\omega/\mathcal{V},\ \hat{\mathbf{e}} = \hat{\mathbf{e}}_\lambda}.} \tag{8.100}$$

This is the content of Einstein's relation, Eqs. (10.84) and (10.85) [1].

8.9.2 The Born bremsstrahlung cross section

(1) With $a_0^2 = e^2E_0^2/m^2\omega^4 = 8\pi\hbar e^2/(\mathcal{V}m^2\omega^3)$, the cross section of Eq. (8.48) for $\ell = -1$ and one mode is

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega'} = \frac{m^2}{4\pi^2\hbar^4}\frac{p'}{p}|\tilde{v}(\mathbf{k})|^2\,\frac{(\mathbf{k}\cdot\hat{\mathbf{e}}_\lambda)^2}{4}\,\frac{8\pi\hbar e^2}{\mathcal{V}m^2\omega^3} = \frac{e^2}{2\pi\hbar^3\mathcal{V}\omega^3}\,\frac{p'}{p}\,|\tilde{v}(\mathbf{k})|^2(\mathbf{k}\cdot\hat{\mathbf{e}}_\lambda)^2, \tag{8.101}$$

with $p' = (p^2 - 2m\hbar\omega)^{1/2}$.
(2) Sum over the modes in vacuum, $\sum_{\mathbf{k}_L,\lambda} \to \mathcal{V}\sum_\lambda \int \omega^2 \mathrm{d}\omega\,\mathrm{d}\Omega_L/(2\pi c)^3$. The two polarizations perpendicular to $\hat{\mathbf{k}}_L$ give $\sum_\lambda(\mathbf{k}\cdot\hat{\mathbf{e}}_\lambda)^2 = k^2 - (\mathbf{k}\cdot\hat{\mathbf{k}}_L)^2$, whose integral over the directions of the photon is $4\pi k^2 - \frac{4\pi}{3}k^2 = \frac{8\pi}{3}k^2$. The emission cross section per unit frequency is

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\omega\,\mathrm{d}\Omega'} = \frac{e^2}{2\pi\hbar^3\omega^3}\,\frac{\omega^2}{8\pi^3c^3}\,\frac{8\pi}{3}\,\frac{p'}{p}\,k^2|\tilde{v}(\mathbf{k})|^2 = \frac{e^2}{6\pi^3\hbar^3c^3\omega}\,\frac{p'}{p}\,k^2|\tilde{v}(\mathbf{k})|^2. \tag{8.102}$$

(3) For a Coulomb ion $k^2|\tilde{v}|^2 = 16\pi^2Z^2e^4/k^2$, and $\int \mathrm{d}\Omega'/k^2 = 2\pi\hbar^2\int_{-1}^{1}\mathrm{d}c/(p^2 + p'^2 - 2pp'c) = (2\pi\hbar^2/pp')L$ with $L = \ln[(p+p')/(p-p')]$, so

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}\omega} = \frac{16}{3}\,\frac{Z^2e^6}{m^2c^3\hbar v^2\omega}\,\ln\frac{p+p'}{p-p'}.} \tag{8.103}$$

This is the nonrelativistic Born cross section of bremsstrahlung, $\mathrm{d}\sigma = \frac{16}{3}Z^2\alpha r_e^2(c^2/v^2)L\,\mathrm{d}\omega/\omega$ with $\alpha = e^2/\hbar c$ and $r_e = e^2/mc^2$.
(4) Kramers' cross section of Eq. (8.12), per unit angular frequency, is $(\mathrm{d}\sigma/\mathrm{d}\nu)/2\pi$ with $h\nu = \hbar\omega$, that is $(16\pi/3\sqrt{3})Z^2e^6/(m^2c^3v^2\hbar\omega)$. The ratio is the Born Gaunt factor,

$$\boxed{g_B(E,\hbar\omega) = \frac{\sqrt{3}}{\pi}\ln\frac{p+p'}{p-p'}, \qquad p = (2mE)^{1/2}, \quad p' = [2m(E-\hbar\omega)]^{1/2}.} \tag{8.104}$$

For soft photons $g_B \to (\sqrt{3}/\pi)\ln(4E/\hbar\omega)$, and at the threshold $p' \to 0$ it vanishes as $(\sqrt{3}/\pi)2p'/p$.

8.9.3 The thermal average

(1) The emission of a Maxwellian plasma is proportional to $\int \mathrm{d}^3v\, f\, v\,\mathrm{d}\sigma/\mathrm{d}\omega \propto \int v\,\mathrm{d}v\, e^{-E/k_BT}g_B = m^{-1}\int \mathrm{d}E\, e^{-E/k_BT}g_B$ over $E > \hbar\omega$, while Kramers' form has $g = 1$. The thermally averaged Gaunt factor is therefore

$$\langle g\rangle = \frac{\int_{\hbar\omega}^{\infty} e^{-E/k_BT}g_B\,\mathrm{d}E}{\int_{\hbar\omega}^{\infty} e^{-E/k_BT}\mathrm{d}E} = \frac{\sqrt{3}}{\pi}\,e^u\int_u^\infty e^{-\epsilon}\ln\frac{\sqrt{\epsilon}+\sqrt{\epsilon-u}}{\sqrt{\epsilon}-\sqrt{\epsilon-u}}\,\mathrm{d}\epsilon, \tag{8.105}$$

with $\epsilon = E/k_BT$.
(2) Substitute $\epsilon = u\cosh^2 t$. Then $\sqrt{\epsilon}\pm\sqrt{\epsilon-u} = \sqrt{u}\,e^{\pm t}$, the logarithm is $2t$, and $\mathrm{d}\epsilon = u\sinh 2t\,\mathrm{d}t$. With $\cosh^2 t = \frac{1}{2}(1+\cosh 2t)$ and $\tau = 2t$,

$$\int_u^\infty e^{-\epsilon}\ln(\cdots)\,\mathrm{d}\epsilon = \frac{u}{2}\,e^{-u/2}\int_0^\infty \tau\sinh\tau\, e^{-(u/2)\cosh\tau}\mathrm{d}\tau. \tag{8.106}$$

(3) Integrate by parts with $\sinh\tau\, e^{-(u/2)\cosh\tau} = \mathrm{d}[-(2/u)e^{-(u/2)\cosh\tau}]/\mathrm{d}\tau$. The boundary terms vanish, and $\int_0^\infty e^{-z\cosh\tau}\mathrm{d}\tau = K_0(z)$, Eq. (10.32.9) [11], leaves $(2/u)K_0(u/2)$. Hence

$$\boxed{\langle g\rangle(u) = \frac{\sqrt{3}}{\pi}\,e^{u/2}K_0\Big(\frac{u}{2}\Big).} \tag{8.107}$$

(4) The power radiated at all frequencies weights the spectrum with e^{-u}, the factor of Kramers' form, so its Gaunt factor is

$$\boxed{\bar{g} = \int_0^\infty \langle g\rangle(u)\, e^{-u}\, \mathrm{d}u = \frac{\sqrt{3}}{\pi}\, 2 \int_0^\infty e^{-x} K_0(x)\, \mathrm{d}x = \frac{2\sqrt{3}}{\pi} = 1.103,} \tag{8.108}$$

using $K_0(x) = \int_0^\infty e^{-x\cosh t}\mathrm{d}t$, so that $\int_0^\infty e^{-x}K_0(x)\mathrm{d}x = \int_0^\infty \mathrm{d}t/(1+\cosh t) = [\tanh(t/2)]_0^\infty = 1$. This is the factor $g \approx 2\sqrt{3}/\pi$ that Atzeni and Meyer-ter-Vehn apply to Eq. (10.93) [1]. It is the Born value exactly.

8.9.4 Kirchhoff's law and the Coulomb logarithm as a Gaunt factor

(1) Compare Eq. (8.107) with Eq. (8.87),

$$\boxed{\Lambda(u) = \frac{\pi}{\sqrt{3}}\, \langle g\rangle(u)\, \frac{1 - e^{-u}}{u}.} \tag{8.109}$$

The quantum Coulomb logarithm of inverse bremsstrahlung is $\pi/\sqrt{3}$ times the thermally averaged Gaunt factor, times the stimulated-emission factor divided by u.
(2) The emission coefficient per unit volume, frequency and solid angle follows from Eq. (8.103) and the steps of the thermal average,

$$\begin{aligned} j_\omega &= \frac{n_e n_i \hbar\omega}{4\pi}\left\langle v\, \frac{\mathrm{d}\sigma}{\mathrm{d}\omega}\right\rangle, \qquad \left\langle \frac{L}{v}\right\rangle = \left(\frac{2}{\pi}\right)^{1/2} \frac{\pi}{\sqrt{3}}\, \frac{\langle g\rangle e^{-u}}{v_t}, \\ j_\omega &= \frac{4}{3\pi}\left(\frac{2}{\pi}\right)^{1/2} \frac{n_e n_i Z^2 e^6}{m^2 c^3 v_t}\, \frac{\pi}{\sqrt{3}} \langle g\rangle e^{-u}. \end{aligned} \tag{8.110}$$

The absorption coefficient of Eq. (8.10) with $n_r = 1$ and $\nu_{\mathrm{eff}} = \nu_T \Lambda$, multiplied by the Planck function per unit angular frequency and solid angle, $B_\omega = (\hbar\omega^3/4\pi^3c^2)/(e^u - 1)$, is, by Eq. (8.109),

$$K\, B_\omega = \frac{\nu_T \omega_p^2}{c\omega^2}\, \frac{\pi}{\sqrt{3}} \langle g\rangle \frac{e^{-u}}{u}\, \frac{\hbar\omega^3}{4\pi^3 c^2} = \frac{4\sqrt{2\pi}}{3\pi^2}\, \frac{n_e n_i Z^2 e^6}{m^{3/2}(k_B T)^{1/2} c^3}\, \frac{\pi}{\sqrt{3}} \langle g\rangle e^{-u}. \tag{8.111}$$

Since $m^2 v_t = m^{3/2}(k_B T)^{1/2}$ and $\frac{4}{3\pi}(2/\pi)^{1/2} = \frac{4\sqrt{2\pi}}{3\pi^2}$,

$$\boxed{j_\omega = K\, B_\omega.} \tag{8.112}$$

Kirchhoff's law holds exactly between the absorption of the master equation and the spontaneous emission of Eq. (8.103), because both come from the same matrix element and the Maxwellian satisfies detailed balance.

In Summary: The free–free emission and the inverse bremsstrahlung absorption of the quantum theory are Kramers' expressions multiplied by the Born Gaunt factor. Its thermal average is $(\sqrt{3}/\pi)e^{u/2}K_0(u/2)$, its power average is $2\sqrt{3}/\pi$, and the Coulomb logarithm of the absorption is the Gaunt factor in disguise, Eq. (8.109).

8.10 The interacting electron gas and dynamic screening

8.10.1 The oscillating frame for many electrons

(1) The Hamiltonian of N electrons in the laser field and the field of the ions is

$$H = \sum_{a=1}^{N} \frac{1}{2m}\Big[\hat{\mathbf{p}}_a + \frac{e}{c}\mathbf{A}(t)\Big]^2 + \sum_{a<b} \frac{e^2}{|\hat{\mathbf{r}}_a - \hat{\mathbf{r}}_b|} + \sum_a V(\hat{\mathbf{r}}_a), \tag{8.113}$$

apart from the constant energy of the fixed ions.
(2) Let $\hat{T} = \exp[-(i/\hbar)\boldsymbol{\alpha}(t)\cdot\hat{\mathbf{P}}]$ with $\hat{\mathbf{P}} = \sum_a \hat{\mathbf{p}}_a$. It translates all electrons together, so the electron–electron interaction, which depends on differences, is unchanged, $\hat{T}^\dagger V_{ee}\hat{T} = V_{ee}$. The steps of Section 8.3.3, with $\dot{S} = Ne^2A^2/2mc^2$, give

$$\boxed{H' = \sum_a \frac{\hat{p}_a^2}{2m} + \sum_{a<b} \frac{e^2}{|\hat{\mathbf{r}}_a - \hat{\mathbf{r}}_b|} + \sum_a V\big(\hat{\mathbf{r}}_a + \boldsymbol{\alpha}(t)\big).} \tag{8.114}$$

In the oscillating frame the electron gas carries no field, and the ions are external charges that move along $\mathbf{R}_j - \boldsymbol{\alpha}(t)$. A uniform field couples only to the center of mass of the electrons [14], which is the content of this transformation.

8.10.2 The absorbed power in the two frames

(1) The power that the laser delivers to the electrons in the laboratory is $P_{\rm lab}(t) = -e\mathbf{E}(t)\cdot\sum_a\langle\mathbf{v}_a\rangle$, with the velocity $\mathbf{v}_a = (\hat{\mathbf{p}}_a + e\mathbf{A}/c)/m$. Since $\hat{T}$ commutes with $\hat{\mathbf{p}}_a$, $\sum_a\langle\mathbf{v}_a\rangle = \langle\hat{\mathbf{P}}\rangle'/m + N\dot{\boldsymbol{\alpha}}$, where the prime marks an average in the oscillating frame. The term $-eN\mathbf{E}\cdot\dot{\boldsymbol{\alpha}} \propto \cos\omega t\sin\omega t$ averages to zero over a period.
(2) In the oscillating frame, $\mathrm{d}\langle H'\rangle/\mathrm{d}t = \langle\partial_t H'\rangle = -\dot{\boldsymbol{\alpha}}\cdot\langle\mathbf{F}\rangle'$, with the force $\mathbf{F} = -\sum_a \nabla V(\hat{\mathbf{r}}_a + \boldsymbol{\alpha})$ of the ions on the electrons. The total momentum obeys $\mathrm{d}\langle\hat{\mathbf{P}}\rangle'/\mathrm{d}t = \langle\mathbf{F}\rangle'$, because the kinetic term commutes with $\hat{\mathbf{P}}$ and the internal forces cancel. With $\ddot{\boldsymbol{\alpha}} = -e\mathbf{E}/m$,

$$\frac{\mathrm{d}\langle H'\rangle}{\mathrm{d}t} = -\dot{\boldsymbol{\alpha}}\cdot\frac{\mathrm{d}\langle\hat{\mathbf{P}}\rangle'}{\mathrm{d}t} = -\frac{\mathrm{d}}{\mathrm{d}t}\big(\dot{\boldsymbol{\alpha}}\cdot\langle\hat{\mathbf{P}}\rangle'\big) - \frac{e}{m}\mathbf{E}\cdot\langle\hat{\mathbf{P}}\rangle'. \tag{8.115}$$

(3) In the steady state $\langle\hat{\mathbf{P}}\rangle'$ is periodic, and the total derivative averages to zero. Hence

$$\boxed{\langle P_{\rm lab}\rangle_t = \Big\langle\frac{\mathrm{d}\langle H'\rangle}{\mathrm{d}t}\Big\rangle_t.} \tag{8.116}$$

The absorbed power is the rate at which the energy of the electron gas grows in the oscillating frame.

8.10.3 Linear response to the oscillating ions

(1) With the density operator $\hat{n}_{\mathbf{k}} = \sum_a e^{-i\mathbf{k}\cdot\hat{\mathbf{r}}_a}$ and Eq. (8.21), the ion term of Eq. (8.114) is

$$\sum_a V(\hat{\mathbf{r}}_a + \boldsymbol{\alpha}) = \frac{1}{\mathcal{V}}\sum_{\mathbf{k}} U_{\mathbf{k}}(t)\,\hat{n}_{-\mathbf{k}}, \qquad U_{\mathbf{k}}(t) = \tilde{V}(\mathbf{k})\,e^{i\mathbf{k}\cdot\boldsymbol{\alpha}(t)} = \tilde{V}(\mathbf{k})\sum_\ell i^\ell J_\ell(\mathbf{k}\cdot\mathbf{a}_0)\,e^{i\ell\omega t}, \tag{8.117}$$

by Eq. (8.31). In this frame the ion potential is a sum of harmonics $\ell\omega$ with the weights J_ℓ.

(2) Let the electron gas start in equilibrium. To first order in U, Kubo's formula gives $\delta\langle\hat n_{\mathbf k}\rangle'(t) = \int \chi(\mathbf k, t-t')U_{\mathbf k}(t')\mathrm dt'$ with the retarded density response

$$\chi(\mathbf k,t) = -\frac{i}{\hbar}\,\theta(t)\,\frac{1}{\mathcal V}\big\langle[\hat n_{\mathbf k}(t),\hat n_{-\mathbf k}(0)]\big\rangle, \qquad \chi(\mathbf k,\Omega) = \int e^{i\Omega t}\chi(\mathbf k,t)\,\mathrm dt, \tag{8.118}$$

which for free electrons is the Lindhard function of Eq. (8.74) [15]. A harmonic $e^{i\ell\omega t} = e^{-i(-\ell\omega)t}$ is answered with the factor $\chi(\mathbf k, -\ell\omega)$.
(3) The power in the oscillating frame is $\mathrm d\langle H'\rangle/\mathrm dt = \mathcal V^{-1}\sum_{\mathbf k}\partial_t U_{\mathbf k}\,\langle\hat n_{-\mathbf k}\rangle'$. The term $\mathbf k = 0$ is constant. With $U_{-\mathbf k} = \tilde V(-\mathbf k)\sum_\ell (-i)^\ell J_\ell(\mathbf k\cdot\mathbf a_0)e^{i\ell\omega t}$, the response $\delta\langle\hat n_{-\mathbf k}\rangle' = \sum_\ell \chi(-\mathbf k,-\ell\omega)U_{-\mathbf k,\ell}e^{i\ell\omega t}$, and $\partial_t U_{\mathbf k} = \sum_{\ell'} i\ell'\omega U_{\mathbf k,\ell'}e^{i\ell'\omega t}$, the period average keeps $\ell' = -\ell$. The product of the coefficients is $\tilde V(\mathbf k)\tilde V(-\mathbf k)\,i^{-\ell}J_{-\ell}(-i)^\ell J_\ell = |\tilde V(\mathbf k)|^2 J_\ell^2$, and with $\chi(-\mathbf k,-\Omega) = \chi(\mathbf k,\Omega)^*$,

$$\left\langle\frac{\mathrm d\langle H'\rangle}{\mathrm dt}\right\rangle_t = \frac{1}{\mathcal V}\sum_{\mathbf k}\sum_\ell (-i\ell\omega)|\tilde V(\mathbf k)|^2 J_\ell^2\,\chi(\mathbf k,\ell\omega)^* = -\frac{1}{\mathcal V}\sum_{\mathbf k}\sum_\ell \ell\omega\,|\tilde V(\mathbf k)|^2 J_\ell^2\,\operatorname{Im}\chi(\mathbf k,\ell\omega), \tag{8.119}$$

where the terms with $\operatorname{Re}\chi$, which is even in the frequency for the isotropic gas, cancel between ℓ and $-\ell$.
(4) After the average over the ion positions and per unit volume, by Eq. (8.116),

$$\boxed{P = -n_i\sum_{\ell=-\infty}^{\infty}\ell\omega\int\frac{\mathrm d^3k}{(2\pi)^3}\,|\tilde v(\mathbf k)|^2 S(\mathbf k)\,J_\ell^2(\mathbf k\cdot\mathbf a_0)\,\operatorname{Im}\chi(\mathbf k,\ell\omega).} \tag{8.120}$$

This is Eq. (8.76) with χ_0 replaced by the density response of the interacting gas. For free electrons the two derivations agree term by term, because the Volkov states of the laboratory are the plane waves of the oscillating frame. The result is exact to second order in the ion potential and exact in the field.

8.10.4 The dielectric form

(1) The longitudinal dielectric function is defined by $1/\varepsilon(\mathbf k,\Omega) = 1 + v_c(k)\chi(\mathbf k,\Omega)$ with $v_c = 4\pi e^2/k^2$. In the random-phase approximation $\varepsilon = 1 - v_c\chi_0$ and $\chi = \chi_0/\varepsilon$. For a Coulomb ion, $|\tilde v|^2\operatorname{Im}\chi = (4\pi Ze^2/k^2)^2(k^2/4\pi e^2)\operatorname{Im}\varepsilon^{-1} = (4\pi Z^2e^2/k^2)\operatorname{Im}\varepsilon^{-1}$, and

$$\boxed{P = -n_iZ^2e^2\sum_{\ell=-\infty}^{\infty}\ell\omega\int\frac{\mathrm d^3k}{(2\pi)^3}\,\frac{4\pi}{k^2}\,S(\mathbf k)\,J_\ell^2(\mathbf k\cdot\mathbf a_0)\,\operatorname{Im}\frac{1}{\varepsilon(\mathbf k,\ell\omega)}.} \tag{8.121}$$

With the classical dielectric function this is Silin's result [9]. With the Lindhard dielectric function it is the quantum form obtained by Kull and Plagne [16] and by Bornath and co-workers [17] from the quantum Vlasov and the quantum kinetic equation.
(2) For weak fields Eq. (8.83) with $k^4|\tilde v|^2\operatorname{Im}\chi = 4\pi Z^2e^2k^2\operatorname{Im}\varepsilon^{-1}$ gives

$$\boxed{\nu_{\mathrm{eff}}(\omega) = -\frac{2Ze^2}{3\pi m\omega}\int_0^\infty k^2S(k)\,\operatorname{Im}\frac{1}{\varepsilon(k,\omega)}\,\mathrm dk,} \tag{8.122}$$

the high-frequency collision frequency of Dawson and Oberman [6].

8.10.5 Dynamic screening for Maxwell electrons

(1) For nondegenerate electrons the Lindhard function follows from Eq. (8.74) with the shift $\mathbf{q} \to \mathbf{q} - \hbar\mathbf{k}$ in its second term and the integral over $\mathbf{q}_\perp$, which leaves the one-dimensional Maxwellian $g_1(v) = (2\pi v_t^2)^{-1/2} e^{-v^2/2v_t^2}$,

$$\chi_0 = \frac{n_e}{\hbar k} \int g_1(v) \Big[\frac{1}{c_- - v + i0^+} - \frac{1}{c_+ - v + i0^+} \Big] \mathrm{d}v = \frac{n_e}{\sqrt{2}\,\hbar k v_t} \big[Z(\zeta_+) - Z(\zeta_-) \big], \qquad \zeta_\pm = \frac{c_\pm}{\sqrt{2} v_t}, \tag{8.123}$$

with $c_\pm = \Omega/k \pm \hbar k/2m$ and the plasma dispersion function $Z(\zeta) = \pi^{-1/2} \int e^{-t^2} (t - \zeta)^{-1} \mathrm{d}t$ for $\mathrm{Im}\,\zeta > 0$ [19]. Its imaginary part on the real axis is $\sqrt{\pi} e^{-\zeta^2}$, which gives back Eq. (8.79).
(2) In the classical limit $Z(\zeta_+) - Z(\zeta_-) \to (\hbar k/\sqrt{2} m v_t) Z'(\zeta)$ with $Z' = -2(1 + \zeta Z)$, so $\chi_0 \to -(n_e/k_B T)[1 + \zeta Z(\zeta)]$ and $\varepsilon = 1 + (k_D^2/k^2)[1 + \zeta Z(\zeta)]$ with $k_D^2 = 4\pi e^2 n_e / k_B T$. This is Eq. (11.19) [1], whose function W is $W(x) = 1 + (x/\sqrt{2}) Z(x/\sqrt{2})$.
(3) With $x = k v_t/\omega$, $r = \omega_p/\omega$, $\hbar k v_t / k_B T = ux$ and $k_D^2/k^2 = r^2/x^2$,

$$\varepsilon(x) = 1 - \frac{r^2}{\sqrt{2}\,ux^3} \big[Z(\zeta_+) - Z(\zeta_-) \big], \qquad \zeta_\pm = \frac{1}{\sqrt{2}} \Big(\frac{1}{x} \pm \frac{ux}{2} \Big). \tag{8.124}$$

Equation (8.122) with $S = 1$, divided by ν_T, becomes

$$\boxed{\nu_{\mathrm{eff}} = \nu_T \Lambda_{\mathrm{RPA}}(r, u), \qquad \Lambda_{\mathrm{RPA}}(r, u) = -\frac{1}{r^2} \Big(\frac{2}{\pi} \Big)^{1/2} \int_0^\infty x^2 \, \mathrm{Im} \, \frac{1}{\varepsilon(x)} \, \mathrm{d}x.} \tag{8.125}$$

Without screening $\mathrm{Im}\,\varepsilon^{-1} \to -\mathrm{Im}\,\varepsilon$, the integral becomes $u^{-1} \int x^{-1} [e^{-\zeta_-^2} - e^{-\zeta_+^2}] \mathrm{d}x$, and Eq. (8.87) returns.
(4) Table 8.3 and Fig. 8.3 give the screening correction. It is negative and nearly independent of u. It is -0.10 at $n_e = n_c/4$ and -0.47 at $n_e = 0.94\, n_c$, and it approaches -0.56 as $\omega \to \omega_p$ for $u \ll 1$, slowly, because the resonance described next carries a weight proportional to $(1 - r^2)^{1/2}$. For $\omega > \omega_p$ the function $\mathrm{Re}\,\varepsilon$ vanishes at the Langmuir wave number, approximately $x_{\mathrm{pl}}^2 = (r^{-2} - 1)/3$, where the ion oscillation excites plasma waves. That resonance is part of the integral. Its weight, estimated with the Bohm–Gross form $\mathrm{Re}\,\varepsilon \simeq 1 - r^2(1 + 3x^2)$, is $\sqrt{2\pi}\, x_{\mathrm{pl}}/6r^4$, of order 0.1 for $r \geq 0.9$. For smaller r the plasma waves at x_{pl} are strongly Landau damped and there is no separate resonance.

Table 8.3: The screening correction $\Lambda_{\mathrm{RPA}} - \Lambda$ of the Coulomb logarithm, Eqs. (8.125) and (8.87), against $r = \omega_p/\omega = (n_e/n_c)^{1/2}$.

u	$r = 0.95$	0.9	0.8	0.7	0.5	0.3	0.1
0.003	−0.440	−0.375	−0.278	−0.204	−0.099	−0.034	−0.004
0.03	−0.439	−0.374	−0.277	−0.203	−0.098	−0.034	−0.004
0.3	−0.393	−0.333	−0.244	−0.178	−0.085	−0.029	−0.003

Remarks:
(1) Without ions H' does not depend on time and nothing is absorbed. Collisions between electrons alone do not absorb light in the dipole approximation. They enter only through χ, by screening, and through the shape of f.
(2) Equation (8.120) holds for any response function. Local-field corrections and strong coupling

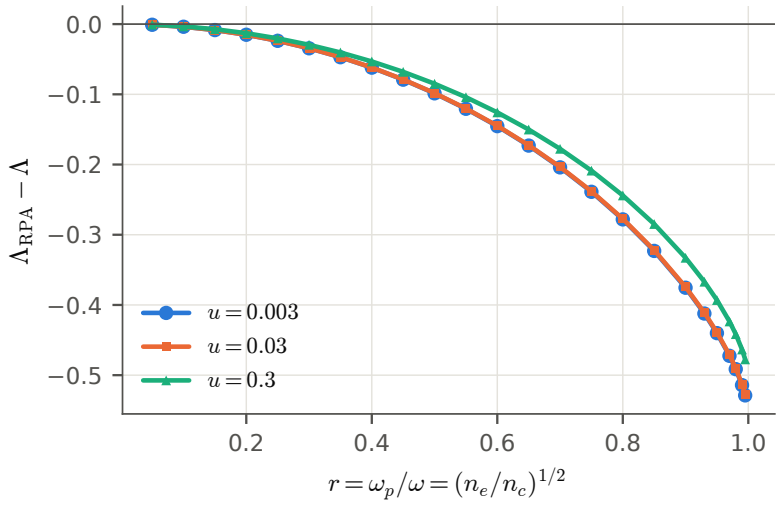


Figure 8.3: The screening correction $\Lambda_{\rm RPA} - \Lambda$ of Table 8.3 against $r = \omega_p/\omega$ for three photon energies.

enter through χ, and correlated ions through $S(k)$.
(3) The same result follows from the Kadanoff–Baym equations with the dynamically screened Born self-energy [18, 17], which is the many-body counterpart of the nonequilibrium Green-function route of Section 3 of the quantum Boltzmann equation paper [4].

8.11 Strong fields

8.11.1 The exact heating rate for Maxwell electrons

(1) Take Eq. (8.76) with bare Coulomb ions, $S = 1$ and an isotropic distribution. At fixed k the direction of $\mathbf{k}$ enters only through $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{e}}$, and $\int \mathrm{d}\Omega_k J_\ell^2(ka_0\mu) = 2\pi B_\ell(ka_0)$ with

$$B_\ell(z) = \int_{-1}^{1} J_\ell^2(z\mu)\,\mathrm{d}\mu = \frac{2}{z}\int_0^z J_\ell^2(t)\,\mathrm{d}t. \tag{8.126}$$

With $|\tilde{v}|^2 = 16\pi^2 Z^2 e^4/k^4$,

$$P = -4n_i Z^2 e^4 \sum_\ell \ell\omega \int_0^\infty \frac{\mathrm{d}k}{k^2}\,\mathrm{Im}\,\chi_0(k, \ell\omega)\,B_\ell(ka_0). \tag{8.127}$$

(2) With $x = kv_t/\omega$ and $\beta = v_0/v_t$, $ka_0 = x\beta$, and Eq. (8.79) at $\Omega = \ell\omega$ has $E_\pm/k_BT = \frac{1}{2}(\ell/x \pm ux/2)^2$ and $\hbar k v_t = ux\,k_BT$. Dividing by $n_e m v_0^2/2$ and by ν_T,

$$\boxed{\nu_{\rm eff} = \nu_T\,\frac{3\,\mathcal{S}(\beta,u)}{u\beta^2}, \qquad \mathcal{S}(\beta,u) = \sum_{\ell=-\infty}^{\infty} \ell \int_0^\infty \frac{\mathrm{d}x}{x^3}\,D_\ell(x)\,B_\ell(x\beta),}$$
$$\boxed{D_\ell(x) = e^{-\frac{1}{2}(\frac{\ell}{x} - \frac{ux}{2})^2} - e^{-\frac{1}{2}(\frac{\ell}{x} + \frac{ux}{2})^2}.} \tag{8.128}$$

The terms $\pm\ell$ are equal. For $\beta \to 0$, $B_{\pm1} \to (x\beta)^2/6$, and Eq. (8.86) gives the weak-field limit on the left below, which is Eq. (8.87). The nonlinearity is measured by the ratio on the right,

$$\mathcal{S} \xrightarrow{\beta\to 0} \frac{\beta^2}{3}\,2\sinh\frac{u}{2}\,K_0\Big(\frac{u}{2}\Big), \qquad R(\beta,u) = \frac{\nu_{\rm eff}(v_0)}{\nu_{\rm eff}(0)} = \frac{3\,\mathcal{S}(\beta,u)}{u\beta^2\Lambda(u)}. \tag{8.129}$$

(3) The functions B_ℓ follow from a recursion. The recurrences of the Bessel functions, Eq. (10.6.2) [11], give $\mathrm{d}[z(J_\ell^2 + J_{\ell+1}^2)]/\mathrm{d}z = (2\ell+1)(J_\ell^2 - J_{\ell+1}^2)$, so

$$\int_0^z J_\ell^2 \, \mathrm{d}t = \int_0^z J_{\ell+1}^2 \, \mathrm{d}t + \frac{z\,[J_\ell^2(z) + J_{\ell+1}^2(z)]}{2\ell+1}, \tag{8.130}$$

which is summed downward from an order $\ell \gg z$ at which the integral vanishes. The terms added are positive, and the recursion is stable.

8.11.2 The quasi-static limit

(1) For $x \gg 1$, that is $k \gg \omega/v_t$, the function $\ell \mapsto \ell D_\ell(x)$ changes on the scale $\Delta\ell \sim x$, much larger than one, and the limit of Eq. (8.58) applies to the sum over ℓ at fixed k,

$$\sum_\ell \ell \, D_\ell(x) \, J_\ell^2(x\beta\mu) \;\longrightarrow\; \frac{1}{2\pi}\int_0^{2\pi} \nu \, D_\nu(x) \, \mathrm{d}\theta, \qquad \nu = x\beta\mu\cos\theta. \tag{8.131}$$

(2) In dimensional form the frequency $\ell\omega$ becomes the instantaneous Doppler frequency $\Omega_t = \mathbf{k}\cdot\mathbf{v}_{\mathrm{osc}}(t)$, up to a sign that does not matter because $\Omega \,\mathrm{Im}\,\chi_0(k,\Omega)$ is even in Ω,

$$\boxed{P \;\longrightarrow\; -n_i \int \frac{\mathrm{d}^3k}{(2\pi)^3} \, |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) \left\langle \Omega_t \,\mathrm{Im}\,\chi_0(\mathbf{k},\Omega_t)\right\rangle_t, \qquad \Omega_t = \mathbf{k}\cdot\mathbf{v}_{\mathrm{osc}}(t).} \tag{8.132}$$

The ions move through the electron gas of the oscillating frame with the velocity $-\mathbf{v}_{\mathrm{osc}}(t)$, and Eq. (8.132) is the stopping power of the dielectric theory of Section 11.5 of [1] at that velocity, times the velocity, averaged over the period. This is Silin's picture.

(3) The limit fails for $k \lesssim \omega/v_t$. There every term of the exact sum carries $e^{-\ell^2/2x^2} \le e^{-1/2x^2}$ and is exponentially small, even when $v_0 \gg v_t$, because the electron gas of the oscillating frame is thermal and the ions supply only the frequencies $\ell\omega$. The exact sum therefore keeps the lower cutoff $k \sim \omega/v_t$ at all field strengths, while the quasi-static formula needs it put in.

8.11.3 The classical evaluation and Silin's formula

(1) In the classical limit, Eq. (8.80) with $\zeta = \Omega_t/\sqrt{2}kv_t = s\mu$, where $s = s_0\cos\theta$ and $s_0 = v_0/\sqrt{2}v_t$, gives $\Omega_t \,\mathrm{Im}\,\chi_0 = -(n_e/k_BT)\sqrt{\pi}\,\sqrt{2}kv_t\,\zeta^2 e^{-\zeta^2}$. The integrand of Eq. (8.132) is then k^{-3} times a function of direction, and the logarithmic integral $\int \mathrm{d}k/k$ between the two cutoffs is the Coulomb logarithm, held fixed here.

(2) The angular integral is

$$\int \mathrm{d}\Omega_k \, \zeta^2 e^{-\zeta^2} = 2\pi \int_{-1}^{1} s^2\mu^2 e^{-s^2\mu^2} \mathrm{d}\mu = \pi g(s), \qquad g(s) = 4s^2 \int_0^1 \tau^2 e^{-s^2\tau^2} \mathrm{d}\tau = \frac{\sqrt{\pi}\,\mathrm{erf}\,s}{s} - 2e^{-s^2}, \tag{8.133}$$

where the last form follows by one integration by parts. For small s, $g = \frac{4}{3}s^2 - \frac{4}{5}s^4 + \frac{2}{7}s^6 + O(s^8)$.

(3) At fixed logarithm the heating rate is proportional to $\langle g(s_0\cos\theta)\rangle_\theta$, and its weak-field value is $\frac{4}{3}s_0^2\langle\cos^2\theta\rangle = \frac{2}{3}s_0^2$. The ratio of the strong-field collision frequency to the weak-field one is therefore

$$\boxed{G(s_0) = \frac{3\,\langle g(s_0\cos\theta)\rangle_\theta}{2s_0^2}, \qquad s_0 = \frac{v_0}{\sqrt{2}\,v_t}.} \tag{8.134}$$

(4) Silin's form follows from the integral representation of g. With $\langle e^{-a\cos^2\theta}\rangle = e^{-a/2}I_0(a/2)$, from Eq. (10.32.1) [11], and $I_0' = I_1$, Eq. (10.29.3) [11],

$$\big\langle \cos^2\theta\, e^{-a\cos^2\theta}\big\rangle = -\frac{\mathrm{d}}{\mathrm{d}a}\Big[e^{-a/2}I_0\Big(\frac{a}{2}\Big)\Big] = \frac{1}{2}e^{-a/2}\Big[I_0\Big(\frac{a}{2}\Big) - I_1\Big(\frac{a}{2}\Big)\Big]. \tag{8.135}$$

With $a = s_0^2\tau^2$ and $z = s_0\tau/\sqrt{2}$,

$$\begin{aligned}\langle g(s_0\cos\theta)\rangle &= 4s_0^2\int_0^1 \tau^2\big\langle \cos^2\theta\, e^{-s_0^2\tau^2\cos^2\theta}\big\rangle \mathrm{d}\tau = \frac{4\sqrt{2}}{s_0}\,Q(r),\\ Q(r) &= \int_0^r z^2 e^{-z^2}\big[I_0(z^2) - I_1(z^2)\big]\mathrm{d}z,\end{aligned} \tag{8.136}$$

with $r = s_0/\sqrt{2} = v_0/2v_t$. Since $s_0^3 = 2\sqrt{2}r^3$,

$$\boxed{G = \frac{3\,Q(r)}{r^3}, \qquad r = \frac{v_0}{2v_t} = \frac{eE_0}{2m\omega v_t},} \tag{8.137}$$

which is Eqs. (3.10) and (3.11) [9]. Silin writes

$$\nu = \frac{32\sqrt{2\pi}N_i e_i^2 e^2}{m^2 v_T^3}\Big(\frac{m\omega v_T}{eE_0}\Big)^3 Q(r)\ln\frac{k_{\max}}{k_{\min}} \tag{8.138}$$

with $e_i = Ze$ and $N_i = n_i$. Since $m\omega v_T/eE_0 = v_t/v_0 = 1/2r$ and $32\sqrt{2\pi}/8r^3 = \frac{4}{3}\sqrt{2\pi}\cdot 3/r^3$, his ν is $\nu_T\ln(k_{\max}/k_{\min})\cdot 3Q(r)/r^3$.
(5) The limits. For small s_0, with $\langle\cos^{2n}\theta\rangle = (2n-1)!!/(2n)!!$, the series of g gives $\langle g\rangle = \frac{2}{3}s_0^2 - \frac{3}{10}s_0^4 + \frac{5}{56}s_0^6$. For large s_0 write $\langle g(s_0\cos\theta)\rangle = (2/\pi)\int_0^{s_0} g(s)(s_0^2 - s^2)^{-1/2}\mathrm{d}s$ and split g into $\sqrt{\pi}\,\mathrm{erf}(s)/s$ and $-2e^{-s^2}$. With $\int_0^S \mathrm{erf}(s)/s\,\mathrm{d}s = \ln S + \gamma/2 + \ln 2 + o(1)$, $\int_0^1[(1-x^2)^{-1/2} - 1]\mathrm{d}x/x = \ln 2$ and $\int_0^\infty 2e^{-s^2}\mathrm{d}s = \sqrt{\pi}$, proved in Appendix 8.B,

$$\langle g(s_0\cos\theta)\rangle = \frac{2}{\sqrt{\pi}\,s_0}\Big[\ln(4s_0) + \frac{\gamma}{2} - 1\Big] + o(s_0^{-1}). \tag{8.139}$$

Hence

$$\boxed{\begin{aligned}&G = 1 - \frac{9}{20}s_0^2 + \frac{15}{112}s_0^4 + O(s_0^6)\quad (s_0 \ll 1), \qquad G \simeq \frac{3}{\sqrt{\pi}s_0^3}\Big[\ln(4s_0) + \frac{\gamma}{2} - 1\Big]\quad (s_0 \gg 1),\\ &Q(r) \simeq \frac{\ln r + c_S}{2\sqrt{2\pi}}, \qquad c_S = \ln(4\sqrt{2}) + \frac{\gamma}{2} - 1 = 1.0215.\end{aligned}} \tag{8.140}$$

Silin found $Q(r) \simeq (\ln r + 1)/2\sqrt{2\pi}$, the 1 being, in his words, the result of numerical integration [9, Eq. (3.13)]. The constant is $c_S = \ln(4\sqrt{2}) + \gamma/2 - 1$.

8.11.4 The quantum strong-field result

(1) Table 8.4 compares the exact ratio $R(\beta, u)$ of Eq. (8.128) with Silin's $G(\beta/\sqrt{2})$ at $u = 0.3$. The two agree for weak fields and part for strong ones, where the quantum result is larger by a factor that grows slowly with β.
(2) The reason is the upper cutoff. In the quantum theory the momentum transfer is limited by backscattering of an electron from the moving ion, $\hbar k \lesssim 2m|\hat{\mathbf{k}}\cdot\mathbf{v}_{\rm osc}(t)|$, which grows with the field, while the lower cutoff stays at $k \sim \omega/v_t$ by step (3) of the quasi-static limit. At fixed logarithm both cutoffs are held. Figure 8.4 shows the ratio for three photon energies.

Table 8.4: The ratio $R = \nu_{\rm eff}(v_0)/\nu_{\rm eff}(0)$ of the exact quantum heating rate, Eq. (8.128) at $u = 0.3$, and Silin's ratio at fixed logarithm, Eq. (8.137), against $\beta = v_0/v_t$.

β	0.05	0.25	1	2	4	8
$R(\beta, 0.3)$	0.9996	0.9909	0.8673	0.5909	0.2102	0.0440
$G(\beta/\sqrt{2})$	0.9994	0.9861	0.8050	0.4604	0.1242	0.0224

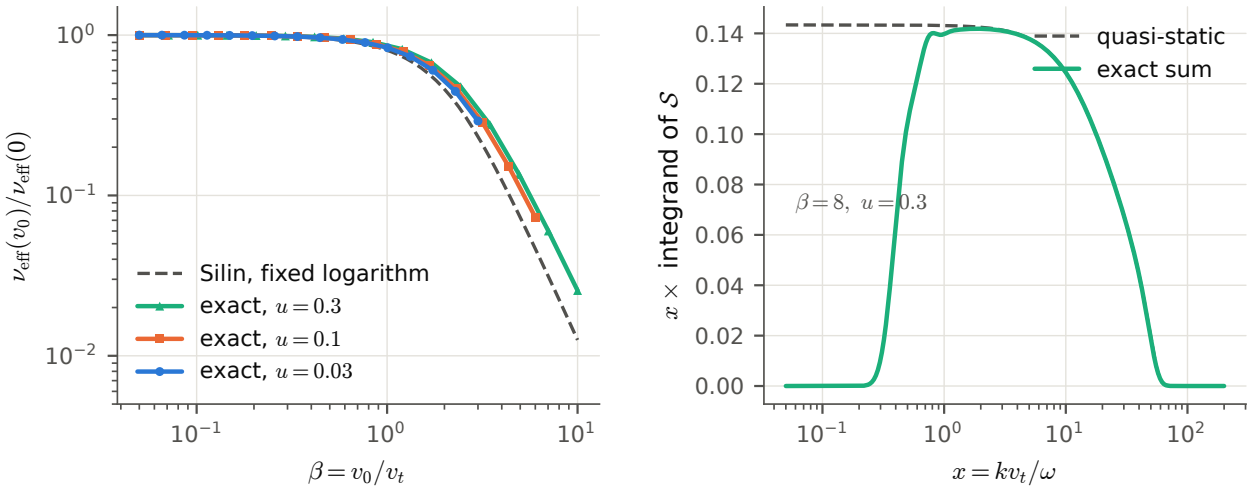


Figure 8.4: Strong fields. Left, the ratio $R = \nu_{\rm eff}(v_0)/\nu_{\rm eff}(0)$ of the exact quantum heating rate of Eq. (8.128) for three photon energies, and Silin's ratio G at fixed logarithm, Eq. (8.137). Right, the integrand of $\mathcal{S}$ times $x = kv_t/\omega$ at $\beta = 8$, $u = 0.3$, exact and in the quasi-static limit of Eq. (8.132). The exact integrand is cut off near $x \approx 0.3$ by the thermal factor, and both are cut off near $x \approx 2\beta/u$ by the quantum limit on the momentum transfer.

8.12 The kinetic equation of an isotropic distribution

This section returns to the master equation and follows the distribution of the electrons, not only its energy. Let $F(v)$ be the isotropic velocity distribution, $n_e = \int 4\pi v^2 F \, dv$, which is $2m^3 f/(2\pi\hbar)^3$ in terms of the occupation numbers.

8.12.1 The averaged master equation for weak fields

(1) For weak fields the master equation (8.69) has the terms $\ell = 0, \pm 1$. The term $\ell = 0$ connects states of equal energy and vanishes for an isotropic distribution. The field generates an anisotropy proportional to $(\hat{\mathbf{p}} \cdot \hat{\mathbf{e}})^2 - \frac{1}{3}$, which elastic collisions remove faster than the field heats, and the isotropic part obeys the master equation averaged over the direction of $\mathbf{p}$,

$$\frac{\partial F(E)}{\partial t} = R_+(E)\big[F(E+\hbar\omega) - F(E)\big] + R_-(E)\big[F(E-\hbar\omega) - F(E)\big], \tag{8.141}$$

where $R_\pm = n_i v \bar{\sigma}_{\pm 1}$ with the cross sections of Eq. (8.55),

$$\boxed{R_\pm(E) = \frac{2\pi}{3}\frac{n_i Z^2 e^4 a_0^2 m}{\hbar^2 p} L_\pm, \qquad L_\pm = \ln\frac{p + p'_\pm}{|p - p'_\pm|}, \qquad R_-(E) = 0 \ \text{ for } \ E < \hbar\omega.} \tag{8.142}$$

(2) The density of states per unit energy and volume is $g(E) = 2 \cdot 4\pi p^2 (2\pi\hbar)^{-3} \mathrm{d}p/\mathrm{d}E = 8\pi m p/(2\pi\hbar)^3$, so $g(E)R_+(E)$ is proportional to $L_+(p)$, which is symmetric in p and p'_+. Since L_- at the momentum p'_+ is the same logarithm,

$$\boxed{g(E)\,R_+(E) = g(E+\hbar\omega)\,R_-(E+\hbar\omega).} \tag{8.143}$$

The upward rate from E and the downward rate from $E+\hbar\omega$ are balanced state by state, which is detailed balance for the rates.

8.12.2 The flux form and the Fokker–Planck limit

(1) Define the net flux of electrons per unit volume across the step from E to $E+\hbar\omega$,

$$\Phi(E) = g(E) R_+(E) \big[F(E) - F(E+\hbar\omega)\big]. \tag{8.144}$$

By Eq. (8.143) the averaged master equation is

$$g(E)\,\frac{\partial F(E)}{\partial t} = \Phi(E-\hbar\omega) - \Phi(E). \tag{8.145}$$

(2) For $\hbar\omega \ll E$ expand, $\Phi(E) \simeq -g R_+ \hbar\omega\,\partial_E F$ and $\Phi(E-\hbar\omega) - \Phi(E) \simeq -\hbar\omega\,\partial_E \Phi$, so $g\,\partial_t F \simeq \partial_E[g R_+ (\hbar\omega)^2 \partial_E F]$. The product $(\hbar\omega)^2 R_+ = \frac{2\pi}{3} n_i Z^2 e^4 v_0^2 L/v$ stays finite as $\hbar\omega \to 0$, because $(\hbar\omega)^2 a_0^2 = \hbar^2 v_0^2$.
(3) Change to the velocity, with $g \propto v$ and $\partial_E = (mv)^{-1}\partial_v$,

$$\frac{\partial F}{\partial t} = \frac{2\pi n_i Z^2 e^4 v_0^2}{3m^2}\,\frac{1}{v^2}\,\frac{\partial}{\partial v}\Big[\frac{L(v)}{v}\,\frac{\partial F}{\partial v}\Big]. \tag{8.146}$$

In terms of the momentum-transfer collision frequency $\nu_m(v) = 4\pi n_i Z^2 e^4 L(v)/m^2 v^3$,

$$\boxed{\frac{\partial F}{\partial t} = \frac{1}{v^2}\,\frac{\partial}{\partial v}\Big[v^2\,\frac{v_0^2}{6}\,\nu_m(v)\,\frac{\partial F}{\partial v}\Big], \qquad \nu_m(v) = \frac{4\pi n_i Z^2 e^4}{m^2 v^3}\,L(v), \qquad L(v) = \ln\frac{2mv^2}{\hbar\omega}.} \tag{8.147}$$

This is the inverse bremsstrahlung operator used by Langdon [20]. The quantum theory supplies its logarithm, $L = \ln(4E/\hbar\omega)$, which is the limit $\hbar\omega \ll E$ of $\ln[(p+p'_+)/(p'_+ - p)]$, the ratio of the largest momentum transfer $2p$ to the smallest $\hbar\omega/v$.

8.12.3 The heating rate of an arbitrary isotropic distribution

(1) Multiply Eq. (8.147) by $4\pi v^2\, mv^2/2$, integrate, and integrate by parts once. With $C = 2\pi n_i Z^2 e^4 v_0^2/3m^2$,

$$P = \int_0^\infty 4\pi v^2 \frac{mv^2}{2}\frac{\partial F}{\partial t}\mathrm{d}v = -4\pi m C \int_0^\infty L(v)\,\frac{\partial F}{\partial v}\,\mathrm{d}v. \tag{8.148}$$

(2) For a fixed logarithm the integral is $-LF(0)$, and

$$\boxed{P = 4\pi m C\,L\,F(0).} \tag{8.149}$$

The absorption is proportional to the number of slow electrons, measured by $F(0)$. For the Maxwellian $F(0) = n_e(2\pi v_t^2)^{-3/2}$, and $P/(n_e m v_0^2/2) = \nu_T L$, which is Eq. (8.87) with $\Lambda \to L$.
(3) With the velocity-dependent logarithm and the Maxwellian, $-\partial_v F = (v/v_t^2)F$, and $y = v^2/2v_t^2$ is exponentially distributed under the weight $(v/v_t^2)e^{-v^2/2v_t^2}$. The average of $L = \ln(4y/u)$ is

$\ln(4/u) - \gamma$, which is Eq. (8.88). The Fokker–Planck limit reproduces the exact Maxwellian result to first order in u.
(4) The operator heats the slow electrons fastest, since $v^2\nu_m \propto v^{-1}$. When this heating outpaces the electron–electron collisions that restore the Maxwellian, the distribution becomes flat-topped [20]. For $F \propto \exp[-(v/v_m)^M]$ at fixed density and kinetic energy, $v_m = v_t[3\Gamma(3/M)/\Gamma(5/M)]^{1/2}$ and $F(0) = Mn_e/4\pi v_m^3\Gamma(3/M)$, so Eq. (8.149) at fixed L gives the ratio to the Maxwellian

$$\boxed{\frac{P_M}{P_2} = \frac{(2\pi)^{3/2}M}{4\pi\Gamma(3/M)}\left[\frac{\Gamma(5/M)}{3\Gamma(3/M)}\right]^{3/2} = 1,\ 0.621,\ 0.501,\ 0.446 \quad \text{for} \quad M = 2, 3, 4, 5,} \tag{8.150}$$

and 0.336 for $M \to \infty$. This is consistent with the reduction by up to a factor of two that Langdon reports [20].

8.13 Summary

In Summary: An electron in the laser field is a Volkov state, Eq. (8.29). Its matrix element with the ion potential is a sum of harmonics weighted by $J_\ell(\mathbf{k}\cdot\mathbf{a}_0)$, Eq. (8.39), and the golden rule gives the multiphoton cross section of Bunkin and Fedorov, Eq. (8.48). The operator method of the quantum Boltzmann equation paper turns these rates into the master equation (8.69) for the occupation numbers, exactly at second order in the ion potential and without Pauli blocking. Its energy moment is the heating rate, which in terms of the Lindhard function is Eq. (8.76), and in the oscillating frame of the interacting electron gas it becomes the dielectric form Eq. (8.121), with Dawson and Oberman for weak fields and Silin for strong ones.

For weak fields and Maxwell electrons the collision frequency of the Drude model is $\nu_T\Lambda(u)$ with $\Lambda(u) = [(1 - e^{-u})/u]e^{u/2}K_0(u/2)$, which is $\ln(4k_BT/e^\gamma\hbar\omega)$ for $\hbar\omega \ll k_BT$. The Coulomb logarithm needs no cutoff. It is $\pi/\sqrt{3}$ times the thermally averaged Born Gaunt factor, Eq. (8.109), Kirchhoff's law holds exactly, and the frequency-integrated Gaunt factor is $2\sqrt{3}/\pi$. The degenerate gas gives Ziman's formula, and every distribution has the same tail for $\hbar\omega \gg k_BT$. Dynamic screening lowers the logarithm by 0.10 at a quarter of the critical density and by at most 0.56 as $\omega \to \omega_p$, Table 8.3. In strong fields the classical limit is Silin's $3Q(r)/r^3$, whose constant is $\ln(4\sqrt{2}) + \gamma/2 - 1$, and the quantum result exceeds it because the largest momentum transfer grows with the quiver velocity. For an isotropic distribution the master equation reduces to Langdon's operator with the quantum logarithm $\ln(2mv^2/\hbar\omega)$, Eq. (8.147).

Three corrections to the classical treatment of [1] follow, Appendix 8.A. The free–free and the Drude absorption coefficients, Eqs. (11.31) and (11.34) [1], differ by a factor of 4.4 as printed, and they agree only when the Gaunt factor is $(\sqrt{3}/\pi)\ln\Lambda$. The current of Eq. (11.8) [1] carries the wrong sign, which Eq. (11.9) [1] does not inherit. The constant of the Coulomb logarithm of Eq. (10.136) [1] is 0.9 below the value of the cutoffs stated with it.

The theory has three limits that the next steps of this project should lift. The Born approximation needs $Ze^2/\hbar v \ll 1$, which fails for high Z and low temperature, where the exact Coulomb Gaunt factors of Sommerfeld [21] and Karzas and Latter [22] and the Elwert factor [23] replace the Born ones [24]. The random-phase response omits local-field corrections and the correlations of a strongly coupled plasma. The anisotropy of the distribution and its relaxation by electron–electron collisions, which decide when Langdon's flat-topped distribution forms, require the full kinetic equation with both collision terms.

8.A Reduction to Atzeni and Meyer-ter-Vehn

This appendix takes the equations of [1] that bear on inverse bremsstrahlung in turn, and states for each what the quantum theory gives. Table 8.5 collects the numbers, which the check script recomputes from the constants of CODATA 2018.

8.A.1 The Drude model, Eqs. (11.7) to (11.12)

Section 8.2.1 derives Eqs. (11.7), (11.9) and (11.12) [1]. The current of Eq. (11.8) [1] as printed is $e^2 n_0 \mathbf{E}_1/[i m_e(\omega + i\nu_e)]$, which equals $-\sigma\mathbf{E}_1$ with the conductivity σ of Eq. (8.4). Inserted into Eq. (11.4) [1] it would give $\varepsilon_T = 1 + \omega_p^2/\omega(\omega + i\nu)$, the wrong sign of the plasma term. Eq. (11.9) [1] has the correct sign, so the slip is confined to Eq. (11.8). The quantum theory replaces ν_e by ν_{eff} of Eq. (8.83).

8.A.2 The collision time and the Coulomb logarithm, Eqs. (10.133) and (10.136)

(1) Eq. (10.133) [1] is $\tau_e = 1/\nu_T \ln\Lambda_e$ with ν_T of Eq. (8.11). Its numerical coefficient, 1.088×10^{-11} s, agrees with the printed 1.09×10^{-11} s.
(2) Eq. (10.136) [1] prints $\ln\Lambda_e = 7.1 - 0.5\ln n_e + \ln T_e$. The text states that it is the logarithm of $\lambda_D/\lambda_{\mathrm{dB}}$ with $\lambda_{\mathrm{dB}} = \hbar/2m_e v$ and $m_e v^2/2 = 3k_B T_e/2$. At $k_B T_e = 1$ keV and $n_e = 10^{21}$ cm^{-3} these cutoffs give $\lambda_D = 7.43\times10^{-7}$ cm, $\lambda_{\mathrm{dB}} = 2.52\times10^{-10}$ cm and $\ln(\lambda_D/\lambda_{\mathrm{dB}}) = 7.99$, so the printed constant is 0.89 lower than its stated definition.
(3) The quantum theory gives the logarithm of Eq. (8.87), whose cutoffs are v_t/ω and $0.445\hbar/mv_t$, Eq. (8.89), and Eq. (8.125) with screening. Table 8.1 compares them with Eq. (10.136).

8.A.3 The critical density and Eqs. (11.30) to (11.33)

(1) Eq. (11.30) [1] is $n_c = \pi m_e c^2/e^2\lambda_L^2 = 1.115\times10^{21}/\lambda_L^2$ cm^{-3} with λ_L in μm, printed as 1.1×10^{21}.
(2) Eq. (11.33) [1] prints $\nu_{ei}/c \approx 25 Z_i/\lambda_L^2 T_e^{3/2}$ cm^{-1} for $\ln\Lambda_e \approx 8$. Equations (10.133) and (11.30) [1] give 27.3 for the same logarithm.
(3) The text estimates $\ln\Lambda_e \approx 8$ from Eq. (10.136) for $k_B T_e = 1$ keV at the critical density of $\lambda_L = 0.5$ μm, $n_c = 4.46\times10^{21}$ cm^{-3}. Eq. (10.136) gives 6.35 there and its stated cutoffs 7.24. The quantum theory gives $\Lambda(u) = 6.81$ at $u = \hbar\omega/k_B T = 0.00248$ without screening.

8.A.4 Kramers' cross section and the free–free opacity, Eqs. (10.89) to (10.93)

(1) The Maxwell average of Kramers' cross section, Eq. (10.89) [1], multiplied by $h\nu/4\pi A m_p$, is

$$\begin{aligned}\eta_\nu &= \frac{h\nu}{4\pi A m_p}\, n_e \Big(\frac{m}{2\pi k_B T}\Big)^{3/2} 4\pi \,\frac{32\pi^2 Z^2 e^6}{3\sqrt{3} m^2 c^3 h\nu} \int_{v_{\min}}^{\infty} v\, e^{-mv^2/2k_B T}\,\mathrm{d}v \\ &= \frac{16\pi}{3\sqrt{6\pi}}\,\frac{e^6 Z^2 n_e}{m^2 c^3 (k_B T/m)^{1/2} A m_p}\, e^{-h\nu/k_B T},\end{aligned} \tag{8.151}$$

which is Eq. (10.91) [1]. Kirchhoff's law with $B_\nu = 2h\nu^3 c^{-2} e^{-h\nu/k_B T}$ for the true absorption gives Eq. (10.92) [1] in its symbolic form. Its numerical form evaluates to $2.74\, Z^3\rho/A^2 T^{1/2}(h\nu)^3$ cm^2 g^{-1} against the printed 2.78, and Eq. (10.93) [1] to 1.74×10^{17} against the printed 1.76×10^{17} W g^{-1}. The two differences are the same 1.4 percent.
(2) The Gaunt factor $g \approx 2\sqrt{3}/\pi$ that the text applies to Eq. (10.93) [1] is the power-averaged Born Gaunt factor of Eq. (8.108), exactly.

8.A.5 The two absorption coefficients, Eqs. (11.31) and (11.34)

(1) By Eq. (8.112) the net absorption coefficient of Kramers' form with a Gaunt factor g is $K = (\nu_T/c)(\omega_p^2/\omega^2)(\pi/\sqrt{3})\, g\,[(1-e^{-u})/u]/n_r$. With $g = 1$ this is Eq. (11.34) [1], exactly. The Drude form Eq. (11.31) [1] is the same with $(\pi/\sqrt{3})g(1-e^{-u})/u$ replaced by $\ln\Lambda_e$. Their ratio is

$$\frac{K_{(11.31)}}{K_{(11.34)}} = \frac{\ln\Lambda_e}{(\pi/\sqrt{3})\, g\,(1-e^{-u})/u}. \tag{8.152}$$

(2) At $\lambda_L = 0.5$ μm and $k_BT = 1$ keV, $u = 0.00248$, and with $g = 1$ and $\ln\Lambda_e = 8$ the ratio is 4.42. The numerical forms give 6.19 and 27.3 $Z/\lambda_L^2 T_e^{3/2}$ cm^{-1} at $n_e = n_c$, the same ratio. The statement below Eq. (11.34) [1], that the two agree within a few percent, holds only for $g = (\sqrt{3}/\pi)\ln\Lambda_e = 4.4$. (3) The quantum theory gives both at once. The Born Gaunt factor at $u = 0.00248$ is $\langle g\rangle = 3.758$, and both forms become $\nu_T\Lambda(u)$ with $\Lambda = 6.81$.

8.A.6 The classical dielectric function, Eq. (11.19)

The classical limit of the Lindhard function of Maxwell electrons is Eq. (11.19) [1], Section 8.10.5 step (2). Its imaginary part is Eq. (8.80).

8.A.7 Bohr and Bethe, Eqs. (11.72) to (11.74)

For stopping, [1] takes the smaller of the classical logarithm of Bohr and the quantum one of Bethe, Eq. (11.74). The same choice arises here. The quantum Coulomb logarithm of Eq. (8.87) is the Bethe-type result and holds for $\eta = Ze^2/\hbar v_t \ll 1$. For $\eta \gtrsim 1$ the lower cutoff becomes the classical distance $r_0 = Ze^2/mv^2$ and the Born result overestimates the absorption, which the exact Coulomb Gaunt factors correct.

8.B Mathematical identities

8.B.1 An integral of the error function

Integrate by parts, $\int_0^S \mathrm{erf}(s)s^{-1}\mathrm{d}s = \mathrm{erf}(S)\ln S - (2/\sqrt{\pi})\int_0^S e^{-s^2}\ln s\,\mathrm{d}s$. Differentiating $\int_0^\infty s^{2a-1}e^{-s^2}\mathrm{d}s = \Gamma(a)/2$ with respect to a at $a = \frac{1}{2}$ gives $\int_0^\infty e^{-s^2}\ln s\,\mathrm{d}s = \Gamma'(\frac{1}{2})/4 = \frac{\sqrt{\pi}}{4}\psi(\frac{1}{2}) = -\frac{\sqrt{\pi}}{4}(\gamma + 2\ln 2)$, with $\psi(\frac{1}{2}) = -\gamma - 2\ln 2$ from Eq. (5.4.13) [11]. Since $1 - \mathrm{erf}\,S$ and the tail of the last integral decay faster than any power,

$$\int_0^S \frac{\mathrm{erf}\,s}{s}\,\mathrm{d}s = \ln S + \frac{\gamma}{2} + \ln 2 + o(1). \tag{8.153}$$

8.B.2 A logarithmic integral

With $x = \sin\varphi$,

$$\int_0^1 \Big[\frac{1}{(1-x^2)^{1/2}} - 1\Big]\frac{\mathrm{d}x}{x} = \int_0^{\pi/2}\frac{1-\cos\varphi}{\sin\varphi}\,\mathrm{d}\varphi = \int_0^{\pi/2}\tan\frac{\varphi}{2}\,\mathrm{d}\varphi = \Big[-2\ln\cos\frac{\varphi}{2}\Big]_0^{\pi/2} = \ln 2. \tag{8.154}$$

Table 8.5: Equations of [1] on inverse bremsstrahlung, as printed and as recomputed or derived here.

Equation	Printed	Recomputed or derived	Status
(11.8)	$e^2 n_0 \mathbf{E}_1/[im_e(\omega + i\nu_e)]$	$ie^2 n_0 \mathbf{E}_1/[m_e(\omega + i\nu_e)]$	sign slip
(11.9), (11.12)	Drude ε_T and K	Eqs. (8.5), (8.10)	confirmed
(10.133)	1.09×10^{-11} s	1.088×10^{-11} s	confirmed
(10.136)	constant 7.1	7.99 from its cutoffs	constant low by 0.89
(11.30)	$1.1 \times 10^{21}/\lambda_L^2$	$1.115 \times 10^{21}/\lambda_L^2$	confirmed
(11.33)	25	27.3	9 percent low
text, $\ln \Lambda_e \approx 8$	from (10.136)	6.35 from (10.136), 6.81 quantum	inconsistent
(10.91)	symbolic form	Maxwell average of (10.89)	confirmed
(10.92)	2.78	2.74	1.4 percent
(10.93)	1.76×10^{17}	1.74×10^{17}	1.4 percent
$g \approx 2\sqrt{3}/\pi$	quoted for (10.93)	Born power average, Eq. (8.108)	exact
(11.31) vs (11.34)	agree within a few percent	ratio 4.42	needs $g = 4.4$
(11.19)	classical ε_L	classical limit of Lindhard	confirmed

8.B.3 The large-field limit of the phase average

(1) With $s = s_0 \sin\theta'$ and the symmetry of the average, $\langle g(s_0 \cos\theta)\rangle = (2/\pi)\int_0^{s_0} g(s)(s_0^2 - s^2)^{-1/2}\mathrm{d}s$.
(2) For the part $\sqrt{\pi}\,\mathrm{erf}(s)/s$ write $(s_0^2 - s^2)^{-1/2} = s_0^{-1}\{1 + [(1 - s^2/s_0^2)^{-1/2} - 1]\}$. The first term gives $s_0^{-1}\sqrt{\pi}[\ln s_0 + \gamma/2 + \ln 2]$ by Eq. (8.153). In the second term $\mathrm{erf}(s) \to 1$ wherever the bracket is not small, and Eq. (8.154) gives $s_0^{-1}\sqrt{\pi}\ln 2$.
(3) The part $-2e^{-s^2}$ is confined to $s \lesssim 1$, where $(s_0^2 - s^2)^{-1/2} \simeq s_0^{-1}$, and gives $-s_0^{-1}\sqrt{\pi}$.
(4) Adding, $\langle g\rangle = (2/\pi s_0)\sqrt{\pi}[\ln s_0 + \gamma/2 + 2\ln 2 - 1] = (2/\sqrt{\pi}s_0)[\ln(4s_0) + \gamma/2 - 1]$, which is the form used in Eq. (8.140).

8.B.4 The phase average of a Gaussian

With $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$ and Eq. (10.32.1) [11], $I_0(z) = \pi^{-1}\int_0^\pi e^{\pm z\cos\phi}\mathrm{d}\phi$,

$$\left\langle e^{-a\cos^2\theta}\right\rangle_\theta = e^{-a/2}\left\langle e^{-(a/2)\cos 2\theta}\right\rangle_\theta = e^{-a/2} I_0\Big(\frac{a}{2}\Big). \tag{8.155}$$

8.C The numerical checks

A check script verifies every boxed result of these notes, in their order, by an independent numerical route. The routes are quadrature of the defining integrals in place of the closed forms, finite differences in place of derivatives, the golden-rule time integral at finite times, the cross-section route against the Lindhard route, an exact computation in the Fock space of three fermion modes for the equation of motion, and the published equations of [3, 9, 1] as printed. It runs with `numpy` and `scipy` under Python 3 in about fifteen seconds. Table 8.6 is written by the script itself. A

quantitative check passes when its deviation is below the tolerance stated in the script, which reflects the precision of the quadrature or of the asymptotic expansion it tests. A printed number passes when the script reproduces it to its last printed digit. The figures are made by a second script.

Table 8.6: The numerical checks of these notes, by the equation, table or section they verify. The second column counts all checks of the row. The third gives the largest deviation among its quantitative checks, relative, or absolute for quantities near zero, and is empty when the row has none. The fourth counts the numbers printed in these notes that the script reproduces to their last printed digit, and the fifth the logical conditions, such as a sign or a monotone decrease.

Equation	Checks	Largest deviation	Printed	Logical
Eq. (8.4)	2	1.3×10^{-16}		
Eq. (8.5)	2	2.9×10^{-16}		
Eq. (8.6)	1	6.4×10^{-16}		
Eq. (8.8)	2	2.9×10^{-11}		
Eq. (8.10)	3	1.6×10^{-7}		
Eq. (8.11)	1		1	
Eq. (8.16)	3	1.6×10^{-12}		
Eq. (8.17)	2	0		
Eq. (8.24)	2	1.9×10^{-16}		
Eq. (8.20)	3	1.8×10^{-16}		
Eq. (8.29)	2	5.0×10^{-13}		
Eq. (8.31)	3	4.6×10^{-16}		
Eq. (8.33)	9	5.6×10^{-16}		
Eq. (8.35)	6	8.5×10^{-15}		
Eq. (8.34)	2	2.5×10^{-7}		
Eq. (8.39)	2	1.2×10^{-16}		
Eq. (8.44)	2	1.1×10^{-3}		
Eq. (8.46)	1	1.4×10^{-3}		
Eq. (8.48)	2	8.6×10^{-8}		
Eq. (8.49)	2	4.0×10^{-16}		
Eq. (8.54)	3	6.1×10^{-15}		
Eq. (8.55)	3	2.0×10^{-15}		
Eq. (8.56)	15	1.5×10^{-6}		1
Eq. (8.58)	2	7.9×10^{-6}		1
Eq. (8.62)	2	0		1
Eq. (8.64)	1	1.8×10^{-16}		
Eq. (8.68)	2	1.3×10^{-11}		
Eq. (8.66)	1	3.0×10^{-13}		
Eq. (8.70)	2	3.6×10^{-16}		
Eq. (8.69)	1	1.4×10^{-14}		
Eq. (8.72)	2	1.5×10^{-15}		1
Eq. (8.77)	3	3.7×10^{-10}		
Eq. (8.79)	3	3.3×10^{-15}		
Eq. (8.78)	6	2.3×10^{-14}		
Eq. (8.80)	4	7.0×10^{-8}		2

Equation	Checks	Largest deviation	Printed	Logical
Eq. (8.86)	2	1.7×10^{-15}		
Eq. (8.87)	10	6.8×10^{-12}		
Eq. (8.88)	4	2.5×10^{-3}		2
Eq. (8.89)	4		4	
Table 8.2	36		36	
Eq. (8.90)	5	4.5×10^{-6}		
Eq. (8.94)	8	2.4×10^{-7}		
Eq. (8.95)	6	5.0×10^{-4}		2
Eq. (8.98)	5	8.4×10^{-5}		
Eq. (8.100)	1	1.1×10^{-16}		
Eq. (8.102)	1	1.7×10^{-15}		
Eq. (8.103)	3	2.4×10^{-12}		
Eq. (8.104)	3	2.2×10^{-16}		
Eq. (8.107)	4	3.5×10^{-12}		
Eq. (8.109)	4	5.2×10^{-15}		
Eq. (8.108)	3	1.1×10^{-13}	1	
Eq. (8.112)	2	6.9×10^{-16}		
Eq. (8.116)	1	6.6×10^{-16}		
Eq. (8.123)	5	1.5×10^{-11}		
Eq. (8.121)	4	1.1×10^{-15}		
Eq. (8.125)	8	1.3×10^{-3}		
Eq. (8.122)	1	3.1×10^{-15}		
Table 8.3	26		26	
Eq. (8.130)	12	9.6×10^{-16}		
Eq. (8.128)	4	1.5×10^{-5}		
Table 8.4	12		12	
Eq. (8.132)	2	3.9×10^{-5}		1
Eq. (8.133)	7	8.7×10^{-10}		
Eq. (8.137)	8	2.4×10^{-14}		
Eq. (8.134)	1	4.5×10^{-7}		
Eq. (8.140)	6	1.0×10^{-4}	1	
Eq. (8.143)	3	0		
Eq. (8.142)	3	0		
Eq. (8.147)	4	2.8×10^{-8}		2
Eq. (8.149)	2	2.2×10^{-13}		
Eq. (8.150)	7	3.0×10^{-16}	5	
Table 8.5	15	3.5×10^{-8}	13	
Sec. 8.A.2	4		4	
Sec. 8.A.5	7	3.3×10^{-16}	5	
Table 8.1	36		36	
Sec. 8.3.4	2		2	
Sec. 8.8	2		2	
Eq. (8.153)	2	1.8×10^{-16}		
Eq. (8.154)	1	4.5×10^{-14}		
Sec. 8.B	6	7.1×10^{-6}		

Equation	Checks	Largest deviation	Printed	Logical
Total	394		148	13

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Chapter 9

Detailed derivation of free–free absorption by electron–atom collisions at terahertz frequencies

The note of October 20, 2026.

These notes derive, with every intermediate result written out, the rate at which an electromagnetic wave of terahertz frequency is absorbed by free electrons that collide with neutral atoms. They ask whether the classical absorption kernel, the Lorentz-gas conductivity built on the momentum-transfer cross section, is accurate at the terahertz frequencies proposed for reheating the electrons of an electron-beam-pumped KrF laser gas. The electron is described by exact scattering states of a local potential that represents the atom, the field is treated to first order, and the rate follows from the golden rule. Every boxed result is checked numerically, as Appendix 9.C describes.

For notation, Gaussian units are used. The electron has charge $-e$ and mass m, and the atom is infinitely heavy and sits at the origin. A state of momentum $\hbar\mathbf{k}$ has energy $E = \hbar^2k^2/2m$ and speed $v = \hbar k/m$. The wave is uniform and linearly polarized along the unit vector $\hat{\mathbf{e}}$, with amplitude E_0 and angular frequency ω, and $f = \omega/2\pi$ is its frequency. The absorption of one photon takes the electron from $\hbar\mathbf{k}$ to $\hbar\mathbf{k}'$ with $k'^2 = k^2 + 2m\omega/\hbar$. The temperature of a Maxwell distribution is T and $u = \hbar\omega/k_BT$. The momentum-transfer cross section is $\sigma_{\mathrm{m}}(E)$, and a cross section with a subscript $+$ or $-$ is the free–free cross section for absorption or stimulated emission per unit photon flux, defined in Sec. 9.3. The numbers are computed in atomic units, $\hbar = m = e = 1$, with the bohr a_0 and the hartree $E_h = 27.211$ eV. A boxed equation is a result that is checked.

9.1 Why the atoms matter

(1) Inverse bremsstrahlung, the absorption of a wave by electrons that scatter on ions, heats laser plasmas [1]. In the gas of an electron-beam-pumped KrF laser the electron density is 10^{19} to 2.7×10^{21} m^{-3} against an atom density of 2.4×10^{25} to 4.9×10^{25} m^{-3}, so that there are 9×10^3 to 5×10^6 atoms for every ion. The Coulomb cross section is larger by orders of magnitude, however, and Sec. 9.10 finds the regime in which each kind of collision dominates the absorption. Away from the coldest and densest afterglow the atoms win.
(2) Free–free absorption on neutral atoms has been computed from phase shifts for the negative hydrogen ion [11] and for other atoms [12], and with the multichannel R-matrix method for argon

[13]. Its inverse, the neutral bremsstrahlung, has been measured [14, 15] and explains part of the electroluminescence of two-phase argon detectors [16].
(3) Proposals to reheat the electrons of the laser gas with a terahertz wave compute its absorption length with the classical kernel of the Lorentz gas. That kernel is the conductivity of electrons that scatter off heavy atoms with the energy dependent collision frequency $\nu(v) = N\sigma_{\rm m}(v)v$, derived in Sec. 9.5.3. It treats the electron as a classical particle that absorbs energy continuously. Two quantum effects are missing from it. The first is the ratio $u = \hbar\omega/k_BT$, which is 0.16 at 1 THz and 4.5 at the 28.3 THz of a CO_2 laser for electrons at the gas temperature of 300 K. The second is the structure of $\sigma_{\rm m}(E)$. Argon and krypton have a Ramsauer–Townsend window in which $\sigma_{\rm m}$ falls from its value at 1 meV by a factor of 82 in argon and 330 in krypton, to minima at 0.25 and 0.51 eV in the LXCat sets, and a photon of 28.3 THz carries 0.117 eV. Table 9.1 lists the photon energy and u at the frequencies used below.

Table 9.1: Photon energy $\hbar\omega$, the ratio $u = \hbar\omega/k_BT$ and the stimulated-emission factor $(1-e^{-u})/u$ at 300 K, and u for electrons at 0.63 eV.

f (THz)	$\hbar\omega$ (meV)	u, 300 K	$(1-e^{-u})/u$, 300 K	u, 0.63 eV
0.3	1.24	0.048	0.976	0.00197
1	4.14	0.16	0.924	0.00656
3	12.4	0.48	0.794	0.0197
10	41.4	1.6	0.499	0.0656
28.3	117	4.53	0.218	0.186

(4) The question is the error of the classical kernel at 3 to 28 THz in two fills of the laser gas, $Ne/Kr/F_2 = 0.948/0.05/0.002$ at 2 atm and $Ar/Kr/F_2 = 0.597/0.40/0.003$ at 1 atm. The kernel is good enough if the error is below 5 % in the band of 0.3 to 4 THz proposed for reheating. Sections 9.2 to 9.4 derive the quantum free–free cross section of one electron and one atom exactly within a one-body model. Section 9.5 finds its soft-photon and classical limits, and Sec. 9.6 two limits in which it is known in closed form at any photon energy. Sections 9.8 and 9.9 evaluate it for model potentials fitted to the measured cross sections of neon, argon and krypton. The answer is in Sec. 9.9 and in the Summary.

9.2 The Hamiltonian

9.2.1 One electron and one atom

(1) The atom is represented by a local central potential $V(r)$ that acts on the scattering electron,

$$H_0 = \frac{p^2}{2m} + V(r), \qquad V(r) \to -\frac{\alpha_d e^2}{2r^4} \quad (r \to \infty), \tag{9.1}$$

where α_d is the static dipole polarizability of the atom. At short range V contains the screened attraction of the nucleus and the repulsion that the Pauli principle produces. In the one-body model the latter appears because the occupied orbitals of the atom are bound states of the same potential, so that the scattering states are orthogonal to them. The models of Sec. 9.8 are of this kind.

(2) The atom is infinitely heavy. Its recoil changes the electron energy by a fraction of order m/M, which is 1.4×10^{-5} for argon, and is neglected. Exchange between the scattering electron and the atomic electrons, and the excitation of the atom, are outside the model. The second needs at least 11.5 eV in argon and 9.9 eV in krypton and is closed for the electrons that absorb terahertz photons.

9.2.2 The wave

(1) The wave has the field $\mathbf{E}(t) = E_0 \hat{\mathbf{e}} \cos \omega t$. At 1 THz its wavelength is 0.3 mm and at 28.3 THz it is 10.6 μm. A collision takes place within a few hundred a_0 of the atom, so the field is uniform over it and the dipole approximation holds. In the length gauge the potential energy of the charge $-e$ in the field is $e\mathbf{E}(t) \cdot \mathbf{r}$, and

$$H(t) = H_0 + eE_0 \cos(\omega t)\, \hat{\mathbf{e}} \cdot \mathbf{r} = H_0 + \frac{eE_0}{2}\, \hat{\mathbf{e}} \cdot \mathbf{r} \left(e^{-i\omega t} + e^{i\omega t}\right). \tag{9.2}$$

The term with $e^{-i\omega t}$ absorbs a photon and the term with $e^{i\omega t}$ emits one.
(2) The field is weak when the quiver amplitude $a = eE_0/m\omega^2$ is small against the wavelength of the electron. The criterion and its numbers are in Sec. 9.7. In the weak field the golden rule to first order in E_0 describes the absorption of one photon.

9.2.3 The response of the atom to the wave

(1) The wave also polarizes the atom. Its induced dipole is $\alpha_d \mathbf{E}(t)$, since $\hbar\omega$ is far below the excitation energies of the atom. The electrostatic potential of a dipole $\mathbf{d}$ at the origin is $\mathbf{d} \cdot \mathbf{r}/r^3$, so the electron acquires the potential energy $-e\alpha_d \mathbf{E}(t) \cdot \mathbf{r}/r^3$. Together with Eq. (9.2) the interaction with the wave becomes

$$e\mathbf{E}(t) \cdot \mathbf{r} \Big(1 - \frac{\alpha_d}{r^3}\Big). \tag{9.3}$$

The second term is the polarization bremsstrahlung of the atom, the photon absorbed through the dipole of the atom rather than through the electron. Inside the atom the field of the dipole is screened, so the term is cut off there. Section 9.8.4 computes its effect with the same cut-off as the polarization potential, and finds it below 0.7 % up to 3 THz. Until then it is left out.

9.3 The golden-rule cross section between exact scattering states

9.3.1 The scattering states

(1) The scattering states of H_0 with energy $E = \hbar^2 k^2/2m$ are normalized to $\langle \psi_{\mathbf{k}} | \psi_{\mathbf{k}'} \rangle = \delta^3(\mathbf{k} - \mathbf{k}')$. In partial waves,

$$\psi_{\mathbf{k}}^{(\pm)}(\mathbf{r}) = (2\pi)^{-3/2} \sum_{l=0}^{\infty} (2l+1)\, i^l e^{\pm i\delta_l}\, \frac{u_{kl}(r)}{kr}\, P_l(\hat{\mathbf{k}} \cdot \hat{\mathbf{r}}), \tag{9.4}$$

where u_{kl} is the real regular solution of the radial equation

$$u_{kl}'' + \Big[k^2 - U(r) - \frac{l(l+1)}{r^2}\Big] u_{kl} = 0, \qquad U(r) = \frac{2mV(r)}{\hbar^2}, \tag{9.5}$$

normalized to unit amplitude, $u_{kl} \to \sin(kr - l\pi/2 + \delta_l)$ as $r \to \infty$, and $\delta_l(k)$ is the phase shift. For $V = 0$ the radial function is the Riccati–Bessel function $\hat{\jmath}_l(kr) = kr\, j_l(kr)$, and Eq. (9.4) is the expansion of the plane wave $(2\pi)^{-3/2} e^{i\mathbf{k}\cdot\mathbf{r}}$. The state $\psi^{(+)}$ has an outgoing scattered wave and is

the initial state. The state $\psi^{(-)}$, with an incoming scattered wave, is the final state of an electron detected in the direction $\hat{\mathbf{k}}'$.
(2) With $P_l(\hat{\mathbf{k}}\cdot\hat{\mathbf{r}}) = \frac{4\pi}{2l+1}\sum_m Y^*_{lm}(\hat{\mathbf{k}})Y_{lm}(\hat{\mathbf{r}})$,

$$\psi^{(\pm)}_{\mathbf{k}}(\mathbf{r}) = (2\pi)^{-3/2}\,\frac{4\pi}{kr}\sum_{lm} i^l e^{\pm i\delta_l} u_{kl}(r)\,Y^*_{lm}(\hat{\mathbf{k}})\,Y_{lm}(\hat{\mathbf{r}}). \tag{9.6}$$

9.3.2 The matrix element

(1) The absorption term of Eq. (9.2) has the matrix element $(eE_0/2)M$ with $M = \langle\psi^{(-)}_{\mathbf{k}'}|\hat{\mathbf{e}}\cdot\mathbf{r}|\psi^{(+)}_{\mathbf{k}}\rangle$. The complex conjugate of Eq. (9.6) for the final state has the factors $(-i)^{l'}e^{+i\delta'_{l'}}Y_{l'm'}(\hat{\mathbf{k}}')Y^*_{l'm'}(\hat{\mathbf{r}})$, with $\delta'_{l'} = \delta_{l'}(k')$. The factor r^2 of d^3r cancels the two factors $1/r$, and

$$M = \frac{(4\pi)^2}{(2\pi)^3 kk'}\sum_{lm,l'm'}(-i)^{l'}i^l e^{i(\delta_l+\delta'_{l'})}\,Y_{l'm'}(\hat{\mathbf{k}}')Y^*_{lm}(\hat{\mathbf{k}})\,\langle l'm'|\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}|lm\rangle\,R_{l\to l'}, \tag{9.7}$$

with the angular matrix element $\langle l'm'|\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}|lm\rangle = \int Y^*_{l'm'}\,\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}\,Y_{lm}\,\mathrm{d}\Omega$ and the radial integral

$$R_{l\to l'} = \int_0^\infty u_{k'l'}(r)\,r\,u_{kl}(r)\,\mathrm{d}r. \tag{9.8}$$

The integrand grows like r at large r, and $R_{l\to l'}$ is defined with the convergence factor $e^{-\eta r}$ and $\eta\to 0^+$. Section 9.3.5 replaces it by an absolutely convergent integral.
(2) The angular matrix element vanishes unless $l' = l\pm 1$, the dipole selection rule. Appendix 9.A sums its square over m and m',

$$\sum_{mm'}|\langle l+1,m'|\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}|lm\rangle|^2 = \frac{l+1}{3}, \qquad \sum_{mm'}|\langle l-1,m'|\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}|lm\rangle|^2 = \frac{l}{3}. \tag{9.9}$$

9.3.3 The rate and its average over directions

(1) The golden rule gives the rate of absorption from the state $\psi^{(+)}_{\mathbf{k}}$ into all final directions,

$$W_+ = \frac{2\pi}{\hbar}\Big(\frac{eE_0}{2}\Big)^2\int \mathrm{d}^3k'\,|M|^2\,\delta\Big(\frac{\hbar^2k'^2}{2m} - \frac{\hbar^2k^2}{2m} - \hbar\omega\Big). \tag{9.10}$$

(2) The electrons of a gas move in all directions, so W_+ is averaged over $\hat{\mathbf{k}}$. The integral over $\hat{\mathbf{k}}'$ and the average over $\hat{\mathbf{k}}$ use the orthonormality of the Y_{lm}, $\int Y_{l'm'}(\hat{\mathbf{k}}')Y^*_{l''m''}(\hat{\mathbf{k}}')\mathrm{d}\Omega' = \delta_{l'l''}\delta_{m'm''}$ and $\frac{1}{4\pi}\int Y^*_{lm}(\hat{\mathbf{k}})Y_{\tilde{l}\tilde{m}}(\hat{\mathbf{k}})\mathrm{d}\Omega = \delta_{l\tilde{l}}\delta_{m\tilde{m}}/4\pi$. The cross terms between different pairs (l,l') vanish, the phases drop out, and with Eq. (9.9)

$$\frac{1}{4\pi}\int\mathrm{d}\Omega\int\mathrm{d}\Omega'\,|M|^2 = \frac{(4\pi)^4}{(2\pi)^6k^2k'^2}\,\frac{1}{4\pi}\,\frac{S}{3} = \frac{S}{3\pi^3k^2k'^2}, \tag{9.11}$$

$$S = \sum_{l=0}^{\infty}\Big[(l+1)\,R^2_{l\to l+1} + l\,R^2_{l\to l-1}\Big]. \tag{9.12}$$

(3) The energy delta function contributes $\int k'^2\mathrm{d}k'\,\delta(\hbar^2k'^2/2m - E - \hbar\omega) = mk'/\hbar^2$. The averaged rate is

$$\langle W_+\rangle = \frac{2\pi}{\hbar}\,\frac{e^2E_0^2}{4}\,\frac{mk'}{\hbar^2}\,\frac{S}{3\pi^3k^2k'^2} = \frac{e^2E_0^2\,m\,S}{6\pi^2\hbar^3k^2k'}. \tag{9.13}$$

9.3.4 The cross section per photon flux

(1) The state $\psi^{(+)}_{\mathbf{k}}$ carries the incident flux $(2\pi)^{-3}\hbar k/m$ of electrons, and the wave carries the flux $\Phi = I/\hbar\omega$ of photons with $I = cE_0^2/8\pi$. The free–free cross section for absorption is the rate divided by both fluxes, $\sigma_+ = \langle W_+\rangle/[(2\pi)^{-3}(\hbar k/m)\Phi]$. With $(2\pi)^3 \cdot 8\pi/6\pi^2 = 32\pi^2/3$,

$$\boxed{\sigma_+(k \to k') = \frac{32\pi^2}{3}\,\frac{e^2 m^2 \omega}{\hbar^3 c}\,\frac{S}{k^3 k'}, \qquad \hbar k' = (\hbar^2 k^2 + 2m\hbar\omega)^{1/2}.} \tag{9.14}$$

It has the dimension length$^4\times$time. An electron of speed v among N atoms per unit volume, in the photon flux Φ, absorbs at the rate $N\Phi v\sigma_+$.
(2) In the first Born approximation the radial functions are $\hat{j}_l$ and Eq. (9.14) must equal the one-photon limit of the cross section of Bunkin and Fedorov [4], with $J_1^2(x) \simeq x^2/4$, divided by Φ. The check script compares the two for a Gaussian potential and finds agreement to 10^{-14}, which tests the factor $32\pi^2/3$ and the angular sums together.

9.3.5 Length, velocity and acceleration forms

(1) For eigenstates of H_0 the commutators $[H_0, \mathbf{r}] = -i\hbar\mathbf{p}/m$ and $[H_0, \mathbf{p}] = i\hbar\nabla V$ give $[H_0, [H_0, \mathbf{r}]] = (\hbar^2/m)\nabla V$. Between states that differ in energy by $\hbar\omega$ the double commutator is $(\hbar\omega)^2$ times the matrix element of $\mathbf{r}$, so

$$\langle f|\mathbf{r}|i\rangle = \frac{-i}{m\omega}\langle f|\mathbf{p}|i\rangle = \frac{1}{m\omega^2}\langle f|\nabla V|i\rangle. \tag{9.15}$$

For scattering states the boundary terms at infinity oscillate and vanish with the convergence factor $e^{-\eta r}$, so the identities hold for the regularized integrals of Eq. (9.8).
(2) Since $\nabla V = V'(r)\hat{\mathbf{r}}$ has the same angular structure as $\mathbf{r} = r\hat{\mathbf{r}}$, the identity holds partial wave by partial wave,

$$\boxed{R_{l\to l'} = \frac{1}{m\omega^2}\int_0^\infty u_{k'l'}(r)\,V'(r)\,u_{kl}(r)\,\mathrm{d}r.} \tag{9.16}$$

The acceleration form converges absolutely when V' falls faster than $1/r$, as the r^{-5} of the polarization tail does, and the numerics use it. The form makes plain that the photon is absorbed where the electron is accelerated, near the atom. The check script verifies Eq. (9.16) against the closed form of the length integral in the zero-range limit of Sec. 9.6.1.

9.4 Emission, detailed balance and the absorption coefficient

9.4.1 Stimulated emission and detailed balance

(1) The term with $e^{i\omega t}$ in Eq. (9.2) takes an electron from $\hbar\mathbf{k}'$ to $\hbar\mathbf{k}$ and adds a photon to the wave. The same steps give its cross section per photon flux, with k and k' exchanged in the prefactor, $\sigma_-(k' \to k) = (32\pi^2 e^2 m^2\omega/3\hbar^3 c)\,S'/(k'^3 k)$. Here S' has the radial integrals $R'_{l\to l'} = \int u_{kl'}\,r\,u_{k'l}\,\mathrm{d}r = R_{l'\to l}$. Shifting the summation index in each of the two sums of Eq. (9.12) shows $S' = S$. Hence

$$\boxed{k^2\,\sigma_+(k \to k') = k'^2\,\sigma_-(k' \to k).} \tag{9.17}$$

This is the principle of detailed balance for free–free transitions.

9.4.2 The absorption coefficient

(1) Let $g(E)\mathrm{d}E$ be the fraction of electrons with energy in $\mathrm{d}E$, for an isotropic distribution normalized to $\int g\,\mathrm{d}E = 1$, and let $\sigma_-(E)$ denote the emission cross section from E to $E-\hbar\omega$, zero for $E < \hbar\omega$. The wave loses photons at the rate $n_e N\Phi \int g\,v(\sigma_+ - \sigma_-)\mathrm{d}E$ per unit volume, so its intensity decays with the coefficient

$$\boxed{\kappa = n_e N \int_0^\infty g(E)\,v(E)\big[\sigma_+(E) - \sigma_-(E)\big]\mathrm{d}E.} \tag{9.18}$$

(2) Shift the emission integral to $E = E'' + \hbar\omega$ and use Eq. (9.17), $\sigma_-(E''+\hbar\omega) = [E''/(E''+\hbar\omega)]\,\sigma_+(E'')$. For the Maxwell distribution $g(E) = 2\pi^{-1/2}(k_BT)^{-3/2}E^{1/2}e^{-E/k_BT}$,

$$g(E''+\hbar\omega)\,v(E''+\hbar\omega)\,\frac{E''}{E''+\hbar\omega} = e^{-u}\,g(E'')\,v(E''), \tag{9.19}$$

since $g\,v$ is proportional to $E\,e^{-E/k_BT}$. Therefore

$$\boxed{\kappa = n_e N\,(1-e^{-u})\int_0^\infty g(E)\,v(E)\,\sigma_+(E)\,\mathrm{d}E \qquad \text{(Maxwell electrons).}} \tag{9.20}$$

Stimulated emission reduces the absorption by the factor $1-e^{-u}$, as it does for ions [3].

9.4.3 The kinetic equation and its Fokker–Planck limit

(1) Let $n(E)\mathrm{d}E$ be the number of electrons per unit volume in $\mathrm{d}E$, and write $n = n_e E^{1/2}F(E)$ with $\int E^{1/2}F\,\mathrm{d}E = 1$, the normalization of the two-term Boltzmann codes [28]. An electron at E climbs to $E+\hbar\omega$ at the rate $W_+(E) = N\Phi v\sigma_+(E)$ and falls to $E-\hbar\omega$ at the rate $W_-(E) = N\Phi v\sigma_-(E)$. The master equation for the distribution is

$$\frac{\partial n(E)}{\partial t} = W_+(E-\hbar\omega)n(E-\hbar\omega) + W_-(E+\hbar\omega)n(E+\hbar\omega) - \big[W_+(E)+W_-(E)\big]n(E). \tag{9.21}$$

(2) Detailed balance, Eq. (9.17), makes the two rates that connect E and $E+\hbar\omega$ equal after weighting with $E^{1/2}$,

$$W_+(E)\,E^{1/2} = W_-(E+\hbar\omega)\,(E+\hbar\omega)^{1/2} = N\Phi\Big(\frac{2}{m}\Big)^{1/2} E\,\sigma_+(E) \equiv \mathcal{Q}\Big(E+\frac{\hbar\omega}{2}\Big). \tag{9.22}$$

Dividing Eq. (9.21) by n_e gives the jump operator in conservation form,

$$\boxed{\frac{\partial(E^{1/2}F)}{\partial t} = \mathcal{Q}\Big(E+\frac{\hbar\omega}{2}\Big)\big[F(E+\hbar\omega)-F(E)\big] - \mathcal{Q}\Big(E-\frac{\hbar\omega}{2}\Big)\big[F(E)-F(E-\hbar\omega)\big].} \tag{9.23}$$

The right side is a difference of two fluxes through the energies $E\pm\hbar\omega/2$, so the operator conserves the number of electrons.

(3) When $\hbar\omega$ is small against the scale on which F and $\mathcal{Q}$ vary, each bracket is $\hbar\omega\,\partial F/\partial E$ at the midpoint, and the difference of the two fluxes is $\hbar\omega$ times their derivative,

$$\frac{\partial(E^{1/2}F)}{\partial t} = \frac{\partial}{\partial E}\Big[\mathcal{D}(E)\,\frac{\partial F}{\partial E}\Big] + O\big((\hbar\omega)^4\big), \qquad \mathcal{D} = (\hbar\omega)^2\mathcal{Q}(E). \tag{9.24}$$

With the soft-photon cross section of Sec. 9.5, $\sigma_+ = 4\pi e^2 v^2 \sigma_\mathrm{m}/3c\hbar\omega^3$, and $\Phi = cE_0^2/8\pi\hbar\omega$,

$$\boxed{\mathcal{D}(E) = \frac{2}{3}\,\frac{e^2 E_\mathrm{rms}^2}{m\omega^2}\,N\Big(\frac{2}{m}\Big)^{1/2} E^2\,\sigma_\mathrm{m}(E), \qquad E_\mathrm{rms}^2 = \frac{E_0^2}{2}.} \tag{9.25}$$

This is the field term of the two-term Boltzmann equation in a high-frequency field [28] in the limit $\omega \gg \nu$. There the field diffusion is $\propto E_\mathrm{rms}^2\,\epsilon\,\sigma/[\sigma^2 + (\omega/N)^2/\gamma^2\epsilon]$ and tends to $E_\mathrm{rms}^2\epsilon^2\sigma/\omega^2$, the dependence $E^2\sigma_\mathrm{m}$ of Eq. (9.25). The check script confirms that both give the same heating rate. Equation (9.23) is therefore the form that a two-term code should use when $\hbar\omega$ is not small against the energies of the electrons.

9.5 The soft-photon limit

9.5.1 The force identity

(1) As $\omega \to 0$ the final state approaches the initial one, and Eq. (9.16) needs the integral of V' between two states of the same energy and neighbouring l. It is fixed by the phase shifts alone. Write $u_l = u_{kl}$, $u_{l+1} = u_{k,l+1}$ and $A_l = l(l+1)/r^2$, so that $u_l'' = (A_l + U - k^2)u_l$ by Eq. (9.5), and define

$$P = u_{l+1}u_l, \qquad h = u_{l+1}'u_l', \qquad g = u_{l+1}'u_l - u_{l+1}u_l'. \tag{9.26}$$

(2) The radial equations give $h' = (U-k^2)P' + A_{l+1}u_{l+1}u_l' + A_l u_{l+1}'u_l$. With $(U-k^2)P' = [(U-k^2)P]' - U'P$ this is

$$U'P = \big[(U-k^2)P\big]' - h' + A_{l+1}u_{l+1}u_l' + A_l u_{l+1}'u_l. \tag{9.27}$$

(3) The two centrifugal terms are $\frac{l+1}{r^2}\big[(l+2)u_{l+1}u_l' + l\,u_{l+1}'u_l\big] = \frac{l+1}{r^2}\big[(l+1)P' - g\big]$. The equations for u_l and u_{l+1} also give $g' = (A_{l+1} - A_l)P = 2(l+1)P/r^2$. Hence

$$\frac{(l+1)^2P'}{r^2} = \Big[\frac{(l+1)^2P}{r^2}\Big]' + \frac{2(l+1)^2P}{r^3} = \Big[\frac{(l+1)^2P}{r^2}\Big]' + \frac{(l+1)g'}{r}, \tag{9.28}$$

and $(l+1)g'/r - (l+1)g/r^2 = (l+1)(g/r)'$. Every term is a derivative,

$$U'(r)\,u_{l+1}u_l = \frac{\mathrm{d}}{\mathrm{d}r}\Big[(U-k^2)P - h + \frac{(l+1)^2P}{r^2} + \frac{(l+1)g}{r}\Big]. \tag{9.29}$$

(4) At $r = 0$ the regular solutions behave as r^{l+1} and r^{l+2}, so $P \sim r^{2l+3}$, h, P/r^2 and g/r all $\sim r^{2l+1}$, and the bracket vanishes for a potential less singular than r^{-3}. Where U has fallen off, $u_l = \sin\varphi$ and $u_{l+1} = -\cos(\varphi+\Delta)$ with $\varphi = kr - l\pi/2 + \delta_l$ and $\Delta = \delta_{l+1} - \delta_l$. Then $g \to k\cos\Delta$ is bounded, the terms with $1/r$ and $1/r^2$ vanish, and

$$-k^2P - h = k^2\big[\sin\varphi\cos(\varphi+\Delta) - \cos\varphi\sin(\varphi+\Delta)\big] = -k^2\sin\Delta, \tag{9.30}$$

a constant. Integrating Eq. (9.29) from 0 to ∞ and returning to $V = \hbar^2U/2m$,

$$\boxed{\int_0^\infty u_{k,l+1}\,V'(r)\,u_{kl}\,\mathrm{d}r = \frac{\hbar^2k^2}{2m}\,\sin(\delta_l - \delta_{l+1}).} \tag{9.31}$$

The check script verifies it for the deep models of argon and krypton of Sec. 9.8 at $ka_0 \geq 0.05$, to 10^{-4} for each $l \leq 2$ and to 4×10^{-6} for the sum over $l \leq 9$ weighted with $l+1$, as the integrals enter the cross sections below. The higher partial waves at small k carry integrals of 10^{-7} and less, which the integration resolves only to a few per cent.

9.5.2 The soft-photon cross section

(1) Insert Eq. (9.31) into Eq. (9.16) at $k' = k$. The terms with $l \to l+1$ and $l \to l-1$ of Eq. (9.12) give the same sum after a shift of the index,

$$S \to \left(\frac{\hbar^2 k^2}{2m^2\omega^2}\right)^2 2\sum_l (l+1)\sin^2(\delta_l - \delta_{l+1}) = \frac{\hbar^4 k^6}{8\pi m^4 \omega^4}\,\sigma_{\rm m}(E), \tag{9.32}$$

with the momentum-transfer cross section of Appendix 9.A,

$$\sigma_{\rm m}(E) = \int (1-\cos\theta)\frac{{\rm d}\sigma}{{\rm d}\Omega}{\rm d}\Omega = \frac{4\pi}{k^2}\sum_{l=0}^{\infty}(l+1)\sin^2(\delta_l - \delta_{l+1}). \tag{9.33}$$

(2) Equation (9.14) becomes

$$\boxed{\sigma_+ \to \sigma_+^{\rm soft} = \frac{4\pi}{3}\,\frac{e^2v^2}{c\hbar\omega^3}\,\sigma_{\rm m}(E) \qquad (\hbar\omega \ll E).} \tag{9.34}$$

This is the content of the low-energy theorem of Low [7] for potential scattering. Kroll and Watson [9] wrote it, angle by angle, as $(k'/k)(\mathbf{Q}\cdot\mathbf{a})^2/4$ times the elastic cross section, where $\mathbf{Q} = \mathbf{k} - \mathbf{k}'$ and $\mathbf{a} = eE_0\hat{\mathbf{e}}/m\omega^2$. The average of $(\mathbf{Q}\cdot\hat{\mathbf{e}})^2$ over directions is $Q^2/3$, and $Q^2 = 2k^2(1-\cos\theta)$ turns the elastic cross section into $\sigma_{\rm m}$, which is Eq. (9.34).
(3) The literature on neutral bremsstrahlung, the emission that corresponds to this absorption, is divided on this point. Buzulutskov et al. [16] report two opinions, the integral elastic cross section, used by Kas'yanov and Starostin [6] among others, and the momentum-transfer cross section, used by Dalgarno and Lane [10] among others, and they adopt the elastic one. The force identity decides for $\sigma_{\rm m}$. The two differ where the scattering is anisotropic, most of all in the Ramsauer windows. At the minimum of the deep models of Sec. 9.8 the elastic cross section is 3.8 times $\sigma_{\rm m}$ in argon and 6.6 times in krypton.

9.5.3 The classical limit is the Lorentz gas

(1) The classical kernel is the conductivity of the Lorentz gas. Expand the distribution of velocities as $f_0(v) + (\mathbf{v}\cdot\hat{\mathbf{e}}/v)f_1(v,t)$. The force on an electron is $-e\mathbf{E}$, and heavy scatterers relax the anisotropic part at the rate $\nu(v) = N\sigma_{\rm m}v$. The Boltzmann equation gives for $f_1 = {\rm Re}[\hat{f}_1 e^{-i\omega t}]$

$$(\nu - i\omega)\hat{f}_1 = \frac{eE_0}{m}\frac{\partial f_0}{\partial v}. \tag{9.35}$$

(2) The current along $\hat{\mathbf{e}}$ is $-e\int v\cos^2\theta\, f_1\,{\rm d}^3v = -(4\pi e/3)\int v^3 f_1\,{\rm d}v$, and with $\int 4\pi v^2 f_0\,{\rm d}v = n_e$ the conductivity is

$$\sigma(\omega) = -\frac{4\pi e^2}{3m}\int_0^\infty \frac{v^3}{\nu(v) - i\omega}\frac{\partial f_0}{\partial v}\,{\rm d}v. \tag{9.36}$$

For a constant ν an integration by parts returns the Drude conductivity $e^2 n_e/m(\nu - i\omega)$ [1].
(3) For $\omega \gg \nu$, ${\rm Re}(\nu - i\omega)^{-1} \to \nu/\omega^2$. For Maxwell electrons $\partial f_0/\partial v = -(mv/k_BT)f_0$, and the absorption coefficient $\kappa = 4\pi\,{\rm Re}\,\sigma/c$ at refractive index one is

$$\boxed{\kappa_{\rm cl} = \frac{4\pi e^2 n_e N}{3c\,k_B T\,\omega^2}\,\langle v^3\sigma_{\rm m}\rangle, \qquad \langle X\rangle = \int_0^\infty g(E)\,X\,{\rm d}E.} \tag{9.37}$$

(4) The quantum result reduces to it. In Eq. (9.20) put the soft-photon cross section, Eq. (9.34), and $1-e^{-u}\to u$,

$$\kappa \to n_e N\,\frac{\hbar\omega}{k_B T}\,\frac{4\pi e^2}{3c\hbar\omega^3}\,\langle v^3\sigma_{\mathrm{m}}\rangle = \kappa_{\mathrm{cl}}. \tag{9.38}$$

The check script confirms the equality numerically with the measured cross sections.

9.5.4 The free–free Gaunt factor and the error of the classical kernel

(1) Define the free–free Gaunt factor of the electron–atom pair by $\sigma_+ = G(E,\omega)\,\sigma_+^{\mathrm{soft}}$. With Eqs. (9.20) and (9.37) the ratio of the quantum to the classical absorption coefficient is

$$\boxed{\mathcal{R}(T,\omega) = \frac{\kappa}{\kappa_{\mathrm{cl}}} = \frac{1-e^{-u}}{u}\,\frac{\langle v^3\sigma_{\mathrm{m}}G\rangle}{\langle v^3\sigma_{\mathrm{m}}\rangle}.} \tag{9.39}$$

The average is over $g\,v^3 \propto E^2 e^{-E/k_BT}$. The quantity $\mathcal{R}-1$ is the error of the classical kernel that Sec. 9.1 asks for. Two effects compete. The factor $(1-e^{-u})/u$ is below one, and G exceeds one for slow electrons because a photon larger than their energy carries them to where σ_{m} is different.

9.6 Exact limits at finite photon energy

9.6.1 The zero-range potential

(1) A potential of zero range scatters only the s wave, with $k\cot\delta_0 = -1/a$ for the scattering length a. Outside the origin $u_{k0} = \sin(kr+\delta_0)$ and $u_{kl} = \hat\jmath_l(kr)$ for $l \ge 1$. The only transitions are $0\to 1$ and $1\to 0$.
(2) Write $\sin(kr+\delta_0) = \cos\delta_0 \sin kr + \sin\delta_0\cos kr$. The part with $\sin kr = \hat\jmath_0(kr)$ is a free–free integral between plane waves and vanishes for $k \neq k'$. Appendix 9.B evaluates the other part with the convergence factor,

$$\int_0^\infty \hat\jmath_1(k'r)\,r\cos(kr)\,\mathrm{d}r = \frac{2k'^2}{(k'^2-k^2)^2}, \tag{9.40}$$

so that, with $k'^2 - k^2 = 2m\omega/\hbar$,

$$R_{0\to1} = \frac{\hbar^2 k'^2}{2m^2\omega^2}\sin\delta_0(k), \qquad R_{1\to0} = \frac{\hbar^2 k^2}{2m^2\omega^2}\sin\delta_0(k'). \tag{9.41}$$

(3) Then $S = R_{0\to1}^2 + R_{1\to0}^2$ and, with $\sigma_{\mathrm{m}} = (4\pi/k^2)\sin^2\delta_0$ from Eq. (9.33), Eq. (9.14) gives

$$\boxed{\sigma_+^{\mathrm{zr}} = \frac{2\pi}{3}\,\frac{e^2\hbar\,kk'}{c\,m^2\omega^3}\Big[\Big(\frac{k'}{k}\Big)^2\sigma_{\mathrm{m}}(E) + \sigma_{\mathrm{m}}(E')\Big], \qquad E' = E+\hbar\omega.} \tag{9.42}$$

The cross section at both energies enters, the initial one weighted by $(k'/k)^2$. As $\omega\to 0$ the bracket tends to $2\sigma_{\mathrm{m}}(E)$ and Eq. (9.34) returns. The check script compares Eq. (9.41) with the acceleration form for a Gaussian well of width $b\to 0$ tuned to $a = -1.46\,a_0$ and finds the deviation falling in proportion to b, from 1.1×10^{-2} at $b = 0.1\,a_0$ to 2.2×10^{-3} at $b = 0.025\,a_0$.
(4) Written for emission with Eq. (9.17) and multiplied by the density $8\pi\nu^2/c^2$ of photon modes per unit frequency, which turns stimulated into spontaneous emission, Eq. (9.42) becomes

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\nu} = \frac{8}{3}\,\frac{r_e}{c}\,\frac{1}{h\nu}\Big(\frac{E-h\nu}{E}\Big)^{1/2}\big[(E-h\nu)\,\sigma_{\mathrm{m}}(E) + E\,\sigma_{\mathrm{m}}(E-h\nu)\big], \qquad r_e = \frac{e^2}{mc^2}, \tag{9.43}$$

for an electron of energy E that emits a photon of frequency ν. This is Eq. (4) [16], the cross section for neutral bremsstrahlung of Buzulutskov et al., with $\sigma_{\rm m}$ in place of their elastic cross section. They attribute it to the Born approximation with one dominant partial wave. Equation (9.42) shows that it is exact for a zero-range potential of any strength, for which the elastic and momentum-transfer cross sections coincide. The check script confirms the prefactors to 10^{-13}.

9.6.2 The Born approximation

(1) In the first Born approximation the one-photon cross section of Bunkin and Fedorov [4] holds for any potential, with the Born amplitude $f_B(Q) = -(m/2\pi\hbar^2)\tilde{V}(Q)$. Dividing by Φ and averaging $(\mathbf{Q}\cdot\hat{\mathbf{e}})^2$ over directions,

$$\sigma_+^B = \frac{2\pi\hbar e^2}{3cm^2\omega^3}\,\frac{k'}{k}\int Q^2 |f_B(Q)|^2\,\mathrm{d}\Omega'. \tag{9.44}$$

(2) At fixed k and k', $Q^2 = k^2 + k'^2 - 2kk'\cos\theta$ gives $\mathrm{d}\Omega' = 2\pi Q\,\mathrm{d}Q/kk'$ with Q from $k'-k$ to $k'+k$. With $M(q) = \int_0^q Q^3|f_B|^2\mathrm{d}Q$,

$$\sigma_+^B = \frac{4\pi^2\hbar e^2}{3cm^2\omega^3k^2}\big[M(k'+k) - M(k'-k)\big]. \tag{9.45}$$

(3) The elastic momentum-transfer cross section at momentum $\hbar\kappa$ is, with $Q^2 = 2\kappa^2(1-\cos\theta)$ and $\mathrm{d}\Omega = 2\pi Q\,\mathrm{d}Q/\kappa^2$, $\sigma_{\rm m}^B(\kappa) = (\pi/\kappa^4)M(2\kappa)$. Hence

$$\boxed{\sigma_+^B = \frac{4\pi\hbar e^2}{3cm^2\omega^3k^2}\Big[\kappa_+^4\,\sigma_{\rm m}^B(\kappa_+) - \kappa_-^4\,\sigma_{\rm m}^B(\kappa_-)\Big], \qquad \kappa_\pm = \frac{k'\pm k}{2}.} \tag{9.46}$$

In the Born approximation the exact one-photon cross section at any photon energy is the momentum-transfer cross section at the mean momentum $(k+k')/2$, less a term at the half difference. The check script confirms Eq. (9.46) to 10^{-14}. Section 9.8 finds that the same form with the true $\sigma_{\rm m}$ is the best of the formulas that use only $\sigma_{\rm m}$, far outside the Born approximation.

9.6.3 The first order in the photon energy

(1) Expand Eqs. (9.42) and (9.46) to first order in $\hbar\omega/E$. With $\bar{E} = E + \hbar\omega/2$ the mean energy, $\kappa_+^2 = k^2 + m\omega/\hbar + O(\omega^2)$, so $\hbar^2\kappa_+^2/2m = \bar{E}$ to first order and κ_-^4 is of fourth order. In Eq. (9.42), $\sigma_{\rm m}(E)$ and $\sigma_{\rm m}(E')$ are $\sigma_{\rm m}(\bar{E}) \mp \frac{1}{2}\hbar\omega\sigma'_{\rm m}$, and the terms with $\sigma'_{\rm m}$ cancel against $k^2 - k'^2$ to second order. Both give

$$\boxed{\sigma_+ = \frac{4\pi}{3}\,\frac{e^2\hbar\,k'^2}{c\,m^2\omega^3}\,\sigma_{\rm m}(\bar{E})\Big[1 + O\Big(\frac{\hbar\omega}{E}\Big)^2\Big].} \tag{9.47}$$

(2) The two exact limits agree to first order. The theorem of Burnett and Kroll [8] states that the term of first order beyond the soft-photon limit is fixed by the elastic cross section alone. For the model potentials of Sec. 9.8, which are neither weak nor of zero range, the check script finds that the residual of Eq. (9.47) falls by a factor between 3.85 and 4.04 each time $\hbar\omega/E$ is halved, the signature of a second-order remainder.

9.7 Strong fields and the weak-field criterion

(1) Kroll and Watson [9] extended the soft-photon result to any field strength. For a collision short against the period of the wave and $\ell\hbar\omega \ll E$, the cross section for the exchange of ℓ photons is

$$\frac{\mathrm{d}\sigma_\ell}{\mathrm{d}\Omega'} = \frac{k_\ell}{k}\,J_\ell^2(\mathbf{Q}\cdot\mathbf{a})\,\frac{\mathrm{d}\sigma_{\rm el}}{\mathrm{d}\Omega}, \tag{9.48}$$

with the elastic cross section at shifted momenta. It has the structure of the Born result of Bunkin and Fedorov [4], with the Born cross section replaced by the exact one. Weingartshofer et al. observed the multiphoton terms directly [17].
(2) One photon dominates when $|\mathbf{Q}\cdot\mathbf{a}| \ll 1$, with $a = eE_0/m\omega^2$ and $E_0 = (8\pi I/c)^{1/2}$. Since $Q \le 2k$, the condition $2ka \le 0.3$ keeps J_1^2 within 2.3 % of $x^2/4$. Table 9.2 gives the intensity at which it is reached for an electron of energy $\hbar\omega$, for which the quantum corrections of Secs. 9.8 and 9.9 matter most, and for electrons of energy $3k_BT/2$.

Table 9.2: Intensity in $\mathrm{W\,cm^{-2}}$ at which $2ka = 0.3$, the edge of the one-photon regime, for an electron of energy $\hbar\omega$ and of energy $3k_BT/2$ at 300 K and at 0.63 eV.

frequency	$E = \hbar\omega$	$E = 3k_BT/2$, 300 K	$E = 3k_BT/2$, 0.63 eV
1 THz	1.4×10^3	1.5×10^2	6.1×10^0
3 THz	3.7×10^4	1.2×10^4	4.9×10^2
10 THz	1.4×10^6	1.5×10^6	6.1×10^4
28.3 THz	3.1×10^7	9.5×10^7	3.9×10^6

(3) Above that intensity the multiphoton terms of Eq. (9.48) appear. For electrons with $\hbar\omega \ll E$ they change how the energy is exchanged but not how fast it is gained. The energy diffusion is proportional to $\sum_\ell \ell^2 J_\ell^2(x)$, and the sum rule $\sum_\ell \ell^2 J_\ell^2(x) = x^2/2$, which follows from the Jacobi–Anger expansion and Parseval's theorem, makes it equal to the one-photon value $2\cdot x^2/4$. The classical kernel therefore holds at any intensity for those electrons. With the field above the collision frequency and only elastic losses the electrons are Maxwellian, and their temperature exceeds that of the gas, T_g, by $k_B(T_e - T_g) = \frac{2}{3}(M/m)U_p$, with the quiver energy $U_p = e^2E_0^2/4m\omega^2$ [5]. A jump of 0.5 eV in pure krypton then needs 53 $\mathrm{W\,cm^{-2}}$ at 0.3 THz and 585 $\mathrm{W\,cm^{-2}}$ at 1 THz, where $2ka$ is 7.1 and 2.1 for an electron of 0.5 eV, and it heats as the classical theory says. The one-photon corrections of these notes need $2ka \le 0.3$ only for the slow electrons, which holds up to 1.4 $\mathrm{kW\,cm^{-2}}$ at 1 THz and 31 $\mathrm{MW\,cm^{-2}}$ at 28.3 THz.

9.8 Realistic atoms

9.8.1 Three families of model potentials

(1) The exact cross section, Eq. (9.14), needs the radial functions of a potential that reproduces the measured scattering. Three local forms are fitted, in atomic units,

$$\begin{aligned}
&\text{A:}\quad V = V_0 e^{-r^2/b^2} - \frac{\alpha_d}{2(r^2+r_c^2)^2},\\
&\text{B:}\quad V = V_0 e^{-r/b} - \frac{\alpha_d\,(1-e^{-(r/r_c)^6})}{2r^4},\\
&\text{D:}\quad V = -\frac{Z}{r}\,\frac{1}{1+H(e^{r/d}-1)} - \frac{\alpha_d}{2(r^2+r_c^2)^2}.
\end{aligned} \tag{9.49}$$

A and B have a repulsive core. They bind no states, except that krypton's model B, whose cut-off $r_c = 0.36\,a_0$ leaves the polarization attraction strong inside the atom, binds one s and one p state. D has the screening function of the independent-particle model of Green, Sellin and Zachor [27] for

the nucleus of charge Z, and is constrained to bind the occupied orbitals of the atom, two s and one p state for neon, three s and two p for argon, four s, three p and one d for krypton. The count is the number of nodes of the zero-energy solution. All three have the polarization tail of Eq. (9.1) with the recommended static polarizabilities, 2.6611, 11.083 and 16.78 atomic units for Ne, Ar and Kr [22].

(2) The parameters minimize $\sum(\ln\sigma_{\mathrm{m}}^{\mathrm{model}} - \ln\sigma_{\mathrm{m}}^{\mathrm{LXCat}})^2$ over 36 energies from 0.003 to 4 eV, against the measured cross sections that the classical kernel uses, the BSR set for neon, the IST-Lisbon set for argon and the Biagi set for krypton, taken from LXCat [30]. Table 9.3 lists the quality of the fits and the scattering lengths, and Fig. 9.1 compares the cross sections. The deep models D fit within 7 to 11 % and reproduce the Ramsauer windows of argon and krypton and the rise toward their d-wave shape resonances. The repulsive cores cannot dig a window as deep as krypton's, and model A misses it by a factor of 1.75 on average. The scattering length of D lies inside the range of the measurements for krypton and $0.024\,a_0$ beyond its edge for argon [24, 25, 23]. For neon the three models give 0.28 to $0.29\,a_0$, between the measured $0.21\,a_0$ [26] and the $0.38\,a_0$ implied by the zero-energy cross section of the BSR set, which they are fitted to. The scattering lengths come from the zero-energy solution, whose polarization tail $-\alpha_d/2r^4$ is followed to $4000\,a_0$ and continued analytically beyond.

Table 9.3: The model potentials of Eq. (9.49). The rms of $\ln\sigma_{\mathrm{m}}^{\mathrm{model}} - \ln\sigma_{\mathrm{m}}^{\mathrm{LXCat}}$ over 0.003 to 4 eV, the scattering length A in a_0, and the range of measured scattering lengths quoted in [23] for argon and krypton and in [26, 23] for neon.

		rms of $\ln\sigma_{\mathrm{m}}$			A (a_0)			
gas	α_d	A	B	D	A	B	D	measured
Ne	2.6611	0.115	0.140	0.107	+0.285	+0.277	+0.288	+0.206 to +0.214
Ar	11.083	0.266	0.316	0.081	−1.450	−1.490	−1.483	−1.459 to −1.365
Kr	16.78	0.559	0.178	0.067	−3.087	−3.205	−3.254	−3.430 to −3.060

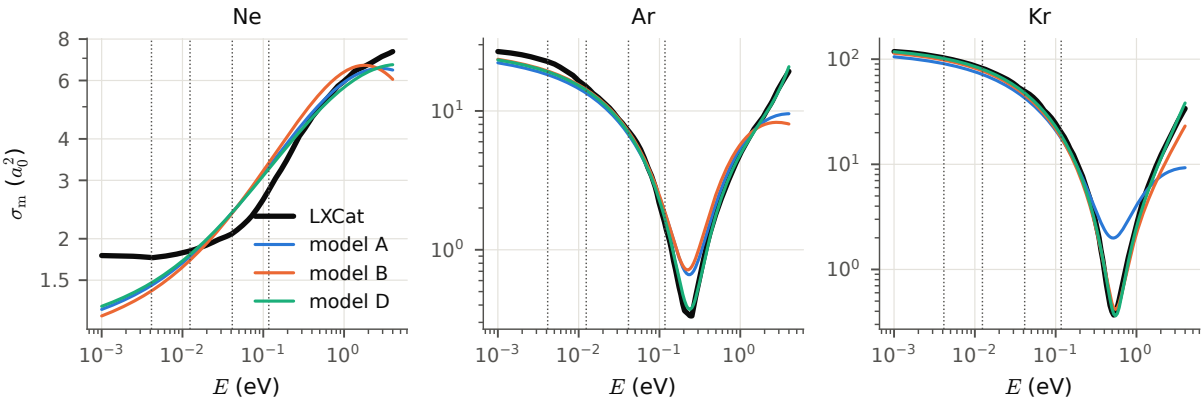


Figure 9.1: Momentum-transfer cross sections of the LXCat sets used by the classical kernel and of the three model families. The vertical lines mark the photon energies of 1, 3, 10 and 28.3 THz.

9.8.2 Numerical method

(1) The radial equation is integrated by the method of Numerov, on a uniform grid of step $0.01\,a_0$ for the smooth cores and on the logarithmic grid $r = e^x$, $u = e^{x/2}y$, with $y'' = [(l+\frac{1}{2})^2 + r^2(2V - k^2)]y$ and step $\Delta x = 5 \times 10^{-4}$, for the Coulomb-like cores of D. At $r_{\max} = 400\,a_0$ the solution is matched to the Riccati–Bessel functions, which fixes δ_l and the amplitude, and the acceleration integrals of Eq. (9.16) are accumulated during the integration and normalized at the end. Partial waves up to $l = 12$ are kept.
(2) For the deep models, halving the step, doubling $r_{\max}$ and raising the largest l to 16 change G by at most 1.4×10^{-5}. The cores of the models A reach 6×10^{10} hartree in krypton, where the recursion of Numerov is unstable on the uniform grid. The outward growth of the solution through the forbidden core filters that error out. Starting instead at a hard wall placed where $h^2 \cdot 2mV/\hbar^2$ last exceeds 2, with half the step, changes G of the models A and B by at most 1×10^{-5}. The force identity holds as stated in Sec. 9.5.1, and detailed balance, Eq. (9.17), to 10^{-13}.

9.8.3 The free–free Gaunt factor

(1) Figure 9.2 shows $G = \sigma_+/\sigma_+^{\text{soft}}$ for the deep models. For $\hbar\omega \ll E$ it tends to one, with the slope $1 + \frac{1}{2}\,\mathrm{d}\ln\sigma_{\mathrm{m}}/\mathrm{d}\ln E$ in $\hbar\omega/E$ that Eq. (9.47) predicts. Below $E \approx \hbar\omega$ it grows as $E^{-3/2}$. There the final state is near $\hbar\omega$ whatever the initial energy, σ_+ varies as $1/k$, and the soft-photon reference vanishes as k^2.

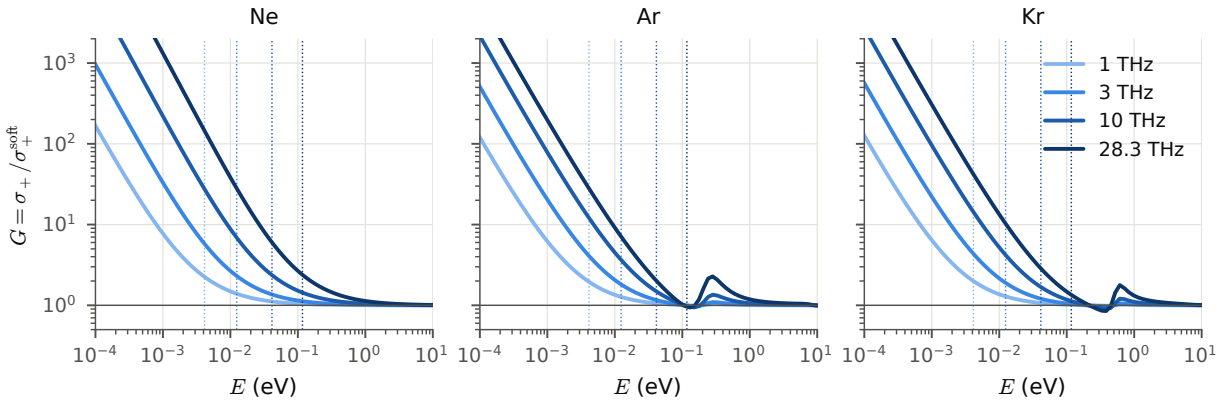


Figure 9.2: The free–free Gaunt factor $G = \sigma_+/\sigma_+^{\text{soft}}$ of the deep models at 1, 3, 10 and 28.3 THz. The dotted lines mark $E = \hbar\omega$.

(2) A code that needs σ_+ without solving the scattering problem can use one of three forms built on σ_{m} alone, the two-energy form of Eq. (9.42), the mean-energy form of Eq. (9.47), or the mean-momentum form of Eq. (9.46) with the true σ_{m}. Table 9.4 compares them with the exact result of each deep model. The two-energy form, exact for the zero-range potential, overestimates σ_+ of argon by a factor 2.1 for thermal electrons at 28.3 THz. The polarization tail makes the p wave matter at low energy, since it raises the p-wave phase shift as $\pi\alpha_d k^2/15$ in atomic units rather than as k^3 [21], a law the check script confirms for the deep models. The mean-energy form falls up to 25 % short there.
(3) The mean-momentum form is the best of the three. For neon it is within 0.5 % of the exact result at every energy above $\hbar\omega$ up to 28.3 THz. For argon and krypton it is within 1.2 % at 3 THz

and 3.3 % at 10 THz above $\hbar\omega$. At 28.3 THz it errs by up to 14 % in argon and 17 % in krypton above $\hbar\omega$, most where the final energy $E+\hbar\omega$ falls at the Ramsauer minimum of the model, 0.24 eV in argon and 0.55 eV in krypton. There $\sigma_{\rm m}$ changes by a large factor across one photon energy. Table 9.4 shows errors of the same size for thermal electrons below $\hbar\omega$. In the thermal averages of Sec. 9.9 up to 4 THz the form is within 1 % of the exact result for every gas.

Table 9.4: The exact Gaunt factor G of the deep models at $E = 26$ meV and 0.3 eV, and the two-energy (TE), mean-energy (ME) and mean-momentum (MM) forms with the model's own $\sigma_{\rm m}$, as ratios to the exact cross section.

		$E = 26$ meV				$E = 0.3$ eV	
gas	f (THz)	G	TE	ME	MM	G	MM
Ne	3	1.588	0.989	1.002	0.999	1.048	1.000
Ne	10	3.345	0.935	1.009	0.996	1.162	1.000
Ne	28.3	10.373	0.805	1.019	0.985	1.479	0.999
Ar	3	1.318	1.030	0.989	0.996	1.095	1.000
Ar	10	1.984	1.223	0.927	0.968	1.350	0.997
Ar	28.3	3.144	2.092	0.747	0.859	2.219	0.987
Kr	3	1.370	1.017	0.989	0.994	0.979	0.999
Kr	10	2.252	1.121	0.927	0.955	0.939	0.986
Kr	28.3	4.478	1.502	0.757	0.834	0.882	0.891

9.8.4 Polarization bremsstrahlung

(1) The term of Eq. (9.3) has the same angular structure as $\mathbf{r}$, so in the acceleration form it adds $-m\omega^2\alpha_d/r^2$ to $V'(r)$. With the cut-off $r^2 \to r^2 + r_c^2$ of the polarization potential of model D, Table 9.5 gives the change of σ_+. It is below 0.7 % up to 3 THz for every gas and energy. At 28.3 THz it reaches -13 % for slow electrons in argon, -8 % in krypton and $+4$ % in neon, so a CO_2-laser calculation for a cold gas has to include it. At 1 eV it stays below 1 % at every frequency. Geltman [20] studied the same field-induced dipole in the laser-assisted elastic scattering of Sec. 9.7.

9.9 Thermal absorption and the error of the classical kernel

9.9.1 The pure gases

(1) Equation (9.39) gives $\mathcal{R}(T,\omega)$ from G and $\sigma_{\rm m}$ of each model. Table 9.6 lists it for the deep models with the range over the three families. Up to 4 THz the classical kernel is right to 2.5 % or better for every gas and temperature, and to 2.8 % with any of the three families. At 10 THz electrons at 300 K absorb 12 % less than the classical kernel says in argon and 7 % less in krypton, and at 28.3 THz 41 % and 25 % less. In neon the two effects of Sec. 9.5 cancel to within 3 %. Above 0.3 eV the classical kernel is right to 1.6 % at every frequency up to 28.3 THz.

9.9.2 The two fills

(1) In a mixture each gas contributes to $\kappa_{\rm cl}$ in proportion to its share $x_g\langle v^3\sigma_{\rm m}\rangle$, and the ratio of the mixture is the share-weighted mean of the ratios of its gases. Fluorine appears only at 0.2 and

Table 9.5: Change of σ_+ in percent when the polarization bremsstrahlung of Eq. (9.3) is added, for the deep models.

gas	E (meV)	1 THz	3 THz	10 THz	28.3 THz
Ne	3	+0.09	+0.43	+1.66	+4.33
Ne	26	+0.01	+0.10	+0.72	+2.84
Ne	300	+0.00	+0.01	+0.06	+0.44
Ne	1000	+0.00	+0.00	+0.01	+0.10
Ar	3	−0.12	−0.63	−3.15	−12.82
Ar	26	−0.03	−0.26	−2.28	−12.68
Ar	300	−0.00	−0.02	−0.07	+0.93
Ar	1000	+0.00	+0.01	+0.09	+0.66
Kr	3	−0.08	−0.41	−2.01	−7.53
Kr	26	−0.02	−0.15	−1.29	−6.68
Kr	300	−0.01	−0.08	−0.95	−7.70
Kr	1000	+0.00	+0.01	+0.12	+0.93

0.3 %, and its molecular cross section is treated with the mean-momentum form. At 300 K krypton carries 44 % of the classical absorption of the neon fill and 83 % of that of the argon fill, because per atom its $\langle v^3 \sigma_\mathrm{m} \rangle$ is 15 times that of neon and 7.6 times that of argon.
(2) Table 9.7 gives the ratio for both fills, from the deep models and, in parentheses, from the mean-momentum form with the measured cross sections alone. Figure 9.3 shows it against frequency.

9.9.3 The answer for the laser gas

(1) In the band of 0.3 to 4 THz proposed for reheating, the classical kernel differs from the quantum result by at most 1.6 % in either fill at any electron temperature from 300 K to 1.08 eV. The mean-momentum form with the measured cross sections gives at most 2.5 %. Both are below the 5 % of the criterion of Sec. 9.1, so the classical kernel holds in that band, and the absorption lengths and reflectances computed with it stand.
(2) For a CO_2 laser on a thermalized fill the classical kernel fails. At 28.3 THz and 300 K the quantum absorption is 10 % below it in the neon fill and 28 % below it in the argon fill, before the polarization bremsstrahlung of Sec. 9.8.4. Above 0.3 eV it is right to 1 % in both fills at every frequency studied. A code that heats electrons at such frequencies should multiply the classical absorption by $\mathcal{R}$, or replace the diffusion term of the two-term equation by the jump operator of Eq. (9.23) with the mean-momentum cross section.

9.10 Electron–ion or electron–atom

(1) The ions contribute the effective collision frequency $\nu_T \Lambda$ [1], with

$$\nu_T = \frac{4\sqrt{2\pi}}{3} \frac{n_i Z^2 e^4}{m^{1/2} (k_B T)^{3/2}}, \qquad \Lambda = \frac{\pi}{\sqrt{3}} \langle g \rangle \frac{1 - e^{-u}}{u},$$

the thermal Gaunt factor $\langle g \rangle$ times the factor of stimulated emission [2, 3]. The atoms contribute

Table 9.6: The ratio $\mathcal{R} = \kappa/\kappa_{\rm cl}$ of Eq. (9.39) for each gas, from the deep model D, and its range over the models A, B and D.

		$T = 300$ K		$T = 0.1$ eV		$T = 0.63$ eV	
gas	f (THz)	D	A–D	D	A–D	D	A–D
Ne	1	1.000	1.000–1.000	1.000	1.000–1.000	1.000	1.000–1.000
Ne	3	0.999	0.999–0.999	1.000	1.000–1.000	1.000	1.000–1.000
Ne	4	0.998	0.998–0.999	1.000	1.000–1.000	1.000	1.000–1.000
Ne	10	0.992	0.992–0.995	1.000	1.000–1.000	1.000	1.000–1.000
Ne	28.3	1.030	1.030–1.043	0.997	0.997–0.998	1.000	1.000–1.000
Ar	1	0.998	0.998–0.998	1.000	1.000–1.000	1.000	1.000–1.000
Ar	3	0.985	0.984–0.985	0.999	0.998–0.999	1.000	1.000–1.000
Ar	4	0.975	0.972–0.975	0.998	0.997–0.998	1.000	1.000–1.000
Ar	10	0.881	0.868–0.881	0.989	0.983–0.989	1.000	1.000–1.000
Ar	28.3	0.591	0.540–0.591	0.933	0.903–0.933	1.001	1.001–1.001
Kr	1	0.999	0.999–0.999	1.000	1.000–1.000	1.000	1.000–1.000
Kr	3	0.992	0.990–0.992	1.000	0.999–1.000	1.000	1.000–1.000
Kr	4	0.985	0.983–0.985	0.999	0.998–0.999	1.000	1.000–1.000
Kr	10	0.928	0.915–0.928	0.994	0.984–0.994	1.001	1.000–1.001
Kr	28.3	0.749	0.697–0.749	0.943	0.887–0.943	1.004	1.000–1.004

κ of Eq. (9.20), which corresponds to the effective collision frequency $(mN/3k_BT)\langle v^3\sigma_{\rm m}\rangle\mathcal{R}$. With singly charged ions, $n_i = n_e$, the ions win above the electron density at which the two are equal.
(2) In the Born approximation the thermal Gaunt factor is $\langle g\rangle = (\sqrt{3}/\pi)e^{u/2}K_0(u/2)$ [3], valid for $Ze^2/\hbar v \ll 1$. For an electron of energy k_BT, $Ze^2/\hbar v$ is 23 at 300 K and 3.5 at 1.08 eV, and the collisions are classical. For $x = \omega Ze^2/mv^3 \ll 1$ and $\hbar\omega \ll k_BT$ the classical thermal Gaunt factor is that of Scheuer [29], $\langle g\rangle = (\sqrt{3}/\pi)\{\ln[(2k_BT)^{3/2}/\pi Ze^2m^{1/2}f] - \frac{5}{2}\gamma\}$, with f in hertz. At 300 K and 1 THz, x is about 5 for $v = (k_BT/m)^{1/2}$, and that formula turns negative. For larger x the classical Gaunt factor tends to Kramers' value one from above. The estimate used here is the larger of Scheuer's value and one. At 1 THz, where $u \le 0.16$, the Born value exceeds this estimate at every temperature and bounds the ionic absorption from above. At 28.3 THz and 300 K, where $u = 4.5$, the Born value falls to 0.44, below Kramers' one, which is then the better estimate.
(3) Table 9.8 and Fig. 9.4 give the density at which the two kinds of absorption are equal at 1 THz. Ions dominate only in a cold and dense plasma. At 300 K that takes 7.0×10^{19} m^{-3} in the neon fill and 1.5×10^{20} m^{-3} in the argon fill with the classical Gaunt factor, and at 0.63 eV more than 10^{22} m^{-3} in both. At 28.3 THz and 300 K, with Kramers' one, the densities are 2.7×10^{20} and 4.5×10^{20} m^{-3}. The plasma of the laser gas with the beam on, 2.7×10^{21} m^{-3} at 1.08 eV, absorbs on atoms, and so does the afterglow while its electrons stay warm. In a thermalized afterglow at 10^{20} m^{-3} and 1 THz the ions absorb 1.4 times as much as the atoms in the neon fill and 0.7 times as much in the argon fill. The atoms dominate because n_e/N is small only away from that corner.

9.11 Summary

(1) The one-photon free–free cross section of an electron in the field of an atom is Eq. (9.14), a sum

Table 9.7: The ratio $\mathcal{R} = \kappa/\kappa_{\mathrm{cl}}$ for the two fills of Sec. 9.1, from the deep models, and in parentheses from the mean-momentum form with the LXCat cross sections.

fill	f (THz)	$T = 300$ K	$T = 0.1$ eV	$T = 0.63$ eV	$T = 1.08$ eV
Ne	0.3	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ne	1	1.000 (0.999)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ne	3	0.995 (0.992)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ne	4	0.992 (0.987)	1.000 (0.999)	1.000 (1.000)	1.000 (1.000)
Ne	5	0.988 (0.981)	1.000 (0.999)	1.000 (1.000)	1.000 (1.000)
Ne	10	0.961 (0.941)	0.999 (0.996)	1.000 (1.000)	1.000 (1.000)
Ne	28.3	0.897 (0.835)	0.991 (0.979)	1.000 (1.000)	1.000 (1.000)
Ar	0.3	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ar	1	0.999 (0.998)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ar	3	0.991 (0.985)	1.000 (0.997)	1.000 (1.000)	1.000 (1.000)
Ar	4	0.984 (0.975)	0.999 (0.995)	1.000 (1.000)	1.000 (1.000)
Ar	5	0.975 (0.963)	0.998 (0.993)	1.000 (1.000)	1.000 (1.000)
Ar	10	0.920 (0.886)	0.993 (0.973)	1.000 (1.000)	1.000 (1.000)
Ar	28.3	0.722 (0.614)	0.941 (0.842)	1.003 (1.000)	1.001 (1.000)

over partial waves of the squared dipole integrals, which the acceleration form, Eq. (9.16), makes absolutely convergent. Emission follows from detailed balance, Eq. (9.17), and the absorption coefficient of Maxwell electrons carries the factor $1 - e^{-u}$, Eq. (9.20). The kinetic equation is the jump operator of Eq. (9.23), whose Fokker–Planck limit, Eq. (9.25), is the field term of the two-term Boltzmann equation at $\omega \gg \nu$.

(2) The force identity, Eq. (9.31), reduces the soft-photon limit to the momentum-transfer cross section, Eq. (9.34), and the classical limit is the Lorentz gas, Eq. (9.37). The cross section that belongs in the classical formula is therefore σ_{m}, which settles the choice between σ_{m} and the elastic cross section that the literature on neutral bremsstrahlung leaves open [16].

(3) At finite photon energy the cross section is known exactly for a zero-range potential, Eq. (9.42), which is also the emission formula of neutral bremsstrahlung, Eq. (9.43), and in Born approximation, Eq. (9.46). Both agree to first order with Eq. (9.47), and the model potentials confirm that this order is universal. Wherever $\hbar\omega \leq E$ the mean-momentum form with the true σ_{m} is accurate to 0.5 % for neon at every frequency and to 3.3 % for argon and krypton up to 10 THz, but errs by up to 17 % at 28.3 THz near the Ramsauer minima.

(4) For the laser gas the quantum correction to the classical kernel is below 1.6 % from 0.3 to 4 THz at every temperature, so the classical kernel holds in that band. It becomes -10 to -28 % at 28.3 THz for electrons at 300 K, where polarization bremsstrahlung changes the cross section of slow electrons by up to -13 %. Electron–ion absorption competes only in a cold afterglow above about 10^{20} m^{-3}.

(5) Left open are the exchange of the scattering electron with the atomic electrons, which a one-body potential only mimics and which the multichannel R-matrix calculation of Bell and Berrington [13] treats for argon, the molecular fluorine, and the laser-assisted measurements on argon that find photon exchange, up to 12 % of the field-free cross section, where the theory of Kroll and Watson predicts none [18, 19]. The measured neutral bremsstrahlung [14, 15] is the experimental test of the models at higher photon energies.

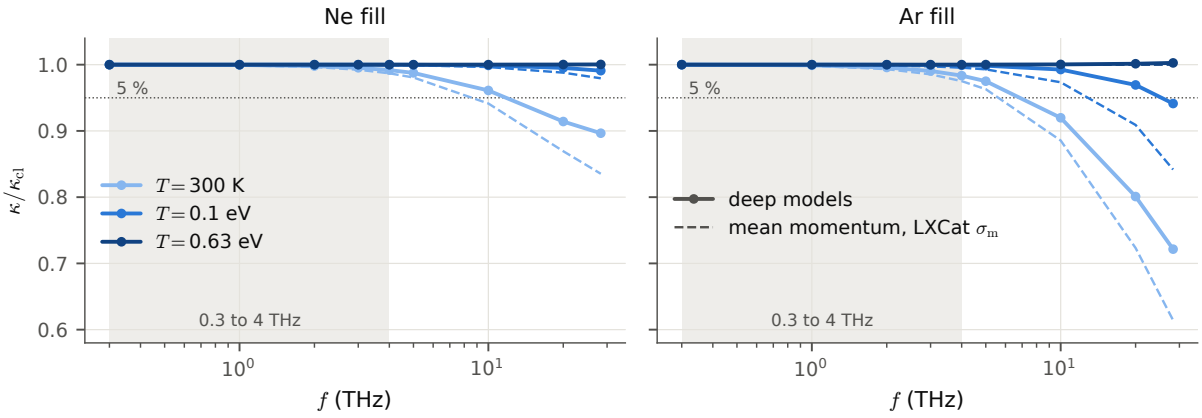


Figure 9.3: The ratio of the quantum to the classical absorption coefficient for the two fills against frequency. Solid lines are the deep models, dashed lines the mean-momentum form with the LXCat cross sections. The shaded band is the band of 0.3 to 4 THz proposed for reheating.

9.A Partial waves

(1) Take $\hat{\mathbf{e}} = \hat{z}$, which the average over the direction of the electron allows. The matrix elements of $\cos\theta$ are $\langle l+1, m|\cos\theta|lm\rangle = [((l+1)^2 - m^2)/((2l+1)(2l+3))]^{1/2}$ and vanish for $m' \neq m$. Summing over m,

$$\sum_{m=-l}^{l} \frac{(l+1)^2 - m^2}{(2l+1)(2l+3)} = \frac{(2l+1)(l+1)^2 - l(l+1)(2l+1)/3}{(2l+1)(2l+3)} = \frac{l+1}{3}, \tag{9.50}$$

and the same steps with $\langle l-1, m|\cos\theta|lm\rangle^2 = (l^2-m^2)/((2l-1)(2l+1))$ give $l/3$, which is Eq. (9.9). (2) The elastic amplitude is $f(\theta) = k^{-1}\sum_l (2l+1)e^{i\delta_l}\sin\delta_l\, P_l(\cos\theta)$. In $\int(1-\cos\theta)|f|^2\mathrm{d}\Omega$ the term with 1 gives $(4\pi/k^2)\sum_l(2l+1)\sin^2\delta_l$, and with $\cos\theta\, P_l = [(l+1)P_{l+1} + lP_{l-1}]/(2l+1)$ the term with $\cos\theta$ gives $(4\pi/k^2)\sum_l 2(l+1)\sin\delta_l\sin\delta_{l+1}\cos(\delta_l - \delta_{l+1})$. Their difference, with $\sin^2\delta_l + \sin^2\delta_{l+1} - 2\sin\delta_l\sin\delta_{l+1}\cos(\delta_l - \delta_{l+1}) = \sin^2(\delta_l - \delta_{l+1})$ and $(2l+1) = (l+1)+l$ regrouped, is Eq. (9.33).

9.B The integrals of the zero-range limit

(1) With $\hat{\jmath}_1(x) = \sin x/x - \cos x$ and the factor $e^{-\eta r}$,

$$\int_0^\infty \hat{\jmath}_1(k'r)\, r\cos(kr)\, e^{-\eta r}\mathrm{d}r = \frac{1}{k'}\int_0^\infty \sin(k'r)\cos(kr)e^{-\eta r}\mathrm{d}r - \int_0^\infty r\cos(k'r)\cos(kr)e^{-\eta r}\mathrm{d}r. \tag{9.51}$$

(2) The first integral is $\frac{1}{2k'}\int[\sin(k'+k)r + \sin(k'-k)r]e^{-\eta r}\mathrm{d}r \to \frac{1}{2k'}\left[\frac{1}{k'+k} + \frac{1}{k'-k}\right] = \frac{1}{k'^2-k^2}$.
(3) With $\int_0^\infty r\cos(qr)e^{-\eta r}\mathrm{d}r = (\eta^2-q^2)/(\eta^2+q^2)^2 \to -1/q^2$, the second integral tends to $-\frac{1}{2}\left[(k'-k)^{-2} + (k'+k)^{-2}\right] = -(k^2+k'^2)/(k'^2-k^2)^2$.
(4) The difference is $[(k'^2-k^2) + (k'^2+k^2)]/(k'^2-k^2)^2 = 2k'^2/(k'^2-k^2)^2$, which is Eq. (9.40). The integral with $\sin kr$ in place of $\cos kr$ tends to zero for $k \neq k'$, since $\int \sin ar \sin br\, e^{-\eta r}\mathrm{d}r$ and $\int r\sin(qr)e^{-\eta r}\mathrm{d}r = 2\eta q/(\eta^2+q^2)^2$ both vanish with η.

Table 9.8: Electron density in m^{-3} at which electron–ion and electron–atom absorption are equal at 1 THz, with the classical estimate of the thermal Gaunt factor and with the Born value [3]. Ions dominate above it.

fill	T (eV)	$\langle g\rangle_{\mathrm{cl}}$	$\langle g\rangle_{\mathrm{Born}}$	n_e, classical	n_e, Born
Ne	0.0259	1.00	1.58	7.0×10^{19}	4.5×10^{19}
Ne	0.1	1.00	2.25	9.9×10^{20}	4.4×10^{20}
Ne	0.3	1.28	2.83	9.1×10^{21}	4.1×10^{21}
Ne	0.63	1.89	3.23	3.3×10^{22}	2.0×10^{22}
Ne	1.08	2.34	3.52	9.3×10^{22}	6.2×10^{22}
Ar	0.0259	1.00	1.58	1.5×10^{20}	9.4×10^{19}
Ar	0.1	1.00	2.25	5.0×10^{20}	2.2×10^{20}
Ar	0.3	1.28	2.83	3.4×10^{21}	1.6×10^{21}
Ar	0.63	1.89	3.23	2.5×10^{22}	1.5×10^{22}
Ar	1.08	2.34	3.52	1.1×10^{23}	7.5×10^{22}

9.C The numerical checks

(1) A check script computes every boxed result of these notes and reproduces every number printed in them, from the tables it writes and from the prose. It needs numpy, scipy and the LXCat cross sections, runs about three minutes, and stops with an error if any check fails. The model potentials are fitted by a second script, and the check script verifies that the stored parameters reproduce the fit quality and the scattering lengths of Table 9.3.
(2) It uses four methods. Identities are verified against independent evaluations, the golden-rule cross section against the plane-wave Born formula and the Born mean-momentum form to 10^{-14}. Limits are verified by their convergence, the zero-range form by narrowing the well and the soft-photon and Fokker–Planck limits by the order of the remainder. The numerics are verified by halving the steps and doubling the ranges. The tables and the numbers in the text are printed from the same computation, and the script recomputes each number of the prose to the digits shown.

Table 9.9: The numerical checks of these notes, by the equation, table or section they verify. The second column counts all checks of the row, the third gives the largest deviation among its quantitative checks, relative except for the slopes of Eq. (9.34), which are absolute, the fourth counts the numbers printed in these notes that the script reproduces to their last digit, and the fifth the logical conditions.

Equation	Checks	Largest deviation	Printed	Logical
Eq. (9.14)	4	2.3×10^{-15}		
Eq. (9.9)	8	4.4×10^{-16}		
Eq. (9.40)	4	4.4×10^{-10}		
Eq. (9.41)	4	1.1×10^{-2}		1
Eq. (9.42)	3	4.7×10^{-16}		
Eq. (9.43)	3	1.2×10^{-16}		

Equation	Checks	Largest deviation	Printed	Logical
Eq. (9.17)	8	8.5×10^{-14}		
Eq. (9.20)	3	1.4×10^{-7}		
Eq. (9.25)	6	6.7×10^{-16}		
Eq. (9.23)	4	6.6×10^{-5}		1
Eq. (9.31)	4	9.2×10^{-5}		
Eq. (9.33)	6	5.1×10^{-8}		
Eq. (9.34)	12	5.9×10^{-3}		6
Eq. (9.37)	4	6.9×10^{-5}		
Eq. (9.46)	4	1.6×10^{-15}		
Eq. (9.47)	6			6
Sec. 9.7	9	2.0×10^{-16}	4	1
Table 9.2	1			1
Sec. 9.8	50	2.6×10^{-5}	6	14
Sec. 9.8.2	6	1.4×10^{-5}	3	1
Sec. 9.8.3	20	2.2×10^{-4}	8	6
Table 9.4	5			5
Table 9.5	1			1
Sec. 9.9	22		15	7
Sec. 9.10	14		10	4
Sec. 9.1	11		11	
Sec. 9.2	5		5	
Sec. 9.2.3	1			1
Sec. 9.3	1			1
Sec. 9.6.2	1			1
Sec. 9.5.1	2		1	1
Sec. 9.5	2		2	
Sec. 9.6.1	2		2	
Sec. 9.6.3	2		2	
Sec. 9.8.4	4		3	1
Total	242		72	59

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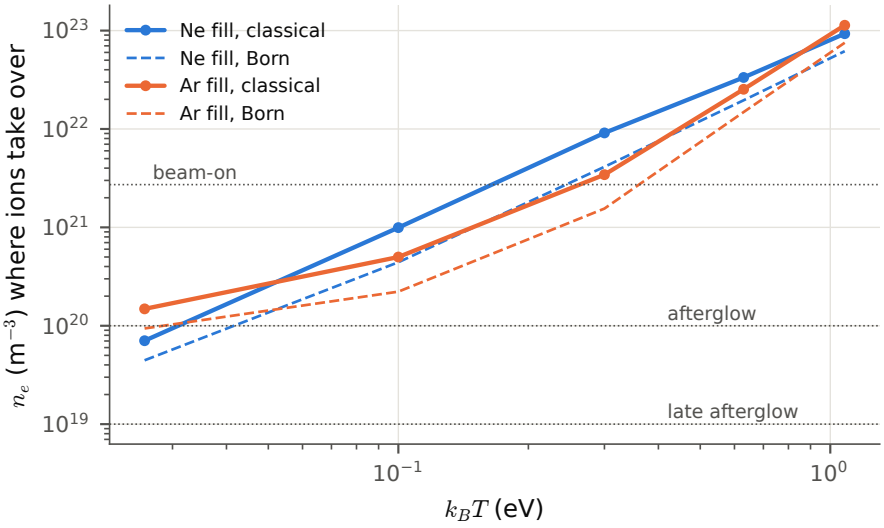


Figure 9.4: The electron density above which electron–ion absorption exceeds electron–atom absorption at 1 THz. Solid lines use the classical estimate of the Gaunt factor, dashed lines the Born value. The horizontal lines mark the electron density of the laser gas with the beam on, 2.7×10^{21} m^{-3}, and in the afterglow, 10^{20} and 10^{19} m^{-3}.

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